Compressing Global Dynamics of Deep Nonlinear Feature Learning

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Abstract

Does large width make deep learning dynamics small? We show that a deep nonlinear network's learning trajectory can be reproduced by an autonomous state with only polylogarithmically many retained coordinates in exact-real arithmetic. The reference is a finite fully connected network with Gaussian hidden weights and zero initial readout, not merely its infinite-width limit. Compression error is negligible relative to actual independent-seed discrepancy, measured over the same query domain and complete training trajectory, including the fitted limit. At fixed problem parameters, Harmonic compression uses $O((\log n)^{3d+2})$ coordinates for all inputs on the sphere, and Taylor uses $O((\log n)^3)$ for a finite input panel declared at setup; n is dense width and d input dimension. Legendre provides an intermediate reduction to $n^{5/4+o(1)}$ moving coordinates while retaining fixed dense mixers. The guarantees hold at arbitrary fixed depth under analytic activation regularity, a positive initial feature-Gram gap, and a width-independent small-label condition. Under additional nondegeneracy assumptions, an early-time theorem separates the compressed predictions from affine predictors and the reference's frozen initial kernel, and proves motion of every dense hidden layer. Thus the reduction preserves genuinely nonlinear feature learning. A separate short-time deep-linear benchmark gives $n^{\Theta(\log m)}$ storage for a literal frozen-top hierarchy with m samples at accuracy comparable to dense-run discrepancy: closure alone need not yield polylogarithmic compression. The common mechanism is regular response memory with stable nonlinear feedback. Numerical implementations and comparisons with small-dense, low-rank, and frozen-feature controls examine this mechanism beyond the theorem's conservative stability regime.

1 Introduction

What does a deep network need to remember in order to keep learning? Backpropagation made it possible to learn internal representations rather than prescribe them [24]. A dynamical explanation of that process must identify the information that governs how those representations will change next. The weights certainly contain such information, but their number need not measure the complexity of learning itself. Two networks can make the same predictions now and respond differently to the next update. Understanding learning therefore calls for more than a compact fitted function: it calls for a compact state sufficient to determine the learner's future behavior. This is our motivation for compression. Removing information while preserving the dynamics is a way to isolate what learning must remember.

Large width offers a particularly sharp test of this idea. Kernel limits revealed regimes in which deep training simplifies to evolution with a fixed tangent kernel [13, 15]. Mean-field theory, tensor programs, and dynamical mean-field theory then furnished descriptions that retain feature learning [20, 28, 3]. These advances make the question more precise: does width merely make learning describable by a limiting law, or does it make the learning dynamics of a finite network representable by a much smaller autonomous state? An evolving distribution or a two-time response kernel can be a powerful theory without yet supplying that finite state.

A small sufficient state for deep learning. We give a positive answer for the fully connected network with Gaussian hidden weights, zero initial readout, and gradient flow specified below, at arbitrary fixed depth with at least two hidden layers. The compression is valid through convergence; under the additional nondegeneracy assumptions stated below, nonlinear feature learning is certified on an early-time interval. For each fixed admissible problem, an autonomous system with polylogarithmically many retained real coordinates reproduces the complete prediction trajectory more accurately than an independent dense initialization does. To our knowledge, this is the first guarantee combining total-state polylogarithmic compression of this dense feature-learning dynamics, fidelity through the fitted limit, and error negligible relative to actual independent-run discrepancy. The advance is a finite sufficient representation of learning, not just a limit equation or a small record of a trajectory already computed.

There are two total-state constructions. Let n denote dense width, d input dimension, and Y the root-mean-square training-label scale. At fixed problem parameters, Harmonic compression uses $O((\log n)^{3d+2})$ coordinates and covers every input on the sphere. Taylor compression uses $O((\log n)^3)$ coordinates on a fixed finite panel of inputs declared at setup, including passive inputs whose labels are never supplied. Both attain complete-trajectory prediction error Y/n. Legendre compression is the first step in the same framework: it retains every neuron but replaces moving matrix information by response-history moments, using $n^{5/4+o(1)}$ moving coordinates and keeping the fixed dense mixers. The main theorem exposes the data and geometry dependence; the polylogarithmic counts include fixed coefficients and working arrays, not just trainable coordinates.

The contract: deep, nonlinear, and feature-learning. The target keeps three ingredients together: at least two hidden layers, nonlinear dependence on the input, and feature learning that does not vanish as width grows. Each matters. A fixed nonlinear feature map can produce a nonlinear predictor without learning features; a deep linear network can change its representations without ever producing a nonlinear input map. The complementary theorem below tests these two distinctions separately. Lazy-training theory explains when linearization is valid [6]; two-layer mean-field theory retains representation learning in a shallow architecture [19]; deep-linear theory reveals nonlinear learning dynamics without nonlinear input maps [25, 2]. These are major simplifications of different parts of the problem. Our contract is to retain all three ingredients.

Why insist on depth? With one hidden layer and fixed input dimension, each neuron has a fixed-dimensional parameter vector. Between two wide hidden layers, learning acts through an n-by-n interaction matrix. Replacing that matrix by an integral operator does not by itself compress its finite representation. Harmonic and Taylor remove this quadratic retained environment while preserving the nonlinear feedback that determines future learning. This is the structural obstacle that a shallow reduction or an assumed low-rank parameterization does not resolve.

Closure alone does not imply compression. The neural tangent hierarchy and finite-width effective theories already provide closed expansions beyond the leading kernel description [12, 9, 23]. A separate theorem here quantifies why closing equations is not enough: for a specified short-time deep-linear family with m training samples, a literal frozen-top hierarchy requires retained size $n^{\Theta(\log m)}$ to attain error within a fixed multiple of actual dense-run discrepancy. Even at fixed sample count this is a power of width whose exponent grows with the number of samples. This is a benchmark for that representation and family, not a lower bound against every encoding or Huang–Yau's original parameterization. Our reductions instead exploit redundancy among responses before closing their dynamics. Section 3 compares this route with population limits, projected models, low-rank training, and earlier dynamical compression.

An intrinsic accuracy benchmark. The reference is a particular finite dense realization, not an assumed ideal population predictor. Independent dense seeds disagree; we require the compressed learner to track its reference below that intrinsic discrepancy. Proving a small absolute error is not enough if the proposed benchmark is only an upper bound on dense variability. We instead prove a lower bound on the discrepancy and make compression error negligible relative to it. Both errors are measured by a supremum over the same query domain and the complete training trajectory, including its fitted limit. The conclusion is a ratio of these two trajectory norms tending to zero in probability, not a ratio bound at every individual time. The dense lower bound is witnessed early, not at the endpoint. No measurement of dense variability is required to choose the compressed state.

One object to compress, two reductions. A learned matrix increment is an integral of paired forward and backward responses. Legendre compression replaces their histories by a small number of polynomial moments, while retaining every neuron and the initialized dense mixers. Projecting both histories onto the same orthogonal space is crucial: mixed retained–omitted terms cancel, leaving a product of the two omitted tails. Thus the approximation benefits from regularity in both histories, not just one.

Harmonic and Taylor go further by also reducing the neuron coordinate. They first approximate the relevant responses in a small function space, then preserve its inner-product geometry on selected neuron coordinates. Selecting coordinates, rather than mixing them arbitrarily, preserves the meaning of coordinatewise nonlinear activation. The two methods share the same reduced nonlinear dynamics; they differ in which responses their source spaces must represent. Harmonic uses joint temporal and spherical regularity to cover the whole input sphere. Taylor represents only the finitely many input curves declared at setup, using temporal pieces adapted to the slowing of learning. The narrower query contract removes the dimension-dependent logarithmic exponent. Passive inputs are supplied without their labels.

Why approximation errors do not accumulate indefinitely. The compressed models are closed learning systems, not playback of a saved dense trajectory. Their current responses determine the next update, which changes their future responses. A generic Lipschitz bound loses the sign of this feedback and can suggest exponential amplification. The comparison instead retains the dissipative training Gram and measures perturbations against the finite amount of remaining learning. For the selected models, a readout correction cancels the leading readout–residual feedback. Infinite physical time is therefore controlled by finite residual activity, not by multiplying a local error by the duration of training.

Nonlinear learning, not a disguised kernel limit. The theorem allows layer-dependent strip-analytic activations with bounded derivative, including unbounded activation values, a positive initial population feature-Gram gap, and sufficiently small nonzero labels. The label cap is fixed independently of width: this is not a limit in which the labels and feature-learning signal shrink to zero as the network widens. The numerical illustrations also include unit-label training, well outside this sufficient cap, to test whether the mechanism is useful beyond its conservative certificate. Under the local additional assumptions of nonaffine activations and no parallel or antiparallel training inputs, Theorem 2.2 proves width-independent early-time separation from every affine input predictor and from the dense reference's own frozen initial-kernel predictor. Every dense hidden layer moves, and the compressed predictions inherit the two separations. Taylor may add up to four predeclared passive witness inputs. Thus the reduced state preserves behavior that neither

deep linear learning nor the reference's frozen features reproduce. This certificate is early-time; the compression guarantee itself extends through convergence.

The foundational conclusion is that, in this regime, deep nonlinear learning does not require retaining its quadratic microscopic description. The proofs identify sufficient response memory and show why its closed evolution remains faithful. Whether those coordinates also reveal simple interpretable mechanisms is the next question, rather than a prerequisite for establishing that such a small dynamical description exists.

2 Setup and main results

There are $m \geq 2$ training pairs (x_a, y_a) with $x_a \in \mathbb{R}^d$, $\|x_a\|_2 = \sqrt{d}$, width n, and a fixed number $L \geq 2$ of hidden layers. Write $v = x/\sqrt{d}$. The first matrix is $W^{(1)} \in \mathbb{R}^{n\times d}$, the hidden matrices are $W^{(\ell)} \in \mathbb{R}^{n\times n}$, and the readout is $w \in \mathbb{R}^n$. The activation may depend on the layer; the superscript is suppressed when the same equation applies to every layer. When a response is indexed by the normalized query v, its network input is $x = \sqrt{d}\, v$. The canonical network is

$$\begin{aligned} z^{(1)}(x) &= W^{(1)}v, & z^{(\ell)}(x) &= W^{(\ell)}h^{(\ell-1)}(x), & h^{(\ell)}(x) &= \phi_\ell(z^{(\ell)}(x)), \\ f_n(x) &= \frac{w^\top h^{(L)}(x)}{n}, & r_a &= f_n(x_a) - y_a, & \mathcal{L} &= \frac{1}{m}\sum_{a=1}^m r_a^2. \end{aligned} \tag{1}$$

The residual-free backward responses are

$$\delta_a^{(L)} = w \odot \phi_L'(z_a^{(L)}), \qquad \delta_a^{(\ell)} = \phi_\ell'(z_a^{(\ell)}) \odot W^{(\ell+1)\top}\delta_a^{(\ell+1)}. \tag{2}$$

At a passive input use the same forward and backward recursions, without supplying a label or adding a training term. Dots denote physical training time. Gradient flow with block mobilities $(n, 1, \ldots, 1, n)$ is exactly

$$\begin{aligned} \dot W^{(1)} &= -\frac{2}{m}\sum_a r_a \delta_a^{(1)} v_a^\top, \\ \dot W^{(\ell)} &= -\frac{2}{nm}\sum_a r_a \delta_a^{(\ell)} h_a^{(\ell-1)\top} \quad (2 \leq \ell \leq L), & \dot w &= -\frac{2}{m}\sum_a r_a h_a^{(L)}. \end{aligned} \tag{3}$$

First-layer entries are independent $\mathcal{N}(0,1)$; hidden entries are independent $\mathcal{N}(0, 1/n)$; all blocks are independent; and $w(0) = 0$. Nothing freezes the hidden features.

Activations, data, and constants. Each ϕ_ℓ is real on the real axis, holomorphic on $|\operatorname{Im} z| < a$ for a common $a > 0$, and has bounded first derivative on that strip. Its values need not be bounded. For explicit estimates put

$$\beta = \max\left\{10, 1 + \max_\ell |\phi_\ell(0)|, 16/a, \max_{\ell,\, j\in\{1,2\}} \sup_{|\operatorname{Im} z| \leq a/2} |\phi_\ell^{(j)}(z)|\right\}.$$

The second derivative is bounded on the smaller strip by Cauchy's formula. Define the uncentered population feature Gram and the label scale by

$$\begin{aligned} Q_{ab}^{(0)} &= v_a^\top v_b, \qquad Q_{ab}^{(\ell)} = \mathbb{E}[\phi_\ell(Z_a)\phi_\ell(Z_b)], \quad Z \sim \mathcal{N}(0, Q^{(\ell-1)}), \\ \gamma &= \lambda_{\min}(Q^{(L)}) > 0, \qquad Y = \|y\|_2/\sqrt{m} > 0. \end{aligned} \tag{4}$$

Assume $m/\gamma \geq 1$ and the same sufficient small-label condition

$$0 < Y \leq \frac{\gamma}{m}\beta^{-30L}. \tag{5}$$

No input-rank, centering, orthogonality, or label-sign hypothesis is imposed. All data, m, d, L, γ, Y, and activations are fixed separately as n grows. Throughout the stated bounds, $C > 0$ depends only on activations and depth, never on n, m, d, γ, Y or a passive-panel size. Its value may change between occurrences. All other constants are defined in the statement or proof using them. Initialization costs use a separate fixed-parameter big-O convention, specified in Proposition 2.3; its constants are not the activation-and-depth-only storage constant C.

Query domains and probability. For a query set $\mathcal{X}$, use

$$\|f - g\|_{\mathcal{X}} = \sup_{t\in[0,\infty]} \sup_{x\in\mathcal{X}} |f(t,x) - g(t,x)|. \tag{6}$$

The endpoint $t = \infty$ means the fitted limit. For Legendre and Harmonic, $\mathcal{X} = \{x : \|x\|_2 = \sqrt{d}\}$. For Taylor, fix $p \geq 0$ additional inputs on this same sphere before initialization and take $\mathcal{X} = \{x_1, \ldots, x_{m+p}\}$. Their labels are never supplied. Let $\widetilde{f}_n$ be an independently initialized dense run on the same training problem.

An eventual assertion at confidence $1-\delta$ means: for every fixed admissible problem and $0 < \delta < 1$, there is a finite width threshold such that the assertion holds at each larger individual width. That threshold may depend on every fixed problem parameter, including the positive Y. The headline theorem quantifies this threshold polynomially for Legendre and Taylor; Harmonic retains the eventual qualification. Polynomial sufficient widths need not be practically small. Each probability bound concerns one fixed problem at one individual width, not a simultaneous event over all datasets or infinitely many independently initialized widths. No unrestricted growing-data limit is asserted. For a ratio below, missing fitted limits or a zero denominator count as failure.

The compression and nonlinear feature-learning statements below inherit this setup and these probability conventions; the quantitative width improvement does not extend to the separate nonlinear feature-learning certificates. The separate NTH benchmark in Section 14 states its own assumptions and probability convention. Every storage count charges a nonprimitive activation evaluator's description and scratch separately.

What must be remembered. The following is the complete persistent notation.

Symbols	Meaning
n, m, d, L, p	width, training count, dimension, depth, passive count
$x, v = x/\sqrt{d}, y, t$	input, normalized input, labels, physical time
$W, w, z, h, r, \delta_a^{(\ell)}$	weights, readout, preactivations, features, residuals, backward responses
$\phi, a, \beta, Q, \gamma, Y$	activations, strip, envelope, population Gram, gap, label RMS
$\mathcal{L}, \mathcal{X}, \|\cdot\|_{\mathcal{X}}$	loss, query domain, trajectory norm
δ, C	failure probability, activation/depth constant
$f_n, \widetilde{f}_n, f_{\text{model}}, f_{\text{NTK},n}$	dense, independent dense, compressed and frozen-kernel predictors

Subscripts naming a compression method specialize f_{model}. Ordinary mathematical operations and bound indices have their usual meanings. Every other symbol is bound to the statement and proof that introduce it: it can be forgotten afterward. In particular, approximation orders, clocks, source spaces, selected metrics, and numerical ledgers are not additional global notation. Later arguments use the stated conclusions, not private coefficients from their proofs.

Theorem 2.1 (Compression of the complete learning trajectory)**.** *Under the preceding assumptions, each construction can be chosen so that*

$$\frac{\|f_{\mathrm{model}} - f_n\|_{\mathcal{X}}}{\left\|f_n - \widetilde{f}_n\right\|_{\mathcal{X}}} \xrightarrow{\mathbb{P}} 0. \tag{7}$$

Here f_n is the construction's coupled ordinary dense reference, and both norms use the same promised query domain and all physical times, including the fitted endpoint. Sufficient real-coordinate counts are

$$\text{Legendre moving state:} \quad Cm\left(\frac{m}{\gamma}\right)^3 n^{5/4}[\log(en)]^2 e^{\sqrt{\log(en)}/2}, \tag{8}$$

$$\text{Harmonic total state:} \quad C(m+d)^2 + \left(\frac{C}{d}\right)^{d+1}\left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^{3d+2}, \tag{9}$$

and the Taylor total state is at most

$$\begin{aligned} C(m+p)^2\Bigg[&\left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^3 \\ &+ \left(\frac{m}{\gamma}\right)^2 [\log(en)]^2[\log(e+\log(en))]^2\Bigg] + C(m+p)d. \end{aligned} \tag{10}$$

Legendre additionally retains $(L-1)n^2$ fixed mixer coordinates. Harmonic and Taylor include the data, selected mixers, metrics, initial copies, and working arrays. Their predictions additionally satisfy

$$\|f_{\mathrm{model}} - f_n\|_{\mathcal{X}} \le Y/n \tag{11}$$

at every fixed confidence for sufficiently large width, with the same storage orders. One theoretical initializer uses the dense initialization, training data, and finitely many initialization derivatives; no later trained weight or dense trajectory is an input. The efficient alternatives in Proposition 2.3 compute disposable numerical continuations from the initial data. Subsequent compressed evolution is an autonomous nonlinear system with the stated retained state in either case.

For Legendre and Taylor, the sufficient width is polynomial: for every $0 < \varepsilon, \delta < 1$, the stated sizes and

$$\|f_{\mathrm{model}} - f_n\|_{\mathcal{X}} \le \varepsilon \left\|f_n - \widetilde{f}_n\right\|_{\mathcal{X}}$$

hold with probability at least $1-\delta$ whenever

$$n \ge \left[c\beta^L(1 + m + d + p + \gamma^{-1} + Y^{-1} + \delta^{-1} + \varepsilon^{-1})\right]^c, \tag{12}$$

for an absolute numerical constant $c > 0$, local to this width bound. For Legendre take $p = 0$ and omit Y^{-1}; Taylor also satisfies (11) *on this event. This strengthens only the width qualification: the storage formulas, label condition, query domains and physical optimizers are unchanged.*

2.1 The compressed dynamics are nonlinear and feature-learning

A moving kernel alone does not distinguish nonlinear learning from a deep linear network. We therefore separate two comparisons. In this subsection, $f_{\mathrm{NTK},n}$ is the predictor obtained by freezing the coupled dense reference's initial tangent kernel. Since $w(0) = 0$, this is exactly its readout-only flow with all hidden features fixed at initialization.

Theorem 2.2 (Compression preserves nonlinear feature learning)**.** *Under the compression theorem's assumptions, suppose additionally that every activation is nonaffine and* $|x_a^\top x_b| < d$ *for* $a \neq b$*. For Taylor, augment* $\mathcal{X}$ *by at most four deterministic passive witness inputs, declared before initialization without their labels. There are* $c_*, t_* > 0$*, depending on the fixed problem but not on width, such that for every fixed* $0 < t \leq t_*$*, each compressed predictor satisfies, with probability tending to one as* $n \to \infty$*,*

$$\inf_{a\in\mathbb{R}^d,\, b\in\mathbb{R}} \sup_{x\in\mathcal{X}} |f_{\mathrm{model}}(t,x) - a^\top x - b| \geq c_* t, \tag{13}$$

$$\sup_{x\in\mathcal{X}} |f_{\mathrm{model}}(t,x) - f_{\mathrm{NTK},n}(t,x)| \geq c_* t^3. \tag{14}$$

The storage orders and approximation conclusions are unchanged; Taylor's bound uses the augmented panel count, at most $p+4$ *instead of* p*.*

In the dense reference, every hidden layer satisfies

$$\left(\frac{1}{n}\sum_{a=1}^{m} \|h^{(\ell)}(t,x_a) - h^{(\ell)}(0,x_a)\|_2^2\right)^{1/2} \geq c_* t^2 \qquad (1 \leq \ell \leq L) \tag{15}$$

with the same probability convention. In the first layer, the nonlinear part of the feature increment itself has positive mean-square size of order t^4*, in the precise sense of* (180)*.*

The first inequality excludes every deep linear predictor, even with biases, an arbitrary optimizer and time-dependent weights. The second excludes the reference network's frozen-feature training. The internal statement further rules out a frozen nonlinear feature component accompanied only by an affine feature change. It does not identify the compressed coordinates with dense neurons. The additional hypotheses belong only to this activity theorem; the broader compression theorem does not require them.

The mechanism has three time scales. The initially zero readout first grows at order t; hidden features move at order t^2; the output departs from frozen-feature training at order t^3. The leading departure in the label direction is a sum of squared hidden-layer accelerations, so contributions from different layers cannot cancel. Separately, the initial feature kernel has nonlinear input components that no nonzero label vector can cancel completely on the sphere. Both signals persist while compression error vanishes. The proof in Section 12 uses only initial Gaussian conditioning and real short-time estimates, not a width-independent complex-time radius or a trained population-flow theorem. These are early-time certificates, not endpoint or test-risk comparisons.

2.2 Efficient initialization

Proposition 2.3 (Efficient initialization)**.** *Under Theorem 2.1, Harmonic and Taylor admit the following exact-real initializers, at every fixed confidence and sufficiently large width.*

Method	*Dense reference*	*Setup work*	*Peak setup memory*
Harmonic	*Supplied matrices*	$O(n^2[\log(en)]^{3d/2+1})$	$O(n^2)$
Taylor	*Supplied matrices*	$O(n^2[\log(en)]^{5/2})$	$O(n^2)$
Harmonic	*Fresh implicit Gaussian*	$O(n[\log(en)]^{9d/2+3})$	$O(n[\log(en)]^{3d/2+1})$
Taylor	*Fresh implicit Gaussian*	$O(n[\log(en)]^5)$	$O(n[\log(en)]^{5/2})$

All four initializers preserve the corresponding retained-state bound, the complete-trajectory error Y/n*, and* (7)*. The constants in this table may depend on the separately fixed* $m, d, p, L, \beta, \gamma, Y$ *and confidence. The implicit version is coupled to a fresh ordinary Gaussian dense reference; it is not a subquadratic conversion of supplied matrices or a prescribed entrywise seed.*

The operation model charges scalar arithmetic, comparisons, square roots, elementary functions, real activation values and Gaussian samples as primitives. Nonprimitive evaluators add their actual work, description and scratch. Activation envelopes and a positive population-gap certificate are supplied; computing them is not included. No high activation derivatives or trained-reference values are inputs. These initializers compute through $T = 32(m/\gamma)\log(en)$, not merely a short physical-time prefix, then discard their working structures. The compressed model starts at time zero. The original initialization-jet compiler remains an alternative without an efficient work bound. The proof in Section 11 and finite-order ledgers in Section 11.4 separate all parameter dependencies, retained storage and setup memory. Taylor uses the polynomial source and accuracy gates above without an additional qualitative source threshold; Harmonic retains its eventual source-width qualification.

These are sufficient sizes, not optimality claims. Dense itself has $(L-1)n^2 + n(d+1)$ weight coordinates. Data storage is additional to the Legendre moving-state count. Coordinate counts alone do not bound precision or preprocessing time; Proposition 2.3 supplies separate arithmetic and peak-memory bounds, not a finite-precision guarantee. The finite-panel theorem makes no accuracy promise on inputs first supplied after initialization. Zero labels give the stationary zero predictor instead. The sharp literal-array cost of a frozen-top neural tangent hierarchy is treated separately in Section 14, for a short-time deep-linear model. It is $n^{\Theta(\log m)}$ uniformly over $4 \le m \le \sqrt{n}$; no growth of m is required, so already a fixed dataset forces a power of n whose exponent grows with m.

3 Related work

The relevant distinction is not whether earlier work describes dynamics, learns features, or compresses a network in some sense. Each has substantial precedents. The question is which work supplies all three together with finite-reference fidelity, a bounded retained state, and a guarantee through fitting. Table 1 summarizes the closest approaches. Its distinctions concern their stated models and theorems, not impossibility claims about their future extensions.

Three instructive simplifications of learning. Lazy training, shallow mean-field dynamics, and deep-linear networks isolate different ingredients of the learning problem and have led to substantial theory. Chizat, Oyallon, and Bach [6] identify a scaling regime in which scaled gradient flow is approximated by its linearization at initialization; under stronger loss and nondegeneracy assumptions their control extends uniformly in time. The simplification is precisely that adaptive feature changes become small. Mei, Montanari, and Nguyen [19] instead retain nonlinear feature motion and describe one-hidden-layer, one-pass SGD by an evolving neuron distribution, with quantitative finite-horizon approximation and additional optimization results. Their symmetry reductions and finite-atom computations are useful reductions, but do not establish a general deep finite-state closure.

Deep-linear models preserve depth and nonlinear parameter dynamics while their input–output maps remain linear. Saxe, McClelland, and Ganguli [25] solve task-aligned mode dynamics under structured input covariance, revealing how representations develop on different time scales. Arora, Cohen, and Hazan [2] obtain an exact autonomous end-to-end matrix flow under balanced initialization. Its state dimension does not grow with hidden width: this is a genuine closed reduction, not merely a description of changing weights. Such results make the target here sharper. We seek small-state dynamics without freezing feature adaptation, removing hidden-layer interactions, or restricting the predictor to a linear function of its input. The additional activity theorem certifies

Approach	What it provides	Relation to our compression target
Kernel limits and finite-width hierarchies [13, 12, 9, 23]	Closed prediction dynamics and finite-order corrections.	Do not establish a small-state, complete-trajectory guarantee with a width-independent nonlinear feature-learning signal.
Deep mean field, tensor programs, and DMFT [20, 28, 3]	Feature-learning population dynamics; finite-time limits and response descriptions.	Do not supply a bounded finite representation coupled to one dense seed, with certified error through convergence.
Projected infinite-width computation [29]	An explicit computable feature-learning limit under structured initialization and projected updates.	Changes the reference; no fidelity to the unrestricted dense flow is proved. Its representation appends directions as training proceeds.
Analytic moment compression [27]	Support reduction for fixed-dimensional symmetric analytic observables; a dynamical extension under the same regularity premise.	Assumes dynamical regularity rather than deriving it uniformly in width and time; leaves the all-wide deep compression rate open.
Dynamical low-rank training [26]	Autonomous factor dynamics and conditional finite-horizon approximation.	Assumes a small normal-gradient defect; does not derive it for the canonical dense flow or give an all-time certificate.
Trajectory transfer and empirical subspaces [18, 16, 17]	Sparse or low-coordinate training; empirical trajectory matching and task-accuracy evidence.	Do not establish a nonlinear, seed-relative complete-trajectory theorem; few trainable coordinates need not mean few retained coordinates.
Harmonic and Taylor (this work)	Polylogarithmic total retained state for the specified finite dense learner.	Whole-sphere (Harmonic) or declared-panel (Taylor) fidelity through fitting, below actual dense-run discrepancy, under Section 2's assumptions.

Table 1: Closest routes to a reduced description of learning. Our total-state counts include fixed data and runtime arrays; the separate Legendre construction reduces moving state while retaining dense mixers. This comparison does not assert that every predecessor addresses the same training regime.

nonaffine compressed predictions and departure from the reference's frozen kernel, alongside motion of every dense hidden layer.

From population dynamics to a finite retained state. Multilayer mean-field theory [20] and Neural Hilbert Ladders [5] describe evolving functions or measures. Tensor Programs IV [28] rigorously identifies feature-learning infinite-width limits for fixed training programs. DMFT [3] gives self-consistent equations for evolving kernels and causal responses. These are genuine dynamical descriptions. Their general representations, however, are function-valued or carry time-indexed response information; writing an operator equation is not yet exhibiting a small finite state that evaluates it.

Nor is seed dependence absent from prior theory. Pham and Nguyen [22] construct a sampled-neuron-coupled fluctuation correction and state quantitative finite-horizon estimates; Bordelon and Pehlevan [4] develop a dynamical fluctuation expansion. Our distinction is not priority for coupling to initialization or studying prediction variance. Their stated finite-horizon estimates do not supply a polylogarithmic autonomous representation with the present all-time approximation guarantee. Taking width to infinity at each fixed time and then studying large time is different from one error bound covering every time at a finite width.

Yang, Santacroce, and Hu [29] address computability directly. Their hidden matrices are built

through fixed-dimensional Gaussian projections, and the optimizer projects updates into the same spaces. The construction yields a nonlinear feature-learning limit, but of this structured network rather than the unrestricted Gaussian reference. Its coefficient arrays append forward and backward directions as examples are processed; retaining the frozen old rows still incurs storage. The source proves a fixed-program limit, not a bound showing that this projected learner shadows the ordinary dense one, nor a fixed-state compression of the accumulating history. Thus computability of a modified limit and faithful compression of a finite reference solve different problems.

Closed expansions are not yet a feature-learning compression theorem. The NTK [13] and parameter-linearized dynamics [15] establish a profound simplification in their scaling regimes. Huang and Yau's NTH [12] goes beyond freezing the kernel: it supplies a hierarchy and finite-order truncation estimates. Dyer and Gur-Ari [9] and Roberts, Yaida, and Hanin [23] also derive initialized finite-order closed systems, including corrections to training and fitted predictions. These are real precedents for autonomous reduced equations, not merely ensemble endpoint formulas.

Their relevant fixed-depth expansions are around kernel scaling, where feature-learning corrections vanish with width at fixed training time. Diagrammatic power counting organizes finite-width corrections systematically; its general deep nonlinear extension is formulated conjecturally [9]. Perturbative endpoint formulas and pathwise remainder estimates uniform through fitting are distinct guarantees. The present result instead couples its small system to a regime with a width-independent feature-learning certificate. Our separate Section 14 isolates another obstacle: a literal frozen-top hierarchy in its specified short-time deep-linear family has storage $n^{\Theta(\log m)}$. This is a representation-specific benchmark, not a lower bound for Huang–Yau's parameterization, every hierarchy implementation, or the nonlinear family of the main theorem.

Assumed compressibility versus compressibility derived from training. Wang et al. [27] identify symmetry and analytic moment matching as mechanisms for polylogarithmic support reduction, with the explicit ambition of preserving learning dynamics rather than only trained functions. Their setting consists of fixed-dimensional objects and a symmetric analytic observable. Extending the reduction through training requires the composed observable to retain this structure and regularity; preservation of that premise and the compressed-width rate for all-wide deep networks remain open in their account [27, Theorem 5 and Section 5]. In an all-wide hidden layer, a neuron's incoming and outgoing coordinates grow with neighboring widths. A fixed-object-dimension logarithmic power therefore does not establish a width-uniform deep-network compression theorem. Likewise, analyticity of each finite-time map does not bound its analytic radius or amplification uniformly through convergence. These are precisely obligations discharged by our response estimates and stability analysis, not consequences of permutation symmetry alone.

Dynamical low-rank training [26] gives a different and substantive reduction: evolve matrix factors using projected gradients and robust low-rank integration. Its approximation analysis assumes a small component of the full gradient normal to the low-rank manifold (supplementary assumption P2). It does not derive that premise from Gaussian-initialized all-wide deep training. Its geometric integrator analysis provides a finite-horizon comparison: the terminal-time dependence introduced by accumulation of local errors must be retained.[1] In contrast, we neither assume low rank of the trained dense matrices nor require the compressed flow to inherit stability automatically. We derive a small response space and separately prove stability of its closed nonlinear evolution.

[1]Although the main statement lists constants without terminal time, the supplementary proof of Theorem 1 accumulates errors by a Lipschitz estimate. Such an estimate does not furnish a time-uniform constant. The finite-horizon robustness result is the relevant comparison here.

Trajectory transfer, low-dimensional training, and learned reductions. Neural Tangent Transfer [18] explicitly seeks a sparse initialization whose later predictions follow a dense teacher, and therefore anticipates our dynamical objective. Its linear-model proposition identifies matching initial outputs and kernels as sufficient for matching trajectories; the deep nonlinear implementation provides empirical support for using this principle in sparse initialization. Exact matching is restrictive in the proposition's binary-mask setting,[2] and the nonlinear method has no width-dependent, complete-trajectory error guarantee. The advance here is to supply such a guarantee while retaining feature learning, not to originate the trajectory-transfer objective.

Intrinsic-dimension experiments [16], concentration in tiny gradient subspaces [11], and training in trajectory-derived subspaces [17] provide evidence that many search directions are redundant. A small coordinate vector may nevertheless require a dense decoder, full-size workspace, and evaluation of the original network. Koopman-based training forecasts [8] evolve learned operators fitted to recorded trajectories, but their displayed implementation retains the full weight state and supplies no complete-trajectory error theorem. Successful task prediction, sparse trainability, and fidelity to a specified dense seed are distinct claims. Our total-state accounting and seed-relative norm make the latter claim explicit.

Feature-learning theory and the role of memory. Deep nonlinear feature learning itself has rigorous precedents. Nichani, Damian, and Lee [21] analyze a feature-learning stage followed by readout training; Allen-Zhu and Li [1] prove hierarchical learning for a structured architecture and staged training procedure with backward feature correction. Adaptive-kernel theories also characterize learned endpoints under their equilibrium and regularization qualifications [14]. These results address representation or sample efficiency, rather than small-state emulation of the dense reference studied here. Our complementary feature-learning theorem establishes that the behavior being compressed has not collapsed to an affine predictor or to its own initial kernel.

The closest physical organizing principle is memory reduction: Mori–Zwanzig and optimal prediction express eliminated degrees of freedom through memory and unresolved forcing [7]. Online polynomial-memory constructions such as HiPPO [10] show how histories can be represented by finite auxiliary coordinates. Here the histories being compressed are generated by the learner reconstructed from those very coordinates. Controlling that feedback, while deriving the required response regularity from dense training, is the additional step that turns memory approximation into a certified learner.

4 Discussion

Compression as a route to understanding. A parameter vector serves two roles: it specifies a current predictor and it stores how that predictor can change. The second role is the one at issue here. For a fixed training problem, a state is sufficient if it can continue the learning process without access to the discarded dense coordinates. The theorem shows that most of the quadratic microscopic description is unnecessary for this purpose, at an accuracy finer than the reference's own seed variability.

This does not identify a unique or minimal memory, and coordinate size is not a ranking of the semantic importance of individual neurons. Rather, the construction isolates one sufficient memory in forward and backward responses: how inputs are represented, how changes propagate through layers,

[2]The removed kernel is a sum of positive-semidefinite outer products of the removed input columns. It can vanish only when those columns vanish on the training set. Thus the exact premise does not yield nontrivial masking for generic data.

and how these responses have accumulated during learning. Compression is therefore a first step toward a mechanism, not a substitute for explaining one. It makes a sharper mechanistic question possible: which interactions among these retained responses govern representation formation? That question is obscured when every dense weight is treated as an equally indispensable state variable.

The sufficiency is also conditional on the training protocol and promised observables. It is not a universal summary for arbitrary future datasets, losses, or interventions. Taylor makes this dependence explicit by specializing to a declared input panel; Harmonic pays a larger logarithmic power to support inputs first queried after setup. This tradeoff is part of the meaning of dynamical memory, not merely an implementation detail.

Why logarithms, and what they mean. Analyticity explains why logarithms enter. Exponentially accurate approximation requires only logarithmically many modes for inverse-width accuracy. The number of input directions, temporal pieces, and the cost of preserving their inner-product geometry determine the final powers. This concerns regularity of the responses that drive learning, not low rank of every dense weight matrix. It also distinguishes a large microscopic system from a large variety of observable behaviors: increasing width need not create proportionally many independent response histories. The stored-state bounds make that distinction quantitative, while the feedback analysis ensures that it survives autonomous evolution rather than only reconstruction of a recorded path.

Intrinsic precision and population interpretations. The finite network is the primary object: no population solution is needed to define its compression or its error benchmark. A lower bound on actual independent-run discrepancy prevents a loose analytic upper envelope from making a poor approximation appear successful. For Harmonic and Taylor, the complete-trajectory upper error is inverse width, while the dense lower bound is inverse square-root width up to logarithms. Thus compression is asymptotically finer than the benchmark, rather than merely within a fixed multiple of it. The comparison remains between trajectory norms; it does not assert a dense lower bound at each time or at the fitted endpoint.

If a population description is independently available in the same norm, the triangle inequality transfers any dense-to-population approximation to the compressed predictor with only the additional compression error. Consequently the result can serve either a finite-realization or a population interpretation without assuming that the population predictor is the scientifically preferred object.

Scope and the next questions. Extending the complete-trajectory result to unrestricted labels or nonsmooth activations such as exact ReLU remains open. So do optimal state counts, practical finite-precision guarantees, and a similarly small Taylor construction supporting arbitrary inputs first supplied after setup. The theorem does not establish improved risk against unknown labels; the final figures provide separately qualified empirical illustrations, not certificates of its asymptotic exponents.

Retained state and the work needed to construct it are distinct. Initialization derivatives establish that later trained weights need not be supplied to the compiler. The efficient alternatives in Proposition 2.3 instead compute and discard numerical continuations from the initial data. At fixed problem parameters their work is quadratic in width times a polylogarithm for supplied matrices, or linear in width times a polylogarithm for a fresh implicitly sampled Gaussian reference. The latter is not a subquadratic conversion of an arbitrary supplied dense matrix. These are exact-real arithmetic bounds, not a claim of automatic practical speedup. They make the representation constructive without conflating a small learner with free preparation.

Proof roadmap. First establish fitting and finite residual activity without a source approximation. A common probabilistic source lemma then controls forward and backward responses on the two analytic domains. Legendre approximates their history integrals. The other two methods share one geometry-preserving selection and one corrected nonlinear optimizer; only their finite source bases differ. A last-layer innovation and a deterministic analytic estimate give a dense lower bound at a training input. Section 13 combines these ingredients. A quantitative source argument and a third-order Legendre history estimate supply the polynomial width refinement. Numerical source construction and a two-direction Gaussian sampler prove Proposition 2.3. Every dependency is supplied below; earlier numerical figures are collected separately at the end.

5 Initialization, fitting, and a signed perturbation estimate

Lemma 5.1 (Initialization and real fitting)**.** *For this lemma put*

$$H = \max_{0\le j\le L} \sqrt{Q_{11}^{(j)}}, \qquad \mathsf{H}(t) = [h^{(L)}(t, v_1), \dots, h^{(L)}(t, v_m)].$$

For the parameter tuple $\theta = (W^{(1)}, \dots, W^{(L)}, w)$ *use the norm*

$$\|\theta\|_{\mathrm{par}}^2 = \|W^{(1)}\|_F^2/n + \sum_{j=2}^{L} \|W^{(j)}\|_F^2 + \|w\|^2/n.$$

The initialized training Grams satisfy

$$\left(\frac{h_0^{(j)}(v_a)^\top h_0^{(j)}(v_b)}{n}\right)_{a,b=1}^{m} \xrightarrow{\mathbb{P}} Q^{(j)} \qquad (1 \le j \le L). \tag{16}$$

With probability tending to one,

$$\begin{gathered} \frac{\|W_0^{(1)}\|_{\mathrm{op}}}{\sqrt{n}} \le 8, \qquad \max_{2\le j\le L} \|W_0^{(j)}\|_{\mathrm{op}} \le 8, \\ \sup_{\|v\|=1,\, 1\le j\le L} \frac{\|h_0^{(j)}(v)\|}{\sqrt{n}} \le \frac{3H}{2}, \qquad \frac{\mathsf{H}_0^\top \mathsf{H}_0}{mn} \succeq \frac{\gamma}{2m} I_m. \end{gathered} \tag{17}$$

On this event the flow is global, converges in parameters, fits the labels, and satisfies, for all $t \ge 0$*,*

$$\begin{gathered} \frac{\|W^{(1)}(t)\|_{\mathrm{op}}}{\sqrt{n}} < 9, \qquad \max_{2\le j\le L} \|W^{(j)}(t)\|_{\mathrm{op}} < 9, \\ \sup_{\|v\|=1,\, 1\le j\le L} \frac{\|h^{(j)}(t, v)\|}{\sqrt{n}} < 2H, \qquad \frac{\mathsf{H}(t)^\top \mathsf{H}(t)}{mn} \succeq \frac{\gamma}{4m} I_m \end{gathered} \tag{18}$$

$$\begin{gathered} \frac{\|r(t)\|_2}{\sqrt{m}} \le Y e^{-\gamma t/(2m)}, \qquad \frac{\|w(t)\|}{\sqrt{n}} \le 2Y\sqrt{\frac{m}{\gamma}}, \\ \int_t^\infty \|\dot\theta(u)\|_{\mathrm{par}}\, du \le \frac{2\|r(t)\|_2}{\sqrt{\gamma}}. \end{gathered} \tag{19}$$

The output converges uniformly on the sphere, with

$$\sup_{\|v\|=1} |f_n(\infty, v) - f_n(t, v)| \le C\frac{m}{\gamma}\frac{\|r(t)\|_2}{\sqrt{m}}. \tag{20}$$

These assertions hold jointly for all rectangular cavities deleting any fixed bounded number of coordinates per layer, with the original width-n variances, flow factors, and denominators unchanged.

Proof. All auxiliary coefficients in this proof are local. Put $\lambda = \gamma/m$, $\rho = \|r\|_2/\sqrt{m}$,

$$s = \max\left\{1, \max_j \sup_{|\operatorname{Im} z| \le a/2} |\phi_j'(z)|\right\}, \qquad b = \max_j |\phi_j(0)|,$$

and retain H as defined in this lemma's statement. Thus $|\phi_j(z)| \le b + s|z|$ on the half strip. Define

$$\begin{aligned} &D_j = s(9s)^{L-j}, && U_1 = D_1, && U_j = 2HD_j \quad (j \ge 2), \\ &F_1 = sU_1, && F_j = s(2HU_j + 9F_{j-1}), && F = F_L. \end{aligned} \tag{21}$$

Equal input norms give $Q_{aa}^{(j)} = Q_{11}^{(j)}$, hence $\lambda \le H^2$. The coefficients satisfy

$$H \le (2\beta)^L, \qquad F = s^2\left[(9s)^{2L-2} + 4H^2\sum_{j=0}^{L-2}(9s)^{2j}\right], \qquad 8H\sqrt{F} \le \beta^{5L}. \tag{22}$$

The first inequality follows from Minkowski's inequality and $\sqrt{Q_{11}^{(j)}} \le b + s\sqrt{Q_{11}^{(j-1)}}$. For the last, summing the geometric series gives $F \le (21/20)H^2s^2(9s)^{2L-2}$, and then $8H\sqrt{F} \le H^2(9s)^L \le 36^L\beta^{3L} \le \beta^{5L}$. Consequently the assumed cap $Y \le \lambda\beta^{-30L}$ implies $Y \le \lambda/(8H\sqrt{F})$, which will close the real-flow estimates.

A maximal $1/4$-separated subset of the unit sphere in $\mathbb{R}^k$ has at most 9^k points: disjoint radius-$1/8$ balls fit inside the radius-$9/8$ ball. Approximating each side of a maximizing bilinear pair gives $\|M\|_{\mathrm{op}} \le 2\max|u^\top Mv|$ over these nets. Each pairing for $W_0^{(\ell)}$, $\ell \ge 2$, or $W_0^{(1)}/\sqrt{n}$ is $N(0, 1/n)$. The exponential Gaussian tail and a union bound prove the operator assertions, with failure bounded by $2(L-1)e^{-(8-2\log 9)n} + 2e^{-(8-\log 9)n + d\log 9}$.

On any fixed finite query list, condition on the preceding initialized layers. Each next empirical Gram is an average of independent Gaussian-row feature products, with conditional mean $\Psi_\ell(C)_{ab} = \mathbb{E}\phi_\ell(Z_a)\phi_\ell(Z_b)$, $Z \sim N(0, C)$. On bounded covariance sets the conditional variances are bounded, by linear growth and Gaussian fourth moments. Moreover Ψ_ℓ is continuous on the entire positive semidefinite cone: write $Z = C^{1/2}G$ with a standard Gaussian vector G of the matching dimension, use continuity of the positive square root, and dominate by a fixed Gaussian polynomial. Conditional Chebyshev and induction therefore prove convergence of each empirical Gram to $Q^{(\ell)}$, including when intermediate covariances are singular. Apply this to the training inputs and a fixed sphere net of mesh $h \le H/[4(8s)^L]$. On the operator event the initialized layer-ℓ feature map, in neuron RMS, is $(8s)^\ell$-Lipschitz. Its norm at net points is eventually at most $\sqrt{5/4}H$, with probability tending to one; the off-net increment is at most $H/4$. This proves the sphere bound and the training Gram assertion in (17).

Here is also a uniform justification for the cavity assertion, without a union bound over deletion sets. On a sphere net of mesh n^{-2}, the preceding RMS bounds and conditional Gaussian tails give $\max_{i,\ell,v}|h_{0,i}^{(\ell)}(v)| = O(\sqrt{\log(en)})$ with probability tending to one. The net has only a fixed power of n points; the coordinate Lipschitz bound is at most $\sqrt{n}(8s)^\ell$, so the same bound holds off the net. Zero-extend each cavity's features into $\mathbb{R}^n$. At a deleted layer, at most a fixed number of original coordinates are removed. At an undeleted coordinate the activation difference is at most s times the preactivation difference. Thus successive RMS differences are bounded by $8s$ times the preceding difference plus $O(\sqrt{\log(en)/n})$. At fixed depth their supremum, simultaneously over all such cavities and sphere queries, tends to zero. Submatrix operator norms do not increase. Taking strictly smaller initial feature and Gram margins before this comparison gives (17) for every cavity. This also shows directly that each cavity's training Gram has the same limit $Q^{(L)}$.

Stop any one of these flows at the first failure of (18). Backward propagation gives $\|\delta_a^{(\ell)}\|/\sqrt{n} \le D_\ell \|w\|/\sqrt{n}$. In Euclidean mobility coordinates $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$, the vector field is minus the gradient of ρ^2. Consequently

$$-\partial_t \rho^2 = \|\dot\theta\|_{\mathrm{par}}^2 \ge \lambda \rho^2.$$

The last inequality uses only the readout contribution and the stopped Gram gap. For any stopped interval $[t, T]$, Cauchy–Schwarz gives

$$\left(\int_t^T \|\dot\theta\|_{\mathrm{par}}\right)^2 \le \left(\int_t^T \frac{\|\dot\theta\|_{\mathrm{par}}^2}{\rho}\right)\left(\int_t^T \rho\right) \le 2\rho(t)\,\frac{2\rho(t)}{\lambda}.$$

If a residual vanishes, all velocities vanish, so the estimate extends without division by zero. In particular $\|w\|/\sqrt{n} \le 2Y/\sqrt{\lambda}$. Sample Cauchy–Schwarz bounds each normalized hidden-block velocity by $2\rho U_\ell \|w\|/\sqrt{n}$. Integration and forward subtraction give

$$\frac{\|W^{(1)} - W_0^{(1)}\|_F}{\sqrt{n}} \le \frac{8U_1 Y^2}{\lambda^{3/2}}, \qquad \|W^{(\ell)} - W_0^{(\ell)}\|_F \le \frac{8U_\ell Y^2}{\lambda^{3/2}},$$
$$\sup_v \frac{\|h^{(\ell)} - h_0^{(\ell)}\|}{\sqrt{n}} \le \frac{8F_\ell Y^2}{\lambda^{3/2}}. \tag{23}$$

Indeed each later feature difference is bounded by s times $2H\|W^{(\ell)} - W_0^{(\ell)}\|_F$ plus nine times the preceding feature difference, which is exactly the recursion defining F_ℓ. The label restriction implies $8FY^2/\lambda^{3/2} \le \sqrt{\lambda}/(8H^2) \le 1/(8H)$. Thus the operators remain below $8 + 1/8$, the features below $3H/2 + H/8$, and the smallest singular value of $\mathsf{H}/\sqrt{mn}$ above $(1/\sqrt{2} - 1/8)\sqrt{\lambda} > \sqrt{\lambda}/2$. Every stop has a strict margin. At fixed width the bounded parameters and locally Lipschitz vector field give global continuation. The finite parameter length proves convergence, and residual decay proves fitting. The argument uses only the retained dimensions and the original denominators n, and hence also proves cavity fitting.

Finally the same forward recursion gives

$$\|\dot w\|/\sqrt{n} \le 4H\rho, \qquad \sup_v \|\dot h^{(L)}(v)\|/\sqrt{n} \le 2F\rho\|w\|/\sqrt{n}.$$

Therefore

$$\sup_{\|v\|=1} |f_n(\infty, v) - f_n(t, v)| \le \frac{16}{\lambda}\left(H^2 + \frac{FY^2}{\lambda}\right)\rho(t).$$

The coefficients H, F depend only on activations and depth, and the headline cap with $\lambda \le 1$ gives $Y^2/\lambda \le 1$. This proves (20), including its uniform endpoint. □

Lemma 5.2 (Signed perturbation estimate)**.** *Let a differentiable prediction map F take Euclidean parameter space to sample space with inner product $\langle u, v\rangle_m = m^{-1}u^\top v$. Write $r = F(\theta) - y$, $r' = F(\theta') - y$, $\rho = \|r\|_m$, $\rho' = \|r'\|_m$, and $J = DF(\theta)$, $J' = DF(\theta')$. Suppose $\dot\theta = -2J^* r + \mathcal{E}$ and $\dot\theta' = -2J'^* r'$. Here J^* and J'^* are adjoints for the stated parameter and sample inner products. If a constant $K \ge 0$ satisfies, for $e = \|\theta - \theta'\|$,*

$$\|r - r' - J'(\theta - \theta')\|_m \le Ke^2/2, \qquad \|(J - J')(\theta - \theta')\|_m \le Ke^2,$$

then, on every finite interval,

$$e(t) \le \left(e(0) + \int_0^t \|\mathcal{E}(u)\|\,du\right)\exp\left\{K\int_0^t (\rho' + 3\rho)\,du\right\}. \tag{24}$$

Proof. Put $d = \theta - \theta'$ and $R = r - r' - J'd$. Exact subtraction gives

$$\tfrac{1}{2}\partial_t e^2 = -2\|r - r'\|_m^2 + 2\langle r - r', R\rangle_m - 2\langle r, (J - J')d\rangle_m + \langle d, \mathcal{E}\rangle \le K(\rho' + 3\rho)e^2 + e\|\mathcal{E}\|.$$

Apply the integrating factor to $\sqrt{e^2 + \epsilon^2}$ and let $\epsilon \downarrow 0$. This proves the assertion also through zeros of the parameter discrepancy. □

6 One source argument on two analytic domains

The following interface supplies the analytic approximation domains and real-time bounds used by the compression constructions. Its proof contains the stopped-cavity argument and all its auxiliary coefficient estimates.

Proposition 6.1 (Analytic sources and real-time response bounds)**.** *Fix the finite query lists below before initialization. Eventually at each fixed confidence, the following hold on one event. In this proposition put*

$$T = 32\frac{m}{\gamma}\log(en), \qquad r_t = \frac{\gamma}{\beta^{30L}Y^2 m\sqrt{(d+3)\log(en)}}, \qquad r_q = \frac{1}{\beta^{3L}\sqrt{(d+3)\log(en)}}.$$

Write $k^{(L)} = w$, $k^{(j)} = W^{(j+1)\top}\delta^{(j+1)}$ $(j < L)$ for the backward responses before the activation derivative. The four source families are

$$h^{(j)}, \qquad W^{(j)}(0)h^{(j-1)}, \qquad \delta^{(j)}, \qquad W^{(j+1)}(0)^\top\delta^{(j+1)}, \tag{25}$$

omitting the forward image for $j = 1$ and reverse image for $j = L$.

(i) The forward and backward fields are holomorphic on a neighborhood of

$$([-r_t, T + r_t] + i[-r_t, r_t]) \times \{v \in \mathbb{C}^d : v^\top v = 1, \|\operatorname{Im} v\|_2 \le \sinh r_q\}.$$

Query holomorphy is relative to the quadric; for $d = 1$ use $\{-1, 1\}$. Here and for $t \in [0, \infty]$ over the real sphere, the first-layer operator divided by $\sqrt{n}$ and all hidden operators have norm at most ten, and

$$\frac{\|h^{(j)}\|_2}{\sqrt{n}} \le \beta^{3L}, \qquad \frac{\max\{\|\delta^{(j)}\|_2, \|k^{(j)}\|_2\}}{\sqrt{n}} \le 16\beta^{4L}\frac{Ym}{\gamma}. \tag{26}$$

Each source coordinate has modulus at most $\beta^{6L}\sqrt{n}$ on the complex domain.

(ii) On each fixed finite query list, the predictions are holomorphic near $[-r, T + r] + i[-r, r]$, where

$$r = \frac{m}{\gamma\sqrt{\log(en)}}, \qquad |f_n(t, \sqrt{d}\, v)| \le 16\beta^{6L}\frac{Ym}{\gamma}. \tag{27}$$

On the entire real training trajectory,

$$\sup_{t\in[0,\infty]} \max_{a,j,i} |k_{a,i}^{(j)}(t)| \le 32\beta^{21L}\frac{Ym}{\gamma}\sqrt{\log(en)}. \tag{28}$$

(iii) For a declared finite panel containing the training inputs, there are $0 = t_0 < \cdots < t_J = T$ and $r_b > 0$, $0 \le b < J$, with $t_{b+1} - t_b \le r_b/2$. The sources are holomorphic near $|t - t_b| \le r_b$, with the operator, RMS and coordinate bounds in (i): forward queries range over the panel, backward queries over training inputs. The partition and radii depend deterministically only on n, m, γ, Y, activations and depth, and

$$J \le 1 + \beta^{30L}\left(\frac{Ym}{\gamma}\right)^2\sqrt{\log(en)} + \beta^{16L}\frac{m}{\gamma}\log\left(1 + \frac{4\log(en)}{\log 2}\right). \tag{29}$$

Proof. We prove stronger intermediate domains with explicitly calculated bounds, then compare them with the displayed interface. All symbols introduced from here through the end of this proof are local to this argument. The quantitative width and initialization proofs explicitly recall the coefficient recurrences they additionally use; other constructions use the proposition's stated bounds. All probability statements concern one sufficiently large width, with the data, depth, activations and confidence fixed. Write $\lambda = \gamma/m$, $Y = \|y\|_2/\sqrt{m}$, $S = 16Y/\lambda$, and $\ell = \log(en)$. We treat $Y > 0$; zero labels give the stationary zero predictor. For each full or cavity trajectory write $\rho = \|r\|_2/\sqrt{m}$, also at complex times using the Euclidean norm of the complex residual vector. Write $W_0^{(j)} = W^{(j)}(0)$ for initialized matrices. Put

$$b = \max_j |\phi_j(0)|, \qquad s = \max\{1, \sup_{j,|\operatorname{Im} z|\le a/2} |\phi_j'(z)|\}, \qquad t_2 = \max\{1, \sup_{j,|\operatorname{Im} z|\le a/2} |\phi_j''(z)|\}.$$

Thus $1 + b, s, t_2, 16/a \le \beta$, and the bound used for activation values is $|\phi_j(z)| \le b + s|z|$, not boundedness. Write $v = x/\sqrt{d}$. Backward fields at a passive input are defined by the same algebraic recursion as at a training input,

$$k^{(L)}(t, v) = w(t), \qquad \delta^{(j)}(t, v) = \phi_j'(z^{(j)}(t, v)) \odot k^{(j)}(t, v), \qquad k^{(j)} = W^{(j+1)\top}\delta^{(j+1)}.$$

This definition uses no passive label and does not add a training term.

1. Local coefficient estimates and working domains. The following scalars are bounds, not coordinates of a model:

$$\begin{aligned}
&H_1 = \max(1, b + 20s), && H_j = \max(1, b + 10sH_{j-1}), && \\
&k_L = H_L, && k_j = 10sk_{j+1}, && \tau_j = sk_j, \\
&P_1 = 3, && P_j = H_{j-1} + 10sP_{j-1} + 1, && f_j = sP_j, \\
&A_* = 2sP_L + 2s^2 \sum_{j=2}^{L} k_j P_{j-1}, && D_* = t_2 \sum_{j=1}^{L} P_j^2, && H_* = A_* + t_2 \sum_{j=1}^{L} P_j^2 k_j.
\end{aligned} \tag{30}$$

Set $f_* = \max_j f_j$, $k_* = \max_j k_j$, $H_{\max} = \max_j H_j$, and define

$$\begin{aligned}
&E_* = t_2 \sum_{j=1}^{L} (10s)^{2(j-1)} k_j, && T_* = 2H_*^2 + 4A_*^2(e^2 - 1), \\
&D_0 = (1 + b + s)(1 + T_* + E_* + s^2 k_*^2), && D_1 = 576e^3(1 + b + s)D_*^3, \\
&C_F = s(8f_*^2 + H_{\max}^2 + 1), && C_{\mathrm{abs}} = 8(D_0 + 1), \\
&V_1^0 = \tau_1, && V_j^0 = \tau_j H_{j-1}^2 + 10sV_{j-1}^0, \\
&G_L^0 = sH_L + 14t_2 V_L^0 + s, && G_j^0 = 10sG_{j+1}^0 + s^3 k_{j+1}^2 H_j + 14t_2 V_j^0 + s, \\
&W_G = 128(1 + H_{\max} + s \max_j V_j^0 + \max_j G_j^0). &&
\end{aligned} \tag{31}$$

Choose the exponential-budget constants in this order:

$$\eta_* = \min\{1, (1024 C_{\mathrm{abs}} W_G)^{-1}\}, \qquad \mathcal{B} = 1024e^2 L, \qquad \Lambda_* = \log(e + \mathcal{B}) + \log(1/\eta_*).$$

$$\begin{aligned}
S_* = \min\Bigg\{ & 1, (4A_*)^{-1}, \sqrt{\frac{\eta_*}{4000 D_*}}, \sqrt{\frac{\eta_*}{16eD_*\sqrt{2\mathcal{B}}}}, \\
& \sqrt{\frac{\eta_*}{\Lambda_*}}, \left(\frac{\eta_*^3}{D_1 \mathcal{B}}\right)^{1/4}, (8D_0 C_F)^{-1/2} \Bigg\}.
\end{aligned} \tag{32}$$

These quantities depend only on the activation bounds and depth. Further put

$$
\begin{aligned}
&C_G = 32\max(1, H_{\max}, \max_j \tau_j), \quad K_{\mathrm{src}} = 16C_{\mathrm{abs}}(1+C_G),\\
&g_* = \tau_1 + \sum_{j=2}^{L} \tau_j H_{j-1}, && r_j^* = P_j g_*, && q_j^* = f_j g_*,\\
&j_j^* = 20(10s)^{j-1}, && b_j^* = s j_j^*,\\
&e_1 = t_2 r_1^* P_1, && e_j = t_2 r_j^* P_j + s(q_{j-1}^* + \tau_j H_{j-1} f_{j-1} + 10 e_{j-1}),\\
&a_1^* = t_2 j_1^* P_1 + 2s, && a_j^* = t_2 j_j^* P_j + s(b_{j-1}^* + 10 a_{j-1}^*),\\
&T_Q = 8 f_* \max_j (f_j H_* + e_j), && T_J = 8 f_* \max_j a_j^*.
\end{aligned} \tag{33}
$$

For a Gaussian multiplier G, define

$$
\begin{aligned}
U_1(G) &= 4sK_{\mathrm{src}},\\
U_j(G) &= 2\{sK_{\mathrm{src}}(H_{j-1}^2 + f_{j-1}^2 + ST_Q + S^2 H_{j-1} q_{j-1}^*) + G q_{j-1}^* + 1\},\\
V_1(G) &= 2(2G + 2sK_{\mathrm{src}} S^2 + 1),\\
V_j(G) &= 2\{G b_{j-1}^* + sK_{\mathrm{src}} S^2 (T_J + H_{j-1} b_{j-1}^*) + 1\}.
\end{aligned} \tag{34}
$$

Use $U = \max_j U_j(16\sqrt{d+3})$, $V = \max_j V_j(16\sqrt{d+3})$ for the sphere, and $U_{\mathrm{fin}} = \max_j U_j(64)$ for a fixed finite list of inputs. Finally let

$$
\mathcal{K} = H_L^2 + S^2 \left(\tau_1^2 + \sum_{j=2}^{L} \tau_j^2 H_{j-1}^2 \right), \qquad D_W = \max\{\tau_1, \max_{j \geq 2} \tau_j H_{j-1}\}. \tag{35}
$$

Claim 1 (Local power bounds)**.** *If $Y/\lambda \leq \beta^{-30L}$, then $S \leq \beta^{-26L} \leq S_*$. Moreover*

$$
H_j \leq 3\beta^{2j}, \quad P_j \leq \beta^{3j-2}, \quad k_j \leq 3\beta^{4L-2j}, \quad \tau_j \leq 3\beta^{4L-2j+1}, \quad K_{\mathrm{src}} \leq \beta^{21L}, \tag{36}
$$

$$
U \leq \beta^{26L}\sqrt{d+3}, \qquad V \leq \beta^{2L}\sqrt{d+3}, \qquad U_{\mathrm{fin}} \leq \beta^{26L}, \qquad \mathcal{K} \leq \beta^{14L}. \tag{37}
$$

Proof. Unroll the affine recurrences using $10s \leq \beta^2$, $4j \leq \beta^j$, $L, L+1 \leq \beta^{L-1}$, and $\sum_{j=1}^{L} \beta^{cj} \leq \beta^{cL}/(1-\beta^{-c})$. They give $H_j \leq 22\beta(10\beta)^{j-1} \leq 3\beta^{2j}$ and $P_j \leq 4j(10\beta)^{j-1} \leq \beta^{3j-2}$. Substitution gives the following bounds; both columns are included to account for the numerical factors.

quantity	intermediate bound	power bound
A_*	$7\beta^{5L-3}$	β^{5L-2}
D_*	$2\beta^{6L-3}$	β^{6L-2}
H_*	$4\beta^{8L-3}$	β^{8L-2}
E_*	$4\beta^{6L-3}$	β^{6L-2}
T_*	$3\beta^{16L-4}$	β^{16L-3}
D_0	$8\beta^{16L-3}$	β^{16L-2}
D_1	$9216e^3\beta^{18L-8}$	β^{18L-2}
C_F	$9\beta^{6L-1}$	β^{6L}
C_{abs}	$65\beta^{16L-3}$	β^{16L-1}

For instance, the summands of A_*'s mixed part, H_*'s curvature part, and E_* are at most $6\beta^{4L+j-3}, 3\beta^{4L+4j-3}, 3\beta^{4L+2j-3}$. The inner sum in D_0 is at most $1 + 3\beta^{16L-4} + 4\beta^{6L-3} + 9\beta^{8L-2} \leq 4\beta^{16L-4}$, and $9216e^3 < \beta^6$. Unrolling the two activity recurrences gives

$$
V_j^0 \leq 27j\beta^{4L+2j-3}, \quad \max_j V_j^0 \leq \beta^{7L-2}, \quad G_j^0 \leq 66L\beta^{8L-2j-1}, \quad W_G \leq \beta^{9L+1}.
$$

In the reverse recurrence the propagated terminal, mixer and curvature contributions are bounded respectively by $380L\beta^{8L-2j-2}, 27L\beta^{8L-2j-1}, 379L\beta^{8L-2j-6}$; the constant contribution is at most $2\beta^{2L-2j-1}$. Hence $\eta_*^{-1} \le \beta^{25L+4}$, $\mathcal{B} \le \beta^{L+3}$, and $\Lambda_* \le \beta^{2L}$. The six nonconstant entries of (32) are at least β^{-c}, with exponents

$$5L-1, \quad (31L+6)/2, \quad (63L+12)/4, \quad (27L+4)/2, \quad (94L+13)/4, \quad (22L-1)/2.$$

All are at most $26L$. Also $16\beta^{-30L} \le \beta^{-26L}$. The remaining recurrences give

$$g_* \le \beta^{5L-1}, \qquad r_j^* \le \beta^{5L+3j-3}, \qquad q_j^* \le \beta^{5L+3j-2},$$
$$e_j \le 2\beta^{5L+6j-4}, \qquad a_j^* \le 3\beta^{5j-2}, \qquad T_Q \le \beta^{14L-3}, \qquad T_J \le \beta^{8L-1}.$$

For e_j, its three forcing exponents are at most $5L+6j-4, 5L+3j-4, 4L+3j-4$, with coefficient nine in the last term; geometric propagation by β^2 gives the displayed factor two. The same calculation gives the factor three for a_j^*. Further $K_{\mathrm{src}} \le 1552\beta^{20L-2} \le \beta^{20L+2}$. Now $ST_Q, S^2H_{j-1}q_{j-1}^* \le 1$, and the bracket in (34) is at most $11\beta^{6L-8}$. Thus for $j \ge 2$

$$U_j(16\sqrt{d+3}) \le 22\beta^{26L-5} + 32\sqrt{d+3}\beta^{8L-5} + 2, \quad U_j(64) \le 22\beta^{26L-5} + 128\beta^{8L-5} + 2.$$

The first-layer bound is $4\beta^{20L+3}$. These give the asserted bounds on U, U_{fin}. In V_j, the term containing S^2 is at most one, giving $V_j \le 64\sqrt{d+3}\beta^{2L-2} + 4$ for $j \ge 2$; the first value is at most $67\sqrt{d+3}$. Finally substitute (36) in (35); $S \le 1$ and the same geometric sum give $\mathcal{K} \le \beta^{14L}$. Every comparison above holds already at $L = 2, \beta = 10$, and improves as either increases. □

Working whole-sphere estimates. The estimates below in fact require only the conclusion of the fitting lemma and $S \le S_*$, both already implied by the headline assumptions. For the remainder of this proof redefine the working radii by

$$T = 32\ell/\lambda, \quad c_t = \frac{a}{64YSU}, \quad c_q = \min\{1/8, a/(8V)\}, \quad r_t = c_t/\sqrt{\ell}, \quad r_q = c_q/\sqrt{\ell}.$$

For every fixed confidence, eventually with that confidence all forward and backward fields extend holomorphically to a neighborhood of

$$([-r_t, T+r_t] + i[-r_t, r_t]) \times \mathcal{Q}_{r_q}, \qquad \mathcal{Q}_r = \{v \in \mathbb{C}^d : v^\top v = 1, \|\operatorname{Im} v\|_2 \le \sinh r\}.$$

For $d = 1$, replace $\mathcal{Q}_r$ by the two points $\{-1, 1\}$. Holomorphy in the query is relative to the complex quadric. Normalized first-layer and hidden operator norms are at most ten, and

$$\|h^{(j)}\|_2/\sqrt{n} \le H_j, \qquad \|k^{(j)}\|_2/\sqrt{n} \le Sk_j, \qquad \|\delta^{(j)}\|_2/\sqrt{n} \le S\tau_j.$$

For training points, $\max|z_{a,i}^{(j)}| \le K_{\mathrm{src}}\sqrt{\ell}$ and $\max|k_{a,i}^{(j)}| \le SK_{\mathrm{src}}\sqrt{\ell}$. In mobility coordinates $\Theta = (W^{(1)}, \sqrt{n}W^{(2)}, \dots, \sqrt{n}W^{(L)}, w)$, put $F_a = nf_n(v_a)$. Then

$$\max_{a,j,i,v} |(D_\Theta z^{(j)}(v)\nabla_\Theta F_a)_i| \le SU\sqrt{\ell}, \qquad \max_{j,i} |\partial_\zeta z_i^{(j)}(u\cos\zeta + v\sin\zeta)| \le V\sqrt{\ell}$$

for real orthonormal frames (u, v) in the query tube. All four source families

$$h^{(j)}, \qquad W_0^{(j)}h^{(j-1)}, \qquad \delta^{(j)}, \qquad W_0^{(j+1)\top}\delta^{(j+1)},$$

omitting boundary images, have coordinate modulus at most $M_0\sqrt{n}$, where $M_0 = 10\max_j(H_j, \tau_j)$. The real fitting trajectory and its endpoint are those of Lemma 5.1. The proof below also establishes the finite-query and expanding-domain variants. Its probability step is shared; the domain verifications are given last.

2. Independent stops and deterministic differentiation. Delete a fixed number u of neurons at one layer, preserving the original normalization n in every rectangular matrix. For each deleted interior neuron, its initialized incoming row transpose y_i and outgoing column x_i are independent $N(0, I/n)$, independent of the retained initialization and other omitted pairs. At the first layer the incoming row instead has covariance I_d and evaluates only its own initial preactivation; at the top there is no outgoing root. Use a separate initial test for each cavity, and set its reference coefficient paths identically zero if it fails. All such paths are defined before integrating an omitted root.

For a nested domain D_s, stop each cavity at its own first exit level. Stop every training sample's budget

$$\mathcal{H}_a = \frac{1}{n}\sum_{j,i} \exp\{\eta_*(Z_{a,i}^{(j)} + K_{a,i}^{(j)}/S)\}, \quad Z_{a,i}^{(j)} = \sup_{D_s} |z_{a,i}^{(j)}|, \quad K_{a,i}^{(j)} = \sup_{D_s} |k_{a,i}^{(j)}|, \tag{38}$$

at $\mathcal{B}$ for the full network and $2\mathcal{B}$ for cavities. Also stop operators at ten, contour activity at $2\int \rho|dt| = S$, and imaginary preactivations at $3a/8$ and $7a/16$, respectively. Impose provisional coordinate and response maxima twice the claimed full bounds and four times the cavity bounds, as well as the passive size and row bounds used below. These have strict margins on the successful prefix.

For rectangles use $D_s = sD_1$ and coordinate clipping P_s. For the query tube also clip its imaginary great-circle coordinate, retaining every real frame. If a cavity stops at s_c, extend a coefficient by $b_c(P_{s_c} z)$. With $a_0(z) = \mathrm{clip}_{[0,T]} \operatorname{Re} z$,

$$P_{s_c} a_0(z) = a_0(P_{s_c} z), \qquad b_c(P_{s_c} z) - b_c(P_{s_c} a_0(z)) = b_c(P_{s_c} z) - b_c(a_0(P_{s_c} z)). \tag{39}$$

Clipping is measurable in retained initialization, is Lipschitz, and fixes the successful prefix. Its real anchors traverse a monotone interval and then freeze. Different cavities are extended separately before subtraction. The expanding domain below has the same properties with a uniformly bounded Lipschitz constant. These extensions need not be holomorphic outside their own domains.

Write $\Theta_0 = \Theta(0)$, $\bar{u} = (\Theta - \Theta_0)/S$, $\bar{F}_a = F_a(\Theta_0 + S\bar{u}, e_a)/S$, $\bar{r}_a = \bar{F}_a/n - y_a/S$, and put bars on backward fields divided by S. Here e_a is an additive forward port. The two actual ports of a deletion are $e_a = \sum_i W_{:,i}^{(j+1)} h_{a,i}^{(j)}$ and $\bar{q}_a = \sum_i W_{i,:}^{(j)\top} \bar{\delta}_{a,i}^{(j)}$. The exact retained equation is

$$\dot{\bar{u}} = -\frac{2}{m}\sum_a \bar{r}_a \left(\nabla_{\bar{u}} \bar{F}_a + D_{\bar{u}} h_a^{(j-1)\top} \bar{q}_a\right). \tag{40}$$

At the top add $n^{-1}\sum_i \bar{u}_{w,i} h_{a,i}^{(L)}$ to $\bar{r}_a$, multiplying both terms. The reverse port does not enter the forward residual. The gradient blocks of $\bar{F}_a$ are $S\bar{\delta}_a^{(1)} v_a^\top$, $S\bar{\delta}_a^{(j)} h_a^{(j-1)\top}/\sqrt{n}$, and $h_a^{(L)}$.

The physical recurrences and linear activation growth give (30)'s RMS bounds whenever the operator and activity stops hold. For augmented mobility/port directions U, write $Z_a^{(j)} U = Dz_a^{(j)}[U]$. The exact derivative recursion is

$$\begin{aligned} Z_a^{(1)} U &= U_{W^{(1)} v_a} + U_{e^{(1)}}, \\ Z_a^{(j)} U &= U_{\sqrt{n} W^{(j)} h_a^{(j-1)}/\sqrt{n}} + W^{(j)} \operatorname{diag}(\phi'_{j-1}) Z_a^{(j-1)} U + U_{e^{(j)}}. \end{aligned} \tag{41}$$

It is bounded by P_j; its activated version is bounded by f_j. The Hessian, with all activations

evaluated at the reference state, is

$$\begin{aligned} D^2F_a[U,V] &= U_w^\top \operatorname{diag}(\phi_L')Z_a^{(L)}V + V_w^\top \operatorname{diag}(\phi_L')Z_a^{(L)}U \\ &\quad + \sum_j (Z_a^{(j)}U)^\top \operatorname{diag}(k_a^{(j)}\phi_j'')Z_a^{(j)}V \\ &\quad + \sum_{j\ge 2} \frac{\delta_a^{(j)\top}}{\sqrt{n}}\{U_{\sqrt{n}W^{(j)}} \operatorname{diag}(\phi_{j-1}')Z_a^{(j-1)}V + (U \leftrightarrow V)\}. \end{aligned} \tag{42}$$

For a rectangular matrix let $\|B\|_{p,n} = n^{-1/p}(\sum_i \sigma_i(B)^p)^{1/p}$. The budget implies $\|k_a^{(j)}\|_p/n^{1/p} \le (S/\eta_*)p(2\mathcal{B})^{1/p}$, because $x^p \le (p/e)^p e^x$. Each noncurvature Hessian block factors through at most n coordinates, and each curvature block contains one carrier diagonal. Matrix ideal inequalities in (42) therefore give

$$\begin{aligned} &\|D^2F_a\|_{p,n} \le A_* + (SD_*/\eta_*)p(2\mathcal{B})^{1/p}, \qquad \|D^2F_a\|_{\mathrm{HS}}/\sqrt{n} \le H_*, \\ &\|D^2F_a\|_{\mathrm{op}} \le A_* + (SD_*/\eta_*)\log(2n\mathcal{B}). \end{aligned} \tag{43}$$

These bounds also cover $D_\Theta\delta$, since $\partial_{e^{(j)}}F_a = \delta_a^{(j)}$. Residual mixtures preserve the bounds by $\sum_a |r_a|/(m\rho) \le 1$.

Suppose for now that products of the negative-Gram propagators on ordered disjoint contour pieces have norm at most two, and $\rho/S \le \lambda/8$. The variational generator is

$$-\frac{2}{mn}\sum_a g_a g_a^\top - \frac{2}{m}\sum_a r_a D^2F_a, \qquad g_a = \nabla_\Theta F_a.$$

Its full propagator obeys

$$\|J(t,s)\| \le 2\exp\{SA_* + (S^2D_*/\eta_*)\log(2n\mathcal{B})\} \le Cn^\kappa, \qquad \kappa = 1/4000. \tag{44}$$

Here and only in qualitative local estimates, C and powers of ℓ may depend on the fixed problem and deletion count. The coefficient of $\log n$ is bounded by (32); it has not been hidden in a width threshold.

Claim 2 (Local maps, remainders, and passive observables)**.** *On the independent stops just specified, with contours of length $O(\ell)$, all first insertion maps have norm $n^\kappa \operatorname{poly}(\ell)$. Their dependence on the supremum norm of scalar controls is Lipschitz with that bound, and their raw terminal/time/frame derivatives are bounded by $\sqrt{n}n^\kappa \operatorname{poly}(\ell)$. On the uniform Gaussian event proved below, the retained full/cavity differences in preactivations, scaled carriers, responses and angular derivatives are at most $n^{-1/30}$ per coordinate and at most $2n^{1/100}$ in Euclidean norm, for every fixed deletion count.*

Proof. For deterministic controls $\alpha_{a,i}, b_{a,i}$, use ports $e_a = \sum_i x_i\alpha_{a,i}$, $\bar{q}_a = \sum_i y_i b_{a,i}$. The three port derivatives of (40) at zero ports are

$$-\frac{2}{mn}\nabla_{\bar{u}}\bar{F}_a\bar{\delta}_a^{(j+1)\top}, \qquad -\frac{2}{m}\bar{r}_a(D_{\bar{u}}\bar{\delta}_a^{(j+1)})^\top, \qquad -\frac{2}{m}\bar{r}_a(D_{\bar{u}}h_a^{(j-1)})^\top.$$

The first term accounts for the changed residual. Integrating them against (44), using gradient norm $O(\sqrt{n})$, response RMS, and contour length $O(\ell)$, gives an insertion-map bound $\sqrt{u}n^\kappa O(\ell^2)$.

The differentiated forward/backward recursions are exactly

$$\begin{aligned}
z_{[1]}^{(j)} &= SV_{\sqrt{n}W^{(j)}}h^{(j-1)}/\sqrt{n} + W^{(j)}h_{[1]}^{(j-1)} + e_{[1]}^{(j)},\\
h_{[1]}^{(j)} &= \phi_j' \odot z_{[1]}^{(j)},\\
\bar{k}_{[1]}^{(j)} &= W^{(j+1)\top}\bar{\delta}_{[1]}^{(j+1)} + SV_{\sqrt{n}W^{(j+1)}}^{\top}\bar{\delta}^{(j+1)}/\sqrt{n} + \bar{q}_{[1]}^{(j)},\\
\bar{\delta}_{[1]}^{(j)} &= \phi_j' \odot \bar{k}_{[1]}^{(j)} + \phi_j'' \odot \bar{k}^{(j)} \odot z_{[1]}^{(j)}.
\end{aligned}$$

At the first layer replace the matrix product by $SV_{W^{(1)}v}$; at the top $\bar{k}_{[1]} = V_w$. A reference carrier costs $O(\ell)$ once in the inhomogeneous last term. Homogeneous propagation costs $10s$. Induction gives the asserted map bounds; subtracting controls gives their Lipschitz bounds because the linear variational equation is linear in these controls. Differentiating these displayed recursions introduces only the terms $\phi'' z_t$, $\phi''' z_t z_{[1]}\bar{k} + \phi'' z_{[1]}\bar{k}_t + \phi''\bar{k}z_{[1],t}$, and ordinary product derivatives. There is at most one raw reference coordinate derivative, bounded by $\sqrt{n}\,\mathrm{poly}(\ell)$. This proves the derivative claim without a width power depending on depth. Cauchy's formula on the strict strip bounds ϕ''' by $4\beta^2$.

Here are the nonlinear estimates needed to justify this linearization. Decompose the augmented displacement into $(V, e) + (U, 0)$, and let N bound its linear Euclidean variations, d_0 their coordinate maxima, and u_0 bound $\|U\|$. Put

$$R = d_0 N + d_0 u_0 + u_0^2, \qquad P = (N + u_0)^2/\sqrt{n}.$$

Assume $N \geq 1$, $d_0, u_0 \leq 1$, $N + u_0 \leq \sqrt{n}$, and $u_0 + R + P \leq 1$. If $E_g = g_1 - g_0 - Dg_0(V + U, e)$ and $g_i = \phi'(z_i)$ in the third line below, the exact identities are

$$\begin{aligned}
E_{z^{(j)}} &= W_1^{(j)} E_{h^{(j-1)}} + S\Delta\bar{u}_{\sqrt{n}W^{(j)}}h_{[1]}^{(j-1)}/\sqrt{n},\\
E_{\bar{k}^{(j)}} &= W_1^{(j+1)\top} E_{\bar{\delta}^{(j+1)}} + S\Delta\bar{u}_{\sqrt{n}W^{(j+1)}}^{\top}\bar{\delta}_{[1]}^{(j+1)}/\sqrt{n},\\
E_{\bar{\delta}} &= g_1 \odot E_{\bar{k}} + (g_1 - g_0) \odot \bar{k}_{[1]} + \bar{k}_0 \odot (g_1 - g_0 - \phi''(z_0)z_{[1]}).
\end{aligned} \tag{45}$$

The first and top boundary remainders vanish. Forward and backward first derivatives are bounded by $J_f = \beta^{10L}$ and $J_b = \beta^{20L}(1 + M)$, where $M \geq 1$ is a common provisional bound for the absolute preactivations and scaled carriers of the reference and full paths, including the deleted-neuron values that define the scalar controls. Such an envelope is $\mathrm{poly}(\ell)$ under the stated stops. The localized product estimate is $\|z_{[1]}^{\odot 2}\|_2 \leq 3J_f^2 R$. Insert $z_0 + z_{[1]}$ into activation subtraction. Taylor's formula and (45), summed by geometric propagation, give

$$\max(\|E_z\|_2, \|E_h\|_2) \leq \beta^{30L}(R + P), \qquad \max(\|E_{\bar{k}}\|_2, \|E_{\bar{\delta}}\|_2) \leq \beta^{60L}(1 + M)(R + P). \tag{46}$$

Indeed the two gate cross terms cost at most $\beta[3J_f J_b R + \beta^{30L}(R + P)(d_0 + J_b u_0)]$ and $M[\beta\|E_z\| + \beta^2\|z_{[1]}^{\odot 2}\|/2]$; the matrix cross term costs $J_b P$. No changed carrier is used as a presumed reference bound.

The hidden gradient remainder is

$$\frac{S}{\sqrt{n}}[E_{\bar{\delta}}h_1^{\top} + \bar{\delta}_{[1]}(h_1 - h_0)^{\top} + \bar{\delta}_0 E_h^{\top}],$$

bounded by $\beta^{70L}(1 + M)(R + P)$. The Hessian identity along partial joining segments bounds the residual difference by $\beta^{13L}(N + u_0)/\sqrt{n}$ and its linearization error by $\beta^{87L}(1 + M)(N + u_0)^2/(2n)$. For a product of residual and gradient, its exact remainder is $\bar{r}_0 E_g + (\Delta\bar{r} - D\bar{r}_0\Delta)g_0 + \Delta\bar{r}(g_1 - g_0)$.

For the reverse force one additionally needs a lower adjoint probe. For an omitted root y_i, include all derivatives of $y_i^\top h^{(j-1)}$ in the Gaussian coordinate event. For $q = \sum_i y_i b_{a,i}$, let $Q = 2u\beta M$, $P_q = u\beta M d_0$. Subtracting its adjoint recursion gives

$$\max_j \|k_{q,1}^{(j)} - k_{q,0}^{(j)}\|_2 \le \beta^{20L}(P_q + Q/\sqrt{n})(N + u_0),$$

since the changed gate multiplies the reference probe maximum, the changed matrix costs $Q(N + u_0)/\sqrt{n}$, and propagation costs $11s$. Integrating the corresponding Hessian gives

$$\|(Dh_1 - Dh_0)^\top q\|_2 \le \beta^{50L}(P_q + Q/\sqrt{n})[N + u_0 + (N + u_0)^2].$$

Combining these estimates in (40) yields the vector-field remainder bound

$$\beta^{240L}(1 + u)(1 + M)\{\bar\rho[R + d_0(N^2 + Nu_0 + u_0^2)] + (1 + \bar\rho)P\}, \qquad \bar\rho = \rho/S. \tag{47}$$

This also holds on the complex joining segments: the sufficient gate $d_0+\beta^{10L}u_0+\beta^{30L}(R+P) < a/32$ keeps the segments and auxiliary Taylor points inside $15a/32$. A first-exit argument proves the gate's use before Taylor expansion; the factor β^{240L} pays for the Cauchy third-derivative bound. Scaling introduced no inverse power of S: in the effective network with readout $\bar u_w$, the affine change of variables multiplies hidden directions by $S \le 1$.

For every existing outgoing column, and for an incoming row at a layer $j \ge 2$, integrating the omitted-parameter equations gives

$$\|\Delta W_{:,i}^{(j+1)}\| \le S^2(\max_r \tau_r)(b + sM)/\sqrt{n}, \qquad \|\Delta W_{i,:}^{(j)}\| \le S^2 H_{\max} sM/\sqrt{n}.$$

At the first layer the incoming-row estimate is instead $\|\Delta W_{i,:}^{(1)}\| \le sS^2M$. There is no retained reverse port below that layer: this row's motion is included in the deleted neuron's scalar controls, rather than in the learned retained-port remainder. Thus every existing learned port entering the retained equation still induces a force at most $S^2\beta^{260L}(1+u)^2(1+M)^3(1+\bar\rho)/\sqrt{n}$. The top residual offset is at most $uM(b+sM)/n$; its product with both retained terms has the same enclosing bound.

It remains to specify all passive quantities in the local claim. For evaluated input q and training driver a, let $C_q^{(j)} = D_\Theta h_q^{(j)}$, $\bar R_{qa}^{(j)} = S^{-1}D_\Theta z_q^{(j)} g_a$, and $\bar Q_{qa}^{(j)} = \phi_j' \odot \bar R_{qa}^{(j)}$. Then

$$\bar R_{qa}^{(1)} = \bar\delta_a^{(1)} v_a^\top q, \qquad \bar R_{qa}^{(j)} = \bar\delta_a^{(j)} c_{aq}^{(j-1)} + W^{(j)}\bar Q_{qa}^{(j-1)}, \qquad c_{aq}^{(j-1)} = h_a^{(j-1)\top} h_q^{(j-1)}/n. \tag{48}$$

Angular fields satisfy $J^{(1)} = W^{(1)}q'$, $J^{(j)} = W^{(j)}I^{(j-1)}$, $I = \phi' \odot J$. Differentiate these product formulas: the four response terms are $\bar\delta_{[1]}c + \bar\delta c_{[1]}$, $SV_{\sqrt{n}W}\bar Q/\sqrt{n}$, and $W\bar Q_{[1]}$, together with the omitted port $\sum_i x_i \bar Q_{qa,i}$. The activated variation adds $\phi''\bar R z_{[1]}$. Angular variations have the same matrix and gate terms, with omitted port $\sum_i x_i I_i$. Below a deletion there is also the direct reverse observable

$$S^{-1}(Q_{qa,\mathrm{full}} - C_q g_a) = C_q C_a^\top \bar q_a. \tag{49}$$

Include the forward images of $C_a^{0\top} y_i$ in the Gaussian event; their coefficient maps are independent of that root.

Product subtraction gives

$$|c_{[1]}| \le \beta^{10L}(N + u_0)/\sqrt{n}, \qquad |E_c| \le \beta^{40L}[(R + P)/\sqrt{n} + (N + u_0)^2/n].$$

The exact response remainder is $E_{\bar\delta}c_1 + \bar\delta_{[1]}(c_1 - c_0) + \bar\delta_0 E_c + W_1 E_{\bar Q} + \Delta W\bar Q_{[1]} + E_{\mathrm{ports}}$; its gate remainder is the third line of (45) with $\bar k$ replaced by $\bar R$. The angular case uses that same identity

with J. Reference response RMS cancels the $1/\sqrt{n}$ in E_c. For $\xi = C_a^{0\top} y_i$, forward directional subtraction gives $\|(C_q^1 - C_q^0)\xi\| \le \beta^{50L}(d_0 + n^{-1/2})(N + u_0)$. The other changed factor in (49) uses the adjoint-probe estimate above. Their sum is bounded by

$$\beta^{70L}(1+u)(1+M)(d_0+n^{-1/2})[N+u_0+(N+u_0)^2].$$

Missing feature products cost $u(b+sM)^2/n$, and learned ports cost $n^{-1/2}\,\mathrm{poly}(\ell)$. These enumerate all additional remainders. Their time and frame derivatives have the same product structure already checked. The probabilistic and first-exit conclusion follows in the next paragraph. □

The uniform Gaussian event and closing the local comparison. For fixed controls, a pairing of an omitted root with a linear variation has the form $g^\top R g$, where $R = (E_i^\top L + L^\top E_i)/2$. Consequently $\|R\| \le \|L\|$, and only its same-root diagonal block has a mean $\operatorname{tr} R/n$. Cross-root forms are centered. On $\|g\| \le 2\sqrt{2u}$, interpolation of a centered form costs at most $10u\|\Delta R\|$, including its trace. All stopped scalar controls have polynomial-logarithmic amplitudes and speeds $\sqrt{n}\,\mathrm{poly}(\ell)$; this follows by differentiating the forward, backward and response recursions above. A scalar control with amplitude M_c, speed $D_c\sqrt{n}$ on a contour of length T_c has an $n^{-1/8}$-net of logarithmic cardinality at most

$$(2 + 8T_cD_cn^{5/8})\log(1+8M_cn^{1/8}).$$

Indeed sample it at spacing $n^{-1/8}/(8D_c\sqrt{n})$, round on mesh $n^{-1/8}/4$, and interpolate. Apply this separately to the finitely many real and imaginary controls. They are deterministic one-dimensional histories on a prescribed contour; no arbitrary two-dimensional analytic continuation of controls is assumed.

After eventual absorption of the fixed logarithmic factors, all linear and quadratic maps have norm at most $n^{1/200}$. A linear coordinate has variance at most $n^{-99/100}$. For a real symmetric form, diagonalization and

$$\mathbb{E}e^{t(G^\top RG - \operatorname{tr} R)} = \prod_i e^{-t\lambda_i}(1-2t\lambda_i)^{-1/2}, \qquad -\log(1-v) - v \le \frac{v^2}{2(1-|v|)}$$

give, for $g \sim N(0, I/n)$,

$$\Pr\{|g^\top Rg - \operatorname{tr} R/n| > t\} \le 2\exp\left[-c\min\left\{\frac{n^2t^2}{\|R\|_{\mathrm{HS}}^2}, \frac{nt}{\|R\|_{\mathrm{op}}}\right\}\right]$$

with an absolute $c > 0$, by applying exponential Markov with $|2t\lambda_i| \le 1/2$ and optimizing the parameter. Here $\|R\|_{\mathrm{HS}} \le \sqrt{2un}\|R\|$. At threshold $n^{-1/10}/8$, real/imaginary splitting gives $4e^{-n^{.78}}$ eventually. Root-norm failure is at most e^{-un}. The control interpolation error is $n^{-1/8+1/200}\,\mathrm{poly}(\ell) = o(n^{-1/10})$. A terminal/time/frame mesh of spacing $n^{-2}/\,\mathrm{poly}(\ell)$ has polynomial size; the sphere frame has $2d$ ambient coordinates. Together with all coordinates, probes and at most uLn^u deletion sets, the logarithmic union cost is eventually less than $n^{.65}$. Thus the uniform event fails with probability at most $e^{-n^{.7}}$. Only after this event is established are the actual adaptive controls substituted. Root-norm bounds also make the total linear vector variation at most $n^{1/100}$, distributing a fixed share among the finitely many components.

Let U be the retained scaled displacement minus its linear variation. It satisfies Duhamel's formula with (47), the learned ports, and the top offset. Set $N = n^{1/100}$, $d_0 = n^{-1/10}$, $u_0 = n^{-1/25}$. On a first-exit prefix $\|U\| \le u_0$,

$$\|U\| \le n^\kappa\,\mathrm{poly}(\ell)[n^{-.08} + u_0^2 + n^{-.48} + n^{-1/2}] = o(u_0). \tag{50}$$

The activity integral bounds the residual-weighted terms and the contour length bounds the others. This strictly improves the stop. The explicit remainders then give coordinate discrepancies $n^{-1/30}$ and vector discrepancies $2n^{1/100}$, including passive responses and angular fields. The joining-strip gate also holds eventually. For every deterministic frozen control history, the retained linear variations are centered Gaussian images conditional on the cavity. After substituting adaptive controls, we use only the uniform bounds already proved; we assert neither Gaussianity nor centering of the resulting variations. Pairing a frozen-control variation again with an omitted root produces a quadratic form. Its nonzero trace is retained before adaptive substitution, as treated next. This completes Claim 2.

3. Trace absorption and Gaussian moments.

Every noncentered contraction. Throughout this paragraph, conditional means and centered forms refer to deterministic frozen controls, conditional on the cavity. The resulting trace bounds hold uniformly over those controls and are then evaluated at the actual adaptive controls. For one interior omitted neuron work in unscaled mobility coordinates, and set $C_a = D_\Theta h_a^{(j-1)}$, $B_a = D_\Theta \delta_a^{(j+1)}$, $E_a = D_{e_a^{(j+1)}} \delta_a^{(j+1)}$, all at the zero-source cavity. With $\alpha_b = h_{b,i}^{(j)}$, $b_b = \delta_{b,i}^{(j)}$, and $d_b = \delta_b^{(j+1)}$, the linear parameter variation is

$$V(t) = -\frac{2}{m} \sum_b \int_0^t J(t,s)\{g_b(s) d_b(s)^\top x_i \alpha_b(s)/n + r_b(s) B_b(s)^\top x_i \alpha_b(s) + r_b(s) C_b(s)^\top y_i b_b(s)\}\, ds.$$

The scalar expansions are

$$z_{a,i} = y_i^\top h_a^0 + y_i^\top C_a V + (\Delta W_{i,:}) h_a + o(1), \qquad k_{a,i} = x_i^\top \delta_a^0 + x_i^\top B_a V + x_i^\top E_a x_i \alpha_a + (\Delta W_{:,i})^\top \delta_a + o(S).$$

The errors follow in scaled coordinates from Claim 2. The four nonzero conditional means are

$$\begin{aligned} &-\frac{2}{m} \sum_b \int r_b b_b \frac{\operatorname{tr}(C_a J C_b^\top)}{n}\, ds, && -\frac{2}{m} \sum_b \int r_b \alpha_b \frac{\operatorname{tr}(B_a J B_b^\top)}{n}\, ds, \\ &\alpha_a \operatorname{tr}(E_a)/n, && -\frac{2}{mn^2} \sum_b \int \alpha_b d_b^\top B_a J g_b\, ds. \end{aligned} \tag{51}$$

All other forms are centered same-root or independent-root forms covered by the uniform Gaussian event. The last displayed mean is $n^{-1+\kappa} \operatorname{poly}(\ell) = o(1)$, since $\|d_b\|, \|g_b\| = O(\sqrt{n})$.

Expand J around its negative-Gram base. A term with h residual-Hessian insertions has ordered activity integral at most $S^h/h!$. The product of its base propagators costs at most two. Normalized Schatten Hölder, with exponent $h+2$ on both endpoint Hessian blocks and the h insertions, gives

$$\frac{2S^h}{h!}[A_* + (SD_*/\eta_*)(h+2)(2\mathcal{B})^{1/(h+2)}]^{h+2}.$$

The zero-insertion term is at most $2H_*^2$. Split the other powers by $(x+y)^{h+2} \le 2^{h+1}(x^{h+2} + y^{h+2})$. The bounded part sums to $4A_*^2(e^{2SA_*} - 1)$; the carrier part is

$$\frac{8e^2 D_*^2 S^2 \mathcal{B}}{\eta_*^2} \sum_{h \ge 1} (2eD_* S^2/\eta_*)^h (h+2)^2 \le \frac{576 e^3 D_*^3 \mathcal{B} S^4}{\eta_*^3}.$$

Indeed $h! \ge (h/e)^h$, $(1 + 2/h)^h \le e^2$, and $\sum_{h \ge 1} q^h (h+2)^2 \le 36q$ for $q \le 1/2$, by differentiating the geometric series. The allowance gives this last condition. For forward endpoints give one exponent

two, the other infinity, and the h Hessians exponent $2h$. The two series are bounded by $e^{2SA_*}-1$ and $\sqrt{2\mathcal{B}}\sum_{h\ge1}(4eD_*S^2/\eta_*)^h$, each at most one by (32). Including the zero term,

$$|\operatorname{tr}(C_aJC_b^\top)|/n \le 8f_*^2. \tag{52}$$

Reciprocal exponents sum to one, giving normalization exactly n^{-1}, independent of parameter-space dimension. The direct port Hessian is $E_a=\sum_{r>j}T_{r,j+1}^\top \operatorname{diag}(k_a^{(r)}\phi_r'')T_{r,j+1}$, with $\|T_{r,j+1}\|\le(10s)^{r-j-1}$; hence carrier RMS gives $|\operatorname{tr}E_a|/n\le SE_*$.

Let $G_{h,a,i}$, $G_{\delta,a,i}$ denote suprema of the corresponding cavity Gaussian pairings. Set $Z_i=(m^{-1}\sum_a Z_{a,i}^2)^{1/2}$ and $U_i=(m^{-1}\sum_a(K_{a,i}/S)^2)^{1/2}$. At every driving time Cauchy–Schwarz gives $m^{-1}\sum_b|r_bh_{b,i}|\le\rho(b+sZ_i)$ and $m^{-1}\sum_b|r_b\delta_{b,i}|\le\rho sSU_i$. Integrating the rank-one updates and pairing retained factors in RMS, the learned outgoing column costs $s^2k_*^2S^3$ times its driving amplitude, and the learned incoming row costs $sS^2U_iH_{j-1}^2$; at the first layer the latter is sS^2U_i. Thus (51) gives

$$\begin{aligned} K_{a,i}/S &\le G_{\delta,a,i}/S+(D_0+D_1\mathcal{B}S^4/\eta_*^3)(1+Z_{a,i}+Z_i)+o(1),\\ Z_{a,i}&\le G_{h,a,i}+C_FS^2U_i+o(1). \end{aligned} \tag{53}$$

The direct external trace keeps the evaluated sample's amplitude $Z_{a,i}$; only integrated terms use the driving-sample RMS. At the first layer $G_{h,a,i}=|W_{0,i}^{(1)}v_a|$; at the top $\sup|w_i|/S\le b+sZ_i$ and there is no outgoing root. These endpoint formulas obey the same enclosing inequalities.

Put $\overline{G}_h=(m^{-1}\sum_a G_{h,a,i}^2)^{1/2}$ and $\overline{G}_\delta=(m^{-1}\sum_a(G_{\delta,a,i}/S)^2)^{1/2}$. The backward coefficient is at most $2D_0$ and $D=C_FS^2\le1/(8D_0)$. Taking sample RMS gives

$$U_i\le\overline{G}_\delta+2D_0(1+2Z_i)+o(1),\qquad Z_i\le\overline{G}_h+DU_i+o(1).$$

Since $4D_0D\le1/2$, solve these two inequalities and substitute back sample by sample to obtain

$$Z_{a,i}+K_{a,i}/S\le C_{\mathrm{abs}}(1+G_{h,a,i}+G_{\delta,a,i}/S+\overline{G}_h+\overline{G}_\delta)+o(1). \tag{54}$$

This coefficient is independent of sample and deletion counts.

Real reference moments. Parameterize a real stopped cavity by its own normalized activity $u=2\int_0^t\rho/S\in[0,1]$, freezing after its endpoint. For $v=|u-u'|$, integrating the updates gives

$$\begin{aligned} \|\Delta w\|_2/\sqrt{n}&\le H_LSv,\qquad \|\Delta W^{(1)}\|_F/\sqrt{n}\le\tau_1S^2v,\\ \|\Delta W^{(j)}\|_F&\le\tau_jH_{j-1}S^2v,\qquad \|\Delta z_a^{(j)}\|_2/\sqrt{n}\le V_j^0S^2v. \end{aligned}$$

The feature difference has the last bound multiplied by s. Split a reference carrier at SR. Its high part satisfies $\|k\mathbf{1}_{|k|>SR}\|_2/\sqrt{n}\le(4S/\eta_*)\sqrt{2\mathcal{B}}e^{-\eta_*R/4}$, by the budget and $x^2e^{-x/2}\le16$. Choose $R=(4/\eta_*)\log[8\sqrt{2\mathcal{B}}/(\eta_*v)]$ when $v>0$. The high changed-gate part is at most sSv; the low part is at most $t_2V_j^0S^3Rv$. Since $R\le\eta_*^{-1}(10\Lambda_*+4\log(1/v))$, $S^2\Lambda_*/\eta_*\le1$, and $v\log(1/v)\le2\sqrt{v}/e$, the low part is bounded by $14t_2V_j^0S\sqrt{v}$. At the top add sH_LSv; below it add the changed-matrix term $s^3k_{j+1}^2H_jS^3v$ and propagated difference. This proves

$$\|\delta_a^{(j)}(u)-\delta_a^{(j)}(u')\|_2/(S\sqrt{n})\le G_j^0\sqrt{|u-u'|}. \tag{55}$$

The top carrier was included; no coordinate readout bound was used.

Conditioned on the cavity, a Gaussian reference has bounded initial variance and this square-root activity modulus. Grids of spacing 4^{-q} have at most $2 \cdot 4^q$ increments of standard deviation at most $2G2^{-q}$. Gaussian tails at thresholds proportional to $2^{-q}\sqrt{z^2 + 4(q+1)}$, summed over levels and the initial value, give

$$\Pr\{X > W_G(1+z) \mid \text{cavity}\} \le 4e^{-z^2}, \qquad \mathbb{E}[e^{X^2/(16W_G^2)} \mid \text{cavity}] < 2.$$

The factor 128 in W_G bounds the geometric threshold sums. For the moment, $(1+z)^2 \le 2(1+z^2)$ and tail integration give $\mathbb{E}e^{Z^2/8} \le 1 + 4/7$, with $e^{1/8}(1+4/7) < 2$. Jensen gives the same bound for sample RMS: $e^{\overline{X}^2/(16W_G^2)} \le m^{-1}\sum_a e^{X_a^2/(16W_G^2)}$, without sample independence. Completing the square and Hölder for the four terms in (54) give the one-neuron moment base

$$2\exp\{qC_{\text{abs}} + 64q^2C_{\text{abs}}^2W_G^2\} < 4, \qquad 0 \le q \le 8\eta_*. \tag{56}$$

The chosen η_* verifies the last inequality.

Claim 3 (Local Gaussian rectangle bound)**.** *Let a deterministic complex coefficient map on a rectangle of sides at most H satisfy $\sup_z \|b_z\| \le D$ and $\|b_z - b_{z'}\| \le K\|z - z'\|$. For a standard real Gaussian vector G, put $X = \sup_z |G^\top b_z|$. Then*

$$M = 256D\sqrt{\log(e + HK/D)}, \qquad \log \mathbb{E}e^{qX} \le qM + 2q^2D^2 \quad (q \ge 0).$$

For sample RMS with the same bounds, replace M by $M + 2D$. If $D = 0$, the process is zero. For covariance I/n, divide coefficient norms by $\sqrt{n}$.

Proof. Use dyadic coefficient meshes with spacing proportional to $D2^{-j}/K$, having at most $4(1 + HK/D)^2 4^j$ points. Connect each point to the preceding mesh; increment standard deviations are at most $3D2^{-j}$. Gaussian exponential moments give $\mathbb{E}\max_{r \le N}|Z_r| \le \sigma\sqrt{2\log(2N)}$ when standard deviations are at most σ. Summing uses $\sum_{j\ge1} 2^{-j} = 1$, $\sum_{j\ge1} 2^{-j}\sqrt{j} \le 2$. A factor 128 covers the real process, and real/imaginary splitting gives 256.

An a-Lipschitz Gaussian function satisfies $\log \mathbb{E}e^{t(F - \mathbb{E}F)} \le a^2t^2/2$ and $\operatorname{Var} F \le a^2$. Here is a derivation. For $P_tg(x) = \mathbb{E}g(e^{-t}x + \sqrt{1 - e^{-2t}}G)$, differentiating invariant entropy and Gaussian integration by parts give

$$\operatorname{Ent}_\gamma(g) = \int_0^\infty \mathbb{E}\frac{|\nabla P_t g|^2}{P_t g}\,dt \le \tfrac{1}{2}\mathbb{E}\frac{|\nabla g|^2}{g}.$$

Use $\nabla P_t g = e^{-t}P_t\nabla g$ and weighted Cauchy–Schwarz for the inequality. Apply it to $g = e^{uF}$ and integrate the resulting derivative inequality for $u^{-1}\log \mathbb{E}e^{uF}$. Truncation and smoothing extend the calculation to Lipschitz functions; their linear growth supplies exponential integrability. Differentiating at zero gives the variance. Here X is at most $2D$-Lipschitz, proving the estimate. The sample RMS is also $2D$-Lipschitz; its mean is at most $M + 2D$ by the variance bound and Cauchy–Schwarz. Independence among sample processes is unnecessary. □

Initial cavities and maximum improvements. The initialization event of Lemma 5.1 applies jointly to fixed-size rectangular cavities. To see why weak covariance estimates need not be union-bounded over deletion sets, zero-embed their features. On the full initial operator event, conditional fresh-row Gaussian tails give initial training maxima $Z_0 = C\sqrt{\ell}$ with probability tending to one. Deleting u coordinates changes each later initialized feature vector by at most

$$D_{\text{init}} = (8s)^L\sqrt{u}(b + sZ_0) = O_u(\sqrt{\ell}).$$

Normalized feature and singular-value changes therefore vanish uniformly over deletion sets, transferring the strict fitting margins. For coordinate localization above a deletion, project the complete lower feature difference onto this deterministic ball before pairing with the independent current row. At threshold $n^{-1/10}$, the tail $2\exp[-n^{4/5}/(2D_{\mathrm{init}}^2)]$ pays for the $O_u(n^{u+1})$ tests. A cavity's covariance retains normalization n; it is not identified with the full covariance. Initial budgets are below $\mathcal{B}/2$ with probability tending to one: their conditional means are below $2e^{\eta_*^2 H_{\max}^2/2} < 3$, their second exponential moments are bounded, and conditional Chebyshev applies layer by layer. The coordinate-small deletion comparison transfers this event to all cavities. Their own initial tests remain part of each globally extended reference's definition.

Under the provisional stops, Gaussian coefficient RMS is bounded by $H_{\max}$ or $S\max_j \tau_j$. A mesh of spacing n^{-2} has polynomial size, with vanishing off-grid error from the normalized coefficient derivative bounds. Complex Gaussian splitting has tail $4e^{-u^2/(4\sigma^2)}$. The multiplier C_G, followed by (54), improves the full training stops to

$$\max|z_{a,i}^{(j)}| \le K_{\mathrm{src}}\sqrt{\ell}, \qquad \max|k_{a,i}^{(j)}| \le SK_{\mathrm{src}}\sqrt{\ell}. \tag{57}$$

There are at most n^7 tests eventually on the original rectangle and n^{10} on the expanding finite-panel rectangle; the factor 32 leaves strict margin. This step uses Gaussian tails on a mesh, not already removed budgets or Gaussian supremum moments.

For response and angular stops, differentiate $Q_{qa} = C_q g_a$: $DQ_{qa} = C_q Dg_a + (DC_q)[\cdot]g_a$. The first term has normalized Hilbert–Schmidt norm at most $f_j H_*$. Curvature, changed matrix, changed feature and propagation in the second give respectively $t_2 r_j^* P_j, sq_{j-1}^*, s\tau_j H_{j-1} f_{j-1}, 10se_{j-1}$. These are the recurrence for e_j. Angular fields give the recurrence for a_j^*, including the first-layer map $U_{W^{(1)}}q'$ of normalized norm at most two. Apply (52) with these endpoints. The four noncentered response terms are exactly the direct feature pairing, the direct reverse term (49), the integrated mixed endpoint, and the learned incoming row. After division by S, their coefficients are

$$sK_{\mathrm{src}}H_{j-1}^2, \quad sK_{\mathrm{src}}f_{j-1}^2, \quad sK_{\mathrm{src}}ST_Q, \quad sK_{\mathrm{src}}S^2H_{j-1}q_{j-1}^*.$$

The centered incoming pairing contributes Gq_{j-1}^*. Angular terms are Gb_{j-1}^* and $sK_{\mathrm{src}}S^2(T_J + H_{j-1}b_{j-1}^*)$. Thus the factor two in (34) strictly improves these stops. For sphere frames, at most n^{4d+10} mesh points require $G = 16\sqrt{d+3}$; polar normalization of nearby frames gives bounded frame derivatives for interpolation. A fixed finite input list has no frame mesh and at most n^{10} tests eventually; $G = 64$ gives failure $4n^{10}e^{-1024\ell}$. The number of passive inputs affects only the qualifying width.

Use the provisional passive-preactivation cap $2(GH_{\max} + 1 + U_0S^2 + a)\ell^2$, where U_0 is the applicable response bound. Its initial maximum is $(GH_{\max}+1)\sqrt{\ell}$, by the same fresh-row argument; real activity adds at most $U_0S^2\sqrt{\ell}$. The domain checks below control the complex displacement and improve this stop without a passive exponential budget.

Claim 4 (Local time derivative under provisional stops)**.** *Let $U_0 = U$ or U_{fin}, as appropriate, and put $N_* = \max_j\{r_j^*, 4U_0\}$, $B_n = 1 + (S^2/\eta_*)\log(e+\ell)$. Define*

$$J_L = 2sH_L + 4et_2N_*, \qquad J_j = 10sJ_{j+1} + 2s^3k_{j+1}^2H_j + 4et_2N_*, \qquad J_* = \max_j J_j.$$

For $\ell \ge \max(e^2, 2\mathcal{B})$, on each independently stopped cavity,

$$\|\dot{h}_a^{(j)}\|_2/\sqrt{n} \le 2\rho Sq_j^*, \qquad \|\dot{\delta}_a^{(j)}\|_2/\sqrt{n} \le J_*B_n\rho.$$

These bounds use the provisional response stops, not their improvement.

Proof. The exact response identity gives $\|\dot z\|_2/\sqrt{n} \le 2\rho S r_j^*$, $\|\dot z\|_\infty \le 8\rho S U_0\sqrt{\ell}$. Interpolate to exponent $2v/(v-2)$ and use the carrier v-norm supplied by (38). For $v \ge 4$,

$$\|k \odot \dot z\|_2/\sqrt{n} \le (2\rho S^2 N_*/\eta_*)v(2\mathcal{B}\ell)^{1/v}.$$

Choose $v = \max(4, \log(2\mathcal{B}\ell))$. The exponential factor is at most e, and $v \le 2\log(e+\ell)$. Backward differentiation now gives the readout term $2sH_L$, propagated derivative, changed-matrix term $2s^3k_{j+1}^2H_j$, and changed-gate term $4et_2N_*$. They are the displayed recursion. Forward differentiation gives the first bound. This growing real exponent is a deterministic budget inequality and does not use growing deletion sets. □

Removing budgets on complete independently stopped paths. The domain checks below will supply a bounding rectangle of side $O(\ell)$, uniformly Lipschitz retractions commuting with real anchors, and normalized complex-minus-real coefficient radius $D_n = O(B_n/\sqrt{\ell})$ with global modulus $O(B_n)$. Claim 3 then gives mean $O((\log(e+\ell))^{3/2}/\sqrt{\ell})$ and vanishing tail radius. Every fixed exponential moment of these corrections tends to one. Fixed squared-exponential moments do also, by integrating the sub-Gaussian tail; sample RMS uses Jensen. Notice that the modulus here is polylogarithmic, not an arbitrary polynomial in n.

For singleton/common-cavity differences, first extend each path by its own retraction, then project their difference onto the deterministic ball of radius $4n^{1/100}$. Both paths omit the paired root. Projection preserves this independence and equals the actual difference on the successful prefix. Its normalized Gaussian radius is $4n^{-49/100}$, and its modulus is at most a fixed power of n, by the already proved derivative bounds. The Gaussian rectangle claim again gives moments tending to one at every fixed order. The two coefficient paths need not be independent of each other.

Fix a positive integer moment order u, a sample and a layer. Distinct roots in an empirical uth moment are independent given their common cavity. Apply Cauchy–Schwarz to separate products of main real references from products of corrections, and then Hölder over the finitely many roots and four reference families for the corrections. Main moments use exponent at most $8\eta_*$ in (56); corrections use fixed orders proportional to u and tend to one. The limiting per-root base is therefore below sixteen. Drop full-event indicators only after dominating by these nonnegative, globally defined cavity bounds. A tuple with $k < u$ distinct roots has normalized multiplicity $O_u(n^{k-u})$. On (57), repeated weights cost at most $e^{2u\eta_* K_{\mathrm{src}}\sqrt{\ell}} = n^{o(1)}$, so collision contributions vanish. Local exceptional events are superpolynomially small, and the stopped budgets are bounded.

Cavities cannot stop first on the successful full prefix: their coordinate differences are $n^{-1/30}$, maxima and poles have strict transferred margins, and

$$\mathcal{H}_a^{\mathrm{cavity}} \le e^{2\eta_* n^{-1/30}}\mathcal{H}_a + O_u(1/n) < 2\mathcal{B}.$$

Operator and activity margins are established in the domain checks. If a full budget is hit, some sample/layer mean is at least $\mathcal{B}/L$. Markov's inequality and the fixed sample/layer union give

$$\limsup_{n\to\infty} \Pr\{\text{a full budget is hit}\} \le mL(16L/\mathcal{B})^u. \tag{58}$$

Take the width limit at each fixed u, then the infimum over positive integers. Since $16L/\mathcal{B} < 1$, the result is zero. The label allowance precedes the moment order and does not shrink with confidence or sample count. This is the shared probability step.

4. Verification of the working domains.

Verification of the working whole-sphere estimates. Lemma 5.1 independently supplies the real trajectory, strict operator caps below nine, and real activity at most $S/2$. The same holds for the independently tested cavities. Impose the eventual deterministic gates

$$n^{-1} \le \min(Y, S, 1), \qquad \ell \ge \max(e^2, 2\mathcal{B}), \qquad \sqrt{\ell} \ge c_t \max\{8, \lambda, 4\mathcal{K}/\log 2, 32YSD_W\}. \tag{59}$$

Follow the real solution to its nearest anchor in $[0, T]$, then at most two short contour pieces of total length $2r_t$. The algebraic residual Gram and the negative-Gram variational generator, divided by their factor two, have norm at most $\mathcal{K}$, by the explicit gradient blocks. No complex positivity is used. Residual growth and products of the base propagators cost at most $e^{4\mathcal{K}r_t} \le 2$. Additional activity is at most $8Yr_t \le S/2$; normalized hidden increments are at most $8YSD_W r_t \le 1/4$. Thus operators have strict margins inside ten and $\rho/S \le \lambda/8$. The total contour length is $O(\ell)$.

These are the premises of the local insertion and trace argument, which improves maxima and response/angular caps. The exact velocity $\partial_t z^{(j)} = -(2/m)\sum_a r_a R_a^{(j)}$ now bounds the complex-time preactivation displacement by $8c_tYSU = a/8$. Every query in the intrinsic tube has the form $u\cosh h + iv\sinh h$ for a real orthonormal frame and $|h| \le r_q$. Its additional displacement is at most $c_qV \le a/8$. Thus the full pole stop improves strictly; the local coordinate comparison transfers this margin to cavities. The passive size stop also improves as already shown.

Under the provisional response stops, Claim 4 bounds normalized complex-minus-real reference radii by $Cr_tB_n = O(B_n/\sqrt{\ell})$, and their global time moduli by CB_n. This holds on complete independently stopped paths: the clipped arguments commute with the real-anchor map as in (39). Their parameter rectangle has side $O(\ell)$. The budget-removal argument therefore applies, with probability of failure tending to zero. The logical order is provisional deterministic bounds, local insertion, maximum/response improvement, strict pole/cavity transfer, and then complete-path Gaussian moments and budget removal. None of these steps conditions an omitted Gaussian root on full-network survival.

Before each stop, finite-width bounded parameters and strict separation from the activation singularities give local holomorphic ODE continuation. Compactness and uniqueness glue the local continuations on the nested rectangles. At a first exit the strict derivative strip allows one-sided estimates; the strict improvements continue the solution across the closed boundary. Query holomorphy follows by forward composition on the quadric. Passive backward fields are then holomorphic by their finite downward recursion, with RMS bounds $Sk_j, S\tau_j$, using the same readout and operator bounds; they require no new probabilistic budget. Initialized operator norm eight proves the four-family coordinate bound. All failures tend to zero, hence each prescribed fixed confidence is attained eventually. □

Claim 5 (Working finite-query time domain and all-time carriers)**.** *Under $Y/\lambda \le \beta^{-30L}$, for any fixed finite list of real unit inputs, eventually with every fixed confidence its predictions are holomorphic on a neighborhood of*

$$[-r, T + r] + i[-r, r], \qquad T = 32\ell/\lambda, \qquad r = 1/(\lambda\sqrt{\ell}),$$

and $|f_n(t, v)| \le SH_L^2$ there. Two independent copies thus satisfy $|f_n - \widetilde{f}_n| \le 32\beta^{6L}Y/\lambda$ on their common event. In particular the Bernstein ellipse of parameter two on $[0, r]$ is contained in this rectangle. On the real all-time trajectory, also

$$\sup_{t\in[0,\infty]} \max_{a,j,i} |k_{a,i}^{(j)}(t)| \le 2SK_{\mathrm{src}}\sqrt{\ell}. \tag{60}$$

The fitted endpoint is included.

Proof. Use the same proof without angular fields or the frame mesh, replacing the Gaussian multiplier by 64. The finite-list response bound is $SU_{\mathrm{fin}}\sqrt{\ell}$. The time-radius coefficient may be any $c \le a/(64YSU_{\mathrm{fin}}) = a/(4\lambda S^2 U_{\mathrm{fin}})$. By Claim 1,

$$\frac{a}{4S^2U_{\mathrm{fin}}} \ge \frac{\beta^{34L-1}}{64} > 1,$$

so $c = 1/\lambda$ is permitted. All other rectangle gates are eventual. Cauchy–Schwarz gives $|f_n| \le SH_L^2$; use $H_L \le \beta^{3L}$ for the difference bound. The parameter-two ellipse on $[0, r]$ has real range $[-r/8, 9r/8]$ and imaginary height $3r/8$, hence lies inside the stated rectangle eventually.

It remains to extend the carrier maximum beyond T; no complex late-time extension is required. Let $\|\Delta\theta\|$ denote the physical parameter norm with blocks $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$. Forward subtraction in the real fitting tube gives feature RMS changes bounded by $P_j\|\Delta\theta\|$. Backward subtraction then gives

$$\max_{a,j} \|k_a^{(j)}(t) - k_a^{(j)}(T)\|_\infty \le C_k n \|\theta(t) - \theta(T)\|,$$

where $C_L^k = 1$, $C_j^k = S\tau_{j+1} + 10sC_{j+1}^k + 10t_2SP_{j+1}k_{j+1}$, and $C_k = \max_j C_j^k \le \beta^{20L}$. The terms are the changed mixer, propagated upper difference, and changed activation gate; RMS-to-coordinate conversions account for the two factors $\sqrt{n}$. The real segment has bounded operators and real activations, so this subtraction needs no complex strip hypothesis. Lemma 5.1 gives $\|\theta(t) - \theta(T)\| \le 2Ye^{-16\ell}/\sqrt{\lambda}$. Consequently the last display is eventually at most $SK_{\mathrm{src}}\sqrt{\ell}$, including at $t = \infty$. Add (57) at T. □

Claim 6 (Working residual-adapted finite-panel domain)**.** *Fix the training inputs and any finite list of passive inputs before initialization. Use the same allowance as the working whole-sphere estimates, and let*

$$\nu = \lambda/4, \quad T = 32\ell/\lambda, \quad r_n = \frac{a}{64YSU_{\mathrm{fin}}\sqrt{\ell}},$$
$$h(t) = \frac{1}{2\mathcal{K}}\log(1 + 2\mathcal{K}r_n e^{\nu t}), \quad d_h = \frac{\nu}{2\mathcal{K}}, \quad T_+ = T + h(T) + r_n.$$

Eventually with every fixed confidence, the required forward and backward fields are holomorphic on a neighborhood of

$$D_1 = \{x + iy : -h(0) \le x \le T_+, \quad |y| \le h(\max(x, 0))\},$$

with the same operator, RMS, training-maximum, and four-family coordinate bounds as in the working whole-sphere estimates; the response bound is $SU_{\mathrm{fin}}\sqrt{\ell}$. In particular it suffices to retain forward families on all declared inputs and backward families only on training inputs. For $0 \le t \le T$, the disk of radius $\varrho(t) = h(t)/(2(1 + d_h))$ about t lies in D_1. The adaptive subdivision $t_0 = 0$, $t_{b+1} = \min(T, t_b + \varrho(t_b)/2)$ has at most

$$J \le 1 + 40960\frac{U_{\mathrm{fin}}}{a}\left(\frac{Y}{\lambda}\right)^2\sqrt{\ell} + 80\frac{\mathcal{K}}{\lambda}\log\left(1 + \frac{4\ell}{\log 2}\right) \tag{61}$$

panels. Here $U_{\mathrm{fin}}/a \le \beta^{27L}$ and $\mathcal{K} \le \beta^{14L}$, independently of the input dimension and the size of the fixed passive list.

Proof. Population RMS and the Gram gap give $\lambda \le H_L^2 \le \mathcal{K}$. Thus $0 \le h' \le d_h \le 1/8$, $h(0) \le r_n$, and $h(t) \le r_n + d_h t$. Define the nested domains by

$$D_s = \{x + iy : -sh(0) \le x \le sT_+, \quad |y| \le sh(\max(x, 0))\}, \quad 0 \le s \le 1.$$

For $X_s(x) = \operatorname{clip}_{[-sh(0), sT_+]} x$, their retractions are

$$P_s(x + iy) = X_s(x) + i \operatorname{clip}_{[-sh(X_s(x)_+), sh(X_s(x)_+)]} y.$$

With $a_0(z) = \operatorname{clip}_{[0,T_+]} \operatorname{Re} z$, $P_s a_0 = a_0 P_s$, and $\operatorname{Lip}(P_s) \le 1 + d_h$. Indeed $|\Delta X_s| \le |\Delta x|$ and the imaginary increment is at most $|\Delta y| + d_h|\Delta x|$; the shear has norm at most $1 + d_h$. Integrate derivatives along the image under P_s of a straight segment to get global coefficient Lipschitz bounds. Convexity of D_s is unnecessary. The real anchors remain monotone and freeze, as required by the moment proof.

We verify the base-propagator premise before Gaussian insertion. Under each cavity's provisional response stop, use Claim 4 with $U_0 = U_{\mathrm{fin}}$. For $G = (mn)^{-1/2}[\nabla_\Theta F_1, \dots, \nabla_\Theta F_m]$, the explicit gradient blocks give

$$\|G\| \le \sqrt{\mathcal{K}}, \qquad \|\dot{G}\| \le C_g B_n \rho, \qquad C_g = J_* \left(1 + \sum_{j=2}^{L} H_{j-1}\right) + 2S^2 \sum_{j=2}^{L} \tau_j q_{j-1}^* + 2S q_L^*. \tag{62}$$

For the second estimate differentiate each column block $m^{-1/2}\delta_a^{(1)} v_a^\top/\sqrt{n}$, $m^{-1/2}\delta_a^{(j)} h_a^{(j-1)\top}/n$, and $m^{-1/2} h_a^{(L)}/\sqrt{n}$, then bound Frobenius norm. The division by $\sqrt{m}$ prevents a sample factor.

The exact residual equation is $\dot{r} = -2G^\top G r$. On a vertical segment based at real $t \ge 0$, norm differentiation and real fitting give

$$\rho(t \pm iv) \le Y e^{-\nu t} e^{2\mathcal{K}v}, \qquad \int_0^{h(t)} \rho(t \pm iv)\, dv \le Y r_n. \tag{63}$$

The second bound is the defining identity for $h(t)$. It also gives $\rho \le Y + 2\mathcal{K} Y r_n \le 2Y$ eventually. The base variational equation is $\dot{P} = -2GG^\top P$, with algebraic transposes. At a real anchor, $A_t = 2G(t)G(t)^\top$ is real symmetric, so $-iA_t$ generates unitary motion. Freeze this generator on the vertical segment and use (62):

$$\|2G(t \pm iv)G(t \pm iv)^\top - A_t\| \le 4\sqrt{\mathcal{K}} C_g B_n \int_0^v \rho(t \pm iu)\, du.$$

Conjugating by the frozen unitary bounds its propagator by

$$\exp\left(4\sqrt{\mathcal{K}} C_g B_n \int_0^{h(t)} (h(t) - v)\rho(t \pm iv)\, dv\right) \le \exp\left(\frac{2C_g}{\sqrt{\mathcal{K}}} B_n Y r_n\right).$$

For the last inequality, integrate the exponential bound in (63); the double integral is at most $Y r_n/(2\mathcal{K})$. Products on ordered disjoint subsegments obey the same bound, since their growth integrals add. Forward real segments contract. The negative-real corner has total exceptional length at most $2r_n$, costing at most $e^{4\mathcal{K} r_n}$. Hence impose the directly checkable eventual gate

$$4\mathcal{K} r_n + (2C_g/\sqrt{\mathcal{K}}) B_n Y r_n < \log(3/2). \tag{64}$$

It supplies the factor-two base premise, without treating a complex Gram matrix as positive. Its left side tends to zero because $B_n r_n = O(\log(e + \ell)/\sqrt{\ell})$.

Also impose (59) with $c_t = a/(64YSU_{\mathrm{fin}})$. Additional activity is at most $8Yr_n$, and normalized parameter increments are at most $8YSD_W r_n \le 1/4$. Thus operators, activity and $\rho/S \le \lambda/8$ meet the shared premises. All canonical anchor contours have length $O(\ell)$, and the bounding rectangle $[-h(0), T_+] + i[-h(T_+), h(T_+)]$ has sides $O(\ell)$. The local map/remainder argument is therefore unchanged: the one-dimensional control entropy is $n^{5/8}\,\mathrm{poly}(\ell)$, endpoint meshes are polynomial, and the same $n^{\cdot 78}$ Gaussian tail dominates them. After the improved response bound, vertical displacement is at most $2YSU_{\mathrm{fin}}\sqrt{\ell}r_n = a/32$; the negative corner costs at most $a/8$. The pole and passive-size stops improve, and cavity transfer is strict.

Finally Claim 4, integrated over these segments, gives the complete-path normalized correction radius

$$D_n = 4Yr_n \max\{2S \max_j q_j^*, J_* B_n/S\} = O(\log(e+\ell)/\sqrt{\ell}).$$

The commuting retractions preserve this estimate and give global modulus $O(B_n)$. Thus the shared Gaussian moment and budget argument applies. Local holomorphic continuation glues because the domain is vertically fibered, retracts onto its real interval, and is simply connected; strict stopped margins continue across its compact boundary. This proves the source event and its confidence.

For the disk claim, if its real coordinate is nonnegative, then $h(x) \ge h(t) - d_h \varrho(t) \ge \varrho(t)$. If it is negative, $t < \varrho(t)$ and $h(t) \le h(0) + d_h t$ imply $\varrho(t) < h(0)$. The right endpoint is below $T + h(T) < T_+$. The radius is nondecreasing with Lipschitz constant $C_r = d_h/[2(1+d_h)]$. Each complete adaptive panel contributes at least $1/(2+C_r)$ to $\int dt/\varrho(t)$, by bounding the radius on that panel by $(1+C_r/2)\varrho(t_b)$. Consequently $J \le 1 + 5\int_0^T dt/h(t)$. Split at $2\mathcal{K}r_n e^{\nu t} = 1$. Before this point use $\log(1+x) \ge x/2$; afterward the logarithm grows at least with slope $\nu/2$. This bounds the integral by

$$\frac{2}{\nu r_n} + \frac{4\mathcal{K}}{\nu} \log\left(1 + \frac{\nu T}{2\log 2}\right).$$

Substitute $\nu = \lambda/4$, $S = 16Y/\lambda$ and T to obtain (61). The coefficient bounds follow from Claim 1 and $a^{-1} \le \beta/16$. □

5. Reduction to the public bounds. We now compare the stronger working estimates with the public statement. The quantitative width and initialization proofs explicitly recall the coefficient recurrences they additionally use. The working whole-sphere time radius is

$$\frac{a}{64YSU\sqrt{\ell}} = \frac{a\lambda}{1024Y^2U\sqrt{\ell}}.$$

Since $U \le \beta^{26L}\sqrt{d+3}$ and $a^{-1} \le \beta/16$,

$$\frac{1024U}{a} \le 64\beta^{26L+1}\sqrt{d+3} \le \beta^{30L}\sqrt{d+3}.$$

The last inequality uses $64 \le \beta^{4L-1}$, valid for $L \ge 2$, $\beta \ge 10$. This gives at least the public time radius. Likewise $V \le \beta^{2L}\sqrt{d+3}$ gives

$$\max\{8, 8V/a\} \le \max\{8, \tfrac{1}{2}\beta^{2L+1}\sqrt{d+3}\} \le \beta^{3L}\sqrt{d+3},$$

so the working query tube contains the stated tube.

The local power bounds yield, for every layer,

$$H_j \le 3\beta^{2j} \le \beta^{3L}, \qquad \max(k_j, \tau_j) \le 3\beta^{4L-1} \le \beta^{4L}.$$

Multiplication by $S = 16Ym/\gamma$ gives the backward bounds. The four-family coordinate envelope obeys

$$M_0 = 10 \max_j (H_j, \tau_j) \le 30\beta^{4L-1} \le \beta^{6L}.$$

For all real times, the fitting statement bounds normalized operators by nine and $\|w\|_2/\sqrt{n} \le 2Y/\sqrt{\lambda}$. To check the remaining population comparison directly, Minkowski and linear activation growth give $\sqrt{Q_{11}^{(j)}} \le b + s\sqrt{Q_{11}^{(j-1)}}$, starting at $Q_{11}^{(0)} = 1$. The local recursion for H_j dominates this recursion, so $\lambda \le Q_{11}^{(L)} \le H_L^2$. Hence $2Y/\sqrt{\lambda} \le SH_L$. The same forward and backward recursions with operator bound nine give $H_j, Sk_j, S\tau_j$ on the whole real trajectory. Their parameter limits give the endpoint bounds as well. This proves (26) in both stated domains.

The working finite-query claim has exactly the public time radius. Its prediction envelope is $SH_L^2 \le 16\beta^{6L}Ym/\gamma$, proving (27). Its all-time training-carrier envelope is $2SK_{\mathrm{src}}\sqrt{\ell} \le 32\beta^{21L}(Ym/\gamma)\sqrt{\ell}$. Substituting the explicit definition of $k^{(j)}$ proves (28), without exporting that auxiliary notation.

For the finite panel, choose $r_b = \varrho(t_b)$ in the working domain's adaptive partition. Its disk containment and half-radius intervals are exactly those in (iii), and its source envelopes have just been bounded in canonical quantities. Its defining coefficients depend only on the width, m, γ, Y, activations and depth, so these choices are deterministic. Finally

$$40960\frac{U_{\mathrm{fin}}}{a} \le 40960\beta^{27L} \le \beta^{30L}, \qquad 80\mathcal{K} \le 80\beta^{14L} \le \beta^{16L},$$

because $40960 \le \beta^{3L}$ and $80 \le \beta^{2L}$. Applying these inequalities to (61) proves (29). For any requested finite collection of the source domains, split the fixed failure allowance and intersect their events. This supplies the single event asserted in the proposition and completes its proof. □

7 Legendre history compression

Lemma 7.1 (Projection identities)**.** *Let $A > 0$ and let $q \ge 1$ be an integer. Set $p_j^A(\xi) = P_j(2\xi/A - 1)$, where P_j is the Legendre polynomial, and let Π_q^A project onto degrees below q in $L^2([0, A])$. For an absolutely continuous Hilbert-valued history u,*

$$\|(I - \Pi_q^A)u\|_{L^2}^2 \le \frac{1}{q(q+1)} \int_0^A \xi(A-\xi)\|u'(\xi)\|^2 \, d\xi, \tag{65}$$

$$\|u(A) - (\Pi_q^A u)(A)\| \le \sqrt{A/(3q)}\, \|u'\|_{L^2}, \qquad \|(\Pi_q^A u)(A)\| \le 64\sqrt{q}\, \|u\|_{L^\infty}. \tag{66}$$

For a fixed absolutely continuous history g, the growing-interval error obeys

$$\frac{d}{dA} \int_0^A \|g - \Pi_q^A g\|^2 \, d\xi = \|g(A) - (\Pi_q^A g)(A)\|^2. \tag{67}$$

Proof. Rodrigues' formula and integration by parts give $\int_0^A p_i^A p_j^A = A\mathbf{1}_{i=j}/(2j+1)$ and $-[\xi(A-\xi)(p_j^A)']' = j(j+1)p_j^A$. The derivative polynomials are orthogonal for the weight $\xi(A-\xi)$. Bessel's inequality followed by the ordinary expansion therefore proves (65); polynomial approximation and increasing finite-dimensional projections give the stated absolutely continuous, Hilbert-valued version. Integration against each lower polynomial also gives $P_j' = \sum_{i<j,\ j-i \text{ odd}} (2i+1)P_i$. Thus the endpoint kernel is

$$K_q^A(A, \xi) = \frac{1}{A} \sum_{j<q} (2j+1)p_j^A(\xi) = \tfrac{1}{2}[(p_q^A)'(\xi) + (p_{q-1}^A)'(\xi)].$$

Its integral is one, and integration by parts gives $u(A) - (\Pi_q^A u)(A) = \frac{1}{2}\int_0^A (p_q^A + p_{q-1}^A)u'$. The squared L^2 norm of this multiplier is $A[(2q+1)^{-1} + (2q-1)^{-1}]/4 \le A/(3q)$.

For completeness the needed kernel L^1 estimate has a short one-dimensional proof. Put $v(\theta) = \sqrt{\sin\theta} P_k(\cos\theta)$ and $a_k = (k+1/2)^2 + (4\sin^2\theta)^{-1}$. The Legendre equation implies $v'' + a_k v = 0$, whence $\partial_\theta(v^2 + v'^2/a_k) = -a_k' v'^2/a_k^2 \ge 0$ on $(0, \pi/2)$. At the midpoint use $P_{2j}(0) = (-1)^j \binom{2j}{j}/4^j$, $P_{2j+1}'(0) = (2j+1)P_{2j}(0)$, and $(\binom{2j}{j}/4^j)^2 \le 1/(j+1)$; the last inequality follows by induction. The midpoint energy is at most $4/k$. Differentiating $P_k = v/\sqrt{\sin\theta}$ now yields

$$|\partial_\theta P_k(\cos\theta)| \le 3\sqrt{k}(\sin\theta)^{-1/2} + 2k^{-1/2}(\sin\theta)^{-3/2}.$$

On $[1/k, \pi/2]$ its integral is at most $3\pi\sqrt{k} + 8$, using $\sin\theta \ge 2\theta/\pi$. On $[0, 1/k]$ use $|P_k'| \le k(k+1)/2$ to obtain at most $1/2$. This derivative bound follows from the derivative expansion and $|P_j| \le 1$, the latter following from the integral formula $P_j(\cos\theta) = \pi^{-1}\int_0^\pi (\cos\theta + i\sin\theta\cos\psi)^j \, d\psi$ (expand its generating function). Reflection gives total variation at most $36\sqrt{k}$. The endpoint kernel consequently has L^1 norm at most $36\sqrt{q} \le 64\sqrt{q}$; $q = 1$ is the averaging kernel. Finally differentiation of the squared projection error proves (67): the interior term vanishes because $\partial_A \Pi_q^A g$ is still a polynomial of degree below q. □

Proposition 7.2 (Legendre approximation at the prescribed size)**.** *Eventually at each fixed confidence, there is an initialization-only Legendre model with*

$$q = \left\lceil C\left(\frac{m}{\gamma}\right)^3 n^{1/4}[\log(en)]^2 e^{\sqrt{\log(en)}/2} \right\rceil \tag{68}$$

such that

$$\sup_{t\in[0,\infty],\ \|x\|=\sqrt{d}} |f_{\mathrm{Leg}}(t,x) - f_n(t,x)| \le \frac{CY}{\sqrt{n}[\log(en)]^3}. \tag{69}$$

Its moving storage is at most

$$Cm(m/\gamma)^3 n^{5/4}[\log(en)]^2 e^{\sqrt{\log(en)}/2};$$

the fixed storage is the additional $(L-1)n^2$.

Proof. **Local notation and construction.** All auxiliary quantities below belong to this proof. Put

$$\lambda = \gamma/m, \qquad H = \max_{0\le j\le L}\sqrt{Q_{11}^{(j)}}, \qquad s = \max\left\{1, \max_j \sup_{|\operatorname{Im} z|\le a/2} |\phi_j'(z)|\right\}, \qquad b = \max_j |\phi_j(0)|,$$

and define $D_j = s(9s)^{L-j}$. These definitions are made here; no coefficient from the proof of Lemma 5.1 is imported. In particular $H \le (2\beta)^L$, by the affine population-RMS recursion, and $s, b \le \beta$.

The Legendre model constructed here shares the realized dense initialization and physical time. Here the whole-sphere version of (6) is abbreviated by

$$\|g\|_* = \sup_{t\in[0,\infty],\,\|x\|=\sqrt{d}} |g(t,x)|,$$

including the fitted limit at $t = \infty$. Write $\widehat{f} = f_{\mathrm{Leg}} = n^{-1}\widehat{w}^\top \widehat{h}^{(L)}$, $\widehat{r}_a = \widehat{f}(x_a) - y_a$ and $\widehat{\rho} = \|\widehat{r}\|_2/\sqrt{m}$. We first construct the model for an arbitrary integer $q \ge 1$ and then take the order in the statement.

At each hidden interface $\ell = 2, \dots, L$, sample a, and order $j = 0, \dots, q-1$, retain $\bar{h}_{a,j}^{(\ell-1)}, \bar{\delta}_{a,j}^{(\ell)} \in \mathbb{R}^n$. Together with $\widehat{W}^{(1)}$, $\widehat{w}$, and a scalar τ, these are the moving state. Reconstruct

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{mn\tau} \sum_{a=1}^{m} \sum_{j=0}^{q-1} (2j+1) \bar{\delta}_{a,j}^{(\ell)} \bar{h}_{a,j}^{(\ell-1)\top}. \tag{70}$$

Use these matrices in the ordinary forward pass and in $\widehat{k}_a^{(L)} = \widehat{w}$, $\widehat{\delta}_a^{(\ell)} = \phi'_\ell(\widehat{z}_a^{(\ell)}) \odot \widehat{k}_a^{(\ell)}$, $\widehat{k}_a^{(\ell)} = \widehat{W}^{(\ell+1)\top} \widehat{\delta}_a^{(\ell+1)}$. The autonomous physical-time equations are

$$\begin{aligned}
\dot{\widehat{W}}^{(1)} &= -\frac{2}{m} \sum_a \widehat{r}_a \widehat{\delta}_a^{(1)} v_a^\top, & \dot{\widehat{w}} &= -\frac{2}{m} \sum_a \widehat{r}_a \widehat{h}_a^{(L)}, \qquad \dot{\tau} = \widehat{\rho}, \\
\dot{\bar{h}}_{a,j}^{(\ell-1)} &= \widehat{\rho} \widehat{h}_a^{(\ell-1)} - \frac{\widehat{\rho}}{\tau} \left(j \bar{h}_{a,j}^{(\ell-1)} + \sum_{i<j} (2i+1) \bar{h}_{a,i}^{(\ell-1)} \right), \\
\dot{\bar{\delta}}_{a,j}^{(\ell)} &= \widehat{r}_a \widehat{\delta}_a^{(\ell)} - \frac{\widehat{\rho}}{\tau} \left(j \bar{\delta}_{a,j}^{(\ell)} + \sum_{i<j} (2i+1) \bar{\delta}_{a,i}^{(\ell)} \right).
\end{aligned} \tag{71}$$

Initially $\tau = 1$, $\widehat{W}^{(1)} = W_0^{(1)}$, $\widehat{w} = 0$, $\bar{h}_{a,0}^{(\ell-1)} = h_{0,a}^{(\ell-1)}$, and all other moments vanish. No equation divides by a residual. At $\widehat{\rho} = 0$ all velocities vanish, and the vector field is locally Lipschitz for $\tau > 0$. The exact inventory is

$$S_{\text{moving}} = n(d+1) + 1 + 2(L-1)mnq, \qquad S_{\text{fixed}} = (L-1)n^2. \tag{72}$$

The latter entries are the initialized mixers. Data storage is additional; reconstructed matrices need not be retained independently.

Moment identities and the physical defect. For use in this proof, let $p_j^A(\xi) = P_j(2\xi/A - 1)$ and let Π_q^A be the degree-below-q orthogonal projection in $L^2([0, A])$, as in Lemma 7.1. The histories in the proof are specified by $h_a(\tau(t)) = \widehat{h}_a(t)$ and $b_a(\tau(t)) = (\widehat{r}_a/\widehat{\rho})\widehat{\delta}_a(t)$. On the prefix $[0, 1]$ they equal the initialized forward feature and zero, respectively. The backward history is continuous at the join because the initial readout is zero. The moments in (71) are exactly $\int_0^\tau p_j^\tau h_a$ and $\int_0^\tau p_j^\tau b_a$; differentiating these integrals proves the equations, including their factors j and $2i+1$. Differentiating their bilinear projection pairing gives the exact defect

$$\dot{\widehat{W}}^{(\ell)} = -\frac{2}{mn} \sum_a \widehat{r}_a \widehat{\delta}_a^{(\ell)} \widehat{h}_a^{(\ell-1)\top} + \mathcal{E}_\ell, \qquad \mathcal{E}_\ell = \frac{2\widehat{\rho}}{mn} \sum_a e_{b,a}^{(\ell)} e_{h,a}^{(\ell-1)\top}, \tag{73}$$

where $e_g = g(\tau) - (\Pi_q^\tau g)(\tau)$. Indeed the derivative of the pairing of the two projected histories equals the full endpoint product minus the product of their endpoint errors, by orthogonality. These identities hold until a residual zero and extend there by stationary continuation. Histories are proof objects, not extra state.

Claim 7 (Order-independent physical bounds)**.** *On* (17)*, under $Y \le \lambda \beta^{-30L}$, every finite order is globally defined, has convergent physical parameters and moments, and fits the labels. Its operator and feature bounds are those of* (18)*, and*

$$\widehat{\rho}(t) \le Y e^{-\lambda t/4}, \qquad \tau(t) \le 1 + 4Y/\lambda, \qquad \|\widehat{w}(t)\|/\sqrt{n} \le 4Y/\sqrt{\lambda}. \tag{74}$$

Verification of the local claim. Let $\sigma = 8Y/\lambda$, $R = \sigma\sqrt{\lambda}$, and $\kappa = \lambda/4$. Define, within the present construction proof,

$$C_1 = 4s^2 D_1^2, \quad C_\ell = 3s^2(64H^4 D_\ell^2 + 82C_{\ell-1}), \quad E_* = 130\sum_{\ell=2}^{L} D_\ell\sqrt{C_{\ell-1}}.$$

It suffices that

$$\sigma \le \min\left\{1, \frac{1}{64H^2 D_1}, \frac{1}{\sqrt{8\sqrt{C_L}}}, \left(\frac{1}{8H^2 D_1 E_*}\right)^{2/7}\right\}. \tag{75}$$

The present label cap implies this: $D_\ell \le \beta^{2L}$, $\sqrt{C_\ell} \le \beta^{8L}$, and $E_* \le \beta^{12L}$. For the middle estimate unroll $C_\ell = 192s^2H^4D_\ell^2 + 246s^2C_{\ell-1}$, bound $192s^2, 246s^2 \le \beta^5$, $H^4D_\ell^2 \le \beta^{10L}$, and sum at most L terms. For the last use $130L \le (\beta^L)^2$. Substitution gives a lower bound β^{-8L} for the minimum in (75), whereas $\sigma \le 8\beta^{-30L}$.

Stop at clock activity σ, readout RMS R, operator norm nine, or feature RMS $2H$. Before the stop, $\|\widehat{\delta}_a^{(\ell)}\|/\sqrt{n} \le D_\ell R$. Projection contraction, sample Cauchy–Schwarz, and the zero backward prefix give

$$\|\widehat{W}^{(\ell)} - W_0^{(\ell)}\|_F \le 4HD_\ell R\sqrt{(1+\sigma)\sigma}, \qquad \|\widehat{W}^{(1)} - W_0^{(1)}\|_F/\sqrt{n} \le 2D_1 R\sigma.$$

These are strictly below one by (75) and $\sqrt{\lambda} \le H$. Let Z_ℓ be the supremum, over queries and stopped horizons, of $n^{-1}\int_1^{\tau(t)} \|\partial_\xi h^{(\ell)}(\xi, v)\|^2 d\xi$. The polynomial endpoint norm on $L^2([0, A])$ is $q/\sqrt{A}$, since the sum of squared normalized basis values at A is $\sum_{j<q}(2j+1)/A = q^2/A$. Thus the aggregate backward endpoint error is at most $(q+1)D_\ell R$. By (67), (65), and (73),

$$\int_0^t \widehat{\rho}\|\mathcal{E}_\ell/\widehat{\rho}\|_F^2 du \le 2(1+\sigma)^2 D_\ell^2 R^2 Z_{\ell-1}.$$

Here the forward prefix error is zero, $\xi(A-\xi) \le A^2/4$, and $(q+1)/q \le 2$. The first-layer chain rule and the three later derivative terms (gradient update, defect, and propagated feature derivative) now give

$$Z_1 \le 4\sigma s^2 D_1^2 R^2, \qquad Z_\ell \le 3s^2\{64\sigma H^4 D_\ell^2 R^2 + [81 + 32H^2 D_\ell^2 R^2]Z_{\ell-1}\} \le C_\ell \sigma R^2.$$

The last step uses $32H^2D_\ell^2R^2 < 1$. Consequently every feature moves in RMS by at most $\sqrt{\sigma Z_\ell} \le \sqrt{C_L}\sigma^2\sqrt{\lambda} \le \sqrt{\lambda}/8$. This excludes the feature boundary and preserves the top Gram gap $\lambda/4$ by singular-value perturbation.

For the pointwise defect use the sharper bounds (66). The two endpoint errors are at most $65\sqrt{q}D_\ell R$ and $\sqrt{(1+\sigma)/(3q)}\sqrt{Z_{\ell-1}}$, respectively. Hence

$$\sum_{\ell=2}^{L} \|\mathcal{E}_\ell\|_F \le E_*\sqrt{\sigma}R^2\widehat{\rho}. \tag{76}$$

The prediction Jacobian applied to this hidden defect has sample RMS at most $2HD_1R\sum\|\mathcal{E}_\ell\|_F \le \lambda\widehat{\rho}/4$, by the last entry in (75). Let J be the Jacobian of the training prediction vector in mobility coordinates $(W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$, evaluated at the reconstructed state, and put $\Gamma = JJ^\top/m$. The mean full tangent Gram then gives $\dot{\widehat{r}} = -2\Gamma\widehat{r} + J\mathcal{E}$ with $\Gamma \succeq \lambda I/4$. Thus $\dot{\widehat{\rho}} \le -\kappa\widehat{\rho}$ and $\int_t^T \widehat{\rho} \le \widehat{\rho}(t)/\kappa$ on every stopped interval. The clock activity is at most $\sigma/2$.

Let $\mathcal{F}$ be the dense gradient vector field at the reconstructed parameters, in the norm of Lemma 5.1. Its squared norm is $4\widehat{r}^\top\Gamma\widehat{r}/m \geq \lambda\widehat{\rho}^2$, and the defect estimate implies $-\partial_t\widehat{\rho}^2 \geq \|\mathcal{F}\|_{\mathrm{par}}^2/2$. Multiplying by $e^{\kappa t}$, integrating by parts, and using $\widehat{\rho} \leq Ye^{-\kappa t}$ gives $\int e^{\kappa t}\|\mathcal{F}\|_{\mathrm{par}}^2 dt \leq 4Y^2$. Weighted Cauchy–Schwarz then gives $\int \|\mathcal{F}\|_{\mathrm{par}} \leq 2Y/\sqrt{\kappa} = R/2$. The readout has no defect, so its boundary is excluded as well. Every stopped inequality improves strictly. The integral formulas bound all moments at finite n, q, proving global continuation. Uniqueness excludes a finite first residual zero when $Y > 0$, since zero residual is an equilibrium of the entire stored-state ODE. The integrable gradient and defect lengths yield physical limits; the moment equations also have integrable velocities. This proves (74) and fitting at the limit. This finishes the local fitting claim. The coefficients C_j just defined remain private to this construction proof.

Comparison with dense training. Intersect the initialization event and the all-time training-carrier bound (28) of Proposition 6.1. Set $X = \beta^L$, $z = Y/\lambda$, $\chi = 1 + \lambda^{-1}$, $u = \sqrt{\log(en)}$, and $M = 32X^{21}zu$. Write $\rho_D = \|r\|_2/\sqrt{m}$ for the dense residual RMS. For the backward comparison at any parameter state, define its pre-gated training responses by $k_a^{(L)} = w$ and $k_a^{(j)} = W^{(j+1)\top}\delta_a^{(j+1)}$ for $j < L$. Only actual dense training carriers, not closure carriers or passive query carriers, are bounded by M. Throughout this proof $z \leq X^{-30}$, $X \geq 100$. Use the Euclidean mobility coordinates $(W^{(1)}/\sqrt{n}, W^{(2)}, \ldots, W^{(L)}, w/\sqrt{n})$, whose Euclidean norm is the parameter norm of Lemma 5.1. Let e be the Euclidean discrepancy and D the sum of its $L+1$ block norms, so $D \leq \sqrt{L+1}e$.

We first verify the signed comparison with no reciprocal-gap factor inserted into its Jacobian coefficient. Put $T_0 = 9s$, $R = 8z\sqrt{\lambda}$, $B_\delta = sT_0^{L-1}R$, and define

$$
\begin{gathered}
Q_1 = b + 9s, \quad Q_\ell = b + 9sQ_{\ell-1}, \quad Q = \max(1, Q_1, \ldots, Q_L), \\
F_{\mathrm{seg}} = QT_0^{L-1}, \quad P = 1 + Q(L-1), \\
d_0 = sT_0^{L-1}(1 + B_\delta), \quad d_1 = Lt_2F_{\mathrm{seg}}T_0^{L-1}, \\
K_0 = \sqrt{L+1}\{Pd_0 + [1 + (L-1)B_\delta]sF_{\mathrm{seg}}\}, \qquad K_1 = \sqrt{L+1}Pd_1,
\end{gathered}
$$

where $t_2 = \max\{1, \max_\ell \sup_{|\operatorname{Im} z| \leq a/2} |\phi_\ell''(z)|\} \leq \beta$. On the straight parameter segment from dense to closure, matrix and readout norms keep their bounds by convexity; linear growth gives feature RMS at most Q. Forward subtraction therefore bounds the preactivation difference at segment fraction $\vartheta \in [0,1]$ by $F_{\mathrm{seg}}\vartheta D$. In backward subtraction use

$$
\delta_\vartheta - \delta_D = \phi'(z_\vartheta) \odot (k_\vartheta - k_D) + [\phi'(z_\vartheta) - \phi'(z_D)] \odot k_D.
$$

Propagation costs T_0, a changed mixer acts on a response of RMS at most B_δ, and the changed gate uses only the dense carrier M. Induction gives response difference at most $(d_0 + d_1M)\vartheta D$. Writing $f_a(\theta) = f_\theta(x_a)$, the gradient blocks of a training output are $\delta_a^{(1)}v_a^\top/\sqrt{n}$, $\delta_a^{(\ell)}h_a^{(\ell-1)\top}/n$, and $h_a^{(L)}/\sqrt{n}$. Subtract each hidden product as $(\delta_\vartheta - \delta_D)h_\vartheta^\top + \delta_D(h_\vartheta - h_D)^\top$. Their block bounds give $\|\nabla f_a(\theta_\vartheta) - \nabla f_a(\theta_D)\| \leq (K_0 + K_1M)\vartheta e$. Segment integration proves both hypotheses of Lemma 5.2 with $K = K_0 + K_1M$. The residual integrals are at most $2z$ and $4z$, so for every finite horizon

$$
\sup_{t \leq T} D(t) \leq \mathcal{A}_n\mathcal{D}_T, \quad \mathcal{A}_n = \sqrt{L+1}e^{14(K_0+K_1M)z}, \quad \mathcal{D}_T = \int_0^T \sum_{\ell=2}^{L} \|\mathcal{E}_\ell\|_F dt. \tag{77}
$$

In particular the carrier contribution is additive in K, not multiplied once per layer.

Here are coefficient bounds needed to control the forcing. Their dependence on the gap and labels is displayed before any simplification. Put $a_0 = 4z$, $\kappa = \lambda/4$, $F_z = 2HT_0^{L-1}$, $P_0 = 1 + 2H(L-1)$,

retain the local coefficients C_ℓ defined above, and define

$$\begin{aligned}
B_d &= sT_0^{L-1}(1+B_\delta) + Lt_2F_zT_0^{L-1}M,\\
V_z &= 2F_zB_\delta P_0,\\
V_\delta &= T_0^{L-1}[4sH + 4sH(L-1)B_\delta^2 + Lt_2MV_z],\\
T_1 &= 2D_1R, \qquad T_\ell = 4HD_\ell R + 130D_\ell\sqrt{8zC_{\ell-1}}R^2,\\
V_1 &= sT_1, \qquad V_\ell = s(2HT_\ell + 9V_{\ell-1}), \qquad V_h = \max_\ell V_\ell,\\
C_c &= 4\left(4H^2 + D_1^2R^2 + 4H^2R^2\sum_{\ell=2}^{L} D_\ell^2\right) + \lambda/2,\\
F_h &= V_ha_0\sqrt{(1+a_0)/2},\\
b_0 &= \sqrt{1+a_0}(Y/\kappa)V_\delta + 2B_\delta\sqrt{a_0}, \qquad b_1 = 2\sqrt{1+a_0}\,C_cB_\delta/\kappa,\\
C_f &= 2H + RsF_z.
\end{aligned}$$

The meanings follow directly from the recursions: B_dD bounds the response discrepancy between the two actual trajectories; $V_z\rho_D$ and $V_\delta\rho_D$ bound dense preactivation and response speeds; $V_h\widehat{\rho}$ bounds closure feature speeds, using (76). Finally $\|\partial_t(\widehat{r}/\widehat{\rho})\|_m \le C_c$ follows by differentiating the normalized residual and bounding its tangent Gram by its trace and its defect by $\lambda\widehat{\rho}/4$.

We spell out sufficient power bounds to fix the exponential rate. The preceding finite recursions imply

$$\begin{aligned}
&K_0 \le X^9, \quad K_1 \le X^{13}, \quad 14(K_0+K_1M)z \le X^{10}z + X^{36}z^2u,\\
&V_h \le X^{20}R, \quad C_c \le X^{11}, \quad B_d \le X^{33}(1+zu), \quad V_\delta \le X^{40}(1+zu),\\
&F_h/Y \le X^{22}z\sqrt{\chi}, \quad b_0 \le X^{42}z(1+zu), \quad b_1 \le X^{16}z\sqrt{\chi}, \quad C_f \le X^7.
\end{aligned} \tag{78}$$

To check the first line, $Q \le X^2$, $F_{\mathrm{seg}} \le X^4$, $d_0 \le 2X^3$, $d_1 \le X^8$, $P \le X^4$ give the first two bounds; insert $M \le 32X^{21}zu$ and use $448 \le X^2$. For the second line, $R \le X^{-27}$, $\sqrt{C_\ell} \le X^8$, and $D_\ell \le X^2$ give $T_\ell \le X^{12}R$. The V_ℓ recursion has forcing at most $2X^{15}R$, propagation at most $T_0^L \le X^2$, and at most $L \le X$ terms. The trace defining C_c gives its stated bound. Also $F_z \le X^5$, $P_0 \le X^4$, and $V_z \le 2X^{12}R$. The non-carrier and carrier terms of V_δ are at most $4LX^5$ and $64LX^{36}Rzu$, respectively; those of B_d are at most $2X^3$ and $32LX^{29}zu$. For the final line retain the normalization: $F_h/Y \le 32X^{20}z/\sqrt{\lambda}$, $b_1 \le 96X^{14}z/\sqrt{\lambda}$, and $b_0 \le 6X^{40}z(1+zu) + 32X^5z^{3/2}$. These inequalities imply (78) because $X \ge 100$, $z \le 1$, and $\lambda^{-1/2} \le \sqrt{\chi}$. In particular

$$\mathcal{A}_n \le eXe^u, \qquad \log\mathcal{A}_n = O_{\mathrm{problem}}(\sqrt{\log(en)}). \tag{79}$$

The universal coefficient of u follows from $X^{36}z^2 \le 1$; no extra power of m/γ occurs in that exponent.

By (65), the forward history tail on $[0,A]$, $A \le 1+a_0$, has aggregate RMS L^2 norm at most F_h/q: use its clock derivative bound V_h and $\int_1^A \xi(A-\xi)d\xi \le A(A-1)^2/2$. For the backward history compare it with $\widetilde{b}_a(\tau(t)) = (\widehat{r}_a/\widehat{\rho})\delta_{D,a}(t)$, with zero prefix. Its physical derivative has aggregate RMS at most $C_cB_\delta + V_\delta\rho_D$. Freeze it after $t_q = 2\log q/\kappa$ (or the current horizon if earlier). Since $A - \tau(t) \le \widehat{\rho}(t)/\kappa$, the weighted derivative energy cancels the inverse clock speed and is at most

$$\frac{1+a_0}{\kappa}\left[2t_qC_c^2B_\delta^2 + 2V_\delta^2\int_0^{t_q}\rho_D(t)^2dt\right].$$

The frozen tail is therefore at most $(b_1\sqrt{\log q}+\sqrt{1+a_0}(Y/\kappa)V_\delta)/q$. The freezing error is at most $2B_\delta\sqrt{a_0}e^{-\kappa t_q/2}=2B_\delta\sqrt{a_0}/q$. The difference between closure and dense response histories contributes at most $B_d\sqrt{a_0}\sup_{t\le T}D(t)$; the normalized residual has sample norm one. All aggregate history norms here use the factor $(mn)^{-1/2}$.

Use Cauchy–Schwarz in samples and then in $dA=\widehat{\rho}\,dt$ in (73). The initial projection errors vanish, so (67) gives

$$\mathcal{D}_T\le 2(L-1)F_h\left[\frac{b_0+b_1\sqrt{\log q}}{q^2}+\frac{B_d\sqrt{a_0}\sup_{t\le T}D(t)}{q}\right]. \tag{80}$$

For $q\ge q_{\mathrm{abs}}:=4(L-1)F_hB_d\sqrt{a_0}\mathcal{A}_n$, (77) absorbs the last term with factor at most one half. Whole-sphere forward/readout subtraction costs C_fD. The coefficients do not depend on T, so increasing horizons and the already constructed physical limits yield

$$\|f_{\mathrm{Leg}}-f_n\|_*\le\frac{4(L-1)C_f\mathcal{A}_nF_h}{q^2}(b_0+b_1\sqrt{\log q}).$$

By the ledger its numerator divided by Y is at most $eX^{75}z^2\chi(1+zu)e^u\sqrt{\log(eq)}$. The cap and $\lambda\le H^2\le X^4$ imply $Y\le X^{-26}\le 1$, so $z\le\lambda^{-1}\le\chi$. Consequently

$$\|f_{\mathrm{Leg}}-f_n\|_*\le CY\chi^6e^{\sqrt{\log(en)}}\frac{\sqrt{\log(en)\log(eq)}}{q^2}\qquad(q\ge q_{\mathrm{abs}}). \tag{81}$$

This enlargement preserves the displayed sample/gap power without spending the label cap to reduce it. All coefficients except $\mathcal{A}_n$ grow at most polynomially in u, hence $q_{\mathrm{abs}}=n^{o(1)}$ for the fixed problem. The prescribed order (68) eventually exceeds it. Since $m/\gamma\ge 1$, $\chi\le 2m/\gamma$, and $\log(eq)\le C\log(en)$ eventually, substitution proves (69). The count follows from (72); nd and rounding are absorbed only after taking sufficiently large width for the fixed problem. One event supports every order used here, so no order-dependent failure union is needed. All initialization data are the specified dense initialization and initialized training features. □

8 An actual dense-trajectory fluctuation

Lemma 8.1 (Derivative to trajectory)**.** *Let $r>0$ and $M\ge 0$. If the scalar function g is holomorphic on a neighborhood of the filled parameter-two Bernstein ellipse for $[0,r]$ and bounded there by M, then for every integer $N\ge 1$,*

$$\sup_{0\le t\le r}|g(t)|\ge\frac{r}{2N^2}|g'(0)|-24M2^{-N}. \tag{82}$$

Proof. Set $t=r(x+1)/2$. The Joukowski map $x=(z+z^{-1})/2$ and Cauchy's coefficient formula give Chebyshev coefficients $|a_k|\le 2M2^{-k}$. The degree-N truncation has uniform tail at most $2M2^{-N}$; its physical derivative tail at zero is at most $4Mr^{-1}2^{-N}(N^2+4N+6)$, by summing $\sum_{k>N}k^22^{-k}$. For a degree-N polynomial, including complex coefficients, $|P'(-1)|\le N^2\|P\|_{[-1,1]}$. Indeed interpolate at $x_j=\cos(j\pi/N)$: the endpoint derivative weights have alternating signs, and their absolute sum equals $|T_N'(-1)|=N^2$ by evaluating on $T_N(x_j)=(-1)^j$. The sign assertion follows directly from the products defining the Lagrange basis; the endpoint weight is $\sum_{j<N}(-1-x_j)^{-1}<0$. Apply this inequality to the truncation, include the physical factor $2/r$, and add the two tails. Rearranging gives error coefficient $4+8/N+12/N^2\le 24$, as required. □

Proposition 8.2 (Positive-time dense variability)**.** *Under the assumptions of Section 2, for every fixed $0 < \delta < 1$, there is a constant $c_{\phi,L,\delta} > 0$, depending only on the activations, depth and this confidence, such that all sufficiently large individual widths satisfy*

$$\Pr\left\{\|f_n - \widetilde{f}_n\|_{\mathcal{X}} \geq c_{\phi,L,\delta}\frac{Y\sqrt{\gamma}}{\sqrt{n}[\log(en)]^{5/2}}\right\} \geq 1 - \delta. \tag{83}$$

This holds for either the whole sphere or any declared panel containing the training inputs. The witness is at a positive training time; no endpoint lower bound is asserted.

Proof. All moments and fluctuation variables introduced here are private to this proof. Let $G \sim N(0,1)$, define

$$\mu_{2,j} = Q_{11}^{(j)} \quad (j = L-1, L), \qquad \mu_4 = \mathbb{E}\phi_L(\sqrt{\mu_{2,L-1}}G)^4,$$

and write Φ for the standard normal distribution function. Let $Z \sim N(0, Q^{(L-1)})$, $U_a = \phi_L(Z_a)$ and $Q = \mathbb{E}UU^\top = Q^{(L)}$. We first establish the innovation and conditional fluctuation needed below, then convert an initial derivative into a positive-time trajectory separation.

Claim 8 (A final-layer innovation)**.** *For $m \geq 2$ and $Y > 0$, some deterministic training index a_* satisfies*

$$\mathrm{Var}\{U_{a_*}(U^\top y)\} \geq \frac{\gamma^3\mu_{2,L}}{16\mu_4}Y^2 > 0. \tag{84}$$

Verification. All fourth moments are finite by linear growth. The positive gap gives $\mu_{2,L} > 0$ and $\mu_4 > 0$. Put $c = Qy$, $A = \|c\|$, $T = \operatorname{tr} Q$, $M_4 = \mathbb{E}\|U\|^4$, and $D = T - c^\top Qc/A^2 > 0$. The last inequality holds because $m \geq 2$ and $Q \succeq \gamma I$. For $V = \mathbb{E}\|U(U^\top y) - c\|^2$, the squared-area identity gives

$$A^2\|U\|^2 - (c^\top U)^2 \leq \|U(U^\top y) - c\|^2\|U\|^2.$$

Splitting at $\|U\| = R_0$ yields $A^2D \leq R_0^2V + A^2M_4/R_0^2$. Choose $R_0^2 = 2M_4/D$ to obtain $V \geq [y^\top Q^2(TI - Q)y]^2/(4M_4\|Qy\|^2)$. Here $M_4 \leq m^2\mu_4$ and $T = m\mu_{2,L}$. Every eigenvalue x of Q lies in $[\gamma, T - (m-1)\gamma]$. Thus, for $N = y^\top Q^2(TI - Q)y$,

$$N \geq (m-1)\gamma\|Qy\|^2, \qquad N \geq \gamma^2(T - \gamma)\|y\|^2.$$

The second uses $x(T - x) - \gamma(T - \gamma) = (x - \gamma)(T - \gamma - x) \geq 0$ and then $x \geq \gamma$. Multiplying these lower bounds gives

$$V \geq \frac{(m-1)\gamma^3(m\mu_{2,L} - \gamma)}{4m^2\mu_4}\|y\|^2 \geq \frac{\gamma^3\mu_{2,L}}{16\mu_4}\|y\|^2,$$

because $\gamma \leq \mu_{2,L}$ and $(m-1)^2/m^2 \geq 1/4$. At least one of the m coordinate variances is at least V/m, which proves (84).

Claim 9 (Conditional scalar fluctuation)**.** *For the index in the preceding local claim, put $g_n(t) = f_n(t, x_{a_*}) - \widetilde{f}_n(t, x_{a_*})$, where the two dense initializations are independent. There is a fixed $s_* > 0$ with*

$$s_*^2 \geq \frac{\gamma^3\mu_{2,L}Y^2}{2m^2\mu_4}, \qquad \liminf_{n\to\infty}\Pr\{\sqrt{n}|g_n'(0)| \geq us_*\} \geq 2[1 - \Phi(u)] \quad (u > 0), \tag{85}$$

where Φ is the standard normal distribution function.

Verification. Let $\mathcal{F}_n$ contain both initializations through layer $L-1$. Their training empirical Grams converge to $Q^{(L-1)}$ by (16) of Lemma 5.1, including when this matrix is singular. Conditionally on $\mathcal{F}_n$, the last-layer preactivation rows of each run are independent Gaussian vectors. Applying ϕ_L gives independent feature vectors, generally non-Gaussian; the two groups are also independent. For either run write $U_{i,a} = h_{0,i}^{(L)}(v_a)$ for its initialized top-layer feature entries. The zero readout gives exactly

$$\dot{f}_n(0, x_a) = \frac{2}{mn} \sum_{i=1}^{n} U_{i,a}(U_i^\top y).$$

Write $v_* = \mathrm{Var}[U_{a_*}(U^\top y)] > 0$ and $s_*^2 = 8v_*/m^2$. Conditional variances of the row products converge to v_*: couple their Gaussian vectors with the continuous covariance square root and dominate the fourth-degree products by Gaussian polynomials. Their centered absolute third moments are uniformly bounded on bounded covariance sets, since linear growth requires only Gaussian sixth moments.

For a centered row product X, Taylor's remainder gives $|\mathbb{E}(e^{itX/\sqrt{n}} \mid \mathcal{F}_n) - 1 + t^2\mathbb{E}(X^2 \mid \mathcal{F}_n)/(2n)| \le |t|^3\mathbb{E}(|X|^3 \mid \mathcal{F}_n)/(6n^{3/2})$. Raise this identity to the nth power in each of the two independent groups. It proves that the conditional law of $Z_n = \sqrt{n}(g_n'(0) - \mathbb{E}[g_n'(0) \mid \mathcal{F}_n])$ converges weakly in probability to $N(0, s_*^2)$. To justify this random-law statement, every subsequence has a further subsequence on which both covariance matrices converge almost surely; the preceding characteristic-function calculation and the continuity theorem then apply deterministically on almost every such outcome. For a continuous limiting CDF, weak convergence is uniform in its argument: choose finite grids with small Gaussian CDF increments and small tails, and use monotonicity between grid points. Therefore the conditional Kolmogorov distance d_n tends to zero in probability, and $\mathbb{E}d_n \to 0$ since $0 \le d_n \le 1$.

For any $\mathcal{F}_n$-measurable shift b_n, a centered Gaussian has its largest interval probability at the interval centered at zero. Consequently

$$\Pr\{|Z_n + b_n| \le us_* \mid \mathcal{F}_n\} \le 2\Phi(u) - 1 + 2d_n.$$

The Gaussian assertion follows by differentiating the interval mass with respect to its center and using the even, decreasing Gaussian density. Take $b_n = \sqrt{n}\,\mathbb{E}[g_n'(0) \mid \mathcal{F}_n]$ and average. This proves (85); previous layers may produce an arbitrary shift and cannot increase the limiting small-ball probability. No recursive matrix central limit theorem is needed.

From the derivative to the trajectory. Use the finite-query conclusion (27) of Proposition 6.1 for the two runs and the finite training query list. Put $\lambda = \gamma/m$, $\ell = \log(en)$ and $r = 1/(\lambda\sqrt{\ell})$. Each run is bounded there by $16\beta^{6L}Y/\lambda$, so their difference has twice this bound. The parameter-two ellipse of $[0, r]$ has real projection $[-r/8, 9r/8]$ and imaginary projection $[-3r/8, 3r/8]$, and hence lies inside the supplied rectangle. On its event, g_n is bounded by $32\beta^{6L}Y/\lambda$. Choose $N = \lceil 2\ell/\log 2 \rceil \le 4\ell$. Lemma 8.1 gives

$$\sup_{0 \le t \le r} |g_n(t)| \ge \frac{|g_n'(0)|}{32\lambda\ell^{5/2}} - \frac{768\beta^{6L}Y}{\lambda n^2}.$$

For fixed $u > 0$, the second term is eventually at most half the first on $|g_n'(0)| \ge us_*/\sqrt{n}$. The two source failures tend to zero; no independence from the derivative event is required. Thus the conditional fluctuation claim gives a limiting lower probability at least $2[1 - \Phi(u)]$ for threshold $us_*/(64\lambda\sqrt{n}\ell^{5/2})$. Insert the lower bound on s_*, use $\lambda = \gamma/m$ and $1/(64\sqrt{2}) \ge 1/128$, and choose $u = \Phi^{-1}(1/2 + \delta/4)$. The limiting probability is at least $1 - \delta/2$, leaving a strict margin to obtain $1 - \delta$ eventually. Since $g_n(0) = 0$, a positive threshold on the compact witness interval requires a strictly positive time. That training query belongs to each promised domain. At its fitted endpoint

the predictions equal the same label, consistently with the asserted scope. The asserted constant can therefore be chosen as the following local quantity:

$$c_{\phi,L,\delta} = \frac{\Phi^{-1}(1/2+\delta/4)}{128}\sqrt{\frac{\mu_{2,L}}{\mu_4}} > 0.$$

Its moment ratio depends only on the activations and depth, as required. □

9 Polynomial sufficient widths for Legendre and Taylor

Use the common setup of Section 2 and, locally, put

$$X = \beta^L, \qquad \lambda = \gamma/m \in (0,1], \qquad \chi = 1+\lambda^{-1}, \qquad S = 16Y/\lambda, \qquad \ell_n = \log(en).$$

Thus $X \geq 100$, $Y/\lambda \leq X^{-30}$ and $S \leq X^{-26}$; S is a normalization, not state. For $0 < \rho < 1/4$ and $N \geq m$ declared inputs define

$$u_\rho = \max\left\{1, \left\lceil \frac{\log(4emL/\rho)}{\log(64e^2)} \right\rceil \right\}, \qquad A_{N,\rho} = X^{4000}\chi^8(N+d+u_\rho+1)^8,$$

$$N_{\mathrm{src}}(N,\rho) = \left\lceil (A_{N,\rho}/\rho)^{40000} \right\rceil. \tag{86}$$

This deliberately loose finite gate contains no Y^{-1}.

Theorem 9.1 (Polynomial onset with the original storage powers)**.** *For $0 < \delta, \varepsilon < 1$, a sufficient width for*

$$\Pr\left\{ \|f_{\mathrm{model}} - f_n\|_{\mathcal{X}} \leq \varepsilon \|f_n - \widetilde{f}_n\|_{\mathcal{X}} \right\} \geq 1-\delta$$

is bounded by

$$\left[c_{\mathrm{wid}} \beta^L (1+m+d+p+\gamma^{-1}+Y^{-1}+\delta^{-1}+\varepsilon^{-1}) \right]^{c_{\mathrm{wid}}}$$

with one absolute finite numerical constant c_{wid}, local to this theorem. For Legendre take $p = 0$ and omit Y^{-1}; its domain is the whole input sphere. For Taylor, p is the number of passive inputs declared before initialization and its domain is that panel together with the training inputs. The Legendre and Taylor storage bounds in (8) *and* (10) *hold with activation/depth coefficients bounded by fixed powers of X. Taylor also has absolute error Y/n.*

The constructions, physical optimizer, initialization-only provenance and label condition are unchanged. Confidence applies to each individual width. The inverse-label gate for Taylor is needed only to simplify its exact source degree to a multiple of $\log(en)$*. This result makes no polynomial onset claim for Harmonic, for the nonlinear feature-learning certificate, or for the stronger coefficient in* (83)*, and gives no pointwise-in-time or endpoint denominator bound.*

The proof quantifies the shared source construction, derives a finite dense lower bound, and improves the Legendre history defect to order q^{-3}.

9.1 Finite initialization and analytic source probability

Use exactly the local coefficient recurrences equations (30) to (35), with

$$b = \max_j |\phi_j(0)|, \quad s = \max\{1, \sup_{j,|\operatorname{Im} z|\le a/2} |\phi_j'(z)|\}, \quad t_2 = \max\{1, \sup_{j,|\operatorname{Im} z|\le a/2} |\phi_j''(z)|\}.$$

In particular C_{abs}, W_G keep (31)'s definitions. The carriers are $k_a^{(L)} = w$, $k_a^{(j)} = W^{(j+1)\top}\delta_a^{(j+1)}$ for $j < L$. Write $\rho(t) = \|r(t)\|_2/\sqrt{m}$ for residual RMS, distinguishing it from the constant failure allowance ρ, and use Euclidean mobility coordinates $\theta = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$. Only finite input lists are used in the complex source argument.

Lemma 9.2 (Finite training and panel sources)**.** *At $n \ge N_{\mathrm{src}}(N, \rho)$, with probability at least $1 - \rho$, the ambient real fitting event holds and the following source conclusions hold. The four families are exactly* (25)*; forward inputs range over the N declared inputs and backward inputs over the m training inputs.*

For the training list alone one may use the closed rectangle

$$[-r, T + r] + i[-r, r], \qquad T = 32\ell_n/\lambda, \qquad r = [u_\rho X^{100}(1 + \lambda)\sqrt{\ell_n}]^{-1}. \tag{87}$$

All training sources and the responses $R_{ba}^{(j)} = D_\Theta z_b^{(j)} \nabla_\Theta(n f_a)$, where

$$\Theta = (W^{(1)}, \sqrt{n}W^{(2)}, \dots, \sqrt{n}W^{(L)}, w),$$

are holomorphic near this rectangle. The source RMS bounds are (26)*, their coordinates are bounded by $X^6\sqrt{n}$, and $|f_n| \le 16X^6 Y/\lambda$. Moreover*

$$\max_{a,b,j,i} |R_{ba,i}^{(j)}| \le S U_{\mathrm{fin}} \sqrt{\ell_n}$$

on that rectangle. On either complex source domain the first-layer operator divided by $\sqrt{n}$ and all hidden operators have norm at most ten, the complex training residual RMS is at most $2Y$, and every declared preactivation has imaginary part below $3a/8$. The training coordinate maxima are bounded by $K_{\mathrm{src}}\sqrt{\ell_n}$ for preactivations and $SK_{\mathrm{src}}\sqrt{\ell_n}$ for carriers. These strict strip and coordinate bounds are included in the event, not inferred from an RMS bound. On the whole real training trajectory, including its limit,

$$\max_{a,j,i} |k_{a,i}^{(j)}| \le 32X^{21}(Y/\lambda)\sqrt{\ell_n}, \qquad \max_{a,b,j,i} |R_{ba,i}^{(j)}| \le Y X^{110}\chi^3(1 + \sqrt{\ell_n}). \tag{88}$$

For the N-input panel put $a_0 = \min(a, 1)$ and

$$b_n = \log(e + \ell_n), \quad B_n = 1 + (S^2/\eta_*)b_n, \quad r_n = \min\left\{\frac{a_0}{64 Y S U_{\mathrm{fin}}\sqrt{\ell_n}}, \frac{1}{X^{300}(1 + \lambda B_n)\sqrt{b_n}}\right\}. \tag{89}$$

There is a deterministic disk cover and half-radius panel partition of $[0, T]$, with the same source envelopes, whose number of panels obeys

$$J \le 1 + X^{400}\left[(Y/\lambda)^2\sqrt{\ell_n} + \lambda^{-1}\log(e + \ell_n)\right]. \tag{90}$$

Neither this radius nor its panel-count coefficient depends on confidence.

Proof. First quantify initialization. Let $H = \max_{0\le j\le L}\sqrt{Q_{11}^{(j)}} \ge 1$ and let b, s have their meanings in the source proof. Define

$$R_0 = 2H, \quad K_0 = b + sR_0, \quad D_0^{\mathrm{cov}} = 2sb + 4s^2R_0, \quad T_1 = \sum_{i=0}^{L-1} s^i, \quad T_2 = \sum_{i=0}^{L-1} s^{2i},$$

$$C_0^{\mathrm{cov}} = T_2(2K_0 + 1 + D_0^{\mathrm{cov}}T_1), \quad e_0 = \min\{(8T_1)^{-1}, \gamma/(4mC_0^{\mathrm{cov}})\}, \quad h_0 = \min\{1/2, H/[4(8s)^L]\}.$$

The Gaussian Lipschitz inequality in Claim 3 gives $\operatorname{Var} F \le a^2$ and $0 \le (\mathbb{E}F^2)^{1/2} - \mathbb{E}F \le a$ for nonnegative a-Lipschitz F, hence

$$\Pr\{|F - (\mathbb{E}F^2)^{1/2}| > e\} \le 2e^{-e^2/(8a^2)} \qquad (e \ge 2a).$$

For iid Gaussian row pairs of marginal deviations at most R_0, the empirical RMS of $\phi(Z_a)$ and $\phi(Z_a) \pm \phi(Z_b)$ has Lipschitz constant $sR_0/\sqrt{n}$ and $2sR_0/\sqrt{n}$, respectively, also for singular pairs. Polarization therefore gives entry error $(2K_0 + 1)e_0$ when the two latter RMS errors are at most $e_0 \le 1$.

To propagate covariance errors without dividing by a variance, consider two Gaussian pairs with standard deviations σ_a, σ_b and σ'_a, σ'_b, all at most R_0, and set $\Delta = \max_i |\sigma_i - \sigma'_i|$. Shrink both pairs to marginal deviations $\tau_i = \min(\sigma_i, \sigma'_i)$, preserving their correlations. Product subtraction and Cauchy–Schwarz bound the total scaling cost by $2sK_0\Delta$. At these common marginals Gaussian integration by parts gives the correlation derivative $\tau_a\tau_b\mathbb{E}[\phi'(Z_a)\phi'(Z_b)]$. The formula follows by differentiating the nonsingular density and integrating its mixed second derivative; linear growth and bounded derivatives justify the integrations. Continuity covers correlations ± 1 and zero marginal variances. Since

$$\tau_a\tau_b|\rho_c - \rho'_c| \le |C_{ab} - C'_{ab}| + 2R_0\Delta,$$

the transformed entry error is at most $s^2|C_{ab} - C'_{ab}| + D_0^{\mathrm{cov}}\Delta$. To check the inequality, put the correlation difference over the larger of $\sigma_a\sigma_b, \sigma'_a\sigma'_b$ and use $|C'_{ab}| \le \sigma'_a\sigma'_b$.

Condition at each layer on the preceding initialized layers. On a sphere h_0-net of size at most $(1 + 2/h_0)^d$, and the training list, let Δ_j denote the largest RMS error and E_j the largest training Gram-entry error. The preceding estimates give

$$\Delta_j \le e_0 + s\Delta_{j-1}, \qquad E_j \le (2K_0 + 1)e_0 + s^2E_{j-1} + D_0^{\mathrm{cov}}\Delta_{j-1}.$$

Thus $\Delta_j \le e_0T_1 \le 1/8$ and $E_j \le e_0C_0^{\mathrm{cov}} \le \gamma/(4m)$. This also keeps all conditional standard deviations below R_0. With

$$\Xi = \log\frac{16L(m+1)^2}{\rho_0} + d\log(1 + 2/h_0),$$

a sufficient initialized fitting width at failure ρ_0 is

$$\left\lceil \max\left\{1, \frac{\log(8L/\rho_0)}{8 - 2\log 9}, \frac{d\log 9 + \log(8/\rho_0)}{8 - \log 9}, \frac{32s^2R_0^2\Xi}{e_0^2}\right\}\right\rceil. \tag{91}$$

The at most $3m^2 + (1 + 2/h_0)^d$ norm tests per layer cost $\rho_0/2$; the two operator-net tails proved in Lemma 5.1 cost the other half. On that operator event the layer-j feature map has RMS Lipschitz

constant $(8s)^j$, so off-net increments are at most $H/4$. Thus (17) holds and its deterministic fitting proof applies. Unrolling gives

$$R_0, K_0 \le X^3, \quad D_0^{\mathrm{cov}} \le X^4, \quad T_1 \le X^2,\ T_2 \le X^3,\ C_0^{\mathrm{cov}} \le X^{10}, \quad e_0^{-1} \le X^{11}\chi.$$

Since $32s^2R_0^2 \le X^8$ and $\log(1+2/h_0) \le 3L\log\beta$, the width is at most $\lceil X^{32}\chi^2[dL\log\beta + \log(16L(m+1)^2/\rho_0)]\rceil$, without ordering m, d.

For source initialization take $\rho_0 = \rho/64$. On the declared list, conditional Gaussian tails give

$$Z_0 = 2H\sqrt{2\log(64nNL/\rho)}.$$

Zero-embedding and forward subtraction give $D_{\mathrm{init}} = (8s)^L\sqrt{u}(b + sZ_0)$ for deletion count $u = u_\rho$. Since the full training Gram starts above $3\gamma/4$ and $D_{\mathrm{init}}/\sqrt{n} < \sqrt{\lambda}/8$, every cavity's normalized least singular value exceeds $(\sqrt{3}/2 - 1/8)\sqrt{\lambda} > \sqrt{\lambda/2}$. Project lower differences before conditioning on the fresh row: their $n^{-1/10}$ tail is $2\exp[-n^{4/5}/(2D_{\mathrm{init}}^2)]$, with at most uLn^uNLn tests. Initial budget means and second moments are below four and fifteen, since $\mathbb{E}e^{\eta_*|Z|} \le 2e^{1/2}$, $\mathbb{E}e^{2\eta_*|Z|} \le 2e^2$ and $\eta_* 2H \le 1$. Conditional Chebyshev costs $15mL/(16n)$ for layer means below eight. All initialization failures total less than $\rho/4$ at the source gate. Each cavity has its own initial test and zero reference paths on failure.

We now quantify the independently stopped source proof section 6 and claim 2. Use $\bar{u} = (\Theta - \Theta_0)/S$, $\bar{F}_a = nf_a/S$ in (40). Since $w(0) = 0$, $\bar{F}_a = \bar{u}_w^\top h_a^{(L)}$ has no inverse-S derivative. Freeze the scalar forward and scaled reverse controls; concatenate omitted interior roots into $g \sim N(0, I_r/n)$, $r \le 2un$. First-layer incoming rows enter only their own scalar controls, with no retained reverse port.

Put $\kappa_0 = 1/4000$, $\Gamma = \sqrt{\mathcal{K}}$, $\tau_* = \max_j \tau_j$, $f_* = \max_j f_j$, and $T_0 = 64/\lambda$. The propagator bound (44) is $\|J(t,s)\| \le J_0 n^{\kappa_0}$ with $J_0 = 2e^{1/4}(2\mathcal{B})^{1/4000}$. Define

$$M_0 = \eta_*^{-1}[1 + \log(2\mathcal{B})], \quad A_0 = b + sM_0, \quad B_0 = sM_0, \quad E_0 = A_* + SD_*M_0.$$

Budgeted scaled carriers are at most $M_0\ell_n$ and controls at most $A_0\ell_n, B_0\ell_n$. The exact product recursions in Claim 2 give the following coefficient ledger; each column omits the common factor $\sqrt{u}n^{\kappa_0}$:

	parameter	forward	backward
map	$X^{80}\chi\ell_n^2$	$X^{85}\chi\ell_n^2$	$X^{125}\chi\ell_n^3$
time derivative	$X^{125}\chi\sqrt{n}\ell_n^4$	$X^{140}\chi\sqrt{n}\ell_n^4$	$X^{190}\chi\sqrt{n}\ell_n^4$

Control differences have coefficient $X^{90}\chi\ell_n^2$ times the supremum control difference. Here is a direct verification, so none of these finite powers comes from a qualitative threshold. The source recurrences and geometric sums give

$$H_j, P_j, f_j \le X^3, \quad k_j, \tau_j \le X^4, \quad V_j^0 \le X^7,$$
$$A_*, D_*, \eta_*^{-1}, \mathcal{B}, \Gamma, J_0 \le X^5, X^6, X^{27}, X^3, X^7, X,$$
$$M_0, A_0, B_0, E_0 \le X^{31}, X^{33}, X^{32}, X^{38}.$$

The parameter map coefficient supplied by the three port derivatives is

$$v_0 = J_0\{A_0(2T_0\Gamma\tau_* + E_0) + Sf_*B_0\} \le X^{80}\chi. \tag{92}$$

Duhamel uses $2\int \rho/S \le 1$ and length $T_0\ell_n$. Forward coefficients are $P_j(Sv_0 + A_0)$, multiplied by s for features. Backward coefficients start at v_0 and satisfy $k_j^0 = 10d_{j+1}^0 + S\tau_{j+1}v_0 + B_0$,

$d_j^0 = sk_j^0 + t_2 M_0 P_j(Sv_0 + A_0)$. Their only carrier factor is additive. Using $(10s)^L \le X^2$ and at most $L \le X$ forcing terms proves the first row. Replacing A_0, B_0 by one in these forcings gives the stated control bound, without changing reference coefficients.

For time moduli, differentiating the reference recursion gives

$$T_L^0 = 2sH_L + 2t_2 M_0 V_L^0, \qquad T_j^0 = 10sT_{j+1}^0 + 2s\tau_{j+1}^2 H_j + 2t_2 M_0 V_j^0.$$

Thus $T_j^0 \le X^{45}$ and the control speed is at most $D_c\sqrt{n}\ell_n$, where

$$D_c = \max\{\lambda s \max_j V_j^0/4, \lambda \max_j T_j^0/8\} \le \lambda X^{45}.$$

All reference derivative coefficients, before the single raw-coordinate factor $\sqrt{n}\ell_n$, are at most λX^{47}. The time equation for the parameter map has coefficient

$$v_t = [2\mathcal{K} + (\lambda/4)SE_0]v_0 + A_0(2\Gamma\tau_* + \lambda E_0/4) + (\lambda/4)Sf_*B_0 \le X^{125}\chi.$$

In the differentiated forward recursion, the largest new coefficient is bounded by $X^{47}X^{85}\chi$; adding the gate derivative and geometric propagation gives $X^{140}\chi$. For backward derivatives the forcing $t_2 M_0 X^{140}\chi$ is at most $X^{172}\chi$. The other gate terms are $t_2 \dot{z}\, k_{[1]}$ and $(\phi'''\bar{k}\dot{z} + \phi''\dot{\bar{k}})z_{[1]}$. With $|\phi'''| \le 4\beta^2$ and the reference speed bound, they are at most $X^{174}\chi$; propagation and all sums fit below $X^{190}\chi$. Differentiating the lower-Jacobian probes uses the same forward products. Every term has at most one raw coordinate derivative, hence one factor $\sqrt{n}$; the log powers are at most four. This proves the second row without differentiating the variational Jacobian. Consequently $C_0 = X^{300}\chi$ bounds every coefficient above. For a root selector E_i, $\mathrm{sym}(E_i^\top L)$ inherits the linear map's moduli; centered interpolation on $\|g\| \le 2\sqrt{2u}$ costs $10u\|\Delta L\|$, including trace change.

For responses use (48), with evaluated input b and training driver a, $\bar{R} = R/S$ and $\bar{Q} = \phi' \odot \bar{R}$. Its linearized products and the direct reverse observable $C_b C_a^\top \bar{q}_a$, $C_a = D_\Theta h_a$, were proved in Claim 2. In particular

$$(C_b^1 C_a^{1\top} - C_b^0 C_a^{0\top})y_i = (C_b^1 - C_b^0)C_a^{0\top} y_i + C_b^1 (C_a^1 - C_a^0)^\top y_i$$

retains $d_0 + n^{-1/2}$ by the reference-probe and lower-adjoint bounds there. For time derivatives the extra product is $\phi''\dot{z}_b\bar{R}$: use the response coordinate cap and z_b's RMS speed before one scalar-control conversion. The raw amplitude coefficient is at most X^{80} and $U_{\mathrm{fin}} \le X^{26}$, so the same product recursions have common envelope

$$X^{100}(1 + X^{80} + 4U_{\mathrm{fin}})^2(1 + C_0)^2 \le X^{1000}\chi^2 =: \mathcal{C}.$$

The exponent count is $100 + 160 + 600$, with the remaining 140 paying all numerical sums and bounded layer propagation. Thus the map norms, control moduli and actual analytic derivatives are bounded by

$$\mathcal{C}\sqrt{u}n^{\kappa_0}\ell_n^5, \qquad \mathcal{C}\sqrt{u}n^{\kappa_0}\ell_n^4, \qquad \mathcal{C}\sqrt{un}n^{\kappa_0}\ell_n^8. \tag{93}$$

There are at most $4u(N+1)^2$ real scalar controls, of amplitude $M = \mathcal{C}\ell_n$ and speed $D = \mathcal{C}\sqrt{n}\ell_n^3$. Every bound concerns frozen one-dimensional histories until the uniform event is established.

We check the deterministic domain premises before the Gaussian event. The training radius is at most

$$\min\{1/8, 1/\lambda, \log 2/(4\mathcal{K}), 1/(32YSD_W), a/(64YSU_{\mathrm{fin}}\sqrt{\ell_n})\}.$$

Thus the residual, activity, operator and pole estimates in the rectangle verification apply without angular fields or an $n^{-1} \le Y$ gate. For the panel use exactly h, d_h, T_+, D_s, P_s of Claim 6, replacing its radius by (89) and taking $\nu = \lambda/4$. That construction gives $\operatorname{Lip} P_s \le 1+d_h$, $d_h \le 1/8$, commuting real anchors, a bounding rectangle of sides at most $H_0 = 128\ell_n/\lambda$, and canonical contours of length at most $H_0/2$. The recurrences claim 4 and equation (62) give $J_* \le X^{34}$, $C_g \le X^{38}$, hence

$$4\mathcal{K} r_n + (2C_g/\sqrt{\mathcal{K}})B_n Y r_n \le 4\mathcal{K}/X^{300} + C_g/(8X^{300}) < 1/100,$$

using $Y = \lambda S/16$ and $\lambda B_n/(1+\lambda B_n) \le 1$. This verifies the complete propagator gate (64): the frozen real generator is unitary in imaginary time, its perturbation costs the second term, and forward real segments contract. The same domain proof gives additional activity $8Yr_n$, normalized operator motion $8YSD_W r_n$, and improved vertical preactivation displacement $2YSU_{\mathrm{fin}}\sqrt{\ell_n} r_n \le a_0/32$. The negative corner costs at most $a_0/8$; all margins are strict.

The uniform event must precede adaptive substitution; endpoints are clamped only by their cavity's own stop. Put $A = A_{N,\rho} = \mathcal{C}^4(N+d+u+1)^8$. The source gate gives

$$A\ell_n^{16} \le n^{1/1000}, \quad u \le n^{1/8000}, \quad \ell_n \le n^{1/16000}. \tag{94}$$

Indeed, with $v = \log(A/\rho) \ge 8000 \log 10$ and $\log n = 40000v$, one has $\log A + 16\log(1+40000v) \le 17v < 40v$; the difference increases thereafter. The other two inequalities use the same gate.

Condition on a cavity's retained initialization, with own stop σ_c. Set $\xi = n^{-2}/(1+A\ell_n^8)$, $J_s = \lceil 16H_0/\xi \rceil$, $s_j = j/J_s$, and let $\mathscr{G}$ be a product grid of spacing at most $\xi/8$ on the bounding rectangle. Test the retained-measurable endpoints

$$z_{j,q,c} = P_{\min(s_j,\sigma_c)} q, \qquad q \in \mathscr{G}, \quad 0 \le j \le J_s. \tag{95}$$

For endpoint $x+iy$ its own-domain contour is $\zeta(t) = \operatorname{sgn}(x)\min(t,|x|) + i\operatorname{sgn}(y)\min((t-|x|)_+, |y|)$ on $[0, H_0]$, then frozen. Solve $V' = \zeta'(\mathsf{A}V + \sum_a \mathsf{B}_a^e e_a + \mathsf{B}_a^q \bar{q}_a)$, $V(0) = 0$, with the preceding port derivatives. Bounded measurable histories suffice, and V is a linear map of omitted roots; its clamped extension is not claimed holomorphic.

For control accuracy $\tau = n^{-1/8}$ sample $[0, H_0]$ in $\lceil 16DH_0/\tau \rceil$ equal intervals, round each real component to mesh $\tau/8$ and interpolate. Every admissible history is within τ and the logarithmic cardinality is at most

$$4u(N+1)^2(2+16DH_0/\tau)\log(1+32M/\tau) \le A\ell_n^8 n^{5/8} \le n^{0.627}.$$

Here $16384u(N+1)^2\mathcal{C}\lambda^{-1} \le A$. The net amplitude is at most $2\mathcal{C}\ell_n$; affine control dependence and (93) give map norm $4\mathcal{C}^2\sqrt{u}n^{\kappa_0}\ell_n^5 \le n^{1/800} < n^{1/200}$.

For $g \sim N(0, I_r/n)$, $r \le 2un$, a real symmetric matrix obeys

$$\Pr\{|g^\top R g - \operatorname{tr} R/n| > t\} \le 2\exp\left[-\frac{1}{8}\min\left\{\frac{n^2t^2}{\|R\|_{\mathrm{HS}}^2}, \frac{nt}{\|R\|}\right\}\right].$$

Diagonalization and $-\log(1-v) - v \le v^2/[2(1-|v|)]$, followed by exponential Markov at parameter at most $n/(4\|R\|)$, prove this inequality as in Claim 2. Since $\|R\|_{\mathrm{HS}}^2 \le 2un\, n^{1/100}$, real/imaginary splitting at $n^{-1/10}/8$ costs at most $4e^{-n^{0.78}}$. Linear-coordinate tails are smaller. The Euclidean image mean is at most $\sqrt{2u}n^{1/200} < n^{1/100}/4$; Gaussian concentration pays the remaining threshold $n^{1/100}/2$. Root-norm failure is at most e^{-un}, and control interpolation costs $10\mathcal{C}u^{3/2}n^{\kappa_0 - 1/8}\ell_n^4 < n^{-1/10}/8$.

For an arbitrary full-network-dependent common prefix D_σ, choose a nearest $q \in \mathscr{G}$ and the largest $s_j \le \sigma$. Since $\sup_z |P_s z - P_t z| \le 3H_0|s-t|$,

$$|z - P_{s_j} q| \le (3/16 + 9\sqrt{2}/64)\xi \qquad (z \in D_\sigma).$$

Its canonical contour lies in D_σ, so actual controls may now be substituted. The domain is vertically fibered and retracts to its real interval, hence is simply connected. Local analytic ODE uniqueness and continuation along compact paths give a single holomorphic variation for these actual controls. The P_σ-image of the joining segment has length below $\xi/2$; the derivative bound (93) therefore costs $10u\mathcal{C}\sqrt{u}n^{\kappa_0+1/2}\ell_n^8\xi < n^{-1/10}/8$. Continuity covers boundary levels, and level zero is tested. There are at most $(2 + 16H_0/\xi)^3 \le n^9$ endpoints, An^3 maps per endpoint and uLn^u deletion sets. Their logarithmic union cost is below $n^{0.65}$, so failure is below $e^{-n^{0.7}} < \rho/16$.

On rectangle prefixes, round each coordinate toward zero to mesh ξ: the grid point and joining segment remain in every common prefix. Clamp each canonical contour only by its cavity's own rectangle. The same conditional tests and interpolation then require at most n^6 endpoints and the same failure allowance.

Apply the exact remainder formulas equations (45) to (47) with stops $N_0 = n^{1/100}$, $d_0 = n^{-1/10}$, $u_0 = n^{-1/25}$ and $R = d_0N_0 + d_0u_0 + u_0^2$, $P = (N_0+u_0)^2/\sqrt{n}$. The force coefficient is $X^{240}(1+u)(1+M)$; learned ports and the top offset cost $X^{260}(1+u)^2(1+M)^3$. Their raw envelope $M \le X^{80}\ell_n$, the map ledger, $2\int \rho/S \le 1$ and length $64\ell_n/\lambda$ give

$$\|U\| \le A\ell_n^8 n^{\kappa_0}[n^{-0.08} + u_0^2 + n^{-0.48} + n^{-1/2}] < u_0/2.$$

The largest power is $n^{-0.07875}$, and the source gate also gives $d_0 + X^{10}u_0 + X^{30}(R+P) < a/32$. Thus the same first-exit proof gives coordinate differences $n^{-1/30}$ and vector differences $2n^{1/100}$, including responses. Its first-layer incoming increment sS^2M belongs to deleted scalar controls, with no retained reverse port.

The four traces (51), the Schatten calculation and scalar feedback solution in the source proof apply before adaptive substitution. The first three constants are unchanged, the residual trace is at most $A\ell_n^8 n^{-1+\kappa_0}$, and finite remainders are below $n^{-1/40}$, so the original C_{abs} applies. The Gaussian maximum argument there gives $K_{\mathrm{src}}\sqrt{\ell_n}$ and $SK_{\mathrm{src}}\sqrt{\ell_n}$. Its four response terms have, after division by S, precisely

$$sK_{\mathrm{src}}H_{j-1}^2, \quad sK_{\mathrm{src}}f_{j-1}^2, \quad sK_{\mathrm{src}}ST_Q, \quad sK_{\mathrm{src}}S^2H_{j-1}q_{j-1}^*.$$

Adding the centered coefficient $64q_{j-1}^*$ gives $U_j(64)$. Multiplier 64 pays the $mNLn$ response tests and polynomial endpoints at the source gate; both maximum failures total less than $\rho/16$. The exact identity $\dot{z}_b = -(2/m)\sum_a r_a R_{ba}$ improves poles and passive maxima, with real displacement $2S^2U_{\mathrm{fin}}\sqrt{\ell_n}$ and no passive budget.

Budget removal must hold at the confidence order $u = u_\rho$. Complete each common-cavity reference by its own stopped retraction before integrating a root. For normalized complex-minus-real Gaussian coefficients the panel derivative estimate gives

$$D \le X^{100}\lambda r_n B_n, \quad K \le X^{100}\lambda B_n, \quad H \le 128\ell_n/\lambda.$$

Use $\bar{D} = X^{-200}\lambda B_n/[(1+\lambda B_n)\sqrt{b_n}]$. By Claim 3, the logarithmic moment is at most $a(M_G + 2\bar{D}) + 2a^2\bar{D}^2$, with $M_G = 256\bar{D}\sqrt{\log(e + HK/\bar{D})}$, and

$$\log(e + HK/\bar{D}) \le 300\log X + 16b_n + \log(1/\lambda).$$

For $b \ge 1, x \ge 0$, $\min(1, be^{-x})\sqrt{x} \le \sqrt{\log b} + 1$: split at $x = \log b$ and use $e^{-s}\sqrt{s} \le 1$ for the excess. Taking $b = B_n$, $x = \log(1/\lambda)$ and $B_n \le 1 + b_n$ gives $M_G + 2\bar{D} \le X^{-190}$ and moment at most $e^{X^{-180}}$ for $0 \le a \le 8$, including sample RMS.

On the training rectangle take $D \le X^{40}\lambda B_n r$, $K \le X^{40}\lambda B_n$, $H \le 64\ell_n/\lambda$. Its entropy is bounded by $\log 128 + \log u + 100\log X + \log\chi + (3/2)\log\ell_n$. Substitution of (87), $u^{-1}\sqrt{\log u} \le 1$ and $\lambda\sqrt{\log\chi} \le 2$ bounds the mean plus $2D$ by X^{-50}. For the remaining logarithms use $\log(e+x) \le 2+\log x$ and maximize $(\log x)^a/\sqrt{x}$, $a = 0, 1/2, 1, 3/2$: this bounds $(1+\log(e+\ell_n))(1+\sqrt{\log\ell_n})/\sqrt{\ell_n}$, with constants absorbed by five of the sixty available powers of X. Thus both geometries have small fixed-exponent corrections.

For each distinct root i, let A_i exponentiate $\eta_* C_{\mathrm{abs}}$ times one plus its four complete common-cavity real scalar/sample-RMS references and four complex-minus-real corrections. Equation (56) and Cauchy–Schwarz give $\mathbb{E}(A_i^2 \mid \text{cavity}) < 3$; Hölder among the four corrections uses $16\eta_* C_{\mathrm{abs}} < 8$. The A_i are conditionally independent because their root pairs are distinct.

Only singleton/common-cavity differences need order u. Extend each path by its own stop and radially project their difference to radius $4n^{1/100}$; both omit the paired root. Their Gaussian radius is $v_n = 4n^{-49/100}$ and Claim 3 gives mean

$$\mu_n = 256 v_n \sqrt{\log(e + X^{400}\chi\ell_n^4/v_n)} + 2v_n, \qquad u\mu_n + u^2 v_n^2 < 10^{-3}.$$

Let E be the successful initialization, map and maximum event above. For $B_i^{\mathrm{wt}} = e^{\eta_*(Z_{a,i}+K_{a,i}/S)}$ and distinct $|I| = k \le u$, scalar absorption, Cauchy–Schwarz between main references and corrections, and Hölder over at most four corrections per root give

$$\mathbb{E}\left[1_E \prod_{i\in I} B_i^{\mathrm{wt}}\right] \le 3^{k/2}\exp\{kn^{-1/40} + 4k\mu_n + 64k^2 v_n^2\} \le 16^k.$$

The indicator is dropped only after domination by complete references; no sample-process or correction-path independence is used.

From the defining constants, $2\eta_* K_{\mathrm{src}} \le 33/4096 < 1/100$, so repeated weights cost at most $W_n = e^{\sqrt{\ell_n}/100}$. Partition an ordered u-tuple by its distinct indices. A partition with j repetitions is encoded by j edges from block minima, giving at most $\binom{u}{2}^j/j!$ possibilities. The preceding distinct-product bound therefore yields

$$\mathbb{E}\left[1_E (n^{-1}\sum_i B_i^{\mathrm{wt}})^u\right] \le 16^u \exp\{\tbinom{u}{2} W_n/(16n)\} \le e16^u,$$

where $n \ge 4u^3$ suffices and is implied by the source gate. Markov and the sample/layer union make the budget-hit probability at most $emL(64e^2)^{-u} \le \rho/4$. Cavity budgets transfer as $\mathcal{H}_a^{\mathrm{cavity}} \le e^{2\eta_* n^{-1/30}}\mathcal{H}_a + u/n < 2\mathcal{B}$; every other stop has a strict margin. For either geometry, initialization, uniform maps, maxima and budgets cost respectively $\rho/4, \rho/16, \rho/16, \rho/4$. Initialization is shared, so doubling only the other three allowances gives total failure below ρ. Holomorphic local existence and uniqueness continue the solution across the compact domain boundary.

For real $t \ge T$, fitting gives $\|\theta(t) - \theta(T)\| \le 2Y\lambda^{-1/2}e^{-16\ell_n}$. The backward subtraction of Claim 5, with coefficient $X^{20}n$, gives the carrier cap. In response subtraction about T, changed gates multiply reference coordinate caps, and changed mixers and pairings use operator/RMS bounds. Its RMS Lipschitz coefficient is $X^{100}\chi^2(1+\sqrt{\ell_n})$; one coordinate conversion leaves tail $YX^{103}\chi^3(1+\sqrt{\ell_n})n^{-31/2} \le Y\chi\sqrt{\ell_n}$ at the source gate. This proves (88), including the limits.

The deterministic disk argument of Claim 6 uses $\varrho(t) = h(t)/[2(1+d_h)]$ and $t_{b+1} = \min(T, t_b + \varrho(t_b)/2)$, giving

$$J \le 1 + 40/(\lambda r_n) + 80\mathcal{K}\lambda^{-1}\log(1 + 4\ell_n/\log 2).$$

For the capped radius,

$$\frac{1}{\lambda r_n} \le \frac{1024 U_{\mathrm{fin}}}{a_0}(Y/\lambda)^2\sqrt{\ell_n} + X^{300}\lambda^{-1}\sqrt{b_n} + X^{300}B_n\sqrt{b_n}.$$

Writing $x = \log \ell_n \ge 0$ gives $b_n \le 2 + x$ and $(2+x)^{3/2}e^{-x/2} \le 4$, so $b_n^{3/2} \le 4\sqrt{\ell_n}$. Together with $S^2 = 256(Y/\lambda)^2$, $a_0^{-1} \le \beta$ and the coefficient bounds this proves (90). □

9.2 A finite lower bound for actual dense variability

For $0 < \tau < 1$, define

$$a_\tau = \frac{\tau\gamma^2}{2^{20} m u_{\tau/8} X^{108}(1+\lambda)},$$

$$N_{\mathrm{low}}(\tau) = \left\lceil \max\left\{ \begin{matrix} N_{\mathrm{src}}(m, \tau/8), \quad 2^{44} m^6 X^{96} \gamma^{-12} \tau^{-2} \log(64/\tau), \\ [2^{30} u_{\tau/8} X^{114} m \chi (1+\gamma^{-1})^2/\tau]^2 \end{matrix} \right\} \right\rceil. \tag{96}$$

Proposition 9.3 (Quantitative trajectory separation)**.** *For each $n \ge N_{\mathrm{low}}(\tau)$, two independent dense runs satisfy*

$$\Pr\left\{ \|f_n - \widetilde{f}_n\|_{\mathcal{X}} \ge a_\tau \frac{Y}{\sqrt{n}\,\ell_n^{5/2}} \right\} \ge 1 - \tau. \tag{97}$$

The domain may be any promised query set containing the training inputs. The lower coefficient is weaker than that in (83)*.*

Proof. We first prove a shift-uniform finite scalar inequality. For iid centered real variables V_i with variance $\sigma^2 > 0$ and third absolute moment at most M_3, put $Z = n^{-1/2}\sum_i V_i$. For every $h > 0$ and real b,

$$\Pr\{|Z - b| \le h\} \le 3h/\sigma + 4\exp[-9h^2\sigma^4 n/(128 M_3^2)]. \tag{98}$$

For an independent copy V', Taylor expansion gives $|\mathbb{E}e^{itV}|^2 = \mathbb{E}\cos(t(V - V')) \le 1 - \sigma^2 t^2 + (4/3)M_3|t|^3$, because $\mathbb{E}|V - V'|^3 \le 8M_3$. Therefore $|\mathbb{E}e^{itV}| \le e^{-\sigma^2 t^2/4}$ for $|t| \le 3\sigma^2/(8M_3)$. The Gaussian majorant $1_{|z-b|\le h} \le \sqrt{e}\, e^{-(z-b)^2/(2h^2)}$ and its Fourier integral bound the probability by

$$\frac{\sqrt{e}\,h}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-h^2 t^2/2} |\mathbb{E}e^{itZ}|\, dt.$$

On $|t| \le 3\sigma^2\sqrt{n}/(8M_3)$ use the preceding Gaussian bound; its full-line integral is below $3h/\sigma$. On the complement use modulus one and $\Pr(|G| > s) \le 2e^{-s^2/2}$. This proves (98). Fourier integrability justifies the exchange of expectation and integration. The argument works conditionally on any sigma-field supplying conditional iid variables and moment bounds; the shift may be measurable in that sigma-field.

Condition on the first run through layer $L - 1$ and the entire independent second run. Its final feature rows $U_i \in \mathbb{R}^m$ are conditionally iid. The initialized covariance calculation above, now choosing

$$e_1 = \min\{(8T_1)^{-1}, \gamma/[4m\max(C_0^{\mathrm{cov}}, s^2 C_0^{\mathrm{cov}} + D_0^{\mathrm{cov}} T_1)]\},$$

gives, except with probability at most $\tau/8$,

$$\bar{Q} := \mathbb{E}(U_i U_i^\top \mid \mathcal{F}) \succeq (\gamma/2) I, \qquad \max_a \|U_{i,a}\|_{L^6(\cdot|\mathcal{F})} \le X^4.$$

Indeed the conditional last-layer mean entry error is at most $s^2 E_{L-1} + D_0^{\mathrm{cov}} \Delta_{L-1} \le \gamma/(4m)$. Its required width is $32 s^2 R_0^2 e_1^{-2} \log[256 L (m+1)^2/\tau]$, which is bounded by the source gate in (96). Linear growth and Gaussian sixth moments prove the L^6 bound.

The squared-area and spectral argument in Claim 8 proves, before its equal-diagonal specialization, that for $\bar{Q} \succeq gI$, $T_q = \operatorname{tr} \bar{Q}$ and $M_4 = \mathbb{E}\|U\|^4$,

$$\mathbb{E}\|U(U^\top y) - \bar{Q} y\|^2 \ge \frac{[y^\top \bar{Q}^2 (T_q I - \bar{Q}) y]^2}{4 M_4 \|\bar{Q} y\|^2} \ge \frac{(m-1) g^3 (T_q - g)}{4 M_4} \|y\|^2.$$

Apply it conditionally with $g = \gamma/2$, $T_q \ge mg$ and $M_4 \le m^2 X^{16}$. Some coordinate then has

$$\operatorname{Var}[U_a(U^\top y) \mid \mathcal{F}] \ge \sigma_0^2 := Y^2 \gamma^4/(256 X^{16}).$$

Choose the smallest maximizing index, which is $\mathcal{F}$-measurable. The centered product has third absolute moment at most $8 m^3 Y^3 X^{24}$: use $\sum_b |y_b| \le mY$, the conditional L^6 bound, and $\mathbb{E}|V - \mathbb{E}V|^3 \le 8\mathbb{E}|V|^3$. Thus $M_3^2/\sigma_0^6 \le 2^{30} m^6 X^{96} \gamma^{-12}$. Zero readout gives the exact identity

$$\dot{f}_n(0, x_a) = \frac{2}{mn} \sum_i U_{i,a} (U_i^\top y).$$

The second run and the conditional means are only a measurable shift. Apply (98) with $h = \tau \sigma_0/24$; its scalar failure is at most $\tau/4$ under the second gate of (96). Hence, allowing the covariance failure, with probability at least $1 - \tau/2$,

$$\max_a |\dot{f}_n(0, x_a) - \dot{\tilde{f}}_n(0, x_a)| \ge \frac{\tau Y \gamma^2}{192 m X^8 \sqrt{n}}.$$

For each run use the training rectangle at failure $\tau/8$. It contains the parameter-two ellipse on $[0, r]$, where the difference is bounded by $32 X^6 Y/\lambda$. Substituting $K = \lceil 2\ell_n/\log 2 \rceil \le 4\ell_n$ in Lemma 8.1 gives

$$\|f_n - \widetilde{f}_n\|_{\mathcal{X}} \ge \frac{\tau Y \gamma^2}{6144 m u_{\tau/8} X^{108} (1+\lambda) \sqrt{n} \ell_n^{5/2}} - \frac{768 X^6 Y}{\lambda n^2}.$$

The tail is at most half the first term when $n^{3/2} \ge 2^{24} u_{\tau/8} X^{114} m (1+\lambda) \ell_n^{5/2}/(\lambda \tau \gamma^2)$. Since $\ell_n^{5/2} \le 64 n^{1/4}$, the third gate in (96) suffices. The union costs less than τ and the remaining coefficient exceeds a_τ. Zero initial predictions force a positive witness time; the training endpoints coincide with the labels. □

9.3 The third-order Legendre history estimate

The Legendre model is exactly (70) and (71), with its constant forward and zero backward prefixes. Let

$$q = \left\lceil \lambda^{-3} n^{1/4} \ell_n^2 e^{\sqrt{\ell_n}/2} \right\rceil, \qquad P = X^{500} \chi^{10}. \tag{99}$$

For fixed positive a, b, define the numerical logarithm coefficient

$$C(a,b)=\max\{1,e^b(a/(eb))^a\}.$$

Maximizing x^ae^{-bx} over $x\ge 1$ proves $\ell_n^a\le C(a,b)n^b$. Define the structural gate

$$N_{\mathrm{Leg}}=\left\lceil \max\left\{(2e^{100}\lambda^{-3})^8,(4P^3)^4,[8P^4C(2,1/64)]^8,d^4\right\}\right\rceil. \tag{100}$$

Proposition 9.4 (Improved error without additional state)**.** *On the finite training-source event, for $n\ge N_{\mathrm{Leg}}$,*

$$\|f_{\mathrm{Leg}}-f_n\|_{\mathcal{X}}\le 4YP^5\ell_n^2n^{-47/64}. \tag{101}$$

The exact moving and fixed inventories remain $n(d+1)+1+2(L-1)mnq$ and $(L-1)n^2$, respectively.

Proof. Use the mobility coordinates θ defined above. For a finite horizon T_0, let

$$d_{T_0}=\int_0^{T_0}\sum_{j=2}^{L}\|\mathcal{E}_j(t)\|_F dt,\qquad D_{T_0}=\sup_{t\le T_0}\sum_{j=1}^{L+1}\|\widehat{\theta}_j(t)-\theta_j(t)\|,\qquad A=\tau(T_0),$$

where $\mathcal{E}_j$ is the exact defect (73). By Claim 7,

$$\widehat{\rho}(t)\le Ye^{-\lambda t/4},\qquad \int_t^{T_0}\widehat{\rho}(s)ds\le 4\widehat{\rho}(t)/\lambda,\qquad A\le 1+4Y/\lambda\le 2.$$

Bootstrap $D_{T_0}\le Y/\sqrt{n}$. In backward and response subtraction use the dense state as reference, so changed gates multiply its coordinate caps (88). Forward and feature-pair differences cost $X^5D_{T_0}$ and $X^9D_{T_0}$; the changed response gate costs $YX^{116}\chi^3(1+\sqrt{\ell_n})D_{T_0}$. Propagation $(9s)^L\le X^2$ and at most $L\le X$ terms give

$$\begin{aligned}\max_{a,j}\|\widehat{k}_a^{(j)}-k_a^{(j)}\|_2/\sqrt{n}&\le X^{40}(1+\sqrt{\ell_n})D_{T_0},\\ \max_{a,b,j}\|\widehat{R}_{ba}^{(j)}-R_{ba}^{(j)}\|_2/\sqrt{n}&\le X^{130}\chi^4(1+\sqrt{\ell_n})D_{T_0}.\end{aligned}$$

One coordinate conversion and the bootstrap therefore yield

$$\max|\widehat{k}|\le YX^{50}\chi(1+\sqrt{\ell_n}),\qquad \max|\widehat{R}|\le YX^{135}\chi^4(1+\sqrt{\ell_n}).$$

Put $\widehat{v}_a=\widehat{r}_a/\widehat{\rho}$ and let $\mathcal{F}$ be the dense vector field at the reconstructed closure state. Finite-time positivity of $\widehat{\rho}$ and stationary continuation at residual zero follow from Claim 7. The ordinary feature velocity per unit activity is

$$a_b^{(j)}=D_\theta h_b^{(j)}(\widehat{\theta})\mathcal{F}(\widehat{\theta})/\widehat{\rho}=-\frac{2}{m}\phi'(\widehat{z}_b^{(j)})\odot\sum_a\widehat{v}_a\widehat{R}_{ba}^{(j)}.$$

The Legendre residual calculation gives $\|\widehat{v}\|_2/\sqrt{m}=1$ and $\|\dot{\widehat{v}}\|_2/\sqrt{m}\le X^{11}$. Differentiating the actual closure trajectory, including its defect, gives

$$\begin{aligned}\max_{b,j}\|\dot{\widehat{z}}_b^{(j)}\|_2/\sqrt{n}&\le YX^{30}\chi^3\widehat{\rho},\\ \max_{a,j}\|\dot{\widehat{\delta}}_a^{(j)}\|_2/\sqrt{n}&\le X^{110}\chi^4(1+\sqrt{\ell_n})\widehat{\rho},\\ \max_{a,b,j}\|\dot{\widehat{R}}_{ba}^{(j)}\|_2/\sqrt{n}&\le X^{220}\chi^8(1+\sqrt{\ell_n})\widehat{\rho},\\ \max_{b,j}(\|a_b^{(j)}\|_2+\|\dot{a}_b^{(j)}\|_2)/\sqrt{n}&\le YX^{230}\chi^8(1+\sqrt{\ell_n}).\end{aligned} \tag{102}$$

Indeed (76) bounds full hidden speed by $YX^{20}\chi^3\widehat{\rho}$, and forward differentiation costs X^5. In $\dot{\delta} = \phi''\dot{z}\,k + \phi'\dot{k}$, $\dot{k} = \dot{W}^\top\delta + W^\top\dot{\delta}$, use the carrier coordinate cap and $\dot{z}$ RMS; the top speed is $\|\dot{w}\|/\sqrt{n} \le X^4\widehat{\rho}$. The forcing is at most $X^{90}\chi^4(1+\sqrt{\ell_n})\widehat{\rho}$, giving the second line after propagation. Differentiating $R = \delta c + W(\phi' R)$ gives $\dot{\delta}c + \delta\dot{c} + \dot{W}(\phi' R) + W(\phi''\dot{z}R) + W(\phi'\dot{R})$. The response cap controls the curvature product; pairing derivatives and operator bounds control the rest, with forcing below $X^{180}\chi^8(1+\sqrt{\ell_n})\widehat{\rho}$. Finally the three terms of $\dot{a}$ use respectively $\dot{z}$ and the response cap, $\dot{\widehat{v}}$ and response RMS, and $\dot{R}$ with $\|\widehat{v}\|_2/\sqrt{m} = 1$; $\widehat{\rho} \le Y$ gives the last line. No coordinate $\dot{z}$ bound or $\sqrt{n}\|\mathcal{E}\|$ is used.

Subtract the accumulated physical feature defect from its history:

$$g_b^{(j)}(\tau(t)) = \widehat{h}_b^{(j)}(t) - \int_0^t D_\theta h_b^{(j)}(\widehat{\theta}(s))\mathcal{E}(s)\,ds,$$

and use its constant initial value on $[0,1]$. Aggregate history norms use $(mn)^{-1/2}$ over samples and neurons. The forward derivative bound gives $\|h-g\|_{L^2}/\sqrt{mn} \le X^{10}d_{T_0}$. Exactly $\partial_\tau g = a$ and $\partial_\tau^2 g = \dot{a}/\widehat{\rho}$ on $(1,A)$. The first derivative vanishes at the prefix join because the initial readout vanishes. Thus no distributional jump occurs there.

For $\mathscr{D}_A g = -\partial_\xi[\xi(A-\xi)\partial_\xi g]$, the inequality $(A-\tau(t))/\widehat{\rho}(t) \le 4/\lambda$ cancels the apparent terminal singularity. By (102),

$$\|\mathscr{D}_A g\|_{L^2}/\sqrt{mn} \le YX^{240}\chi^9(1+\sqrt{\ell_n}).$$

The shifted Legendre modes have eigenvalues $j(j+1)$. Two integrations by parts have zero endpoint flux and no prefix jump; Bessel's inequality then gives $\|(I-\Pi_q^A)g\|_{L^2} \le [q(q+1)]^{-1}\|\mathscr{D}_A g\|_{L^2}$. Projection contraction on $h-g$ yields the forward tail

$$H_{T_0} := \max_j \|(I-\Pi_q^A)h^{(j)}\|_{L^2}/\sqrt{mn} \le YP\ell_n/q^2 + Pd_{T_0}.$$

The backward-history freezing proof following (78) uses the same closure residual and dense response reference, hence applies unchanged. Its bounds $b_0 \le X^{42}(Y/\lambda)(1+(Y/\lambda)\sqrt{\ell_n})$, $b_1 \le X^{16}(Y/\lambda)\sqrt{\chi}$ and $B_d\sqrt{4Y/\lambda} \le P\ell_n$ give, for $b_a = (\widehat{r}_a/\widehat{\rho})\widehat{\delta}_a$,

$$B_{T_0} := \max_j \|(I-\Pi_q^A)b^{(j)}\|_{L^2}/\sqrt{mn} \le YP\ell_n/q + P\ell_n D_{T_0} \le 2YP\ell_n/q \quad (q \le \sqrt{n}).$$

The growing-projection identity (67) and vanishing initial errors convert squared endpoint errors into final squared history tails. Sample/activity Cauchy–Schwarz in the exact defect gives $d_{T_0} \le 2(L-1)H_{T_0}B_{T_0}$. The signed comparison and ledger equations (77) and (78) give

$$D_{T_0} \le \mathcal{A}_n d_{T_0}, \qquad \mathcal{A}_n \le eXe^{X^{36}(Y/\lambda)^2\sqrt{\ell_n}} \le X^2 n^{1/64} \le Pn^{1/64}.$$

Here $c = X^{36}(Y/\lambda)^2 \le X^{-24}$ and $c\sqrt{\ell_n} \le \ell_n/64 + 16c^2$. For $q \ge 4P^3\ell_n$, absorb the term linear in d_{T_0}, using $Y \le 1$ and $2(L-1) \le P$, to obtain

$$d_{T_0} \le 4YP^3\ell_n^2/q^3, \qquad D_{T_0} \le 4YP^4\ell_n^2 n^{1/64}/q^3. \tag{103}$$

The gate (100) ensures $q \le \sqrt{n}$, $q \ge 4P^3\ell_n$, and $8P^4\ell_n^2 \le n^{15/64}$. For the first use $\ell_n^2 e^{\sqrt{\ell_n}/2} \le e^{100}n^{1/8}$ and absorb the ceiling by two. The second follows from $n^{1/4} \ge 4P^3$. For the third use $\ell_n^2 \le C(2,1/64)n^{1/64}$. Consequently the bootstrap improves to $D_{T_0} \le Y/(2\sqrt{n})$. Continuity excludes a first exit. All constants are independent of T_0, so the estimate extends to all times and to the fitted limits. Real whole-sphere prediction subtraction costs at most $X^7 D_{T_0}$, proving (101). The exact inventory is (72); $n \ge d^4$ absorbs the first-matrix term in the original headline count.

□

9.4 Taylor accuracy, storage and final width gates

Use Proposition 10.2's exact-metric selected dynamics. The finite source lemma supplies its real fitting, RMS, carrier and approximation premises. Its explicit ledger gives $F \le X^6$, $K_1 \le X^{18}$, $H_C \le 2X^3$, $F_c \le X^{15}$ and $\mathcal{B}_n/(Y/\lambda) \le X^{22}(1+\sqrt{\ell_n})$. Thus $e^v - 1 \le ve^v$ permits $C_{\mathrm{sel}} = X^{60}$ for both coefficients of (109). Choose its exact tolerance

$$\eta = \min\left\{1, Y, 16Y/\lambda, \frac{\lambda}{2C_{\mathrm{sel}} n(1+\lambda^{-1/2})(1+\sqrt{\ell_n})e^{\sqrt{\ell_n}}}\right\}.$$

The finite-time error is at most $Y/(2n)$; the fitted tail is at most $Y/(2n)$ under the label-independent gate

$$n^7 \ge 2C_{\mathrm{sel}}\lambda^{-1}(1+\lambda^{-1/2}). \tag{104}$$

since $ne^{-8\ell_n} \le n^{-7}$.

Apply the source construction in the proof of Proposition 10.6 with our disk cover and $M_n = X^6\sqrt{n}$. Its real interval lies strictly inside the open disk union, so compactness supplies the rectangle required by Proposition 10.3. For declared-input span dimension k, use

$$K = \max\{0, \lceil \log_2(8M_n/\eta)\rceil\}, \qquad R = 2m + k + 1 + 2(2m+p)J(K+1).$$

The proof there gives Taylor tail $\eta/8$, base coefficient accuracy $\eta/128$, summed base error $\eta/64$, and image error $\eta/8$ using exact initialized images and operator norm eight. Its input-span reduction preserves panel trajectories and Gaussian first-row laws without passive labels. The exact retained count is

$$1020(L+1)R^2 + 36R + 10m(k+1) + pk + (m+p) + 3(m+p)d.$$

The additional gate

$$n \ge N_{\mathrm{count}} := [32X^6(1+C_{\mathrm{sel}})\lambda^{-3/2}(1+Y^{-1})]^4 \tag{105}$$

implies $K+1 \le 8\ell_n$. To check it, the logarithm defining K is at most $(3/2)\log n + \sqrt{\ell_n} + \log(1+\sqrt{\ell_n})$ plus the logarithm of the bracket in this gate. The latter is at most $(1/4)\log n$, and the source gate gives $\ell_n \ge 16$, so each intervening term is at most $\ell_n/4$. Division by $\log 2$ and the ceiling leave the stated bound. The branches $\eta = Y$ and $\eta = 1$ are smaller. Squaring the rank bound with (90) and $k \le m+p$ yields

$$R^2 \le X^{810}(m+p)^2\left[(Y/\lambda)^4\ell_n^3 + \lambda^{-2}\ell_n^2 b_n^2\right].$$

The exact inventory is therefore bounded by the Taylor headline with, for example, coefficient X^{820}. The inverse-label gate belongs to this simplification only. Without it the exact K displays the extra $\log(1/Y)$ dependence. Dense arrays, jets, coefficients and proof-panel anchors are discarded after initialization, exactly as before.

We finally prove Theorem 9.1. Allocate $\tau = \delta/2$ to the dense lower event and $\rho = \delta/4$ to the relevant source event. On their intersection the Legendre ratio is at most

$$4P^5 a_\tau^{-1}\ell_n^{9/2} n^{-15/64} \le 4P^5 a_\tau^{-1} C(9/2, 7/64) n^{-1/8}.$$

Consequently the following fully specified maximum is a sufficient Legendre width:

$$\left\lceil \max\left\{N_{\mathrm{src}}(m, \delta/4), N_{\mathrm{low}}(\delta/2), N_{\mathrm{Leg}}, [4P^5 C(9/2, 7/64)/(a_{\delta/2}\varepsilon)]^8\right\}\right\rceil. \tag{106}$$

For Taylor the absolute error Y/n and the same lower event give ratio at most $a_\tau^{-1}\ell_n^{5/2}/\sqrt{n} \le 64a_\tau^{-1}n^{-1/4}$. A sufficient width is

$$\left\lceil \max\left\{ N_{\mathrm{src}}(m+p,\delta/4), N_{\mathrm{low}}(\delta/2), N_{\mathrm{count}}, [2C_{\mathrm{sel}}\lambda^{-1}(1+\lambda^{-1/2})]^{1/7}, [64/(a_{\delta/2}\varepsilon)]^4 \right\} \right\rceil . \tag{107}$$

The probability union costs less than δ; no independence between the source and comparison events is required. Since $\lambda^{-1} = m/\gamma$, u_ρ is logarithmic in mL/ρ, $L \le X$, and a_τ^{-1} has only the displayed polynomial factors, these maxima are bounded by the polynomial asserted in the theorem. The proof counts retained real coordinates and exact-arithmetic error; it does not infer efficient preprocessing or finite precision from a polynomial sufficient width.

10 One selected nonlinear dynamics for both source constructions

The sparse-metric lemma first supplies exact geometry on any finite source space. The transfer proposition then constructs and analyzes one autonomous selected model, with its optimizer and comparison coefficients local to that proof. Two later propositions supply its source spaces from whole-sphere polynomials or finite-panel Taylor expansions. They use only the transfer statement's error and storage interface.

Lemma 10.1 (Sparse coordinates with an exact source metric)**.** *Let $E \subset \mathbb{R}^n$ be a linear subspace containing $\mathbf{1}_n$, with $r = \dim E$. There are a set I of at most $9r$ coordinates, a positive diagonal matrix D on I, and a positive definite matrix M such that*

$$u_I^\top M v_I = u^\top v/n \quad (u,v \in E), \qquad D/4 \preceq M \preceq D, \qquad \mathbf{1}^\top M \mathbf{1} = 1, \quad \mathbf{1}^\top D \mathbf{1} \le 4.$$

Write $\|u\|_M = (u^\top M u)^{1/2}$. Consequently a coordinate multiplication operator B satisfies $\|B\|_{M\to M} \le 2\max_i |B_{ii}|$, and an activation with $|\phi'| \le s$, $|\phi(0)| \le b$ satisfies $\|\phi(u)-\phi(v)\|_M \le 2s\|u-v\|_M$ and $\|\phi(u)\|_M \le 2b + 2s\|u\|_M$.

Proof. We give the finite sparsification argument. For vectors $v_i \in \mathbb{R}^r$ with $\sum_i v_i v_i^\top = I$, start with $A = 0$, lower barrier $l = -3r$ and upper barrier $u = 6r$. Maintain

$$lI \prec A \prec uI, \qquad \Phi_l(A) = \mathrm{tr}(A - lI)^{-1} \le \tfrac{1}{3}, \quad \Phi^u(A) = \mathrm{tr}(uI - A)^{-1} \le \tfrac{1}{6}.$$

Set $l' = l+1$, $u' = u+2$, and define

$$U(v) = \frac{v^\top (u'I - A)^{-2} v}{\Phi^u(A) - \Phi^{u'}(A)} + v^\top (u'I - A)^{-1} v, \qquad L(v) = \frac{v^\top (A - l'I)^{-2} v}{\Phi_{l'}(A) - \Phi_l(A)} - v^\top (A - l'I)^{-1} v.$$

The lower potential implies that every eigenvalue is at least three above l, so both inverse matrices and both denominators are positive. The identity $(B + tvv^\top)^{-1} = B^{-1} - B^{-1}vv^\top B^{-1}/(t^{-1} + v^\top B^{-1} v)$ shows that $U(v) \le t^{-1} \le L(v)$ preserves the two potential inequalities at the shifted barriers. For the upper potential, the positive first term of $U(v)$ also preserves the strict spectral barrier.

Such a nonzero v_i exists. If a_j are the eigenvalues of A, then $(u - a_j)^{-1} - (u' - a_j)^{-1} \ge 2(u' - a_j)^{-2}$, hence $\sum_i U(v_i) \le 1/2 + \Phi^{u'}(A) \le 2/3$. For the lower sum put $a'_j = (a_j - l')^{-1}$, $b'_j = (a_j - l)^{-1} = a'_j/(1 + a'_j)$ and $\Delta = \sum_j a'_j b'_j = \Phi_{l'} - \Phi_l$. Weighted Cauchy–Schwarz gives

$$\Delta^2 \le \Big(\sum_j (a'_j)^2 b'_j \Big) \sum_j b'_j \le \sum_j (a'_j)^2 - \Delta.$$

Therefore $\sum_i L(v_i) \ge 1 - \Phi_l(A) \ge 2/3$. Choose $t > 0$ with reciprocal in the resulting nonempty interval and add $tv_iv_i^\top$. After $9r$ iterations the barriers are $6r, 24r$. Dividing all accumulated weights by $6r$ gives nonnegative weights s_i, supported on at most $9r$ indices, with $I \preceq \sum_i s_i v_i v_i^\top \preceq 4I$.

Choose U with range E and $U^\top U/n = I$, apply this result to $v_i = U_{i,:}^\top/\sqrt{n}$, and set $P = U_I$, $D = \operatorname{diag}(s_i/n)$, $G = P^\top DP$ and $Z = D^{1/2}P$. Define

$$M = D^{1/2}\{ZG^{-2}Z^\top + I - ZG^{-1}Z^\top\}D^{1/2}. \tag{108}$$

On $\operatorname{ran} Z$ the middle matrix has eigenvalues those of G^{-1}, and on its orthogonal complement it is the identity. Thus $D/4 \preceq M \preceq D$, while multiplication gives $P^\top MP = I$. The constant source proves the two mass assertions. Finally $\|Bu\|_M \le \|Bu\|_D \le \max_i |B_{ii}|\|u\|_D \le 2\max_i |B_{ii}|\|u\|_M$. Apply this to coordinate divided differences and use $\|\mathbf{1}\|_M = 1$ for the activation claims. □

10.1 A portable source-to-runtime construction

Proposition 10.2 (Source-to-runtime transfer)**.** *Work on the fitting and analytic-source event of Lemma 5.1 and proposition 6.1, or its quantitative finite-panel version in Lemma 9.2. Write $W_0^{(j)} = W^{(j)}(0)$ for the initialized matrices in this statement. Fix*

$$T = 32\frac{m}{\gamma}\log(en), \qquad 0 < \eta \le \min\{1, Y, 16Ym/\gamma\}, \qquad q \ge \max\{m, d\}.$$

For each layer let $E_j \subset \mathbb{R}^n$ have dimension at most q and contain the constant, exact initialized training features and their initialized forward images when present, and (at the first layer) the columns of $W_0^{(1)}$. Suppose these spaces approximate the families (25) to coordinate accuracy η on $[0, T]$, with forward queries in $\mathcal{X}$ and backward queries at training inputs. Require exact initialized-image pairing: each image approximant is the corresponding initialized map applied to its base approximant. There is an autonomous selected nonlinear predictor satisfying

$$\begin{aligned}\|f_{\text{model}} - f_n\|_{\mathcal{X}} \le{}& C\frac{Ym}{\gamma}\left(1 + \sqrt{\frac{m}{\gamma}}\right)(1 + \sqrt{\log(en)})e^{\sqrt{\log(en)}}\eta \\ &+ C\frac{Ym}{\gamma}\left(1 + \sqrt{\frac{m}{\gamma}}\right)e^{-\gamma T/(4m)}.\end{aligned} \tag{109}$$

It fits, has a fitted limit, and retains at most

$$1020(L+1)q^2 + 36q + 10m(d+1) \tag{110}$$

real coordinates, including data, selected indices, metrics, initial copies and working arrays. Here $C \ge 1$. In particular, the permissible choice

$$\eta = \min\left\{1, Y, \frac{16Ym}{\gamma}, \frac{\gamma}{2Cmn(1 + \sqrt{m/\gamma})(1 + \sqrt{\log(en)})e^{\sqrt{\log(en)}}}\right\} \tag{111}$$

gives error at most Y/n eventually. Source bases, dense coefficients and compilation scratch are discarded after the selected initial arrays are formed.

Proof. All auxiliary quantities in this proof belong to this construction. Put $\lambda = \gamma/m$, $z = Y/\lambda$, $\alpha = Y/\sqrt{\lambda}$, $S = 16z$, and $\ell = \log(en)$, so $0 < \lambda \le 1$ and $0 < z \le \beta^{-30L}$. For activation estimates use the envelopes $b = \beta - 1$ and $s = t_2 = \beta$; thus $|\phi_j(0)| \le b$, $|\phi_j'| \le s$ and $|\phi_j''| \le t_2$ on the real line.

Write $c_n = (y_a - f_n(t, x_a))_{a=1}^m$ for the dense training deficits and $\rho_n = \|c_n\|_2/\sqrt{m}$ for their RMS. The public source bounds permit the local envelopes

$$u_0 = \beta^{3L}, \qquad \tau = \beta^{4L}, \qquad K_{\mathrm{src}} = \beta^{21L}.$$

Dense feature norms are at most u_0 and dense training response norms at most $S\tau$ in neuron RMS. Dense training carriers on $[0,T]$ are the local fields $k_{n,a}^{(L)} = w$ and $k_{n,a}^{(j)} = W^{(j+1)\top}\delta_{n,a}^{(j+1)}$; by (28) they are bounded coordinatewise by $32zK_{\mathrm{src}}\sqrt{\ell}$. Dense vector norms below are normalized by $\sqrt{n}$; selected norms are specified by their metrics. Every query v denotes $x/\sqrt{d}$ for an input x in the promised domain.

Autonomous selected model. Apply Lemma 10.1 to obtain (I_j, M_j, D_j) and let $q_j = |I_j| \le 9q$. The selected moving variables are $W_C^{(1)} \in \mathbb{R}^{q_1\times d}$, $B_C^{(j)} \in \mathbb{R}^{q_j\times q_{j-1}}$ $(2 \le j \le L)$, $w_C \in \mathbb{R}^{q_L}$ and $c_C \in \mathbb{R}^m$. For $v = x/\sqrt{d}$ the forward pass is

$$z_C^{(1)} = W_C^{(1)} v, \qquad z_C^{(j)} = B_C^{(j)} h_C^{(j-1)}, \qquad h_C^{(j)} = \phi_j(z_C^{(j)}).$$

All layer norms and adjoints below use M_j; in particular $(B_C^{(j)})^* = M_{j-1}^{-1} B_C^{(j)\top} M_j$. Let V_C have columns $h_C^{(L)}(v_a)/\sqrt{m}$ and define

$$Q_C = V_C^* V_C, \qquad \widehat{w}_C = w_C + V_C Q_C^{-1}\{(y - c_C)/\sqrt{m} - V_C^* w_C\}, \qquad f_C(v) = \langle \widehat{w}_C, h_C^{(L)}(v)\rangle_{M_L}. \tag{112}$$

The statement's predictor is $f_{\mathrm{model}}(t,x) = f_C(t, x/\sqrt{d})$; physical time is suppressed in the local forward formulas. Thus $f_C(v_a) = y_a - c_{C,a}$ exactly. Define training signals by $k_{C,a}^{(L)} = \widehat{w}_C$, $\delta_{C,a}^{(j)} = \phi_j'(z_{C,a}^{(j)}) \odot k_{C,a}^{(j)}$ and $k_{C,a}^{(j)} = (B_C^{(j+1)})^* \delta_{C,a}^{(j+1)}$. The unnormalized sample matrix is

$$\begin{aligned} K_{C,ab} = {} & \langle h_{C,a}^{(L)}, h_{C,b}^{(L)}\rangle_{M_L} + \langle \delta_{C,a}^{(1)}, \delta_{C,b}^{(1)}\rangle_{M_1} v_a^\top v_b \\ & + \sum_{j=2}^{L} \langle \delta_{C,a}^{(j)}, \delta_{C,b}^{(j)}\rangle_{M_j} \langle h_{C,a}^{(j-1)}, h_{C,b}^{(j-1)}\rangle_{M_{j-1}}. \end{aligned} \tag{113}$$

The autonomous optimizer is

$$\begin{aligned} \dot{W}_C^{(1)} &= \frac{2}{m}\sum_a c_{C,a}\delta_{C,a}^{(1)} v_a^\top, & \dot{B}_C^{(j)} &= \frac{2}{m}\sum_a c_{C,a}\delta_{C,a}^{(j)} h_{C,a}^{(j-1)\top} M_{j-1}, \\ \dot{w}_C &= \frac{2}{m}\sum_a c_{C,a} h_{C,a}^{(L)}, & \dot{c}_C &= -2K_C c_C/m. \end{aligned} \tag{114}$$

These are prescribed directions; coordinate gates need not be self-adjoint in M_j, and no gradient interpretation of the corrected predictor is used.

For bases U_j of E_j with $U_j^\top U_j/n = I$, set $P_j = (U_j)_{I_j}$. Initialize

$$W_C^{(1)}(0) = (W_0^{(1)})_{I_1}, \quad w_C(0) = 0, \quad c_C(0) = y, \qquad B_C^{(j)}(0) = P_j \frac{U_j^\top W_0^{(j)} U_{j-1}}{n} P_{j-1}^\top M_{j-1}. \tag{115}$$

Since $P_j^\top M_j P_j = I$, these operators have norms at most eight. They reproduce an initialized forward image exactly whenever both members belong to the source spaces; their metric adjoints reproduce paired reverse images. The exact additions therefore preserve the entire initialized training forward pass and Gram. All bases, dense arrays and source coefficients are discarded after these finite initial arrays are formed.

Claim 10 (Independent fitting and endpoint tails)**.** *With the notation of this construction, the following local bounds hold. The selected dynamics exist for every $t \geq 0$, have $Q_C \succeq \lambda I/4$, fit the training labels and converge in parameter space. Write*

$$H_1^c = 2b + 16s, \quad H_j^c = 2b + 18sH_{j-1}^c, \quad H_c = H_L^c, \quad d_j^c = 2s(18s)^{L-j}, \quad F_c = (d_1^c)^2 + 4H_c^2 \sum_{j=2}^{L} (d_j^c)^2.$$

For the selected parameter tuple $\theta_C = (W_C^{(1)}, (B_C^{(j)})_{j=2}^L, w_C)$ use

$$\|\theta_C\|_{\mathrm{par}}^2 = \mathrm{tr}(W_C^{(1)\top} M_1 W_C^{(1)}) + \sum_{j=2}^{L} \|M_j^{1/2} B_C^{(j)} M_{j-1}^{-1/2}\|_F^2 + \|w_C\|_{M_L}^2.$$

Then $\|h_C^{(j)}\| \leq 2H_c$, $\|\widehat{w}_C\| \leq 5\alpha$, $\rho_C := \|c_C\|/\sqrt{m} \leq Ye^{-\lambda t/2}$ and $\int_0^\infty \|\dot{\theta}_C\|_{\mathrm{par}} \leq 2\alpha$. There is a coefficient $\mathcal{D} \leq Cz(1+\lambda^{-1/2})$ such that comparison on $[0,T]$ extends to all times and the fitted endpoint with additional error at most $\mathcal{D}e^{-\lambda T/4}$. Here and below C depends only on activations and depth.

Proof of the claim. Stop at operator norm nine, feature norm $2H_c$, or Gram margin $\lambda/4$. Expanding the velocities, with all cross-sample terms, gives

$$-\frac{d}{dt}\rho_C^2 = 4c_C^\top K_C c_C/m^2 = \|\dot{\theta}_C\|_{\mathrm{par}}^2, \qquad K_C/m \succeq Q_C. \tag{116}$$

Thus $-\dot{\rho}_C \geq \lambda\rho_C/2$ and $\|\dot{\theta}_C\| \leq 2(-\dot{\rho}_C)/\sqrt{\lambda}$. Integration gives the asserted residual and path bounds, as well as $\int \rho_C \leq 2z$ and $\|w_C\| \leq 2\alpha$. With $T_C = V_C Q_C^{-1}$ and $P_C = T_C V_C^*$, the readout is the orthogonal sum $(I - P_C)w_C + T_C(y - c_C)/\sqrt{m}$. Its squared norm is at most $4\alpha^2 + 16\alpha^2$.

Set $R_c = 5\alpha$, $U_1^c = d_1^c$, $U_j^c = 2H_c d_j^c$ for $j \geq 2$, and $F_1^c = 2sU_1^c$, $F_j^c = 2s(2H_c U_j^c + 9F_{j-1}^c)$. The latter recurrence has $F_L^c = F_c$. Backward propagation bounds $\|\delta_C^{(j)}\| \leq d_j^c R_c$, and integration bounds each hidden parameter displacement by $20U_j^c Y^2/\lambda^{3/2}$ and each feature displacement by $20F_j^c Y^2/\lambda^{3/2}$. Under the stated cap, $H_c \leq \beta^{3L}$, $F_c \leq \beta^{15L}$ and $z \leq (16H_c\sqrt{F_c})^{-1}$; hence the last bound is at most $5\sqrt{\lambda}/(64H_c^2) < \min\{1/(8H_c), \sqrt{\lambda}/8\}$. The operator and feature stops improve strictly. The normalized training feature matrix starts with smallest singular value at least $\sqrt{\lambda/2}$ and moves by less than $\sqrt{\lambda}/8$, so the Gram stop improves too. Smoothness on this strict Gram-positive domain and bounded parameters give global existence. Finite path length and (112) give convergence; $c_C \to 0$ gives fitting.

For quantitative tails, differentiating the inverse gives

$$\dot{T}_C = (I - P_C)\dot{V}_C Q_C^{-1} - T_C \dot{V}_C^* T_C, \qquad \dot{P}_C = (I - P_C)\dot{V}_C T_C^* + T_C \dot{V}_C^* (I - P_C).$$

Thus $\|\dot{T}_C\| \leq 8\|\dot{V}_C\|/\lambda$, $\|\dot{P}_C\| \leq 4\|\dot{V}_C\|/\sqrt{\lambda}$ and $\|\dot{V}_C\| \leq 2R_c F_c \rho_C$. Put $G_c = 4H_c^2 + R_c^2 F_c$ and

$$B_w = 4H_c + 48R_c F_c z + 4G_c/\sqrt{\lambda}, \qquad B_f = 2H_c B_w + 2R_c^2 F_c.$$

Differentiating the orthogonal readout formula proves $\|\dot{\widehat{w}}_C\| \leq B_w \rho_C$ and $|\dot{f}_C(v)| \leq B_f \rho_C$ uniformly over unit inputs. Therefore $|f_C(\infty, v) - f_C(t, v)| \leq 2B_f z e^{-\lambda t/2}$. For the dense tail, the RMS bounds hold at every real time, not merely on the finite complex source domain. Indeed Lemma 5.1 bounds all normalized operators by nine and gives $\|w\|/\sqrt{n} \leq 2Y/\sqrt{\lambda} \leq Su_0$. The all-time RMS bounds of Proposition 6.1, or the same real forward/backward induction, give $\|h^{(j)}\|/\sqrt{n} \leq u_0$ and

$\|\delta^{(j)}\|/\sqrt{n} \le S\tau$ for all real times. The flow equations and sample Cauchy–Schwarz therefore imply $|\dot f_n(v)| \le 2\mathcal{K}\rho_n$, where

$$\mathcal{K} = u_0^2 + S^2\tau^2[1 + (L-1)u_0^2].$$

Comparing each flow with its value at T proves the claimed extension with $\mathcal{D} = z(16\mathcal{K} + 4B_f)$. The small cap gives $\mathcal{K} \le C$, $B_f \le C(1+\lambda^{-1/2})$, which establishes the stated parameter dependence. □

Comparison coefficients. Here is an explicit coefficient ledger for the comparison. Its size is independent of the number of sources. Use the local envelopes $u_0, \tau, K_{\mathrm{src}}$ fixed at this proof's start, and put

$$H_r = u_0 + 3, \quad H_C = 2H_c, \quad P_h = 6u_0 + 9, \quad P_\delta = 6\tau + 9,$$
$$A_f = 18 + S^2(\tau+3)P_h, \qquad A_b = 18 + S^2 H_r P_\delta.$$

Define the forward-error recursion and the reference-response bounds by

$$\begin{aligned}
&F_1^z = 1, \quad F_1^h = 2s,\\
&F_j^z = 9F_{j-1}^h + H_r + A_f, \quad F_j^h = 2sF_j^z \quad (j \ge 2), \quad F = \max_j F_j^h,\\
&d_j^E = 2s(9s)^{L-j} + 3H_c, \quad J_C = 5\sqrt{F_c},\\
&J_R = \left[(d_1^E)^2 + H_r^2 \sum_{j=2}^{L} (d_j^E)^2\right]^{1/2},\\
&D_s = P_h + 16zP_\delta\{1 + (L-1)H_r^2\} + (L-1)(16z)^2\tau^2 P_h.
\end{aligned}$$

The backward-error recursion is

$$\begin{aligned}
B_L &= 2s(1 + 6zF + 32zP_h) + 2t_2F_L^z,\\
B_j &= 18sB_{j+1} + 2s(d_{j+1}^E zH_C + A_b) + 2t_2F_j^z \quad (j < L),\\
B_h &= \sqrt{L}\{H_C \max_j B_j + (\max_j d_j^E) zH_C F\},\\
K_1 &= 12F + 2B_h + 20z(J_C + J_R)B_h + 20D_s.
\end{aligned}$$

Increasing any of these positive coefficients is harmless. Finally set

$$\begin{aligned}
\mathcal{B}_n &= 400z^2F_c + 2zK_1 + 40zF + 64z^2K_1K_{\mathrm{src}}\sqrt{\ell},\\
\mathcal{A}_n &= [H_C(1+6zF) + 3\alpha F](1+\lambda^{-1/2})(e^{\mathcal{B}_n} - 1)\\
&\quad + 6H_CzF + 32H_CzP_h/\sqrt{\lambda} + 3\alpha F + 16zP_h.
\end{aligned} \tag{117}$$

Claim 11 (Finite-horizon comparison)**.** *With the local coefficients* (117)*,*

$$\sup_{0\le t\le T}\sup_{x\in\mathcal{X}} |f_C(t, x/\sqrt{d}) - f_n(t,x)| \le \mathcal{A}_n\eta.$$

Proof of the comparison claim. We record both the source-energy estimate and the cancellation that make the parameter dependence possible. For one layer abbreviate $E = E_j$, $I = I_j$ and $M = M_j$. If $p \in E$ approximates u to coordinate accuracy e_u, exact isometry and the mass bound give

$$\|u - p\|_n \le e_u, \quad \|u_I - p_I\|_M \le 2e_u, \quad \|u_I\|_M \le \|u\|_n + 3e_u.$$

Expanding two inner products about $p, q \in E$ therefore gives

$$|\langle u_I, v_I\rangle_M - \langle u, v\rangle_n| \le 3e_u\|v\|_n + 3e_v\|u\|_n + 9e_u e_v. \tag{118}$$

Thus feature-pair defects are at most $P_h\eta$, and response-pair defects at most $SP_\delta\eta$; here $\eta \le 1, S$ is used.

Let the proof reference have first layer and readout equal to the coordinate restrictions of the dense ones, and hidden operators

$$B_R^{(j)}(t) = B_C^{(j)}(0) + \frac{2}{m}\int_0^t \sum_a c_{n,a}(u)(\delta_{n,a}^{(j)}(u))_{I_j}(h_{n,a}^{(j-1)}(u))_{I_{j-1}}^\top M_{j-1}\,du.$$

Its preactivation, feature and response symbols below mean the restricted dense values, not a fresh forward pass; its carrier symbols mean $k_{R,a}^{(j)} = (k_{n,a}^{(j)})_{I_j}$ as well. The exact initialized pairing and operator bound eight make an initialized action defect at most 18η. In each learned action, subtract the dense outer-product integral and use (118) together with $\int \rho_n \le 2z$, $\|\delta_n^{(j)}\|_n \le S\tau$ and $\|h_n^{(j)}\|_n \le u_0$. The resulting forward and reverse defects are at most $A_f\eta$ and $A_b\eta$. The identical calculation for the integrated raw readout gives

$$|\langle w_R, h_R^{(L)}(v)\rangle - f_n(v)| \le 16zP_h\eta. \tag{119}$$

Write V_n, V_R for the dense and restricted top feature maps, with columns divided by $\sqrt{m}$. Their common approximants and exact isometry give, for every $a \in \mathbb{R}^m$, $\|V_R a\| \le \|V_n a\| + 3\eta\|a\|$. For $\nu = \|V_R c_n/\sqrt{m}\|$, the dense energy identity gives

$$\int_0^T \nu \le \alpha + 6\eta z \le 2\alpha. \tag{120}$$

Here let $\theta_n = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n})$ with its Euclidean block norm. Indeed $2\|V_n c_n/\sqrt{m}\| \le \|\dot\theta_n\|$, whose integral is at most 2α. Integrating source approximants for $\dot w_n = 2V_n c_n/\sqrt{m}$ gives a vector in E_L at coordinate error at most $4\eta z$. Consequently $\|w_R\| \le 2\alpha + 12\eta z \le 3\alpha$. Likewise the restricted responses have norm at most $d_j^E\alpha$. These arguments never differentiate an approximation error.

Use the hidden-parameter Hilbert norms of the preceding local fitting claim, and let $\theta_{h,C}, \theta_{h,R}$ denote the selected and reference tuples of first-layer and hidden-layer matrices. Let $\mathcal{J}_C, \mathcal{J}_R$ have sample columns, divided by $\sqrt{m}$, equal respectively to the tuples

$$(\delta^{(1)} v_a^\top, (\delta^{(j)} h_a^{(j-1)\top} M_{j-1})_{j=2}^L).$$

Thus $K_C/m = V_C^* V_C + \mathcal{J}_C^*\mathcal{J}_C$ and $\dot\theta_{h,C} = 2\mathcal{J}_C c_C/\sqrt{m}$; the reference satisfies the latter identity with R, n. Moreover $\|\mathcal{J}_C\| \le J_C\alpha$, $\|\mathcal{J}_R\| \le J_R\alpha$. Put

$$a_h = \|\theta_{h,C} - \theta_{h,R}\|, \quad e = (c_C - c_n)/\sqrt{m}, \quad u = \|e\|, \quad z_w = w_C - w_R,$$

$$T_C = V_C Q_C^{-1}, \quad P_C = T_C V_C^*, \quad p = T_C e, \quad \zeta = z_w + p, \quad b_e = \|p\| + \|\zeta\|, \quad E = a_h + b_e.$$

All these error variables vanish at zero. Forward induction, first subtracting the reference action and then the nonlinear gate, gives

$$\|h_C^{(j)} - h_R^{(j)}\| \le F_j^h(a_h + \eta), \qquad \|V_C - V_R\| \le F(a_h + \eta). \tag{121}$$

The first-layer preactivation error is a_h; at layer j it is at most $9F_{j-1}^h(a_h+\eta)+H_r a_h + A_f\eta$, which proves the displayed recursion rather than presuming an ambient Lipschitz bound.

Set $d_R = V_R^* w_R - (y-c_n)/\sqrt{m}$ and $\Delta V = V_C - V_R$. Subtracting the exact readout formula yields

$$\widehat{w}_C - w_R = (I-P_C)z_w - p - T_C\Delta V^* w_R - T_C d_R. \tag{122}$$

Since $p \in \operatorname{ran} V_C$, $(I-P_C)z_w = (I-P_C)\zeta$. Equations (119)–(121) and $\|T_C\| \le 2/\sqrt{\lambda}$ imply

$$\|\widehat{w}_C - w_R\| \le b_e + 6zF(a_h+\eta) + 32zP_h\eta/\sqrt{\lambda}, \tag{123}$$

$$|f_C(v) - f_n(v)| \le H_C\{b_e + 6zF(a_h+\eta) + 32zP_h\eta/\sqrt{\lambda}\} + 3\alpha F(a_h+\eta) + 16zP_h\eta. \tag{124}$$

Let K_n be the dense training tangent Gram: it is the formula (113) with dense fields and the inner products $n^{-1}u^\top v$ in each neuron layer. The dense deficit therefore obeys $\dot{c}_n = -2K_n c_n/m$. The source-only Gram defect $D = V_R^* V_R + \mathcal{J}_R^*\mathcal{J}_R - K_n/m$ satisfies $\|D\| \le D_s\eta$. To see this without an extra factor m, split each product of two pairings, use (118), and recall that the sample matrix entries carry the factor $1/m$; a matrix all of whose entries have magnitude at most a/m has operator norm at most a. With $\Delta\mathcal{J} = \mathcal{J}_C - \mathcal{J}_R$, write

$$\Delta\mathcal{K} = V_C^*\Delta V + \Delta V^* V_R + \mathcal{J}_C^*\Delta\mathcal{J} + \Delta\mathcal{J}^*\mathcal{J}_R + D.$$

The error equations are exactly

$$\begin{aligned}
\dot{e} &= -2(Q_C + \mathcal{J}_C^*\mathcal{J}_C)e - 2\Delta\mathcal{K}c_n/\sqrt{m},\\
\dot{z}_w &= 2V_C e + 2\Delta V c_n/\sqrt{m},\\
\dot{\zeta} &= 2\Delta V c_n/\sqrt{m} + \dot{T}_C e - 2T_C\mathcal{J}_C^*\mathcal{J}_C e - 2T_C\Delta\mathcal{K}c_n/\sqrt{m}.
\end{aligned}$$

The term $2V_C e$ cancels because $T_C Q_C = V_C$. This is why estimating $c_C - c_n$ by an unconstrained Gronwall inequality would lose the result.

Here are the needed norm estimates, including the dissipative one. The derivative formula for T_C, and $\|\dot{V}_C\| \le 10\alpha F_c\rho_C$, give $\|\dot{T}_C e\| \le 40zF_c\rho_C\|p\|$. In the derivative of $\|p\|^2/2$, the Q_C term is $-2u^2$. The possible absolute contribution from the hidden Gram is at most $8\|\mathcal{J}_C\|^2u^2/\lambda \le 200z^2F_c u^2$; the label cap bounds this by $25u^2/(32H_c^2)$, leaving at least $39u^2/32$. Define

$$\mathfrak{f} = 2\|T_C\Delta\mathcal{K}c_n/\sqrt{m}\|, \qquad I(t) = \int_0^t \{40zF_c\rho_C b_e + \mathfrak{f}\}.$$

Dividing the dissipative inequality by $\|p\|$ gives

$$\|p(t)\| + \int_0^t u^2/\|p\| \le I(t), \qquad \int_0^t u \le 2I(t)/\sqrt{\lambda}.$$

At zeros use $(\|p\|^2+\epsilon^2)^{1/2}$ and let $\epsilon \downarrow 0$; $p=0$ implies $e=0$ because Q_C is invertible. The second inequality also follows directly from $\|p\| \le 2u/\sqrt{\lambda}$. Integrating the ζ equation bounds its hidden-Gram term by $200z^2F_cI \le 25I/32$. Therefore

$$b_e \le 2F\int_0^t \rho_n(a_h+\eta) + 3I(t). \tag{125}$$

For completeness, the remaining response estimate can be checked layer by layer. The dense training preactivation carriers are bounded coordinatewise by $M = 1+32zK_{\mathrm{src}}\sqrt{\ell}$ on $[0,T]$. At each gate split $\phi'(z_C)k_C - \phi'(z_R)k_R$ as $\phi'(z_C)(k_C-k_R) + (\phi'(z_C)-\phi'(z_R))k_R$. The first multiplication

costs $2s$, and the second at most $2t_2MF_j^z(a_h+\eta)$. Subtracting the initialized and learned reverse actions costs $A_b\eta$, and the hidden parameter difference costs $d_{j+1}^E\alpha a_h$. Starting with (123), the displayed B_j recursion, followed by the outer-product split, gives

$$\|\Delta\mathcal{J}\| \le B_h\{b_e + M(a_h+\eta) + \eta/\sqrt{\lambda}\}.$$

Only the carrier factor contains M; it is not multiplied once per layer. Applying this to the four terms of $\Delta\mathcal{K}$ gives

$$\begin{aligned}\mathfrak{f} \le{}& 2F\rho_n(a_h+\eta) + 4F\nu(a_h+\eta)/\sqrt{\lambda} + 4D_s\eta\rho_n/\sqrt{\lambda}\\ &+ 4z(J_C+J_R)B_h\rho_n\{b_e + M(a_h+\eta) + \eta/\sqrt{\lambda}\}.\end{aligned}$$

The hidden-parameter equation, $\int u \le 2I/\sqrt{\lambda}$ and $4zJ_C \le 5/4$ similarly yield

$$a_h \le \tfrac{5}{4}I + 2B_h\int_0^t \rho_n\{b_e + M(a_h+\eta) + \eta/\sqrt{\lambda}\}.$$

Combining with (125) proves the scalar inequality

$$E(t) \le \int_0^t \{200zF_c\rho_C + K_1M\rho_n + 20F\nu/\sqrt{\lambda}\}\{E + \eta(1+\lambda^{-1/2})\}.$$

The integral of its coefficient is at most $\mathcal{B}_n$, by $\int\rho_n, \int\rho_C \le 2z$ and (120). The integrating-factor formula gives $E \le \eta(1+\lambda^{-1/2})(e^{\mathcal{B}_n}-1)$. Substitution in (124) proves the finite-horizon claim with precisely (117). □

Canonical error bound and retained inventory. The local fitting claim supplies the tails and endpoint. Thus

$$\|f_{\text{model}} - f_n\|_{\mathcal{X}} \le \mathcal{A}_n\eta + \mathcal{D}e^{-\lambda T/4}.$$

We verify that the explicit ledger has the promised dependence. Put $X = \beta^L \ge 100$. The local envelopes and fitting bounds give $u_0, H_c \le X^3$, $\tau \le X^4$ and $K_{\text{src}} \le X^{21}$. For example $16z \le 16X^{-30}$ makes both additional terms in A_f, A_b less than one; hence $A_f, A_b \le 19$. The forward recursion then gives $F \le X^6$. For further detail, $d_j^E \le X^4$, $H_C \le 2X^3$ and $P_h, P_\delta \le X^4, X^5$. In the backward recursion the additive term is at most X^7, and $18s \le \beta^3$. Geometric summation over at most L layers gives $\max_j B_j \le X^{11}$ and then $B_h \le X^{15}$. Also $J_C, J_R \le X^9$ and $D_s \le X^5$. Substitution gives $K_1 \le X^{18}$; here the generous powers absorb $L \le X$, all numerical factors, and $H_C = 2H_c$. The selected fitting recursion already gives $F_c \le X^{15}$. Consequently

$$\mathcal{B}_n \le 400X^{-45} + 2X^{-12} + 40X^{-24} + 64X^{-21}\sqrt{\ell} \le 1 + \sqrt{\ell},$$

and $\mathcal{B}_n/z \le C(1+\sqrt{\ell})$ with C depending only on X. Since $e^u - 1 \le ue^u$, (117) gives

$$\mathcal{A}_n \le Cz(1+\lambda^{-1/2})(1+\sqrt{\ell})e^{\sqrt{\ell}}, \qquad \mathcal{D} \le Cz(1+\lambda^{-1/2}),$$

after increasing the activation/depth constant to absorb e. These are exactly the canonical factors in (109). This is a small-cap proof, not a claim on a larger recurrence-defined label interval. Finally $T = 32\ell/\lambda$ makes the tail $\mathcal{D}e^{-8\ell}$. For fixed positive Y it is at most $Y/(2n)$ eventually; the other half is (111).

Retained inventory. Retain both moving and initial copies of the first matrix, mixers and raw readout; the selected indices; each M_j, its inverse or factorization, and D_j; the training data and labels; current training forward and backward arrays; c_C, V_C, Q_C and its inverse or factorization; the corrected readout; direction arrays; and one-query forward workspace. Every selected width is at most $9q$. Each mixer or metric therefore costs at most $81q^2$, each first matrix at most $9qd$, each layer's training arrays at most a fixed multiple of $9mq$, and each sample matrix at most m^2. Since $m, d \le q$, these arrays total at most $1020(L+1)q^2 + 36q + 10m(d+1)$, including simultaneous initial copies and working arrays. A nonprimitive activation evaluator is charged separately. Dense mixers, source bases, coefficient vectors and coefficient-compilation scratch are not retained by the runtime. The construction itself uses the finite source spaces only through its initial arrays. The next compiler establishes initialization-jet provenance; Proposition 2.3 supplies separate efficient numerical implementations of the same source-to-runtime interface. □

10.2 Finite initialized jets suffice

Proposition 10.3 (Initialization-only coefficient compiler)**.** *On the initialized operator event of Lemma 5.1, fix $T, r_0, M > 0$. Let finitely many sources from Proposition 6.1 extend coordinatewise holomorphically near $[-r_0, T + r_0] + i[-r_0, r_0]$, bounded by M uniformly over their compact query domain. For sphere queries assume also the joint holomorphic extension of Proposition 6.1. Any finite list of their integrals against specified polynomial time–sphere bases, or derivatives at real anchors in $[0, T]$, can be approximated to arbitrary positive accuracy using finitely many derivatives of the dense ODE and passive recursions at time zero, with exact initialized-image pairing. No later dense state or trajectory sample is needed; no bound on initialization time or working storage is asserted.*

Proof. We first prove the continuation assertion that makes initialized jets sufficient. For any one scalar source g, use its common bound M on the closed rectangle $\{-r_0 \le \Re t \le T + r_0,\ |\Im t| \le r_0\}$, with a slightly larger open domain. The source rectangle, or a finite closed-disk cover with the real interval strictly inside its union, supplies such a positive r_0 by compactness. Put

$$\chi = \frac{\pi T}{8r_0}, \quad b_0 = \tanh\chi, \quad b_1 = \tanh(\chi + \pi/4), \quad \theta = b_0/b_1, \quad \psi(\xi) = \frac{\xi - \theta}{1 - \theta\xi},$$

$$\mathfrak{t}(\xi) = \frac{T}{2} + \frac{2r_0}{\pi} \log \frac{1 + b_1\psi(\xi)}{1 - b_1\psi(\xi)}, \qquad \xi_* = \frac{2\theta}{1 + \theta^2} < 1.$$

The logarithm is the branch equal to a real number on the real interval. For $|\xi| < 1$, $|\psi(\xi)| < 1$; the ratio in the logarithm has positive real part and modulus between $(1 - b_1)/(1 + b_1)$ and its reciprocal. Thus $|\Im\mathfrak{t}| < r_0$ and $-r_0 < \Re\mathfrak{t} < T + r_0$. Direct substitution gives $\mathfrak{t}(0) = 0$ and $\mathfrak{t}(\xi_*) = T$, and the derivative is nonzero on $[0, \xi_*]$. This proves continuation from the origin onto the entire physical interval, without stepping a dense trajectory.

The Taylor coefficients of $g \circ \mathfrak{t}$ are

$$a_j = \sum_{k=0}^{j} \frac{g^{(k)}(0)}{k!} [\xi^j]\mathfrak{t}(\xi)^k.$$

Cauchy's coefficient formula gives $|a_j| \le M$, so truncation at N has error at most $M\xi_*^{N+1}/(1 - \xi_*)$ on the whole interval. This bound is uniform over all passive inputs in the compact domain and over all source coordinates. Derivatives at finitely many later anchors are also recovered: choose $\xi_* < \rho_* < 1$ and $0 < \epsilon_* < \rho_* - \xi_*$. On $|\xi| \le \rho_*$ the tail is $M\rho_*^{N+1}/(1 - \rho_*)$, hence its kth derivative

at any real anchor is at most $k!M\rho_*^{N+1}/\{\epsilon_*^k(1-\rho_*)\}$. Repeated application of $(\mathfrak{t}')^{-1}\partial_\xi$ converts these to the finitely many required time derivatives. All its coefficients are bounded on that compact interval. A finite N therefore achieves every prescribed finite list of accuracies.

Let $\Theta = (W^{(1)}, \dots, W^{(L)}, w)$ be the dense parameter tuple and $\mathcal{F}$ its autonomous vector field. For the initialized ODE $\dot{\Theta} = \mathcal{F}(\Theta)$ the formal coefficients are computed successively by

$$\Theta_{j+1} = \frac{1}{j+1}[t^j]\mathcal{F}\left(\sum_{i=0}^{j}\Theta_i t^i\right).$$

The analytic local solution has precisely these coefficients, by coefficient comparison and uniqueness of the local ODE. The forward and passive backward recursions then give their source jets by finite addition, multiplication and composition at time zero. Any scalar activation derivatives required here are part of the fixed analytic activation evaluator; their description, evaluation and scratch are not asserted to have unit cost unless taken as primitives.

For time–sphere integral coefficients, insert the uniformly convergent continued polynomials into their defining integrals. Each integrand is continuous on the compact time interval times sphere. Finite partitions and weighted Riemann sums converge uniformly for the finite collection of integrands. A finite mesh can be chosen from their modulus of continuity: the source strip gives derivative bounds by Cauchy's formula, and the finitely many basis functions are known polynomials. Thus the required integral approximations use only finitely many evaluations of the continued origin-jet polynomial. For $d = 1$ the spatial integral is simply the two-point average.

Finally perform every scalar linear coefficient operation on a base source and set its image coefficient equal to the initialized matrix times that vector. Linear operations commute with this fixed matrix. Since its Euclidean operator norm is at most eight, approximating a base coefficient in Euclidean norm to $\epsilon/8$ approximates both the base and its image to ϵ. There are only finitely many coefficients and a finite sum of their basis suprema; choose each coefficient tolerance smaller than the desired final error divided by this sum. This imposes exact pairing while retaining the chosen uniform source tolerance. No coefficient is retained after the selected initial arrays are formed. □

10.3 Whole-sphere polynomial sources

Lemma 10.4 (Joint time–sphere approximation and its dimension)**.** *Let $T, r_t, r_q, M_n > 0$ and $0 < \eta \le \min\{1, M_n\}$, and put $\alpha_T = r_t/(4T)$. Suppose a real $\mathbb{R}^n$-valued source on $[0,T] \times S^{d-1}$ extends coordinatewise holomorphically to a neighborhood of*

$$([-r_t, T+r_t] + i[-r_t, r_t]) \times \{v \in \mathbb{C}^d : v^\top v = 1,\ \|\operatorname{Im} v\|_2 \le \sinh r_q\},$$

with coordinate modulus at most M_n on this set. Query holomorphy is relative to the quadric; for $d = 1$ use its two real points. For $d \ge 2$ put

$$b_d = 2d-2, \quad D_d = 2^{d+1}d^{d-2}, \quad P_T = \frac{18 D_d b_d!\, 2^{b_d+1}}{\alpha_T r_q^{b_d+1}}, \quad H_* = 2\log\frac{16 M_n P_T}{\eta}.$$

For $\alpha_T, r_q \le 1$, a real expansion in Chebyshev time polynomials and real spherical harmonics with $\alpha_T k + r_q j \le H_*$ approximates the source to coordinate error $\eta/8$ using at most

$$N \le \frac{2\{H_* + \alpha_T + (d-1)r_q\}^d}{d!\,\alpha_T r_q^{d-1}} \tag{126}$$

coefficient vectors. For the dense sources of Proposition 6.1, Proposition 10.3 gives initialized-jet coefficients with total coordinate error at most η and exact initialized-image pairing. For $d = 1$, use the two points of S^0 and at most $2N_1$ vectors, where $N_1 = 1 + \lfloor H_1/\alpha_T \rfloor$ and $H_1 = \log(64 M_n/(\alpha_T \eta))$.

Proof. Fix one coordinate and call its scalar source g. We supply the spatial decay argument. For $d \ge 2$ let $\mathcal{H}_j$ be the restrictions of homogeneous harmonic polynomials of degree j to S^{d-1}, with probability surface measure. Its dimension is

$$h_j = \binom{j+d-1}{d-1} - \binom{j+d-3}{d-1} \le 2\binom{j+d-2}{d-2},$$

where a binomial coefficient with negative upper index is zero. Indeed, applying the Laplacian to $|x|^{2k}H_l(x)$ gives $2k(2l+2k+d-2)|x|^{2k-2}H_l(x)$; induction decomposes every homogeneous polynomial into these summands and gives the dimension formula by subtraction. The spherical Laplacian eigenvalues $-j(j+d-2)$ give orthogonality of different degrees by integration by parts. Polynomials contain constants and separate points on the compact sphere, so their uniform density (the real Stone–Weierstrass theorem) proves completeness. For an orthonormal basis $Y_{j\nu}$, rotational invariance and integration give the addition identity $\sum_\nu Y_{j\nu}(v)^2 = h_j$; consequently $|Y_{j\nu}(v)| \le \sqrt{h_j}$ and the projection Π_j has supremum operator norm at most h_j.

For fixed v, average an analytic function over the complex tangent circle $\cos\zeta\, v + \sin\zeta\, u$, $u \perp v$, $|u| = 1$. On $\mathcal{H}_j$ this averaging is scalar: rotations fixing v leave a one-dimensional zonal eigenspace, and its normalized eigenfunction solves $(1-x^2)p'' - (d-1)xp' + j(j+d-2)p = 0$, $p(1) = 1$. Coefficient comparison gives $p(x) = C_j^{(d-2)/2}(x)/C_j^{(d-2)/2}(1)$ for $d > 2$. For real ζ the averaging is self-adjoint because the joint law of its two endpoints is symmetric. Apply this to each harmonic coefficient and continue analytically in ζ; it follows that the averaged analytic source has jth projection equal to $p(\cos\zeta)\Pi_j g$. At $\zeta = ir_q$ its arguments satisfy $z^\top z = 1$, $|\Im z| \le \sinh r_q$, so its supremum is at most M_n. Thus $\|\Pi_j g\|_\infty \le h_j M_n / p(\cosh r_q)$.

For $\nu = (d-2)/2 > 0$, the generating function $(1-2xw+w^2)^{-\nu} = \sum_j C_j^\nu(x)w^j$ at $x = \cosh r$, for any $r > 0$, factors as $(1-e^r w)^{-\nu}(1-e^{-r}w)^{-\nu}$ with nonnegative coefficients. It gives $C_j^\nu(\cosh r) \ge e^{rj}(\nu)_j/j!$ and $C_j^\nu(1) = (2\nu)_j/j!$. For $j \ge 1$ their ratio is $2\prod_{i=1}^{j-1}(1+\nu/(\nu+i))$. Its logarithm is at most $\log 2 + \nu\int_0^{j-1}(\nu+x)^{-1}\,dx$. Since $\nu \ge 1/2$, the ratio is at most $2(2j)^\nu$, and hence at most $2^d(j+1)^d$; the $j = 0$ inequality is immediate. Together with $h_j \le 2d^{d-2}(j+1)^{d-2}$ this proves

$$\|\Pi_j g\|_\infty \le M_n D_d (j+1)^{b_d} e^{-r_q j}. \tag{127}$$

For $d = 2$, the basis is the constant and the sine–cosine pair at each positive integer, and the averaging multiplier is $\cosh(jr) \ge e^{jr}/2$; the same loose bound with $D_2 = 8$, $b_2 = 2$ holds directly.

Set $t = T(1+\cos u)/2$. For $|\Im u| \le \alpha_T \le 1$ this lies inside the time rectangle, since its displacement from $[0,T]$ is at most $T\alpha_T$. The resulting even periodic function has cosine coefficients whose jth harmonic block is bounded by $2M_nD_d(j+1)^{b_d}e^{-\alpha_T k - r_q j}$: shift the Fourier contour to $\Im u = \pm\alpha_T$ and use (127) there. The series is absolutely uniformly convergent. Discarding indices with $\alpha_T k + r_q j > H_*$ bounds the tail by

$$2M_nD_d e^{-H_*/2}\sum_{k\ge0} e^{-\alpha_T k/2}\sum_{j\ge0}(j+1)^{b_d}e^{-r_q j/2} \le M_n P_T e^{-H_*/2} \le \eta/16.$$

Here $\sum_k e^{-\alpha_T k/2} \le 3/\alpha_T$ and $\sum_j (j+1)^b e^{-r_q j/2} \le 3b!(2/r_q)^{b+1}$, obtained by comparison with the gamma integral after shifting each unit interval.

The number of retained coefficients is $N = \sum_{r_q j \le H_*} h_j(1 + \lfloor (H_* - r_q j)/\alpha_T \rfloor)$. Replace h_j by twice the number of weak compositions of j into $d-1$ parts. The corresponding nonnegative lattice points have weighted sum $\alpha_T k + r_q \sum l_i \le H_*$. Their disjoint unit cubes lie in the simplex with right-hand side $H_* + \alpha_T + (d-1)r_q$, whose volume is exactly the right-hand side of (126). This proves the count. For the finite collection of coefficients, choose the Euclidean approximation

tolerance in Proposition 10.3 so that its product with $8\sum_{k,j,\nu}\|Y_{j\nu}\|_\infty$ is at most $\eta/16$. Both a base source and its paired image then have total error below η, without increasing the coefficient count.

If $d = 1$, there is no spatial approximation: apply the preceding one-variable Fourier contour argument separately at the two points. The geometric tail after degree $\lfloor H_1/\alpha_T \rfloor$ is at most $6M_n e^{-H_1}/\alpha_T < \eta/8$; the same finite coefficient accuracy argument finishes this case. □

Proposition 10.5 (Harmonic: whole-sphere accuracy and total state)**.** *Eventually at each fixed confidence, the initialized-jet construction gives a selected nonlinear model with* $\|f_{\mathrm{Harm}} - f_n\|_{\mathcal{X}} \le Y/n$ *on the whole sphere and total retained real-coordinate count at most*

$$C(m+d)^2 + \left(\frac{C}{d}\right)^{d+1}\left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^{3d+2}.$$

Proof. For this whole-sphere construction put $\lambda = \gamma/m$, $z = Y/\lambda$, $\ell = \log(en)$, and $T = 32\ell/\lambda$. Choose η by the canonical tolerance formula (111). The public source proposition supplies the radii and coordinate bound

$$r_t = \frac{\gamma}{\beta^{30L}Y^2 m\sqrt{(d+3)\ell}}, \qquad r_q = \frac{1}{\beta^{3L}\sqrt{(d+3)\ell}}, \qquad M_n = \beta^{6L}\sqrt{n}.$$

Set $\alpha_T = r_t/(4T)$. For $d \ge 2$, let N be the coefficient count in Lemma 10.4, with its explicitly defined P_T, H_* evaluated at these local parameters. For $d = 1$, let N_1 be the two-point time count defined in that same statement. Apply the approximation lemma to the four joint time–sphere families from Proposition 6.1, including its passive backward family; hence the same coefficient vectors serve all training inputs. In layer j include the approximate feature and response coefficients and the paired initialized forward and reverse image coefficients. Add $\mathbf{1}_n$, the at most $2m$ exact initialized training vectors needed for forward-image pairing, and the d first-layer columns. Each source space has rank at most

$$R = 2m + d + 1 + 4N \quad (d \ge 2), \qquad R = 2m + d + 1 + 8N_1 \quad (d = 1).$$

The source contract is now satisfied. Invoke Proposition 10.2 with its source-dimension bound $q = R$; it selects at most $9R$ coordinates per layer and gives the claimed absolute error. All choices use initialized jets only.

We spell out the asymptotic count because dropping the label factor here would change the theorem. The exact source radius gives

$$\alpha_T^{-1} = \frac{4T}{r_t} = 128\beta^{30L}z^2\sqrt{d+3}\,\ell^{3/2}, \qquad r_q^{-1} = \beta^{3L}\sqrt{(d+3)\ell}. \tag{128}$$

For each fixed admissible problem, (109) and (111) imply $\log(1/\eta) \le 2\ell$ eventually. Every other fixed parameter inside the logarithm defining H_* is additive, whereas $\log P_T = O_{d,\mathrm{problem}}(\log \ell)$. Thus $H_* + \alpha_T + (d-1)r_q \le 8\ell$ eventually. The dimension bound gives

$$N \le C^d\beta^{CLd}\frac{(d+3)^{d/2}}{d!}\,z^2\ell^{3d/2+1}.$$

For $d = 1$, the direct two-point count gives the same form with $N_1 \le C\beta^{CL}z^2\ell^{5/2}$.

The public inventory (110), with $q = R$, is $1020(L+1)R^2 + 36R + 10m(d+1)$, plus the separately charged activation evaluator. It already includes selected mixers, metrics, initial copies, data and working arrays. No dense mixer, source basis, initialized jet, quadrature table or coefficient vector is retained.

Finally, for $d \geq 2$, $R^2 \leq C(m+d)^2 + CN^2$ and $d! \geq (d/e)^d$ imply

$$\frac{(d+3)^d}{(d!)^2} \leq \frac{(8e^2)^d}{d^{d+1}}.$$

Indeed, for $d \geq 2$, $(1+3/d)^d \leq (5/2)^d$ and $d \leq 2^d$ give the stronger numerator $(5e^2)^d$. Absorb activation/depth powers into C^d to obtain the asserted count. For $d = 1$, use $R^2 \leq C(m+1)^2 + CN_1^2$ and the already proved bound on N_1; this gives the stated logarithmic exponent five. If a finite small width makes $9R \geq n$, retaining all coordinates is a valid fallback; the claimed polylogarithmic bound concerns the eventual range only. □

10.4 Finite-panel sources on an adaptive time partition

Proposition 10.6 (Taylor: panel accuracy and total state)**.** *On the declared $(m+p)$-input panel of Section 2, eventually at each fixed confidence there is an initialization-only selected model with $\|f_{\text{Taylor}} - f_n\|_{\mathcal{X}} \leq Y/n$ and total state at most*

$$C(m+p)^2 \left[\left(\frac{Ym}{\gamma}\right)^4 [\log(en)]^3 + \left(\frac{m}{\gamma}\right)^2 [\log(en)]^2 \{\log(e + \log(en))\}^2 \right] + C(m+p)d.$$

Only training inputs enter the deficit equation and Gram inverse. Accuracy outside the panel is not asserted.

Proof. For this finite-panel construction put $\lambda = \gamma/m$, $z = Y/\lambda$, $\ell = \log(en)$ and $T = 32\ell/\lambda$. Use the public disk-and-partition interface of Proposition 6.1: there are $0 = t_0 < \cdots < t_J = T$ and radii $r_b > 0$ such that each source is holomorphic near $|t - t_b| \leq r_b$ and $t_{b+1} - t_b \leq r_b/2$. Its count (29) gives

$$J \leq 1 + \beta^{30L} z^2 \sqrt{\ell} + \frac{\beta^{16L}}{\lambda} \log(1 + 4\ell/\log 2). \tag{129}$$

The centers and radii are deterministic functions of the problem parameters, as that public interface guarantees, so they require no later dense training data.

There are at most $2(2m+p)$ source curves per layer: forward features and their initialized images at all declared inputs, and backward responses and their initialized transpose images at training inputs. Each coordinate is bounded by $M_n = \beta^{6L}\sqrt{n}$ on its disks. Choose the source tolerance (111) and

$$K = \max\{0, \lceil \log_2(8M_n/\eta) \rceil\}.$$

On panel b, retain coefficient vectors $r_b^k u^{(k)}(t_b)/k!$ for $0 \leq k \leq K$, where u denotes one of the source curves. They have coordinate magnitude at most M_n. Since $|t - t_b|/r_b \leq 1/2$, the discarded Taylor tail is at most $M_n 2^{-K} \leq \eta/8$. The vectors from all panels span one time-independent space of dimension at most $J(K+1)$ per curve. The local approximants may differ at a panel boundary: both approximate the same source there, and the comparison theorem uses values, never their derivatives or a boundary matching rule.

Apply Proposition 10.3 to approximate every scaled base coefficient in Euclidean norm to $\eta/128$, and form its initialized image exactly. The base reconstruction error is at most $(\eta/128)\sum_{k\geq 0} 2^{-k} = \eta/64$; initialized operator norm eight gives image error at most $\eta/8$. Adding their independently controlled Taylor tails leaves both below η. Include the constant, exact initial training vectors and paired images, and first-weight columns required by Proposition 10.2. Before input reduction the rank per layer is at most $2m + d + 1 + 2(2m+p)J(K+1)$. This verifies the identical source contract, so the same selected runtime, fitting, cancellation, and endpoint proofs apply. In particular taking a

maximum over the passive inputs does not introduce p into any comparison coefficient. No passive deficit or additional sample Gram is introduced.

There is an exact reduction of the first-layer dimension. Let $U_0 \in \mathbb{R}^{d\times k}$ have orthonormal columns spanning all declared normalized inputs, where $k \le \min(d, m+p)$. Replace v_i by $U_0^\top v_i$ and the first matrix by $W^{(1)}U_0$. Every first-layer derivative is a sum of outer products with v_a, hence the component on the orthogonal complement of this input span is constant. Induction through the layers proves equality of the original and reduced coupled trajectories on the panel. The reduced inputs remain unit vectors; there is no new normalization factor. Each initialized Gaussian row multiplied by U_0 is exactly standard Gaussian in $\mathbb{R}^k$, and the sample inner products and population feature Gram are unchanged. Thus the same source event applies to the reduced system, whose public panel-count coefficients depend only on activations and depth, not on k or d. The source rank is now

$$R = 2m + k + 1 + 2(2m+p)J(K+1).$$

Invoke Proposition 10.2 with source-dimension bound $q = R$ and input dimension k. To its public inventory (110), add the passive inputs and output buffer and retain the input map. A safe explicit count is

$$1020(L+1)R^2 + 36R + 10m(k+1) + pk + (m+p) + 3(m+p)d,$$

plus the separately charged nonprimitive activation evaluator. This includes original and transformed data, the map, all metrics and inverse caches, initial and moving weights, direction arrays, training feature/response and sample-matrix buffers, and query workspace. Discard the dense arrays, source coefficients, origin jets, panel anchors and partition schedule. The optimizer is autonomous; a partition used to prove source approximation is not a retained forcing table.

For each fixed problem, (111) implies $K + 1 \le 8\ell$ eventually. The coefficients β^{30L} and β^{16L} in (129) depend only on activations and depth. Squaring that bound with $(a+b+c)^2 \le 3(a^2+b^2+c^2)$, and using $k \le m+p$, yields

$$R^2 \le C(m+p)^2\left[z^4\ell^3 + \lambda^{-2}\ell^2\{\log(e+\ell)\}^2\right].$$

The second term absorbs the initial rank and plain ℓ^2 terms because $\lambda \le 1$. Crucially, the z^2 in the first term of the panel count was retained before squaring, so its storage contribution remains $z^4 = (Ym/\gamma)^4$; it was not replaced by a label-independent constant. This proves the stated inventory and finishes the construction. □

11 Efficient initialization of the selected models

We prove Proposition 2.3 by computing disposable source values on $[0,T]$, $T = 32(m/\gamma)\log(en)$, then applying Proposition 10.2 to real-node interpolation (Taylor) or positive coefficient quadrature (Harmonic). The autonomous model starts at time zero with original activations, exact initialized training features and (112). Supplied matrices specify its dense reference; the implicit algorithm instead samples a fresh Gaussian reference through consistent forward and transpose actions. It does not convert supplied matrices or a prescribed entrywise seed in subquadratic work.

11.1 Numerical source values from initialization

All constants here are local to the numerical proof. Put $\lambda = \gamma/m$, $S = 16Y/\lambda$, $\ell = \log(en)$, $T = 32\ell/\lambda$, and $W_0^{(j)} = W^{(j)}(0)$. The normalized parameter vector and its norm are

$$\theta = (W^{(1)}/\sqrt{n}, W^{(2)}, \dots, W^{(L)}, w/\sqrt{n}), \qquad \|\theta\|^2 = \|W^{(1)}\|_F^2/n + \sum_{j=2}^{L} \|W^{(j)}\|_F^2 + \|w\|_2^2/n.$$

For the predictions and residuals in (1), set $\xi_a = \nabla_\theta f_n(x_a)$ and $\rho = \|r\|_2/\sqrt{m}$. The mobilities in (3) give precisely

$$\dot\theta = F(\theta), \qquad F(\theta) = -\frac{2}{m}\sum_a r_a \xi_a, \qquad \xi_a = \left(\frac{\delta_a^{(1)} v_a^\top}{\sqrt{n}}, \left(\frac{\delta_a^{(j)} h_a^{(j-1)\top}}{n} \right)_{j=2}^{L}, \frac{h_a^{(L)}}{\sqrt{n}} \right). \tag{130}$$

Complex products use algebraic transposes and ordinary complex Euclidean/Frobenius norms; no complex Gram positivity is used.

Condition on Lemma 5.1 and proposition 6.1 for Harmonic and Lemma 9.2 for Taylor. These events give real operator caps below nine, $\|w\|_2/\sqrt{n} \le 2Y/\sqrt{\lambda}$, $\rho(t) \le Y e^{-\lambda t/2}$ and $\int_0^\infty \rho \le 2Y/\lambda$. For centers in $[0,T]$, the respective source proofs supply complex disks of radius $r_t/2$ with preactivation imaginary parts at most $3a/8$, operator caps ten, residual RMS at most $2Y$, feature RMS at most β^{3L}, and training carrier coordinates bounded by

$$r_t = \frac{\lambda}{\beta^{30L} Y^2 \sqrt{(d+3)\ell}}, \qquad M_c = 2S\beta^{21L}\sqrt{\ell}. \tag{131}$$

Here $k^{(L)} = w$, $k^{(j)} = W^{(j+1)\top}\delta^{(j+1)}$. The displayed r_t is for Harmonic. Taylor uses $r_t = h(0)/2$ from (144); its radius-$h(0)/4$ disk is certified at every real center. Only declared Taylor queries, or real unit Harmonic queries, use these complex domains. For Taylor, apply the input reduction below and use its reduced dimension in the preactivation envelope.

11.1.1 Real signed stability and its constants

Use $b = \beta - 1$, $s = t_2 = \beta$, and define in order

$$\begin{gathered}
H_1^{\mathrm{r}} = \max(1, b + 10s), \quad H_j^{\mathrm{r}} = \max(1, b + 10sH_{j-1}^{\mathrm{r}}), \quad H^{\mathrm{r}} = H_L^{\mathrm{r}}, \quad R^{\mathrm{r}} = 1 + 2Y/\sqrt{\lambda}, \\
B^{\mathrm{r}} = R^{\mathrm{r}} s (10s)^{L-1}, \quad F_z^{\mathrm{r}} = H^{\mathrm{r}}(10s)^{L-1}, \quad P_{\mathrm{layer}}^{\mathrm{r}} = 1 + (L-1)H^{\mathrm{r}}, \\
D^{\mathrm{r}}(M) = (10s)^{L-1}[s(1+B^{\mathrm{r}}) + Lt_2 F_z^{\mathrm{r}} M], \\
J^{\mathrm{r}}(M) = \sqrt{L+1}[P_{\mathrm{layer}}^{\mathrm{r}} D^{\mathrm{r}}(M) + (1 + (L-1)B^{\mathrm{r}}) s F_z^{\mathrm{r}}], \\
C^{\mathrm{r}}(M) = sF_z^{\mathrm{r}}\sqrt{L} + LB^{\mathrm{r}} s F_z^{\mathrm{r}} + \tfrac{1}{2} L^2 t_2 (F_z^{\mathrm{r}})^2 M.
\end{gathered} \tag{132}$$

Evaluate $J^{\mathrm{r}}, C^{\mathrm{r}}$ at $M = M_c$, and put

$$\begin{gathered}
A_n = 4J^{\mathrm{r}} Y/\lambda, \quad K_{\mathrm{qry}}^{\mathrm{r}} = R^{\mathrm{r}}(10s)^{L-1}, \\
C_{\mathrm{src}} = 8\sqrt{L+1}\max\left\{ sF_z^{\mathrm{r}}, (10s)^{L-1}[s(1+B^{\mathrm{r}}) + Lt_2 F_z^{\mathrm{r}} K_{\mathrm{qry}}^{\mathrm{r}}] \right\}.
\end{gathered} \tag{133}$$

Lemma 11.1 (Stability of an approximate dense path)**.** *Let u be an absolutely continuous path on $[0,T]$, let $d_u = u' - F(u)$, and set $E_0 = \|u(0) - \theta(0)\| + \int_0^T \|d_u\|$. If*

$$E_0 \le e^{-A_n}/4, \qquad 8(C^{\mathrm{r}} + J^{\mathrm{r}})^2 T e^{2A_n} E_0^2 \le 1,$$

then $\sup_{[0,T]} \|u - \theta\| \le 2e^{A_n} E_0$. Every real-query source in (25) has coordinate error at most $C_{\mathrm{src}} n \|u(t) - \theta(t)\|$ when evaluated at $u(t)$ with the original activations.

Proof. Stop at $E = \|u - \theta\| = 1$, so both states have operator cap ten and readout RMS cap R^{r}. For the sum D_h of hidden-block discrepancies and readout RMS discrepancy D_w, $D_h \le \sqrt{L}E$, $D_h + D_w \le \sqrt{L+1}E$. Forward subtraction bounds preactivation and feature RMS errors by $F_z^{\mathrm{r}} D_h$ and $sF_z^{\mathrm{r}} D_h$. Backward subtraction is

$$\delta^{(j)}(u) - \delta^{(j)}(\theta) = \phi'(z^{(j)}(u)) \odot [k^{(j)}(u) - k^{(j)}(\theta)] + [\phi'(z^{(j)}(u)) - \phi'(z^{(j)}(\theta))] \odot k^{(j)}(\theta).$$

The second term uses M_c; propagation, changed mixers and top readout cost $10s$, $sB^{\mathrm{r}}\|\Delta W\|_F$ and sD_w. Their sum gives response RMS error $D^{\mathrm{r}}(M_c)(D_h + D_w)$ and $\|\xi_a(u) - \xi_a(\theta)\| \le J^{\mathrm{r}} E$. The endpoint remainder obeys

$$|f_n(u, x_a) - f_n(\theta, x_a) - \langle \xi_a(\theta), u - \theta \rangle| \le C^{\mathrm{r}} E^2.$$

Indeed, put $a^{(j)} = \Delta h^{(j)} - \phi'(z^{(j)}(\theta)) \odot \Delta z^{(j)}$. Scalar Taylor's formula gives $|a_i^{(j)}| \le t_2 |\Delta z_i^{(j)}|^2/2$. Backward substitution expresses the remainder as

$$\frac{1}{n} \sum_j k_a^{(j)}(\theta)^\top a_a^{(j)} + \frac{1}{n} \sum_{j=2}^{L} \delta_a^{(j)}(\theta)^\top \Delta W^{(j)} \Delta h_a^{(j-1)} + \frac{1}{n} \Delta w^\top \Delta h_a^{(L)}.$$

Its three bounds are $L^2 t_2 M_c (F_z^{\mathrm{r}})^2 E^2/2$, $LB^{\mathrm{r}} sF_z^{\mathrm{r}} E^2$, and $sF_z^{\mathrm{r}} \sqrt{L} E^2$. For $e_{f,a} = f_n(u, x_a) - f_n(\theta, x_a)$, the exact signed identity in the proof of Lemma 5.2, with these two remainder bounds, gives

$$\tfrac{1}{2}(E^2)' \le -2\|e_f\|_2^2/m + 2(C^{\mathrm{r}} + J^{\mathrm{r}})\|e_f\|_2 E^2/\sqrt{m} + 2J^{\mathrm{r}} \rho E^2 + E\|d_u\|.$$

Maximizing over $\|e_f\|_2/\sqrt{m}$ gives

$$E' \le 2J^{\mathrm{r}} \rho E + \tfrac{1}{2}(C^{\mathrm{r}} + J^{\mathrm{r}})^2 E^3 + \|d_u\|. \tag{134}$$

The regularization in Lemma 5.2 covers $E = 0$. Multiplying by $\exp(-2J^{\mathrm{r}} \int_0^t \rho)$, whose inverse is at most e^{A_n}, bounds the transformed error before its first $2E_0$ crossing by $E_0 + 4(C^{\mathrm{r}} + J^{\mathrm{r}})^2 T e^{2A_n} E_0^3 \le 3E_0/2$. This excludes the crossing and gives $E \le 1/2$, removing the stop; the same inequality covers $E_0 = 0$. For any real query, replace the carrier cap by $\sqrt{n} K_{\mathrm{qry}}^{\mathrm{r}}$ in the same subtraction and convert RMS to coordinates. These two $\sqrt{n}$ factors and initialized operator cap eight give $C_{\mathrm{src}} nE$, also between any two states in this real neighborhood. Finally $A_n = O(\sqrt{\ell})$ at fixed parameters. □

11.1.2 Complex restarts and scalar activation approximation

For complex neighborhoods of computed anchors, define separately

$$\begin{gathered}
R_{\mathrm{loc}} = 1 + S\beta^{3L}, \quad H_1 = \max(1, b + 11s), \quad H_j = \max(1, b + 11sH_{j-1}), \\
B_j = R_{\mathrm{loc}} s(11s)^{L-j}, \quad P_1 = 1, \quad P_j = H_{j-1} + 11sP_{j-1}, \quad P_* = \max_j P_j, \\
\kappa_L(M) = 1, \quad \kappa_j(M) = 11s\kappa_{j+1}(M) + 11t_2 M P_{j+1} + B_{j+1}, \\
D_j(M) = s\kappa_j(M) + t_2 M P_j, \quad \kappa_*(M) = \max_j \kappa_j(M), \\
G = H_L + B_1 + \sum_{j=2}^{L} B_j H_{j-1}, \\
J_{\mathrm{loc}}(M) = sP_L + D_1(M) + \sum_{j=2}^{L} [H_{j-1} D_j(M) + B_j sP_{j-1}].
\end{gathered} \tag{135}$$

The same endpoint subtraction, now at operator cap eleven and readout RMS cap R_{loc}, gives preactivation, carrier and response RMS errors $P_jE, \kappa_j(M)E, D_j(M)E$ when the reference carrier maximum is M. The gradient blocks then give $\|\xi_a\| \le G$, $\|\Delta\xi_a\| \le J_{\mathrm{loc}}(M)E$, and

$$\|F(u) - F(v)\| \le [2G^2 + 2\rho(v)J_{\mathrm{loc}}(M)]\|u - v\|.$$

Here integration along the parameter segment bounds the residual change by GE; all scalar preactivation segments lie in the half-strip below.

Define

$$\begin{gathered} b_n = \min\{1/4, a/(16\sqrt{n}P_*), 1/(\sqrt{n}\kappa_*(M_c))\}, \\ L_n = 2G^2 + 2(2Y+G)J_{\mathrm{loc}}(M_c+1), \quad V = 2G(2Y+G), \quad \widehat{R}_n = \min\{r_t/2, (8L_n)^{-1}\}. \end{gathered} \tag{136}$$

In a radius-b_n ball about the true complex trajectory, first-exit subtraction gives preactivation change at most $a/16$, hence imaginary part below $7a/16$, operator cap eleven, carrier maximum $M_c + 1$ and residual RMS $2Y + G$. Thus L_n is a proved pairwise Lipschitz bound there. The source proofs also give the center envelope

$$Z_* = [\beta^{21L} + 16\sqrt{d+3}\,\beta^{3L} + 1 + \beta^{26L}\sqrt{d+3}\,S^2]\sqrt{\ell} + a/8.$$

by integrating $2\rho S\beta^{26L}\sqrt{d+3}\sqrt{\ell}$ and adding the initial Gaussian and short complex-time bounds. For Taylor use its finite-list maximum and $SU_{\mathrm{fin}}\sqrt{\ell}$, $U_{\mathrm{fin}} \le \beta^{26L}$. The moving-ball rectangle is therefore $|\Re z| \le Z_* + a/16$, $|\Im z| \le 7a/16$. For source-vector tolerance $\varepsilon_v > 0$, put $\tau = \varepsilon_v/(32\sqrt{n})$ and define

$$\begin{gathered} e_d = e^{-A_n}\min\left\{\frac{1}{4}, \frac{1}{\sqrt{8}(C^{\mathrm{r}} + J^{\mathrm{r}})\sqrt{T}}, \frac{\tau}{4C_{\mathrm{src}}n}, \frac{b_n}{16}\right\}, \quad \zeta_F = \min\{e_d/(2T), b_nL_n/4, V/2\}, \\ U_0 = 0, \quad U_j = 1 + 11sU_{j-1}, \quad U_* = \max_j U_j, \quad K_j^{\mathrm{act}} = R_{\mathrm{loc}}(11s)^{L-j}, \\ Q_{L+1}(M) = 0, \quad Q_j(M) = 11(s+1)Q_{j+1}(M) + K_j^{\mathrm{act}} + 11t_2MU_{j-1}, \\ Q_\xi = U_L + Q_1(M_c+1) + \sum_{j=2}^{L}[Q_j(M_c+1)(H_{j-1}+1) + B_jU_{j-1}], \\ Q_f = R_{\mathrm{loc}}U_L, \quad C_F = 2Q_f(G+1) + 2(2Y+G)Q_\xi, \\ M_q = \sqrt{n}\max_j K_j^{\mathrm{act}}, \quad C_p = 8\sqrt{n}\max\{U_*, \max_j Q_j(M_q)\}, \\ \epsilon = \min\{1, U_*^{-1}, a/(5632\sqrt{n}U_*), Q_\xi^{-1}, \zeta_F/C_F, \tau/(2C_p)\}. \end{gathered} \tag{137}$$

Every denominator is positive. Set

$$\begin{gathered} B = 8(Z_* + a + 1), \quad \tau_o = \operatorname{arsinh}(31a/(64B)), \quad \tau_i = \operatorname{arsinh}(29a/(64B)), \quad \Delta = \tau_o - \tau_i, \\ M_\phi = b + s(B+a), \quad C_a = \max\{1, 512/a, 2(512/a)^2\}, \quad \epsilon_p = \epsilon/C_a, \\ D = \max\left\{1, \left\lceil \Delta^{-1}\log\max\{e, 24M_\phi/(\epsilon_p\Delta)\}\right\rceil\right\}, \qquad N_\phi = 4(D+1). \end{gathered} \tag{138}$$

With $x_i = B\cos(2\pi i/N_\phi)$, compute for each layer

$$c_{j,0} = N_\phi^{-1} \sum_i \phi_j(x_i), \qquad c_{j,k} = 2N_\phi^{-1} \sum_i \phi_j(x_i) \cos(2\pi ki/N_\phi)(k > 0),$$
$$\psi_j(z) = \sum_{k=0}^{D} c_{j,k} T_k(z/B). \tag{139}$$

Finite value-call errors of magnitude at most $\epsilon_p/[4(D+1)e^{D\tau_i}]$ are allowed in this formula only. Initialized training features below still use exact original values.

Lemma 11.2 (Certified numerical source polynomials)**.** *Let $C_g = \max_j\{H_j + 1, B_j + 1\}$ and choose*

$$\begin{aligned} K = \max\Big\{ & 8, \lceil 2\log_2 \max\{1, 16TV/e_d\}\rceil, \left\lceil \log_2 \max\{1, 8V\widehat{R}_n/b_n\}\right\rceil, \\ & \left\lceil \log_2 \max\{1, 128V\widehat{R}_n C_{\mathrm{src}} n^{3/2}/\varepsilon_v\}\right\rceil, \lceil \log_2 \max\{1, 64\sqrt{n}C_g/\varepsilon_v\}\rceil \Big\}, \\ J = & \lceil 2T/\widehat{R}_n\rceil = \lceil \max\{4T/r_t, 16TL_n\}\rceil. \end{aligned} \tag{140}$$

Use J equal physical-time panels. Starting at the original initialization, compute Taylor coefficients through degree K for the polynomial-activation field $\widetilde{F}$ by

$$[u]_{k+1} = \frac{1}{k+1}[h^k]\widetilde{F}\left(\sum_{i=0}^{k}[u]_i h^i\right), \qquad 0 \le k < K, \tag{141}$$

and advance to each polynomial endpoint. The forward and backward source jets through degree K, computed with the same coefficients and ψ_j, approximate every base source $h^{(j)}$ or $\delta^{(j)}$ in Euclidean norm by at most $\varepsilon_v/16$ at every real time in its numerical panel. Queries are all real unit inputs for Harmonic and the declared input list for Taylor. If $\log(1/\varepsilon_v) = O(\ell)$ at fixed parameters, then

$$J = O(\ell^{3/2}), \qquad K = O(\ell), \qquad D = O(\ell^{3/2}). \tag{142}$$

Proof. The outer ellipse $z = B(\zeta + \zeta^{-1})/2$, $|\zeta| = e^{\tau_o}$ lies in the half-strip and has $|\phi| \le M_\phi$. The inner ellipse contains radius-$a/512$ disks about the $a/512$ neighborhood of the moving-ball rectangle, since $(1/8)^2 + (226/232)^2 < 1$; also $a/(64B) \le \Delta < 1$. The Fourier contour argument of Lemma 10.4, applied to $\phi(B\cos u)$, gives coefficients $2M_\phi e^{-k\tau_o}$ and inner-ellipse tail $4M_\phi\Delta^{-1}e^{-(D+1)\Delta}$. Discrete sampling aliases frequencies $k + qN_\phi$; their geometric sum for $k \le D$ is at most $8M_\phi(D+1)e^{-2(D+1)\tau_o} \le 8M_\phi\Delta^{-1}e^{-(D+1)\Delta}$. The prescribed D makes these errors at most $\epsilon_p/2$; summing the allowed value-call errors adds at most $\epsilon_p/2$. Cauchy on the verified disks gives value and first-two-derivative errors at most ϵ.

At a fixed moving-ball state, forward substitution errors are $U_j\epsilon$ in feature RMS and $11U_{j-1}\epsilon$ in preactivation RMS. Since $11\sqrt{n}U_*\epsilon \le a/512$, all polynomial arguments remain in that domain. Backward subtraction has the three terms $11(s+1)Q_{j+1}$, K_j^{act} and $11t_2MU_{j-1}$, proving the response bound $Q_j(M)\epsilon$. The prediction and gradient bounds are $Q_f\epsilon, Q_\xi\epsilon$; product subtraction in (130) gives $\|\widetilde{F} - F\| \le C_F\epsilon \le \zeta_F$. For passive sources and initialized images the coordinate error is $C_p\epsilon \le \tau/2$. On the concentric half-ball, directional Cauchy gives derivative error at most $2\zeta_F/b_n \le L_n/2$, so $\widetilde{F}$ is $2L_n$-Lipschitz.

For $\|u_0 - \theta(t)\| \le b_n/8$, consider on $|h| \le \widehat{R}_n$ the holomorphic-error map

$$e(h) \longmapsto u_0 - \theta(t) + \int_0^h [\widetilde{F}(\theta(t+z) + e(z)) - F(\theta(t+z))]\, dz.$$

On the supremum ball of radius $b_n/2$ it contracts by at most $1/4$ and has image norm at most $9b_n/32$. Uniform Picard convergence and Cauchy's formula give a holomorphic solution v, real by conjugation and uniqueness, with discrepancy at most $5b_n/24 < b_n/4$ and velocity at most $2V$. Its degree-K truncation u on a half-radius panel has parameter and derivative tails $2V\widehat{R}_n 2^{-K}$ and $2V(2K+4)2^{-K}$. It remains in the smaller moving ball, with defect at most $2V(2K+5)2^{-K} + \zeta_F$. Because $2K + 5 \le 4\,2^{K/2}$ for $K \ge 8$, the prescribed order bounds total integrated defect by e_d. Lemma 11.1 on every completed prefix gives

$$\|u - \theta\| \le 2e^{A_n} e_d \le \min\{b_n/8, \tau/(2C_{\mathrm{src}} n)\}.$$

This closes finite restart induction from exact initialization, without unknown-norm tests or rejected panels. For a base source let g^0, g^p denote original and polynomial activation evaluation and g_K the degree-K source polynomial along v. Its error splits as

$$g_K - g^0(\theta) = [g_K - g^p(v)] + [g^p(v) - g^0(v)] + [g^0(v) - g^0(u)] + [g^0(u) - g^0(\theta)].$$

Each term is at most $\varepsilon_v/64$: the first uses $\|g^p(v)\|_2 \le \sqrt{n}C_g$ and its geometric Taylor tail; the second and fourth use $\sqrt{n}\tau/2$; the third uses $2V\widehat{R}_n C_{\mathrm{src}} n^{3/2} 2^{-K}$. Here the bound C_g follows from H_j, B_j and substitution error below one. This proves the asserted $\varepsilon_v/16$. Finally $M_c, L_n, A_n = O(\sqrt{\ell})$, $b_n^{-1} = O(\sqrt{n\ell})$, $\widehat{R}_n^{-1} = O(\sqrt{\ell})$, $C_p = O(n)$ and $Z_* = O(\sqrt{\ell})$ at fixed parameters. Thus all inverse-tolerance logarithms are $O(\ell)$ and $\Delta^{-1} = O(\sqrt{\ell})$, proving (142). The causal recursion differentiates only the finite polynomials ψ_j. □

11.2 Two source spaces with the original retained sizes

Let $\mathcal{A}_n$ be the comparison coefficient in (117), and let $\mathcal{D}$ be the explicit endpoint coefficient in the proof of Claim 10. Use

$$\eta = \min\{1, Y, S, Y/(2n\mathcal{A}_n)\}, \qquad \mathcal{D}e^{-8\ell} \le Y/(2n). \tag{143}$$

Then Proposition 10.2 gives all-time error Y/n once its source contract holds. At fixed positive parameters, $\log(1/\eta) \le 2\ell$ eventually. Taylor uses $C_{\mathrm{sel}} = \beta^{60L}$ from the polynomial-width assembly: its conservative tolerance is no larger than this η, and (104) gives the tail gate above. We use exactly (105) for the degree, so Taylor incurs no additional qualitative threshold. Harmonic retains its eventual source and scalar gates.

11.2.1 Taylor from real source nodes

Put $M = m + p$ and apply the exact input reduction proved in Proposition 10.6: $U_0 \in \mathbb{R}^{d\times k}$ spans the declared normalized inputs, $U_0^\top U_0 = I$, $k \le \min(d, M)$, and the reduced inputs and first matrix are $U_0^\top v_i, W_0^{(1)} U_0$. This preserves their Gram, Gaussian law and all declared trajectories; the retained map costs at most Md. Use the deterministic finite-panel domain of Lemma 9.2, with $\eta_*, U_{\mathrm{fin}}, \mathcal{K}$ defined by the source recurrences, (34) and (35), without observed trajectory norms:

$$\nu=\lambda/4,\quad a_0=\min(a,1),\quad b_{\mathrm{cap}}=\log(e+\ell),\quad B_{\mathrm{cap}}=1+(S^2/\eta_*)b_{\mathrm{cap}},$$
$$r_{\mathrm{cap}}=\min\left\{\frac{a_0}{64YSU_{\mathrm{fin}}\sqrt{\ell}},\frac{1}{\beta^{300L}(1+\lambda B_{\mathrm{cap}})\sqrt{b_{\mathrm{cap}}}}\right\},\tag{144}$$
$$h(t)=\frac{\log(1+2\mathcal{K}r_{\mathrm{cap}}e^{\nu t})}{2\mathcal{K}},\qquad d_h=\lambda/(8\mathcal{K}).$$

Set $t_0=0$, $r_b=h(t_b)/[2(1+d_h)]$ and $t_{b+1}=\min\{T,t_b+r_b/2\}$. These certified disks have radius at least $4h(0)/9>h(0)/4$, since h is nondecreasing and $d_h\le 1/8$, verifying the numerical disk used above. Identifying $X=\beta^L$, $\ell_n=\ell$, $r_n=r_{\mathrm{cap}}$ in (90) gives the source-panel count

$$J_s\le 1+\beta^{400L}\left[(Y/\lambda)^2\sqrt{\ell}+\lambda^{-1}\log(e+\ell)\right].\tag{145}$$

Here also $b_n=b_{\mathrm{cap}}$ and $B_n=B_{\mathrm{cap}}$ in that proof; its confidence-free count preserves the $(Y/\lambda)^2$ factor. Since $2\mathcal{K}r_{\mathrm{cap}}\le 1$, $h(0)\ge r_{\mathrm{cap}}/2$ and $h(0)^{-1}=O(\sqrt{\ell})$ at fixed parameters. Thus (142) holds even while the cap is active.

Put $M_n=\beta^{6L}\sqrt{n}$ and

$$K_s=\max\{8,\lceil 2\log_2(64M_n/\eta)\rceil\},\quad N_s=J_s(K_s+1),\quad \varepsilon_v=\eta/[128(K_s+1)].\tag{146}$$

On each $[t_b,t_{b+1}]$, evaluate the K_s+1 Chebyshev roots $t_b+(t_{b+1}-t_b)[1+\cos((i+1/2)\pi/(K_s+1))]/2$ by Lemma 11.2, forward on all M inputs and backward on training inputs, and insert exact initialized images. The nodes generate the space without computed interpolation coefficients. Discrete cosine orthogonality and $|T_j|\le 1$ give $\|I_{K_s}\|\le 2K_s+1$, in coordinate or Euclidean norm. Reproduction of the degree-K_s Taylor polynomial, whose tail on the half-radius panel is $M_n2^{-K_s}$, gives

$$\|g-I_{K_s}g\|_\infty\le(2K_s+3)M_n2^{-K_s}\le 4M_n2^{-K_s/2}\le\eta/16$$

for each exact source and image. Numerical nodes and images have Euclidean errors at most $\varepsilon_v,8\varepsilon_v$, so their interpolated perturbation is at most $16(K_s+1)\varepsilon_v=\eta/8$. Both families therefore meet accuracy η, with exact pairing $I_{K_s}(W_0\widetilde{g})=W_0I_{K_s}\widetilde{g}$ and its transpose counterpart.

Add the constant, exact initialized training features and images, and first-layer columns. The rank and actual selected width obey

$$r\le R=2m+k+1+2(M+m)N_s,\qquad q\le 9r.\tag{147}$$

Apply Proposition 10.2 with source-dimension bound R. To verify this numerical degree at the stated polynomial gate, put $B_0=32\beta^{6L}(1+C_{\mathrm{sel}})\lambda^{-3/2}(1+Y^{-1})$. At $n\ge B_0^4$ and the source gate, $\ell\ge 16$, $\log B_0\le(\log n)/4$. On the active comparison branch, $1+\lambda^{-1/2}\le 2\lambda^{-1/2}$ gives

$$\log(64M_n/\eta)\le\tfrac{3}{2}\log n+\log B_0+\log 8+\sqrt{\ell}+\log(1+\sqrt{\ell})\le\tfrac{9}{4}\ell+\log 8.$$

The Y and 1 branches are smaller; $S\ge 16Y$ is inactive. Thus $K_s+1\le\max\{9,(9/(2\log 2))\ell+8\}\le 8\ell$. The rank squaring in Proposition 10.6, now using (145), proves (10) with its $(Y/\lambda)^4\ell^3$ factor. The map and data add $O(Md)$; no passive labels enter.

11.2.2 Harmonic from positive coefficient quadrature

Use the current source radii r_t in (131) and $r_q=1/[\beta^{3L}\sqrt{(d+3)\ell}]$, with $M_n=\beta^{6L}\sqrt{n}$ and $\alpha=r_t/(4T)$. Require the eventual scalar gates $\alpha,r_q\le 1$. For $d\ge 2$, take exactly the mode set of Lemma 10.4:

$$b_d = 2d-2, \quad D_d = 2^{d+1}d^{d-2}, \quad P_T = 18D_d b_d! 2^{b_d+1}/(\alpha r_q^{b_d+1}), \quad H_* = 2\log(16M_nP_T/\eta),$$
$$\Lambda = \{(k,j,b) : \alpha k + r_q j \le H_*,\ 1 \le b \le h_j\}, \quad h_j = \binom{j+d-1}{d-1} - \binom{j+d-3}{d-1}, \tag{148}$$
$$N = |\Lambda|, \quad p_t = \lfloor H_*/\alpha \rfloor, \quad j_* = \lfloor H_*/r_q \rfloor, \quad H_{\mathrm{sph}} = \sum_{j \le j_*} h_j, \quad \mathcal{Y} = \max_{j \le j_*} \sqrt{h_j}.$$

Impossible binomials are zero. For a scalar source g, its coefficient is

$$c_{kjb} = \gamma_k \int_0^{2\pi} \int_{S^{d-1}} g(T(1+\cos u)/2, v) \cos(ku) Y_{jb}(v)\, d\sigma(v) \frac{du}{2\pi}, \quad \gamma_0 = 1, \quad \gamma_k = 2\ (k>0), \tag{149}$$

where Y_{jb} are real orthonormal harmonics for probability surface measure. The proof of Lemma 10.4 bounds the omitted-mode tail by $\eta/16$.

Here are finite quadrature orders and their construction. For $d \ge 3$ use polar angles $\theta_1, \dots, \theta_{d-2}$ and azimuth φ in the usual recursive coordinates $v = (\cos\theta_1, \sin\theta_1 u)$; for $d = 2$ use only $v = (\cos\varphi, \sin\varphi)$. Define

$$I_0 = \pi, \quad I_1 = 2, \quad I_b = (b-1)I_{b-2}/b, \quad A_d = \prod_{b=1}^{d-2} \pi/I_b,$$
$$a_t = \alpha/2, \quad h_{\mathrm{ang}} = r_q/[2(d-1)], \quad \sigma = \operatorname{arsinh}(2h_{\mathrm{ang}}/\pi),$$
$$\epsilon_c = \eta/(16N\mathcal{Y}), \quad \delta_{\mathrm{node}} = \eta/(64NA_d\mathcal{Y}^2),$$
$$B_{\mathrm{int}} = 2M_n\mathcal{Y}A_d e^{a_t p_t + j_*(d-1)h_{\mathrm{ang}}} (\cosh h_{\mathrm{ang}})^{(d-2)(d-1)/2}, \tag{150}$$
$$N_t = \max\{1, \lceil a_t^{-1} \log(1 + 4dB_{\mathrm{int}}/\epsilon_c) \rceil\},$$
$$N_\varphi = \max\{1, \lceil h_{\mathrm{ang}}^{-1} \log(1 + 4dB_{\mathrm{int}}/\epsilon_c) \rceil\},$$
$$N_\theta = \max\{1, \lceil \sigma^{-1} \log[8dB_{\mathrm{int}}/(\epsilon_c(1-e^{-\sigma}))] \rceil\} \quad (d \ge 3),$$
$$N_x = N_\varphi N_\theta^{d-2} \quad (d \ge 3), \qquad N_x = N_\varphi \quad (d = 2).$$

Empty products equal one. Apply equal-weight periodic trapezoids with N_t, N_φ nodes to u, φ. For each polar coordinate, transport to $\theta = \pi(1+x)/2$ the nodes and weights

$$\omega_i = (2i+1)\pi/(2N_\theta), \quad x_i = \cos\omega_i, \quad w_i = \frac{1}{N_\theta}\left[1 - 2\sum_{k=1}^{\lfloor (N_\theta-1)/2 \rfloor} \frac{\cos(2k\omega_i)}{4k^2-1}\right]. \tag{151}$$

Multiply spatial weights by the normalized spherical Jacobian $\prod_{i=1}^{d-2} (\pi/I_{d-1-i}) \sin^{d-1-i}\theta_i$; integration by parts verifies $I_b = \int_0^\pi \sin^b\theta\, d\theta$. The Fourier estimate already used above gives trapezoid error $2B/(e^{aN}-1)$ for strip half-width a and bound B, since the rule retains exactly the frequencies divisible by N. Discrete cosine orthogonality gives the polar interpolant $N_\theta^{-1}\sum_i g_i[1+2\sum_{k=1}^{N_\theta-1}\cos(k\omega_i)T_k(x)]$. Its normalized integral is (151): odd modes integrate to zero and T_{2k} to $-1/(4k^2-1)$. The rule is exact through $N_\theta-1$; its weights sum to one and are positive because $2\sum_{k=1}^{q}(4k^2-1)^{-1} = 1-(2q+1)^{-1} < 1$. Twice the ellipse Chebyshev tail therefore bounds its error by $4Be^{-N_\theta\sigma}/(1-e^{-\sigma})$, since both positive rule and normalized integral have norm one.

The domain checks hold jointly. A complex rotation of imaginary angle y sends imaginary norm $\sinh r$ to at most $\sinh(r+|y|)$: the real and imaginary parts of a quadric point have norms $\cosh r, \sinh r$, and the rotation costs $\cosh|y|, \sinh|y|$. Thus the $d-1$ rotations stay in the radius-$r_q/2$

tube. A harmonic has trigonometric degree at most j_* in each angle, so Laurent maximum modulus bounds it by $\mathcal{Y}e^{j_*(d-1)h_{\mathrm{ang}}}$; the Jacobian contributes $A_d(\cosh h_{\mathrm{ang}})^{(d-2)(d-1)/2}$, giving B_{int}. The polar ellipse has height $(\pi/2)\sinh\sigma = h_{\mathrm{ang}}$. Telescoping the d integral/rule factors bounds coefficient error by

$$B_{\mathrm{int}}\left[\frac{2}{e^{a_t N_t}-1}+\frac{2}{e^{h_{\mathrm{ang}}N_\varphi}-1}+\frac{4(d-2)e^{-\sigma N_\theta}}{1-e^{-\sigma}}\right]\le \epsilon_c/2.$$

Run Lemma 11.2 at $\varepsilon_v = \delta_{\mathrm{node}}$. Base Euclidean error is below $\delta_{\mathrm{node}}/8$, hence image coordinate error is at most δ_{node}. Spatial weight mass is at most A_d, so each coefficient perturbation is at most $2A_d\mathcal{Y}\delta_{\mathrm{node}} = \epsilon_c/2$. Adding the analytic tail gives $\eta/16 + N\mathcal{Y}\epsilon_c = \eta/8$ for sources and images separately. Accumulate only base coefficients and apply initialized matrices afterward: the identical scalar sums ensure $\widetilde{c}^{W_0 g} = W_0\widetilde{c}^{g}$ exactly. Thus no coordinate-error propagation through a matrix norm is needed; Proposition 10.2 applies with $R = 2m + d + 1 + 4N$.

For $d = 1$, use the two points, $N_x = H_{\mathrm{sph}} = 2$, $H_1 = \log(64M_n/(\alpha\eta))$, $N_1 = 1 + \lfloor H_1/\alpha \rfloor$, $p_t = N_1 - 1$, $N = 2N_1$, and $R = 2m + d + 1 + 8N_1$. Take $\epsilon_c = \eta/(16N_1)$, $\delta_{\mathrm{node}} = \eta/(64N_1)$, $B_{\mathrm{int}} = 2M_n e^{a_t p_t}$, and $N_t = \max\{1, \lceil a_t^{-1}\log(1 + 4B_{\mathrm{int}}/\epsilon_c)\rceil\}$. Only the time trapezoid is used. Its error allocation is identical; the one-dimensional analytic tail is at most $\eta/8$ by Lemma 10.4, still leaving total error below η.

The count proof of Proposition 10.5 applies to this identical mode set and the same logarithmic tolerance. In particular its factor $\alpha^{-1} = 128\beta^{30L}(Y/\lambda)^2\sqrt{d+3}\ell^{3/2}$ is preserved before squaring the rank. This proves the complete retained bound (9), with its label and dimension factors.

11.3 Exact implicit Gaussian execution

The next finite-dimensional sampler is useful because the numerical construction accesses an initialized hidden matrix only through its actions. All its zero tests and normalizations are exact.

Lemma 11.3 (Adaptive queries to both directions of a Gaussian matrix)**.** *For each of finitely many layers let the reference matrix have independent $N(0, 1/n)$ entries, independently across layers and of the client's independent randomness. The client may choose the next layer, direction and query vector measurably from that randomness and its complete past transcript, and may stop within deterministic finite query budgets. There is an online sampler whose entire transcript has exactly the dense reference law. On an extension of its probability space, independent Gaussian reference matrices can be completed so that every returned reply is simultaneously the requested product with that one matrix.*

For Q requests to one layer, the sampler uses at most $4nQ^2+6nQ+2$ scalar arithmetic operations, at most $n\min\{Q, 2n-1\}$ standard normal samples, and $2n\min\{Q, 2n\}+8n+5\min\{Q, 2n\}+64$ real or address words. These counts exclude the client's own work and retained arrays. The completion matrices are proof objects, not arrays constructed by the algorithm.

Proof. For one layer store $V, Y \in \mathbb{R}^{n\times p}$ and $U, R \in \mathbb{R}^{n\times q}$, initially empty, with

$$V^\top V = I, \qquad U^\top U = I, \qquad U^\top Y = R^\top V,$$

representing $Y = WV$, $R = W^\top U$. For a forward query v, compute $a = V^\top v$, $v_\perp = v - Va$, $\alpha = \|v_\perp\|_2$. If $\alpha = 0$ return Ya; otherwise set $e = v_\perp/\alpha$ and

$$h = UR^\top e + n^{-1/2}[g - U(U^\top g)], \tag{152}$$

return $Ya + \alpha h$ and append (e, h) to (V, Y). Here g is a fresh standard Gaussian vector if $q < n$; if $q = n$ omit the Gaussian term. Transpose queries use this identical algorithm with $(V, Y, p) \leftrightarrow (U, R, q)$, hence innovation

$$h = VY^\top f + n^{-1/2}[g - V(V^\top g)] \tag{153}$$

for a normalized new direction f, and no draw if $p = n$. Each layer has independent draws and its own four column lists; no dense projector is formed.

For proof only put $P = VV^\top$, $Q_0 = UU^\top$ and $A = YV^\top + UR^\top(I - P)$. Compatibility implies $AV = Y$, $A^\top U = R$. The homogeneous observation space is exactly $\{(I-Q_0)B(I-P) : B \in \mathbb{R}^{n\times n}\}$, orthogonal to A. Inductively the conditional law given the full transcript is

$$W = A + (I - Q_0)G(I - P), \qquad G_{ij} \text{ independent } N(0, 1/n). \tag{154}$$

It holds initially. Given the past, the adaptive query is fixed. For a new forward direction set $P' = P + ee^\top$ and split

$$(I - Q_0)G(I - P) = ((I - Q_0)Ge)e^\top + (I - Q_0)G(I - P').$$

The two terms are independent: their cross-covariances are $n^{-1}(I - Q_0)_{ia}[e^\top(I - P')]_b = 0$. This remains true for singular Gaussians, since the joint characteristic function of linear images factors when their cross-covariance vanishes. Thus We has the sampler law, and conditioning updates the mean to $A + (h - UR^\top e)e^\top$, precisely the new stored-state expression, while $U^\top h = R^\top e$ preserves compatibility. Transposition exchanges P, Q_0 and $(V, Y), (U, R)$ in this calculation: with $Q_0' = Q_0 + ff^\top$, the independent split is $f((I-P)G^\top f)^\top + (I - Q_0')G(I - P)$ and the mean update is $A + f(h - VY^\top f)^\top$. Dependent queries leave the kernel unchanged; full opposite rank removes the innovation.

Conditional layer kernels remain a product: a transcript-measurable choice of layer or direction reveals nothing further, and its reply conditions only that layer. Finite induction, partitioned by the finitely many stopping events if needed, proves equality of transcript laws. At stopping, use fresh independent G in (154) for each layer. Integrating these conditional kernels gives the ordinary joint Gaussian matrix law, including independence from client randomness. The completion agrees on all final basis columns, hence on every earlier reply, pathwise.

At ranks p, q, direct dot-product counts give at most $6np + 8nq + 9n + 1 \le 8n(p + q) + 10n$ operations per forward request; transpose swaps the ranks and dependent queries cost less. Since $p + q \le j - 1$ before request j, summation gives $4nQ^2 + 6nQ$, plus two operations for $1/\sqrt{n}$. There are at most $2n - 1$ drawing increments before a rank reaches n; afterward every new opposite action is determined. The four lists occupy $2n(p + q)$ words, with $p + q \le \min(Q, 2n)$. Eight length-n scratch vectors, five coefficient/address words per column and 64 scalar words give the stated memory. At most two comparisons per request suffice. Exact zero tests require neither a condition-number bound nor a nonsingular query Gram. □

11.3.1 Factored numerical anchors and output coupling

Fix all orders, scalar routines, generator ordering, orthogonalization signs and selector tie-breaking. On panel b of length Δ_b, use unscaled coefficients in $t - \tau_b$. In the polynomial network put $U_i^{(j)} = ([r_a \delta_a^{(j)}]_i)_{a=1}^m$ and $H_k^{(j)} = ([h_a^{(j)}]_k)_{a=1}^m$, both n by m matrices. Coefficient comparison in the actual gradient equations gives, for $s \ge 1$,

$$
\begin{aligned}
[W^{(1)}]_s &= -\frac{2}{ms} U^{(1)}_{s-1} (v_a)_{a=1}^{m\ \top}, \\
[W^{(j)}]_s &= -\frac{2}{mns} \sum_{i+k=s-1} U^{(j)}_i H^{(j-1)\top}_k, \\
[w]_s &= -\frac{2}{ms} \sum_a [r_a h^{(L)}_a]_{s-1}.
\end{aligned}
\tag{155}
$$

Only lower orders determine order s. Acting on a query series X gives

$$
-\frac{2}{mn} \sum_{i+k+c=s-1} \frac{U^{(j)}_i (H^{(j-1)\top}_k X_c)}{i+k+1}. \tag{156}
$$

Cache $H_k^\top X_c$: for b' queries, vector work is $O(nmb'K^2)$ and scalar sums cost $O(mb'K^3)$ per layer. Transposition exchanges the factors. Training coefficients are causal; passive jets use the completed training jets.

The endpoint update groups into mK outer products per hidden layer:

$$
W^{(j)}_{b+1} - W^{(j)}_b = -\frac{2}{mn} \sum_{i=0}^{K-1} U^{(j)}_i V^{(j-1)\top}_i, \qquad V^{(j-1)}_i = \sum_{k=0}^{K-1-i} \frac{\Delta_b^{i+k+1}}{i+k+1} H^{(j-1)}_k. \tag{157}
$$

Explicit execution adds these outer products to the dense anchor; implicit execution appends them, with the scalar in A_b, to $A^{(j)}_b, B^{(j)}_b \in \mathbb{R}^{n \times bmK}$:

$$
W^{(j)}_b = W^{(j)}_0 + A^{(j)}_b B^{(j)\top}_b, \qquad W^{(j)}_b X_c = \mathrm{Action}_j(X_c) + A^{(j)}_b (B^{(j)\top}_b X_c). \tag{158}
$$

Transpose actions exchange the factors. This exactly represents numerical anchors, without asserting low rank of the continuous dense trajectory. The first matrix is explicit and its higher coefficients use training input pairings in (155), without K dense first-matrix arrays; the readout is a vector. Hidden initialized-matrix requests are exactly:

1. m forward actions per mixer for exact initialized training features/images using ϕ_j; initial backward fields vanish.
2. Constant-anchor actions on source/training coefficients through K per panel; all increments use the displayed factors.
3. Images of completed Harmonic coefficients or Taylor nodes.
4. Images $W^{(j)}_0 U_{j-1}$ of the final source basis, at most r per mixer, including any not spanned by earlier image requests.

Harmonic uses both directions on $m + N_x$ inputs; Taylor uses M forward and m backward inputs. Sufficient per-mixer counts are

$$
Q_{\mathrm{H}} = m + 2J(m+N_x)(K+1) + 2N + r, \qquad Q_{\mathrm{T}} = m + J(M+m)(K+1) + (M+m)N_s + r. \tag{159}
$$

Use deterministic $\min(n, R)$ in place of r to invoke the sampler; the realized cost may use actual r. No other hidden-matrix information (entries, norms or singular values) is accessed. Under completion, explicit and implicit executions agree by induction through layers, coefficients and panels, using equations (156) and (158). Identical generator lists then give identical ranks, bases and selector decisions. For source bases $U_j^\top U_j / n = I$, selected row matrices $P_j = (U_j)_{I_j}$, and source metrics M_j, item 4 gives the same core matrix $U_j^\top W^{(j)}_0 U_{j-1}/n$ and thus the same initialized mixer

$$B_C^{(j)}(0) = P_j \frac{U_j^\top W_0^{(j)} U_{j-1}}{n} P_{j-1}^\top M_{j-1}.$$

The first weights, zero raw readout, $c_C = y$, exact initial Gram and corrected-readout caches also agree. Hence retained arrays and autonomous futures agree pathwise, with the same source/fitting probability and no extra failure allowance. Transcripts and trained factors are discarded.

11.4 Complete operation and storage ledgers

Finite-order big-O constants below are numerical: no problem parameter, order, scalar-evaluator cost or confidence is hidden. Assume $n \geq \max(m, d)$; full-width retention is available otherwise when the source budget reaches n. Let R bound generators, r be maximum actual rank and q maximum selected support, so $r \leq \min(n, R)$ and $r \leq q \leq 9r$. These are local cost quantities, distinct from the source-dimension argument of Proposition 10.2.

11.4.1 Scalar geometry, global coefficients, and selection

For Harmonic, assign each time node $t_a = T(1 + \cos u_a)/2$ to one numerical panel, retaining multiplicity if two time angles have the same physical time. Let I_b be those node indices. Form

$$\mathcal{B}_{kc}^{(b)} = \frac{\gamma_k}{N_t} \sum_{a \in I_b} \cos(k u_a)((t_a - \tau_b)/\Delta_b)^c.$$

If $[g]_c(v_s)$ is an unscaled local source coefficient, first multiply it by Δ_b^c to obtain $G_{b,c}(v_s)$. Its global coefficient is exactly

$$\widetilde{c}_{kjb'} = \sum_b \sum_s \omega_s Y_{jb'}(v_s) \sum_{c=0}^{K} \mathcal{B}_{kc}^{(b)} G_{b,c}(v_s). \tag{160}$$

Process each panel's training jets, spatial queries and anchor advance in that order; accumulate spatial coefficients before transforming retained mode pairs. Scalar tables cost $O((N_t + J)(p_t + 1)(K + 1))$ work, and projection costs $O(LnJ(K + 1)(N_x H_{\mathrm{sph}} + N))$ work with $O(Ln(K + 1)H_{\mathrm{sph}})$ extra vector words. Only $O(LnR)$ global coefficient words persist across panels. Local panels do not create separate retained Harmonic source blocks.

For finite basis evaluation, use the circle basis $1, \sqrt{2}\cos(j\varphi), \sqrt{2}\sin(j\varphi)$ and angle addition. For $d' \geq 3$, at $v = (\cos\theta, \sin\theta\, u)$ use normalized

$$(\sin\theta)^l C_{j-l}^{l+(d'-2)/2}(\cos\theta) Y_{lb}(u), \qquad 0 \leq l \leq j,$$

Sphere-Laplacian substitution gives eigenvalue $-j(j+d'-2)$. For each orthogonal lower harmonic, the remaining weight is $(1 - z^2)^{l+(d'-3)/2}$. Multiplying the Gegenbauer equation $(1 - z^2)y'' - (2\mu + 1)zy' + s(s + 2\mu)y = 0$ by $(1 - z^2)^{\mu - 1/2}$ and integrating by parts proves orthogonality: $\mu \geq 1/2$ kills the boundary term. The dimension decomposition in Lemma 10.4 proves completeness. Use

$$(s + 1)C_{s+1}^\mu(z) = 2(s + \mu)zC_s^\mu(z) - (s + 2\mu - 1)C_{s-1}^\mu(z), \quad C_0^\mu = 1, \quad C_1^\mu = 2\mu z.$$

from the generating function $(1 - 2zt + t^2)^{-\mu}$. For squared angular norm $N_{l,s}$ under probability-sphere measure, $\mu = l + (d' - 2)/2$ gives

$$N_{0,0} = 1, \quad \frac{N_{l+1,0}}{N_{l,0}} = \frac{2l + d' - 1}{2l + d'}, \qquad \frac{N_{l,s+1}}{N_{l,s}} = \frac{(s + 2\mu)(s + \mu)}{(s + 1)(s + \mu + 1)}.$$

The first ratio follows by integration by parts on sine powers; the second follows by pairing the recurrence for zC_s^μ with C_{s+1}^μ and using symmetry of multiplication by z. Only recurrences and inverse square roots are required. Per intermediate dimension there are $O((j_*+1)^2)$ degree pairs and $O(H_{\mathrm{sph}})$ lower-basis products. Recompute spatial values per panel, giving

$$\mathcal{G}_{\mathrm{time}} = \begin{cases} N_t + J, & d = 1, \\ N_t + N_x + JN_x(j_*+1), & d = 2, \\ N_\theta^2 + dN_\theta + N_t + N_\varphi + d + d(j_*+1)^2 + JdN_x[1 + (j_*+1)^2 + H_{\mathrm{sph}}], & d \geq 3, \end{cases}$$

$$\mathcal{G}_{\mathrm{memory}} = \begin{cases} N_t + 1, & d = 1, \\ N_t + N_x + j_* + 1, & d = 2, \\ dN_\theta + N_t + N_\varphi + d[1 + (j_*+1)^2 + H_{\mathrm{sph}}], & d \geq 3. \end{cases} \tag{161}$$

These bounds include $O(N_\theta^2)$ for (151), $O(dN_\theta)$ for sine powers, and $O(d)$ per node for coordinates and Jacobians, as well as the displayed basis recurrence.

For either model, orthogonalizing R generators costs $O(LnRr)$. Each of the at most $9r$ selector steps from Lemma 10.1 costs $O(r^3 + nr^2)$ for shifted inverses and all-row quadratic forms. Thus selection costs $O(Lnr^3)$ since $r \leq n$.

For the final metric, suppress layer indices and let P be the selected basis rows, D the positive diagonal selector weights and $G = P^\top \mathsf{D} P$. Formula (108) and direct multiplication give

$$M = \mathsf{D} + \mathsf{D}P(G^{-2} - G^{-1})P^\top \mathsf{D}, \quad M^{-1} = \mathsf{D}^{-1} + P(I - G^{-1})P^\top, \quad P^\top M = G^{-1}P^\top \mathsf{D}.$$

The r by r Gram has spectrum in $[1, 4]$. Metric caches and core mixers cost $O(L(qr^2 + r^3 + q^2 r)) = O(Lq^2 r)$. Basis images cost $O(n^2 r)$ per explicit mixer, or requests in Q plus $O(nr^2)$ for implicit contraction. The initial training Gram, identical by exact source isometry, costs $O(nm^2 + m^3)$ work and $O(m^2)$ words including factorization.

Degree-D activation composition with degree-K series costs $O(DK^2)$ work and $O(DK)$ scratch: the $O(D)$ Chebyshev products each cost $O(k+1)$ at coefficient k. Preparing coefficients and derivatives costs $O(LD^2)$ once. Training and passive sources use this same routine.

11.4.2 Shared finite execution costs

The four ledgers share the same continuation, orthogonalization and selection routines. To expose that reuse without suppressing parameters, let e be the explicit input dimension and B, F local query-batch sizes, and put $P_e = (L-1)n^2 + n(e+1)$. Define the exact sums

$$\begin{aligned} \mathcal{W}_{\mathrm{ex}}(P, B) &= LD^2 + mP + nm^2 + m^3 + J\{PBK + LmBK^2(n+K) + LnBDK^2\} \\ &\quad + LnRr + Lnr^3 + Lq^2 r, \\ \mathcal{M}_{\mathrm{ex}} &= P_d + LmnDK + LnR + Lq^2 + LD + J + m^2 + L, \\ \mathcal{W}_{\mathrm{im}}(e, F, Q) &= ne(1+m) + em^2 + nm^2 + m^3 + L + e + m + 1 + L(D+1)^2 \\ &\quad + (L-1)nQ^2 + (L-1)nm(m+F)J(J-1)K(K+1) \\ &\quad + J[ne(m+F) + emF] + JLm(m+F)(K+1)^2(n+K+1) \\ &\quad + JLn(m+F)(D+1)(K+1)^2 + LnRr + Lnr^3 \\ &\quad + (L-1)nr^2 + L(qr^2 + r^3 + q^2 r) + qe, \end{aligned}$$

$$\mathcal{M}_{\mathrm{im}}(e,Q) = ne + (L-1)n[Q + JmK] + Lmn(D+1)(K+1) \\ + LnR + Lq^2 + q(e + Lm + 1) + m^2 + L(D+1) + J + L + e + m + 1.$$

The substitutions are $(e, F) = (d, N_x)$ for Harmonic and (k, M) for Taylor; in the latter M includes training and the extra m pays backward queries. These symbols are local cost arguments. In each display below the work or peak memory is bounded by O of the entire right-hand sum, with numerical constants only.

For supplied matrices the Harmonic and Taylor work bounds are

$$\mathcal{W}_{\mathrm{ex}}(P_d, m + N_x) + P_d + L + d + m + 1 + (N_t + J)(p_t + 1)K \\ + LnJK(N_x H_{\mathrm{sph}} + N) + Ln^2(N + r) + \mathcal{G}_{\mathrm{time}} + N_t \log(2 + N_t), \tag{162}$$

$$\mathcal{W}_{\mathrm{ex}}(P_k, M + m) + P_d + dM^2 + ndk + L + k + m + 1 \\ + LnMKN_s + Ln^2 R + N_s \log(2 + N_s) + J_s + N_s, \tag{163}$$

and their respective peak-memory sums are

$$\mathcal{M}_{\mathrm{ex}} + Lm^2K^2 + LnKH_{\mathrm{sph}} + (p_t + 1)K + N_t + m(d+1) + d + \mathcal{G}_{\mathrm{memory}}, \tag{164}$$

$$\mathcal{M}_{\mathrm{ex}} + LnMK + LmMK^2 + N_s + J_s + Md + dk + k + m + 1. \tag{165}$$

Constant-anchor products cost $JPBK$; cached higher coefficients and activation compositions are the next two terms of $\mathcal{W}_{\mathrm{ex}}$. Grouping (157) costs $O(nmK^2)$ and adding its mK outer products costs $O(n^2mK)$ per layer and panel, already covered. Harmonic separately pays Ln^2N for source images and Ln^2r for basis images; Taylor bounds both by Ln^2R. The Taylor reduction costs $dM^2 + ndk$. Sort source nodes into numerical panels and evaluate them in one pass for $O(LnMKN_s)$. One anchor, original matrices, current jets and global coefficients are live; neither a JP_d trajectory nor KP_d dense jets are stored.

For fresh implicit matrices set $Q = Q_{\mathrm{H}}$ or Q_{T} from (159). The corresponding work sums are

$$\mathcal{W}_{\mathrm{im}}(d, N_x, Q_{\mathrm{H}}) + (N_t + J)(p_t + 1)(K + 1) \\ + LnJ(K+1)(N_x H_{\mathrm{sph}} + N) + \mathcal{G}_{\mathrm{time}} + N_t \log(2 + N_t), \tag{166}$$

$$\mathcal{W}_{\mathrm{im}}(k, M, Q_{\mathrm{T}}) + dM^2 + Ln(M + m)(K+1)N_s \\ + N_s \log(2 + N_s) + N_s + J + J_s, \tag{167}$$

and their peak-memory sums are

$$\mathcal{M}_{\mathrm{im}}(d, Q_{\mathrm{H}}) + Lm^2(K+1)^2 + Ln(K+1)H_{\mathrm{sph}} \\ + m(d+1) + (p_t + 1)(K+1) + N_t + \mathcal{G}_{\mathrm{memory}}, \tag{168}$$

$$\mathcal{M}_{\mathrm{im}}(k, Q_{\mathrm{T}}) + Md + dk + LnM(K+1) + LmM(K+1)^2 + Mk + N_s + J_s. \tag{169}$$

The sampler contributes $(L-1)nQ^2$ work and $(L-1)nQ$ memory. At panel b, the bmK trained-factor columns cost $O(nbmK)$ per coefficient action. Summing at most $2(m + F)(K + 1)$ actions and $\sum_{b<J} b = J(J-1)/2$ gives the second work term; the factor memory is $(L-1)nJmK$. Their grouped formation is covered by current-panel contractions. All image queries, the nr^3 selector, and final Gram and metric solves are charged explicitly.

Implicit Taylor samples only the n by k reduced first matrix. Its law is $W_0^{(1)}U_0$; an independent Gaussian component on the orthogonal input complement is completed only in the proof and is never queried. These finite sums therefore charge every accessed matrix action and every trained-factor product, independently of the eventual rank.

11.4.3 Primitive costs and fixed-parameter bounds

Each model uses $4L(D+1)+Lnm$ original activation-value calls, plus at most L unsupplied values at zero. Only the first $4L(D+1)$ permit the tolerance of (139). Setup needs no original derivatives; runtime uses original values and first derivatives.

Supplied matrices need no draws; fresh explicit initialization adds $nd+(L-1)n^2$. Implicit Harmonic and Taylor use at most $nd+(L-1)nQ_{\mathrm{H}}$ and $nk+(L-1)nQ_{\mathrm{T}}$ respectively, with no readout or proof-completion draws. Harmonic's elementary-function call count is at most

$$O(L+d+m+1+L(D+1)+N_t+A_{\mathrm{ang}}+(L-1)Q+Lr),\quad A_{\mathrm{ang}}=\begin{cases}0, & d=1,\\ N_x, & d=2,\\ N_\theta+N_\varphi+d(j_*+1)^2, & d\ge 3.\end{cases}$$

Taylor replaces time/angular terms by N_s+J_s and adds $O(M)$ input-basis square roots; explicit execution omits Q. These counts cover certificates, orders, geometry and normalizations; rounding/enumeration fit the finite tables. Nonprimitive scalar calls add their actual work, description and peak scratch. Envelopes and population gap certificates are supplied, not computed from analyticity.

For separately fixed admissible parameters, (142) and the source formulas give

	Harmonic	Taylor
J, D	$O(\ell^{3/2})$	$O(\ell^{3/2})$
K	$O(\ell)$	$O(\ell)$
source sizes	$N_t, p_t+1=O(\ell^{5/2}),\quad N_x, H_{\mathrm{sph}}=O(\ell^{3(d-1)/2})$	$J_s=O(\ell^{1/2}),\quad K_s+1=O(\ell)$
R, r, q	$O(\ell^{3d/2+1})$	$O(\ell^{3/2})$
Q	$O(\ell^{3d/2+1})$	$O(\ell^{5/2})$.

(170)

The two-point convention gives the same Harmonic exponents at $d=1$. For the supplied dense variants the full ledgers respectively imply

$$O(n^2\ell^{3d/2+1}+n\ell^{9d/2+3}+\mathrm{polylog}(n)),\qquad O(n^2\ell^{5/2}+n\ell^5+\mathrm{polylog}(n)).$$

At fixed parameters the leading terms dominate eventually, and memory is $O(n^2)$. For implicit Harmonic, sampler, trained-factor/activation and projection exponents are respectively $3d+2$, $3d/2+7/2$, and at most $\max(3d-1/2, 3d/2+7/2)$. For $d\ge 1$ all are dominated by the selector $O(nr^3)=O(n\ell^{9d/2+3})$; non-width terms are fixed powers of ℓ and memory is $O(n\ell^{3d/2+1})$. For implicit Taylor,

$$nQ_{\mathrm{T}}^2=O(n\ell^5),\qquad nm(M+m)J^2K^2=O(n\ell^5),\qquad Jn(M+m)DK^2=O(n\ell^5).$$

The selector costs $O(n\ell^{9/2})$ and all remaining work fits $O(n\ell^5)$, while transcript, factors, compositions and other buffers fit $O(n\ell^{5/2})$. This proves all four rows of Proposition 2.3 including final assembly.

Discard anchors, generators, bases, schedules, activation polynomials, transcripts and factors; retain exactly the inventory and optimizer of Proposition 10.2. Selection uses the computed span and bound R, so accidental rank deficiencies need not survive approximation. Exact initialized additions preserve the Gram and pairing. Finite-precision values, ranks, barrier comparisons or infinite-time evolution would need new perturbation results: these are real-coordinate and operation

counts. The setup covers $T = 32\ell/\lambda$, not merely a short physical prefix. Dense Euler at step h costs $\Theta(mP_dT/h)$ before assembly; the fixed-parameter Taylor/Harmonic setup ratios are $O(h\ell^{3/2})$, $O(h\ell^{3d/2})$. This neither proves that h is necessary nor compares against every high-order dense solver.

12 Nonlinearity and feature learning

We prove Theorem 2.2 using the real bounds already established in Lemma 5.1. All auxiliary notation below is local; constants and $O(\cdot)$ bounds may depend on the entire fixed problem, but not on width. The key simplification is to test feature increments against their initial adjoints, rather than follow every feature acceleration.

Lemma 12.1 (Nonlinear initial velocity)**.** *Every $Q^{(\ell)}$, $\ell \geq 1$, is positive definite. The limiting initial prediction velocity is nonaffine on the sphere, witnessed by an affine-annihilating combination of four deterministic inputs chosen before initialization.*

Proof. Write $v = x/\sqrt{d}$. Gaussian covariance recursion gives $Q^{(\ell)}_{ab} = F_\ell(v_a^\top v_b)$, $F_0(s) = s$, with

$$F_\ell(s) = \sum_{k\geq 0} c_{\ell,k} s^k, \qquad c_{\ell,k} \geq 0, \quad \sum_k c_{\ell,k} < \infty, \tag{171}$$

and unbounded positive coefficient support. Here is a direct justification. The orthonormal Gaussian Hermite expansion of $\phi_\ell(\sqrt{q}G)$ has coefficients α_k and covariance $\sum_k \alpha_k^2 r^k$ for Gaussian correlation r. The covariance identity follows by comparing coefficients in $\mathbb{E}[e^{zG-z^2/2}e^{wG'-w^2/2}] = e^{rzw}$. Completeness follows because a function orthogonal to all polynomials has entire transform $\mathbb{E}[h(G)e^{zG}]$ identically zero, hence zero Fourier transform. Linear growth gives Gaussian L^2 integrability. Finite Hermite support would make the activation a polynomial, hence affine by its bounded derivative. Composing the covariance series with $F_{\ell-1}(s)/F_{\ell-1}(1)$ proves (171): Tonelli at $s = 1$ gives summability, and a positive inner coefficient of degree $j \geq 1$ and outer degree k produces degree jk. These arguments allow nonzero means and justify uniform absolute convergence on $[-1, 1]$. The tensor Grams $[(v_a^\top v_b)^k]$ are positive semidefinite and converge to I_m by the nonparallel-input hypothesis. Thus every $Q^{(\ell)} \succ 0$.

Let P be the orthogonal projection onto affine functions in spherical L^2, with normalized surface measure σ: $Ph(v) = \int h\, d\sigma + d\, v^\top \int uh(u)\, d\sigma(u)$. Projecting the tensor feature in both variables gives the positive kernel

$$R_k(v, u) = (v^\top u)^k - a_k - b_k v^\top u, \quad a_k = \int (z^\top u)^k\, d\sigma(z), \quad b_k = d \int (z^\top u)^{k+1}\, d\sigma(z).$$

Here $a_k, b_k \to 0$ by dominated convergence, since $d \geq 2$, so $[R_k(v_a, v_b)] \to I_m$. The limiting velocity $g(v) = (2/m)\sum_a y_a F_L(v^\top v_a)$ is not affine: otherwise

$$0 = \sum_a y_a[(I - P)g](v_a) = \frac{2}{m} \sum_k c_{L,k}\, y^\top [R_k(v_a, v_b)]y > 0.$$

Every term is nonnegative and some large-degree term is positive. Uniform convergence justifies the projection and sums.

Some great circle witnesses this nonaffinity. Otherwise the antipodal average of g is constant, and the odd remainder extends homogeneously to a function linear on every plane, hence globally

linear. On such a circle, interpolate three distinct points by an affine function and take a fourth where it fails. The resulting dependence can be signed and normalized so that, for unit vectors u_i,

$$\sum_i |c_i| = 1, \qquad \sum_i c_i = 0, \qquad \sum_i c_i u_i = 0, \qquad \sum_i c_i g(u_i) > 0. \tag{172}$$

These inputs depend on the fixed problem only. The finite-query Gram convergence proved in Lemma 5.1, applied to this augmented list, and $\dot{f}_n(0,x) = (2/mn)\sum_a y_a h_a^{(L)}(0)^\top h^{(L)}(0,x)$ give convergence of their initial velocities. □

Lemma 12.2 (Initial backward fields and hidden forces)**.** *The initialized forward fields and first backward derivatives, including the vectors before derivative gates, have deterministic joint empirical row-law limits with second moments. This includes the full first-layer weight and acceleration rows. Every hidden block's squared initial acceleration in inverse-mobility norm converges to a positive constant.*

Proof. All unmarked fields in this proof are at zero. Put $k = 2/m$ and define the pre-gated vectors P and their gated versions B by

$$P_a^{(L)} = \sum_b y_b h_b^{(L)}, \qquad B_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}) \odot P_a^{(\ell)}, \qquad P_a^{(\ell-1)} = W^{(\ell)\top} B_a^{(\ell)}.$$

Thus $\dot{\delta}_a^{(\ell)}(0) = kB_a^{(\ell)}$. Differentiating (3), since all hidden velocities initially vanish, gives

$$\ddot{W}^{(1)}(0) = k^2 \sum_a y_a B_a^{(1)} v_a^\top, \qquad \ddot{W}^{(\ell)}(0) = \frac{k^2}{n} \sum_a y_a B_a^{(\ell)} h_a^{(\ell-1)\top}. \tag{173}$$

Only a forward pass and a backward pass are needed for their joint limits. For one hidden Gaussian matrix W, after revealing $WH = Z$, its conditional law is $Z(H^\top H)^{-1}H^\top + \widetilde{W}(I - \Pi_H)$, where $\Pi_H = H(H^\top H)^{-1}H^\top$ and $\widetilde{W}$ is fresh with entries $N(0, 1/n)$. This is orthogonal Gaussian conditioning on the linear constraint. For a reverse query U, known from the previous answers,

$$W^\top U \overset{d}{=} HQ_n^{-1}C_n + (I - \Pi_H)\Xi D_n^{1/2}, \quad Q_n = H^\top H/n,\ C_n = Z^\top U/n,\ D_n = U^\top U/n, \tag{174}$$

with independent standard Gaussian rows in Ξ. The removed projection has conditional squared RMS $\operatorname{rank}(H)\operatorname{tr}(D_n)/n$, hence vanishes in probability. Adaptive queries cause no extra dependence: conditional on the global transcript, the next query is fixed; revealing one Gaussian action leaves the other matrices' residuals independent. In forward-then-backward order no matrix is queried again after its reverse answer.

For precision, the induction uses empirical quadratic Wasserstein convergence. Appending independent Gaussian rows preserves a deterministic such limit: conditional variances for bounded tests are $O(1/n)$, and the appended second moment obeys the law of large numbers. Continuous coordinate maps of at most linear growth preserve this convergence, by an L^2 coupling and uniform integrability. These include $\phi(z)$ and $\phi'(z)u$. Same-layer products consequently converge; inverse Grams and positive covariance square roots converge when their limiting Grams are, respectively, positive definite or positive semidefinite. The vanishing projected term in (174) does not change the row-law limit. Starting with the full iid first-layer rows, these rules prove the forward limits and then the backward limits, including all pre-gated fields. The forward Grams being inverted are positive by Lemma 12.1; $Q^{(0)}$ is never inverted. The first-row acceleration is a linear combination of the B fields, so its joint limit is included.

Let $D^{(\ell)}$ be the limiting Gram of $B^{(\ell)}$. At the top it is positive definite: a null vector c would give

$$\Big(\sum_b y_b \phi_L(z_b)\Big)\Big(\sum_a c_a \phi_L'(z_a)\Big) = 0 \qquad (z \in \mathbb{R}^m).$$

Indeed $L \geq 2$ gives full Gaussian support. The first factor is not identically zero since $y^\top Q^{(L)} y > 0$; analyticity forces the second to vanish everywhere. Varying each coordinate, the nonconstant derivative gives $c = 0$. Downward, the fresh reverse innovation in (174) is independent of the lower-layer fields, with covariance $D^{(\ell+1)}$, so

$$c^\top D^{(\ell)} c \geq \lambda_{\min}(D^{(\ell+1)}) \sum_a c_a^2 \mathbb{E}[\phi_\ell'(Z_a^{(\ell)})^2] > 0.$$

Each Gaussian marginal is nondegenerate and a nonzero analytic derivative has a zero set of measure zero. Finally, expanding (173), the layer-ℓ squared acceleration limit is $k^4 y^\top (Q^{(\ell-1)} \circ D^{(\ell)}) y > 0$: for $Q \succeq 0$, $D \succ 0$, $Q \circ D \succeq \lambda_{\min}(D)\,\mathrm{diag}(Q)$ by decomposing Q into rank-one terms, and every relevant Q has positive diagonal. This also covers singular input Grams. □

Lemma 12.3 (Adjoint test of the actual motion)**.** *Every hidden layer satisfies* (15)*. On the training set,*

$$y^\top (f_n(t) - f_{\mathrm{NTK},n}(t)) = \frac{m}{3} t^3 \left[\frac{\|\ddot{W}^{(1)}(0)\|_F^2}{n} + \sum_{\ell=2}^{L} \|\ddot{W}^{(\ell)}(0)\|_F^2 \right] + o_*(t^3). \tag{175}$$

Here $o_*(t^j)$ means: for every $\varepsilon > 0$, some deterministic $T_\varepsilon > 0$ makes the probability that the normalized remainder exceeds ε anywhere on $0 < t \leq T_\varepsilon$ tend to zero.

Proof. Use $\|u\|_n = \|u\|_2/\sqrt{n}$ and the inverse-mobility norm $\|U\|_{\mathrm{mob}}^2 = \|U^{(1)}\|_F^2/n + \sum_{\ell \geq 2} \|U^{(\ell)}\|_F^2$ on hidden matrices. On the event of Lemma 5.1, all relevant operators, feature RMS and residuals have deterministic bounds. The readout equation therefore gives $\|w(t)\|_n = O(t)$; backpropagation gives $\|\delta_a^{(\ell)}(t)\|_n = O(t)$; integration of the hidden updates and forward subtraction give, uniformly on the sphere for $0 \leq t \leq 1$,

$$\|W(t) - W(0)\|_{\mathrm{mob}} = O(t^2), \qquad \|\Delta z^{(\ell)}\|_n + \|\Delta h^{(\ell)}\|_n = O(t^2), \qquad f_n(t,x) = t\dot{f}_n(0,x) + O(t^2). \tag{176}$$

Here Δ means the increment from zero. For the last assertion, the readout equation gives $\|\dot{w}(t) - \dot{w}(0)\|_n = O(t)$; integrate and use $f = w^\top h/n$. No new bootstrap is needed.

The remainder control uses one elementary estimate. For bounded Lipschitz g and an initial vector u,

$$\|[g(z) - g(z_0)] \odot u\|_n^2 \leq \mathrm{Lip}(g)^2 D^2 \|z - z_0\|_n^2 + 4\|g\|_\infty^2 \|u \mathbf{1}_{|u|>D}\|_n^2. \tag{177}$$

Split at $|u_i| = D$. The deterministic row-law limits with second moments in Lemma 12.2 give, for every $\eta > 0$, a fixed cutoff D for which the probability that any needed initial squared tail exceeds η tends to zero. Choose D first, then a small time in (176). Thus gate differences times initial fields are $o_*(1)$, uniformly also along the segment from z_0 to z.

For this proof define $k = 2/m$, $P_a^{(L)} = \sum_b y_b h_b^{(L)}(0)$, $B_a^{(\ell)} = \phi_\ell'(z_a^{(\ell)}(0)) \odot P_a^{(\ell)}$, and $P_a^{(\ell-1)} = W^{(\ell)}(0)^\top B_a^{(\ell)}$. Since $w(t)/t = \dot{w}(0) + O_{\|\cdot\|_n}(t)$, downward subtraction of the backward recursion yields

$$\delta_a^{(\ell)}(t) = kt B_a^{(\ell)} + o_{*,\|\cdot\|_n}(t), \qquad \Delta W = \tfrac{1}{2} t^2 \ddot{W}(0) + o_{*,\|\cdot\|_{\mathrm{mob}}}(t^2). \tag{178}$$

Indeed each backward difference is the next-layer error through bounded operators, an $O(t^2)$ operator increment acting on B, and a gate difference acting on P, controlled by (177). Insert this into the

hidden updates with $r = -y + O(t)$ and $h = h(0) + O_{\|\cdot\|_n}(t^2)$, then integrate. The normalized outer-product identities $\|uv^\top\|_F/\sqrt{n} = \|u\|_n\|v\|_2$ and $\|uh^\top\|_F/n = \|u\|_n\|h\|_n$ justify the matrix-norm remainder.

Instead of expanding each forward feature, test its increment against P. Put $E_{1,n} = \|\ddot{W}^{(1)}(0)\|_F^2/n$ and $E_{\ell,n} = \|\ddot{W}^{(\ell)}(0)\|_F^2$ for $\ell \geq 2$. The integral formula for an activation increment and (177) give

$$\langle P_a^{(\ell)}, \Delta h_a^{(\ell)}\rangle_n = \langle B_a^{(\ell)}, \Delta z_a^{(\ell)}\rangle_n + o_*(t^2), \qquad \langle u, v\rangle_n = u^\top v/n.$$

The error is bounded by the RMS gate difference times P, multiplied by $\|\Delta z\|_n = O(t^2)$. In $\Delta z = \Delta W h(0) + W(0)\Delta h + \Delta W \Delta h$, the last term is $O(t^4)$ in RMS. Summing the first term's pairings with $y_a B_a$ and using (173) gives $t^2 E_{\ell,n}/(2k^2) + o_*(t^2)$; the second pairs the lower feature increment with $P_a^{(\ell-1)}$. Telescoping to the first layer yields

$$\sum_a y_a \langle P_a^{(\ell)}, \Delta h_a^{(\ell)}\rangle_n = \frac{t^2}{2k^2} \sum_{j\leq \ell} E_{j,n} + o_*(t^2). \tag{179}$$

All energies have positive limits. Cauchy–Schwarz, with bounded $\sum_a y_a^2 \|P_a^{(\ell)}\|_n^2$, therefore proves the feature-motion lower bound in every layer on one deterministic small interval.

The tangent kernel is the readout Gram plus the hidden terms

$$K_{ab}(t) = \langle h_a^{(L)}, h_b^{(L)}\rangle_n + (v_a^\top v_b)\langle \delta_a^{(1)}, \delta_b^{(1)}\rangle_n + \sum_{\ell=2}^{L} \langle \delta_a^{(\ell)}, \delta_b^{(\ell)}\rangle_n \langle h_a^{(\ell-1)}, h_b^{(\ell-1)}\rangle_n.$$

Equation (176) gives $K(t) - K(0) = O(t^2)$. Its readout part, contracted with y, is twice the top pairing in (179), plus $O(t^4)$. By (178) and the squared outer products in (173), the hidden contribution has the same coefficient. Consequently

$$y^\top [K(t) - K(0)] y = \frac{2t^2}{k^2} \sum_\ell E_{\ell,n} + o_*(t^2).$$

Finally $\dot{f}_n = kK(t)(y - f_n)$, whereas the frozen flow uses $K(0)$. Variation of constants gives their difference as $k\int_0^t e^{-kK(0)(t-s)}[K(s) - K(0)](y - f_n(s))\, ds$. Here $f_n(s) = O(s)$ and the exponential is $I + O(t-s)$ because $K(0)$ is bounded. Pair with y and integrate the last display. The additional terms are $O(t^4)$, giving $2/(3k) = m/3$ in (175). □

Lemma 12.4 (The nonlinear feature component moves)**.** *Let $V = \operatorname{span}\{x_1, \ldots, x_m\}$, with normalized surface measure σ_V on its unit sphere. For sufficiently small fixed $t > 0$, with probability tending to one,*

$$\frac{1}{n}\sum_i \inf_{a\in V,\, b\in\mathbb{R}} \int_{S(V)} |h_i^{(1)}(t, \sqrt{d}\, v) - h_i^{(1)}(0, \sqrt{d}\, v) - a^\top v - b|^2 \, d\sigma_V(v) \geq c_* t^4. \tag{180}$$

Proof. Since $\dim V \geq 2$, for nonzero $g, r \in V$ the function $\phi_1'(g^\top v)(r^\top v)$ is not affine on $S(V)$. Otherwise its values on $r^\perp$ and antipodes force the affine function to be $\lambda r^\top v$. Off the equator, division and then continuity make ϕ_1' constant on $[-\|g\|, \|g\|]$, contradicting analyticity and nonaffineness. Project the initial first-layer row onto V and call it g_i; call its acceleration row $r_i \in V$. By Lemma 12.2, their joint empirical law converges with second moments; g is nondegenerate Gaussian and $\mathbb{E}\|r\|^2 > 0$. The squared distance of $\phi_1'(g^\top v)(r^\top v)$ from affine functions in $L^2(\sigma_V)$ is continuous

in (g, r), bounded by $\|\phi_1'\|_\infty^2\|r\|^2$, and positive when both are nonzero. Its empirical mean therefore tends to a positive constant.

The first-layer expansion in (178) and bounded ϕ_1' reduce the actual increment to the fixed row direction $t^2 r_i/2$. On $\|r_i\| \le D$, its Taylor remainder divided by t^2 is at most $\|\phi_1''\|_\infty t^2 D^2/8$; on the complement it is at most $\|\phi_1'\|_\infty\|r_i\|$. The deterministic second-moment limit, choosing D and then time, gives $o_*(t^2)$ in neuron-index-times-sphere L^2. Orthogonal projection off affine functions is contractive, so the positive limiting projected acceleration proves the claim. □

Proof of Theorem 2.2. The four-point combination (172), initial velocity convergence and (176) give a positive multiple of t for the dense predictor on a width-independent small interval. It annihilates every affine predictor and has coefficient absolute sum one, so it bounds their uniform distance below. The positive energy limit in (175), divided by $\|y\|_1$, similarly gives a training-input gap of order t^3 from the frozen kernel. The two preceding lemmas supply all internal feature conclusions.

For each fixed positive time, the vanishing absolute compression errors in Propositions 7.2, 10.5 and 10.6 transfer both prediction gaps by the triangle inequality; eventual estimates at every confidence imply convergence in probability. Choose common smaller $c_*, t_* > 0$. Legendre and Harmonic already cover the witnesses. Taylor appends them before initialization, without labels. Its bound replaces p by at most $p + 4$, multiplying the quadratic panel factor by at most 9 and the linear data factor by at most 3, since $m + p \ge 2$. No endpoint gap or identification of compressed coordinates with dense neurons is used or asserted. □

13 Assembly of the headline theorem

Proof of Theorem 2.1. The preceding constructions provide the stated retained counts and the initialization-jet provenance alternative; the efficient alternatives have their own execution proofs. Here we first verify the original asymptotic comparison with actual dense variability, not an upper certificate. The quantitative width refinement uses its separately proved finite dense lower bound rather than claiming polynomial onset for the sharper coefficient used in this first argument. Fix a confidence budget $0 < \delta_0 < 1$. By Proposition 8.2, for sufficiently large width,

$$\Pr\left\{\left\|f_n - \widetilde{f}_n\right\|_{\mathcal{X}} \ge c_{\phi,L,\delta_0}\frac{Y\sqrt{\gamma}}{\sqrt{n}[\log(en)]^{5/2}}\right\} \ge 1 - \delta_0.$$

Its witness is a training input, so the same lower bound applies on either promised domain. The source and fitting events give all required limits. The Legendre order of Proposition 7.2 gives error at most $CY/(\sqrt{n}[\log(en)]^3)$; its ratio to the displayed lower bound is at most a fixed-problem, fixed-confidence multiple of $[\log(en)]^{-1/2}$. For the two selected models, Propositions 10.5 and 10.6 give error at most Y/n, and the corresponding ratio is at most

$$\frac{[\log(en)]^{5/2}}{c_{\phi,L,\delta_0}\sqrt{\gamma n}} \longrightarrow 0.$$

Intersecting each approximation event with the lower event costs at most $2\delta_0$ by the union bound, without any independence requirement. For every fixed positive ratio tolerance the limiting superior of its failure probability is therefore at most $2\delta_0$. Letting $\delta_0 \downarrow 0$ proves (7), including the stated failure convention. For (11), use the selected-model event at the requested confidence. The label assumption stays fixed throughout; no label inequality is used to reduce a displayed sample/gap power. For the polynomial sufficient width (12), apply Theorem 9.1 with the explicit maxima in equations (106) and (107). That argument supplies its own quantitative actual-dense lower event

and retains exactly equations (8) and (10); it does not attach polynomial onset to the sharper coefficient in the display above. □

14 Literal storage of a frozen-top neural tangent hierarchy

Literal neural tangent hierarchy (NTH) arrays provide a separate benchmark for how a prescribed representation affects storage. We study the ordered hierarchy associated with Huang and Yau [12], with its highest retained rank frozen, in the same feature-learning metric and zero-readout initialization as Section 2. On a correlated dataset of m samples, its necessary and sufficient retained size on a fixed short interval is $\exp[\Theta(\log m \log n)] = n^{\Theta(\log m)}$, uniformly over $4 \le m \le \sqrt{n}$. No growth of m is required: for each fixed m the size is a power of n whose exponent grows with m, hence above every power of $\log n$. When m grows as a power of n, it is $\exp[\Theta((\log n)^2)]$.

This section has its own assumptions and probability convention. The small-label, Gram-gap and complete-trajectory hypotheses of Theorem 2.1 are not imposed here; the result is an independent benchmark, and gives no same-family storage separation from the preceding compression constructions. No theorem for another parameterization or readout initialization is transferred.

14.1 Model, closure, and storage

There are $m \ge 1$ deterministic samples, independent of initialization, $v_a = x_a/\sqrt{d} \in \mathbb{R}^d$, with $\|v_a\|_2 \le 1$ and $|y_a| \le 1$. Predictions indexed by v mean the physical network input $x = \sqrt{d}\, v$. We specialize (1) to two identity hidden activations and use its readout w. Thus $W^{(1)} \in \mathbb{R}^{n\times d}$, $W^{(2)} \in \mathbb{R}^{n\times n}$, $w \in \mathbb{R}^n$, and

$$\begin{aligned} h^{(1)} &= W^{(1)} v, \quad h^{(2)} = W^{(2)} h^{(1)}, \\ f_n(v) &= \frac{w^\top h^{(2)}}{n}, \qquad r_a = f_n(v_a) - y_a, \\ \mathcal{L}_{1/2} &= \frac{1}{2m} \sum_a r_a^2. \end{aligned} \tag{181}$$

Initially the first two matrices have independent entries with laws $\mathcal{N}(0,1)$ and $\mathcal{N}(0,1/n)$, respectively, and $w(0) = 0$. All blocks train with mobilities $(n, 1, n)$. The half-MSE is deliberate: the full-MSE flow (3) at physical time t equals this flow at time $2t$. Throughout this section, t and dots refer to half-MSE physical time. Thus $T = 2^{-15}$ here corresponds to $[0, T/2] = [0, 2^{-16}]$ in the paper's full-MSE time.

Put

$$\begin{aligned} B = W^{(1)}/\sqrt{n}, \qquad W = W^{(2)}, \qquad c = w/\sqrt{n}, \\ \theta = (B, W, c). \end{aligned} \tag{182}$$

Then $f_n(v) = c^\top W B v$ and training is ordinary Euclidean/Frobenius gradient flow in θ. In particular B_0, W_0 have independent $\mathcal{N}(0, 1/n)$ entries. Define the fixed input second moment $\Sigma \in \mathbb{R}^{d\times d}$, label-input mean $b \in \mathbb{R}^d$, and prediction coefficient $u \in \mathbb{R}^d$ by

$$\begin{aligned} \Sigma = \frac{1}{m} \sum_a v_a v_a^\top, \qquad b = \frac{1}{m} \sum_a y_a v_a, \\ u = B^\top W^\top c, \qquad f_n(v) = v^\top u. \end{aligned} \tag{183}$$

Here Σ is an input-space second moment, distinct from the sample-space population Grams $Q^{(\ell)}$ in (4). We have $\|\Sigma\|_{\mathrm{op}} \le 1$, $\|b\|_2 \le 1$, and the exact dense equations are

$$\begin{aligned} \dot{B} = W^\top c(b - \Sigma u)^\top, \qquad \dot{W} = c[B(b - \Sigma u)]^\top, \\ \dot{c} = WB(b - \Sigma u). \end{aligned} \tag{184}$$

For arbitrary input vectors define the source field and ordered tensors by

$$\begin{aligned} V(v;\theta) = \nabla_\theta f_n(v) = (W^\top c v^\top,\ c(Bv)^\top,\ WBv), \\ K_1(v;\theta) = f_n(v), \\ K_{r+1}(v_0,\dots,v_r;\theta) = D_\theta K_r(v_0,\dots,v_{r-1};\theta)[V(v_r;\theta)]. \end{aligned} \tag{185}$$

Every input slot is linear; permutation symmetry is not assumed. $K_r(\mathbf{v};0)$ means evaluation at θ_0. For an integer $q \ge 2$, copy these initialized tensors through rank q, freeze rank q, and evolve all lower ranks by

$$\begin{aligned} \dot{K}_r^{(q)}(\mathbf{v};t) = -\frac{1}{m}\sum_a r_a^{(q)}(t) K_{r+1}^{(q)}(\mathbf{v}, v_a; t), \\ r_a^{(q)} = f_{n,q}(v_a) - y_a, \\ f_{n,q}(v) = K_1^{(q)}(v). \end{aligned} \tag{186}$$

The closure uses its own residual. In each array the first slot is a training input or a passive coordinate query e_i, $1 \le i \le d$; all later slots are training indices. Passive queries have no labels. Linearity gives the exact decoder $f_{n,q}(t,v) = v^\top u_q(t)$ for any later query, where $u_q(t) = (f_{n,q}(t,e_i))_{i=1}^d$. The moving and frozen array counts are, respectively,

$$\begin{aligned} N_{\mathrm{mov}}(q) = (m+d)\sum_{r=1}^{q-1} m^{r-1}, \\ N_{\mathrm{top}}(q) = (m+d)m^{q-1}, \qquad N_{\mathrm{arr}} = N_{\mathrm{mov}} + N_{\mathrm{top}}. \end{aligned} \tag{187}$$

The dataset adds at most $md + m$ fixed scalars. These are retained real coordinates, excluding coefficient-generation work, initialization peak memory, numerical integration and bit precision. They do not charge a factorized or implicit encoding. Once the initial tensors have been formed from B_0, W_0 and the data, those dense matrices need not be retained.

Fix throughout

$$\begin{aligned} T = 2^{-15}, \\ E_n(q) = \sup_{0\le t\le T}\sup_{\|v\|_2=1} |f_n(t,v) - f_{n,q}(t,v)|, \\ D_n = \sup_{0\le t\le T}\sup_{\|v\|_2=1} |f_n(t,v) - \widetilde{f}_n(t,v)|, \end{aligned} \tag{188}$$

where $\widetilde{f}_n$ is an independent dense initialization on the same dataset. Both errors use the same physical window and sphere. In probability statements, nonexistence on the window counts as failure; $E_n(q) \le AD_n$ includes $E_n(q) = D_n = 0$ without taking a ratio.

Theorem 14.1 (Matched literal storage)**.** *Fix $0 < Y \le 1$, a success probability $0 < \pi < 1$, and an accuracy factor $A > 0$. For $4 \mid m$, $4 \le m \le \sqrt{n}$, take $d = m+1$, orthonormal coordinates $e_0, \dots, e_m$, and*

$$\begin{aligned} v_a = \frac{e_0 + e_a}{\sqrt{2}}, \qquad y_a = Y s_a, \\ s_a = \begin{cases} 1, & a \le 3m/4, \\ -1, & a > 3m/4. \end{cases} \end{aligned} \tag{189}$$

Let $S_n(\pi, A)$ be the minimum of $N_{\mathrm{arr}}(q) + md + m$ over deterministic orders $q \geq 2$ for which $\Pr\{E_n(q) \leq AD_n\} \geq \pi$. For sufficiently large n, uniformly over the displayed sample range,

$$\exp[c_Y \log m \log n] \ \leq S_n(\pi, A) \leq \exp[C_Y \log m \log n] \tag{190}$$

for positive constants depending only on Y; the width threshold may also depend on π, A. The same order holds for the minimum moving-array count, and survives omission of a zero odd frozen top. The lower bound excludes all insufficient orders simultaneously, including initialization-dependent order selection under a deterministic budget. For $m = 4\lfloor n^a/4 \rfloor$, fixed $0 < a \leq 1/2$, the size is $\exp[\Theta_{Y,a}((\log n)^2)]$.

Corollary 14.2 (A uniform local upper bound for deterministic datasets)**.** *For every dataset in the setup, $n \geq 4$ and $1 \leq d \leq n$, put*

$$q_n = \left\lceil \frac{\log(2^{37} n^3)}{\log 1024} \right\rceil, \qquad c_* = 4 - \tfrac{3}{2} \log 9 > 0. \tag{191}$$

With probability at least $1 - 4e/\sqrt{n} - 4e^{-c_* n}$, the dense and order-$q_n$ solutions exist and $E_n(q_n) \leq D_n/n$. For $m \geq 2$ their arrays satisfy $N_{\mathrm{mov}} \leq 2(m+d)m^{q_n-2}$ and $N_{\mathrm{arr}} \leq 2(m+d)m^{q_n-1}$; for $m = 1$ the exact counts are $(1+d)(q_n - 1)$ moving and $1 + d$ frozen. No Gram gap or positive label mean is required. If $b = 0$, both dense predictors and the closure vanish identically. The probability is for each deterministic dataset, not one simultaneous anti-concentration event for all datasets.

The proof separates three issues. Ordered source integration controls the unchanged closure with its own residual. A nonzero omitted derivative and analytic continuation force a lower error at every finite order. Upper and lower probability bounds on the *actual* dense-pair discrepancy then give the two storage inequalities.

Lemma 14.3 (Ordered source convergence and a common analytic disk)**.** *Suppose deterministically that $\|B_0\|_{\mathrm{op}}, \|W_0\|_{\mathrm{op}} \leq 4$. The dense solution and every finite closure exist on $[0, T]$. Their prediction coefficients extend holomorphically to $|t| < R_0 = 2^{-12}$, with norms at most one there; the dense coefficient satisfies $\|u(t)\|_2 \leq \|b\|_2(e^{1875|t|} - 1) < \|b\|_2$ when $b \neq 0$. Simultaneously for every $q \geq 2$, with $j = 2\lfloor q/2 \rfloor + 1$,*

$$E_n(q) \leq 64(j+1)(j+2)(\|b\|_2/4096)^j \leq 64\|b\|_2\, 1024^{-q}. \tag{192}$$

Proof. In the maximum block norm, a source derivative replaces one of the three parameter leaves of $c^\top W B v_0$ by a quadratic expression. After k derivatives there are at most $(k+2)!/2$ terms, each with $k+3$ parameter leaves and one occurrence of each input. Products, transposes, outer products and scalar contractions are bounded by their operator/vector norm products; this construction introduces no free trace. Consequently, on a block-norm ball of radius s,

$$\begin{aligned} |K_{k+1}(v_0, \dots, v_k; \theta)| &\leq \tfrac{1}{2}(k+2)!\, s^{k+3} \prod_i \|v_i\|_2, \\ |K_{k+1}(\mathbf{v}; 0)| &\leq 32\, 4^k (k+2)! \prod_i \|v_i\|_2. \end{aligned} \tag{193}$$

Readout reflection $J(B, W, c) = (B, W, -c)$ obeys $f_n \circ J = -f_n$ and $V(v; J\theta) = -JV(v; \theta)$. Induction gives $K_r(\mathbf{v}; J\theta) = (-1)^r K_r(\mathbf{v}; \theta)$. Thus all odd initialized ranks vanish.

Extend (184) over complex coordinates using ordinary transposes. Within distance one of θ_0, all block norms are at most five, $\|u\|_2 \leq 125$, and $\|b - \Sigma u\|_2 \leq 126$. Each block velocity is at most

$25 \cdot 126 = 3150$. A unit variation changes u by at most 75, so each velocity derivative is at most $5 \cdot 126 + 5 \cdot 126 + 25 \cdot 75 = 3135$. Picard integration on $|t| \le R_0$ maps this path ball into itself and contracts, since $3150R_0, 3135R_0 < 1$. Its uniform iterates give a holomorphic solution, unique also as a real polynomial ODE solution. Differentiating $u = B^\top W^\top c$ gives $\|u'\|_2 \le 1875(\|b\|_2 + \|u\|_2)$. Integration along a radial segment and scalar Gronwall give the stated bound, using $1875R_0 < \log 2$.

For a vector path z, put $a_z(t) = b - \Sigma z(t)$. Define a vector-valued map by its pairing with every v:

$$v^\top \mathcal{F}_q[z](t) = \sum_{k=1}^{q-1} \int_{0<s_k<\cdots<s_1<t} K_{k+1}(v, a_z(s_1), \dots, a_z(s_k); 0)\, ds_k \cdots ds_1. \tag{194}$$

The complex integrals follow the straight segment to t; $\mathcal{F}_\infty$ uses all $k \ge 1$. The first appended slot is always at the latest integration time. For $\zeta_a(t) = (y_a - v_a^\top z(t))/m$, define $I_{a_1 \dots a_k}[\zeta]$ as the ordered integral of $\prod_i \zeta_{a_i}(s_i)$ over that simplex, and $I_\varnothing = 1$. For each retained rank the exact reconstruction is

$$K_r^{(q)}(\mathbf{v}; t) = \sum_{k=0}^{q-r} \sum_{a_1, \dots, a_k} K_{r+k}(\mathbf{v}, v_{a_1}, \dots, v_{a_k}; 0) I_{a_1 \dots a_k}[\zeta](t). \tag{195}$$

Differentiating the outer integration limit gives (186) whenever $z = \mathcal{F}_q[z]$; multilinearity identifies its rank-one equation with (194). Thus a fixed point supplies all actual arrays, their initialized values, and their frozen top. Polynomial ODE uniqueness identifies this solution with the prescribed closure.

On the unit ball of holomorphic coefficient paths on $|t| \le R_0$, $\|a_z\| \le 2$. The simplex volume and (193) bound every finite map by $32 \sum_{k \ge 1} (k+1)(k+2)(8R_0)^k < 0.38$. Telescoping its k source slots bounds its Lipschitz constant by $16 \sum_{k \ge 1} k(k+1)(k+2)(8R_0)^k < 0.2$. Contraction gives all the asserted analytic closure paths.

On real $[0, T]$, if $\|b\|_2 > 0$, use instead the path ball of radius $\|b\|_2$ and put $x = 8T = 1/4096$. Uniformly over finite and infinite maps,

$$\begin{aligned} \|\mathcal{F}_q[z]\|_\infty &\le 64\|b\|_2[(1-x)^{-3} - 1] < \|b\|_2/16, \\ L &\le \frac{96x}{(1-x)^4} < \frac{1}{32}. \end{aligned} \tag{196}$$

These follow from $\|a_z\| \le 2\|b\|_2$ and $\|a_z - a_{\widetilde{z}}\| \le \|z - \widetilde{z}\|$; the series converge uniformly. The dense path is the infinite fixed point: repeated integration of its exact hierarchy leaves, after N steps, an N-fold integral of $K_{N+1}(v, a_u(s_1), \dots, a_u(s_N); \theta(s_N))$. Previously selected source vectors are held fixed during each integration. By the dense bound and (193), this remainder is at most $\frac{125}{2}(N+1)(N+2)(10\|b\|_2 T)^N$, which tends to zero.

Parity makes $j = 2\lfloor q/2 \rfloor + 1$ the first possible omitted index. For $z_0 = x\|b\|_2$, the omitted tail is at most

$$32 \sum_{k \ge j} (k+1)(k+2) z_0^k \le 32(j+1)(j+2) z_0^j \frac{1 + z_0}{(1 - z_0)^3}.$$

Here $(j + \ell + 1)(j + \ell + 2) \le (j+1)(j+2)(\ell+1)^2$ and $\sum_{\ell \ge 0} (\ell+1)^2 z^\ell = (1+z)/(1-z)^3$. The difference of the two own-residual fixed points is bounded by this tail divided by $1 - L$. Since $(1+x)/[(1-x)^3(1-L)] < 2$ and $(j+1)(j+2) \le 4^j$ for $j \ge 2$, (192) follows. If $b = 0$, the dense initial state is stationary and the zero path is a fixed point of every finite map; uniqueness proves the omitted case. □

Lemma 14.4 (Gaussian events and dense concentration)**.** *For $d \le n$ the matrix-norm event in Lemma 14.3 has probability at least $1 - 2e^{-c_* n}$. For the hard family, with probability at least $1 - n^{-1} - Ce^{-cn}$,*

$$D_n \le C\sqrt{\frac{m + \log(en)}{n}}. \tag{197}$$

Here and below unspecified $c, C > 0$ are absolute.

Proof. A maximal 1/4-separated sphere set has at most 9^d elements by comparing disjoint radius-1/8 balls with the radius-9/8 ball, and is a 1/4 net. It gives $\|M\|_{\mathrm{op}} \le (4/3)\max_v \|Mv\|_2$. For a fixed unit v, $n\|B_0 v\|_2^2$ is χ^2_n. The Gaussian integral gives $\mathbb{E}e^{t\chi^2_n} = (1-2t)^{-n/2}$. Exponential Markov at $t = 4/9$ yields $\Pr(\|B_0 v\|_2 > 3) \le e^{-4n} 9^{n/2}$. The net union bound proves the asserted $e^{-c_* n}$ failure for B_0; the same calculation proves it for W_0.

We supply the Gaussian Lipschitz concentration used here. For standard Gaussian measure γ in any finite dimension define $P_s h(x) = \mathbb{E}h(e^{-s}x + \sqrt{1-e^{-2s}}\, Z)$, $Z \sim \gamma$. Write $\mathscr{A} = \sum_i \partial^2_{x_i} - x \cdot \nabla$. For positive bounded smooth h with bounded derivatives and bounded away from zero, Gaussian integration by parts gives $\partial_s P_s h = \mathscr{A} P_s h$ and

$$-\frac{d}{ds}\operatorname{Ent}_\gamma(P_s h) = \mathbb{E}_\gamma \frac{|\nabla P_s h|^2}{P_s h},$$
$$\operatorname{Ent}_\gamma(h) = \mathbb{E}_\gamma(h \log h) - \mathbb{E}_\gamma h \log \mathbb{E}_\gamma h.$$

Indeed $\mathbb{E}_\gamma[\zeta \mathscr{A} v] = -\mathbb{E}_\gamma \nabla\zeta \cdot \nabla v$, and P_s preserves γ. Since $\nabla P_s h = e^{-s} P_s \nabla h$, weighted Cauchy–Schwarz gives $|\nabla P_s h|^2 / P_s h \le e^{-2s} P_s(|\nabla h|^2/h)$. As $s \to \infty$, bounded convergence gives $P_s h \to \mathbb{E}_\gamma h$ and entropy tending to zero. Integrating proves $\operatorname{Ent}_\gamma(h) \le \frac{1}{2}\mathbb{E}_\gamma(|\nabla h|^2/h)$. For bounded smooth L-Lipschitz F, with $L > 0$, substitute $h = e^{\lambda F}$. With $\psi(\lambda) = \log \mathbb{E}e^{\lambda F}$ this gives $\lambda\psi'(\lambda) - \psi(\lambda) \le \lambda^2 L^2/2$. Integrating $(\psi/\lambda)'$ from zero yields $\log \mathbb{E}e^{\lambda(F - \mathbb{E}F)} \le \lambda^2 L^2/2$. Markov, optimized at $\lambda = s/L^2$, and the same argument for $-F$ give

$$\Pr\{|F - \mathbb{E}F| \ge s\} \le 2e^{-s^2/(2L^2)}. \tag{198}$$

For an arbitrary Lipschitz F, truncate its values and smooth by convolution, preserving the Lipschitz bound. Pass first through smoothing and then truncation by dominated convergence: $|F(x)| \le |F(0)| + L|x|$, whose exponential is Gaussian-integrable. The case $L = 0$ is constant. This also justifies the preceding smooth calculations by bounded approximations.

On the matrix-norm event, all real states have block norms at most five. For two such trajectories, measure parameter differences by the sum of Frobenius/vector norms. Their u values differ by at most 25 times this sum. Expanding (184) bounds each velocity difference by 1255 times the sum: the two direct-product differences each cost 630 on distinct blocks, while the residual difference costs 625 on each block. Gronwall gives factor e^{3765t} in the sum norm. Therefore a fixed unit-query output at any $t \le T$ is $L_T = 50e^{4000T}$-Lipschitz in the Euclidean list of initial entries. Each such scalar map extends from the good set by $F(x) = \inf_{z\ \mathrm{good}}\{F(z) + L_T\|x - z\|_2\}$, with unchanged values and Lipschitz constant. Apply (198) to the variance-$1/n$ Gaussian entries. Two independent copies differ by at most $2s$ except with probability $4e^{-ns^2/(2L_T^2)}$, since their extension means agree.

The coefficient path has time Lipschitz constant at most $3 \cdot 25 \cdot 3150 < 250000$. Take a 1/2 sphere net with at most 5^d points and a time grid with $M = \lceil nT \rceil + 1$ points and mesh at most $1/n$. A linear functional's sphere norm is at most twice its net maximum. For $s = L_T\sqrt{2\log(4\,5^d M/\delta)/n}$ a union bound and the time Lipschitz estimate give $D_n \le 4s + 500000/n$, except with probability $\delta + 4e^{-c_* n}$. Set $\delta = 1/n$, $d = m + 1$ to obtain (197). □

Lemma 14.5 (A nonzero omitted derivative at every effective order)**.** *For the hard family put* $\bar v = m^{-1}\sum_a s_a v_a$, $\lambda = \|\bar v\|_2$, *and* $v_* = \bar v/\lambda$. *With probability at least* $1 - Ce^{-cn}$, *simultaneously for every* $q \ge 2$, *the signed prediction difference* $g_q(t) = m^{-1}\sum_a s_a[f_n(t, v_a) - f_{n,q}(t, v_a)]$ *satisfies*

$$\begin{aligned} g_q^{(k)}(0) &= 0 \ (k < j), \\ g_q^{(j)}(0) &\ge j!(Y/64)^j, \qquad j = 2\lfloor q/2\rfloor + 1. \end{aligned} \tag{199}$$

Proof. Directly, $G = (v_a^\top v_b)_{a,b} = \frac{1}{2}(I_m + \mathbf{1}\mathbf{1}^\top)$, $\lambda^2 = 1/8 + 1/(2m)$, label RMS is Y, and label variance is $3Y^2/4$. Identity activations preserve this population feature Gram. For $\xi_0 = B_0 v_*$, the Gaussian norm ratios $\|\xi_0\|_2^2$ and $\|W_0\xi_0\|_2^2/\|\xi_0\|_2^2$ are independent χ^2_n/n. Each is at least $3/4$ except with exponentially small probability: Markov applied to $e^{-\chi^2_n/6}$ gives failure at most $\exp[n/8 - (n/2)\log(4/3)]$. Intersecting with the matrix-norm event gives $\|\xi_0\|_2^2, \|W_0\xi_0\|_2^2 \ge 1/2$.

An odd frozen top is zero, so its predictor equals that of the preceding even top: for $h = 2\lfloor q/2\rfloor$, parity gives $\mathcal{F}_q = \mathcal{F}_h$. Put $e = u - u_h$ and $j = h + 1$. The first omitted index h vanishes, so the ordered series gives

$$\begin{aligned} v^\top(\mathcal{F}_\infty[u] - \mathcal{F}_h[u])(t) = \frac{t^j}{j!}K_{j+1}(v, b, \dots, b; 0) \\ + O(t^{j+1}). \end{aligned}$$

If $e(t) = O(t^\ell)$, telescoping source slots in the finite map gives $\mathcal{F}_h[u] - \mathcal{F}_h[u_h] = O(t^{\ell+1})$: one slot contributes $-\Sigma e$, and there is at least one time integration. Since $e(0) = 0$, repeated substitution in the difference of the two fixed-point equations proves $e = O(t^j)$; the finite-map difference is then $O(t^{j+1})$. This proves both the vanishing derivatives and the displayed leading coefficient, with the closure's own feedback retained. Contracting $v = \bar v$ and using $b = Y\bar v$ gives exactly

$$g_q^{(j)}(0) = Y^j K_{j+1}(\bar v, \dots, \bar v; 0). \tag{200}$$

For source ascent of $F = c^\top W B\bar v$, write $\xi = Bv_*$ and use the source clock $\tau = \lambda s$. The exact equations are $\xi' = W^\top c$, $W' = c\xi^\top$, $c' = W\xi$. Their invariants $WW^\top - cc^\top = W_0W_0^\top$ and $\|\xi\|_2^2 - \|c\|_2^2 = \|\xi_0\|_2^2$ imply

$$\begin{aligned} c'' &= (\|\xi_0\|_2^2 I + W_0W_0^\top)c \\ &+ 2\|c\|_2^2 c, \qquad c(0) = 0, \quad c'(0) = W_0\xi_0. \end{aligned} \tag{201}$$

Diagonalize the positive semidefinite matrix, choosing eigenvector signs so that $a = W_0\xi_0$ is coordinatewise nonnegative. The Taylor recurrence then has nonnegative coefficients. With $A_0 = \|\xi_0\|_2^2$, $H = \|a\|_2^2$, coefficientwise induction compares c from below with $a\zeta$, where $\zeta'' = A_0\zeta + 2H\zeta^3$, $\zeta(0) = 0$, $\zeta'(0) = 1$. Indeed the linear term dominates $A_0 a_i\zeta$ in coordinate i, and $\|c\|_2^2 c_i$ dominates $Ha_i\zeta^3$ at every successive coefficient. Since $A_0, H \ge 1/2$, it also compares ζ from below with $2\tan(\tau/2)$, which solves $\zeta'' = \frac{1}{2}\zeta + \frac{1}{8}\zeta^3$ with the same initial values. Writing $\tan z = \sum_{k\ge0} t_k z^{2k+1}$, $(2k+1)t_k = \sum_{i=0}^{k-1} t_i t_{k-1-i}$ for $k \ge 1$, and $t_0 = 1$, give $t_k \ge 3^{-k}$ by induction. Hence $\zeta \ge \tau/(1 - \tau^2/12)$ coefficientwise. The degree-$j = 2k+1$ coefficient of $c^\top W\xi = c^\top c'$ is at least $H(k+1)^2 12^{-k} \ge 8^{-j}$. Returning to F and clock s multiplies its jth derivative by λ^{j+1}. Since $\lambda \ge 2^{-3/2}$, $K_{j+1}(\bar v, \dots, \bar v; 0) \ge j!\lambda^{j+1}8^{-j} \ge j!64^{-j}$, proving the claim on one event for every order. □

Lemma 14.6 (From an omitted derivative to error on a real interval)**.** *If g is holomorphic on a neighborhood of $|t| \le R$, bounded there by two, and $|g^{(j)}(0)| \ge j!\rho^j$, where $j \ge 1$ and $0 < \vartheta = \rho R/8 < 1$, then, for $\kappa = 16[1 + \log(1/\vartheta) + \log(8/3)]$,*

$$\|g\|_{[0,R/4]} \ge \frac{1}{8\kappa}\left(\frac{\vartheta}{8e^2\kappa^2}\right)^j. \tag{202}$$

Proof. For a degree-N polynomial P, map $[0, R/4]$ to $[-1, 1]$. Its nonconstant Chebyshev coefficients have modulus at most $2\|P\|$, by their cosine integral formula, and

$$|T_k^{(j)}(-1)| = \prod_{r=0}^{j-1} \frac{k^2 - r^2}{2r+1} \le k^{2j}/j!,$$
$$|P^{(j)}(0)| \le \frac{2N}{j!}(8N^2/R)^j \|P\|_{[0,R/4]}.$$

For $k < j$ the derivative is zero; for $k \ge j$ the product formula follows by successively differentiating $(1-x^2)T_k'' - xT_k' + k^2T_k = 0$ at $x = -1$. Take the degree-$N = \lceil \kappa j \rceil$ Taylor polynomial of g. Cauchy's coefficient bound gives remainder at most $(2/3)4^{-N}$. Since its jth derivative agrees with that of g,

$$\|g\|_{[0,R/4]} \ge \frac{\vartheta^j (j!)^2}{2N^{2j+1}} - \frac{2}{3}\,4^{-N}.$$

To verify that the tail is at most half the first term, put $\zeta = \log(1/\vartheta)$, $v = \log(8/3)$, $D = 1 + \zeta + v$. Using $N \le 2\kappa j$, $\log(j!) \ge j\log j - j$, and $\log j \le j$,

$$j\zeta + (2j+1)\log N - 2\log(j!) + v \le j[\zeta + v + 3\log(2\kappa) + 3] \le 15jD < \kappa j \log 4 \le N \log 4.$$

The second inequality uses $\kappa = 16D$, $\log 32 < 4$ and $\log D \le D - 1$. Exponentiation proves the tail comparison. The resulting lower bound $\vartheta^j (j!)^2/(4N^{2j+1})$, with the same factorial bound, $N \le 2\kappa j$ and $j \le 2^j$, yields (202). □

Lemma 14.7 (An actual dense-pair lower bound)**.** *For every fixed deterministic dataset in the setup with $n \ge 4$, $1 \le d \le n$, and $b \ne 0$, with probability at least $1 - 4e/\sqrt{n} - 4e^{-c_* n}$,*

$$D_n \ge \frac{\|b\|_2}{2^{31} n^2}. \tag{203}$$

Proof. Use $v_* = b/\|b\|_2$ and $g(t) = f_n(t, v_*) - \widetilde{f}_n(t, v_*)$. On the pair's matrix-norm event, Lemma 14.3 bounds $|g| \le 2\|b\|_2$ on $|t| < R_0$. Set $R = 2^{-13}$, so $T = R/4$. For $0 \le t \le T$ the radius-$R/2$ circle centered at t lies in this disk; Cauchy's formula gives $|g''(t)| \le 2^{30}\|b\|_2$. At initialization,

$$f_n'(0, v_*) = \|b\|_2 H,$$
$$H = \|W_0 B_0 v_*\|_2^2 \overset{\text{law}}{=} UV,$$

where U, V are independent χ_n^2/n variables. Indeed $U = \|B_0 v_*\|_2^2$, and conditional on $B_0 v_*$ the ratio H/U has that same chi-square law independently of the conditioning vector.

For $n \ge 4$, the density $p(\zeta) = C_n \zeta^{n/2-1} e^{-n\zeta/2}$ is maximal at $\zeta_0 = 1 - 2/n \ge 1/2$. On $[\zeta_0, \zeta_0 + n^{-1/2}]$ its logarithm has zero initial derivative and second derivative at least $-2n$. Thus $p(\zeta) \ge e^{-1} p(\zeta_0)$ there and $\|p\|_\infty \le e\sqrt{n}$. Integration by parts in the same density gives $\mathbb{E}V^{-1} =$

$n/(n-2) \le 2$. The product density satisfies $\|p_H\|_\infty \le 2e\sqrt{n}$, and convolution gives the same bound for $H - \widetilde{H}$. Therefore $\Pr(|H - \widetilde{H}| \le a) \le 4e\sqrt{n}\, a$. Taking $a = 1/n$ gives $|g'(0)| \ge \|b\|_2/n$ outside an event of probability $4e/\sqrt{n}$, without conditioning on the matrix norms. Intersect afterward with their good event. At $t_n = 1/(2^{30}n) \le T$, Taylor's integral remainder now gives $|g(t_n)| \ge \|b\|_2/(2^{31}n^2)$, proving the assertion. □

Proof of Theorem 14.1 and corollary 14.2. For the general upper, combine (192) and (203) to get $E_n(q)/D_n \le 2^{37}n^2\, 1024^{-q}$ when $b \ne 0$. The choice (191) proves Corollary 14.2, including its probability. For $b = 0$, uniqueness gives the claimed exact zero predictions instead. The counts follow by summing (187).

For the lower, apply Lemma 14.6 to g_q from Lemma 14.5 with $R = 2^{-13}$ and $\rho = Y/64$; the common analytic bound is two. Since $g_q = \bar{v}^\top(u - u_q)$ and $\lambda \le 1$, $E_n(q) \ge \|g_q\|_{[0,T]}$. Writing

$$\kappa_Y = 16[1 + \log(2^{22}/Y) + \log(8/3)],$$
$$a_Y = (8\kappa_Y)^{-1}, \qquad b_Y = \log(2^{25}e^2\kappa_Y^2/Y),$$

gives, on one event of probability $1 - Ce^{-cn}$, $E_n(q) \ge a_Y e^{-b_Y j}$ simultaneously for every $q \ge 2$. Put $R_n = n/(m + \log(en))$. If $q \le (4b_Y)^{-1}\log R_n$, then $j \le q + 1$ gives $E_n(q) \ge a_Y e^{-b_Y} R_n^{-1/4}$, whereas (197) gives $D_n \le CR_n^{-1/2}$. Because $R_n \to \infty$ uniformly in $4 \le m \le \sqrt{n}$, these bounds imply $E_n(q) > AD_n$ simultaneously over these orders. Thus their success probability for any fixed factor A tends to zero, and eventually is below π.

Literal training arrays alone have at least m^{q-1} entries, even after omitting a zero odd top; nonfrozen arrays have at least m^{q-2}. The necessary order therefore costs $\exp[c_Y \log m \log R_n]$ in either convention. Uniformly in the sample range $\log R_n$ is bounded above and below by positive absolute multiples of $\log n$ for sufficiently large n. For the upper, $q_n = O(\log n)$ has success tending to one and error at most D_n/n. With $d = m + 1$, its count is at most $5m^{q_n} + md + m$. These prove Theorem 14.1 and its moving-array assertion. For an adaptive order under a deterministic smaller budget, every permitted order lies in the simultaneously excluded range, proving the stated budget interpretation. □

The matched result concerns a deep-linear model with $4 \le m \le \sqrt{n}$ samples, a specified literal tensor representation, and the window $[0, 2^{-15}]$ in half-MSE physical time. All layers are trainable, but the input-output function is linear. No longer-time, nonlinear-activation, arbitrary-encoding, or finite-precision conclusion is asserted. Fixed m, d already give size $n^{\Theta(\log m)}$; growing m and d together with width gives the superpolynomial example. Here Y is fixed independently of m, n. The small-label hypothesis of Theorem 2.1 is therefore not an assumption of this result. The preceding fitting and dense-variability theorems are not used, and this result establishes no same-family storage separation from those compression constructions.

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15 Figures and numerical illustrations

This collection preserves the static figures already accompanying the paper, including alternate layouts and all seven pages of the illustrated gallery. Format copies and interactive explainers are omitted. These are earlier illustrations under their recorded protocols, not new experiments or certificates of the headline theorem. Unless stated otherwise, RMS means disagreement with dense predictions on sampled queries, not error against labels. Recorded times do not certify a continuous-time supremum, and separately fitted endpoints need not share a physical time. Historical artwork sometimes writes P for the history order q. In radial plots, $f(\theta)$ denotes the prediction at input angle θ.

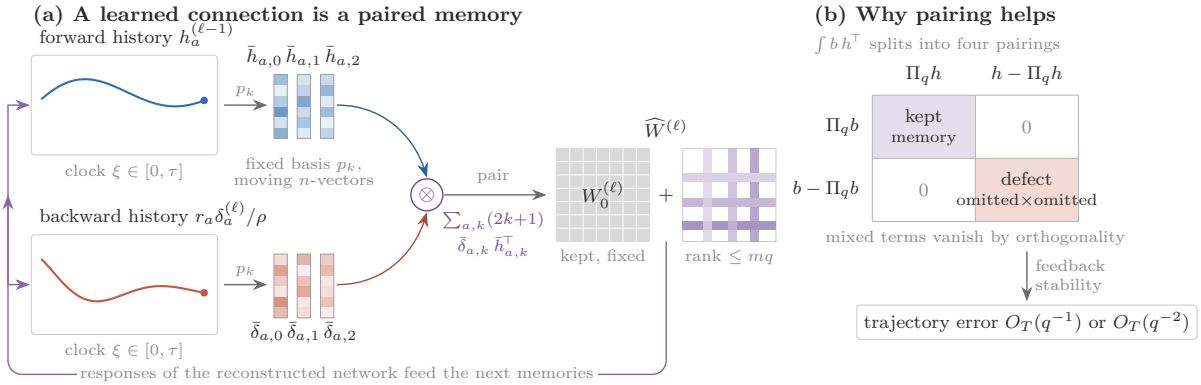


Figure 1: Response-memory reconstruction and the product-of-tails identity. Both histories use the same Legendre modes; the initialized mixer remains fixed and retained. The printed finite-horizon rates are schematic, not the present complete-trajectory estimates.

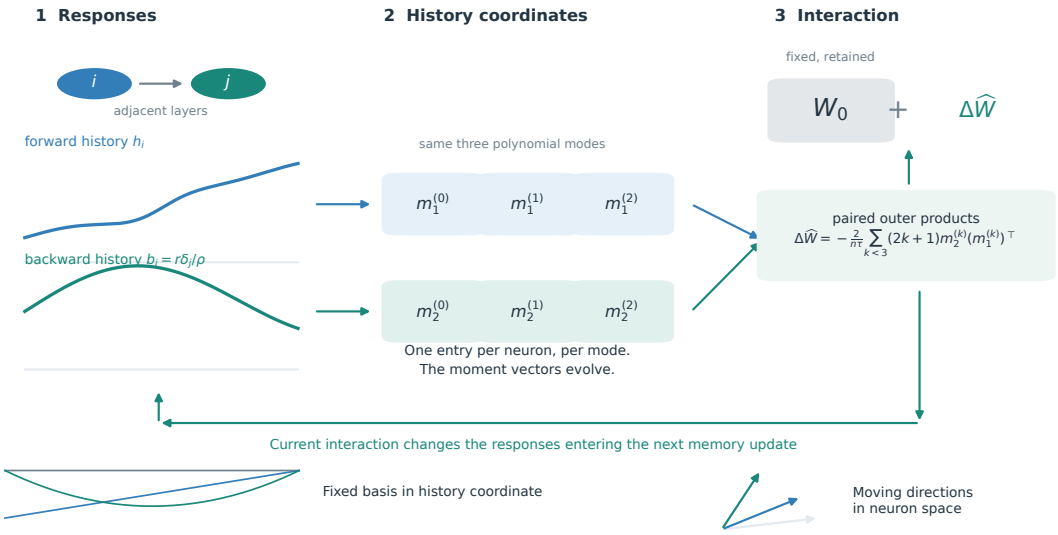


Figure 2: Single-sample, three-mode illustration: paired forward and residual-weighted backward moments reconstruct the learned connection. The reconstructed network supplies the next responses; the polynomial basis is fixed while its neuron-valued coefficients evolve.

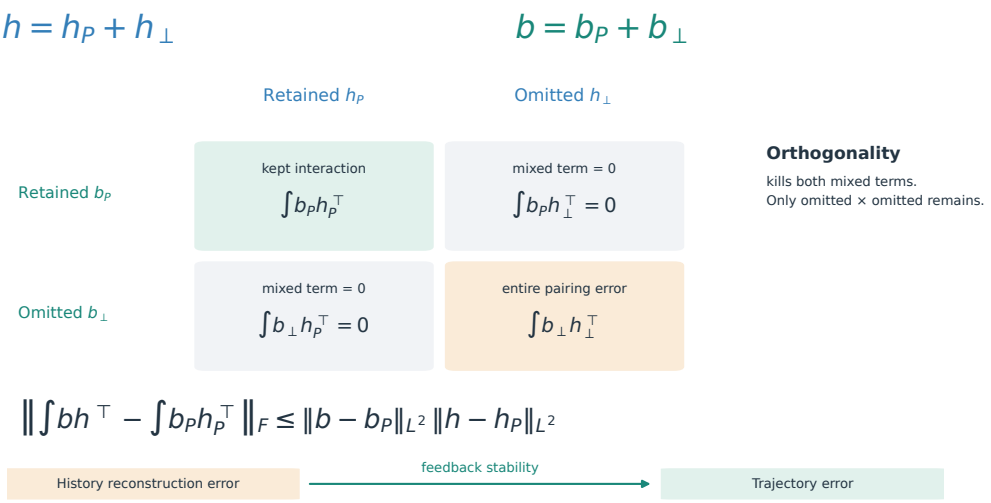


Figure 3: Projecting both histories onto the same orthogonal history space cancels the two mixed terms, leaving the pairing of the omitted histories. Trajectory control additionally requires regularity and feedback stability.

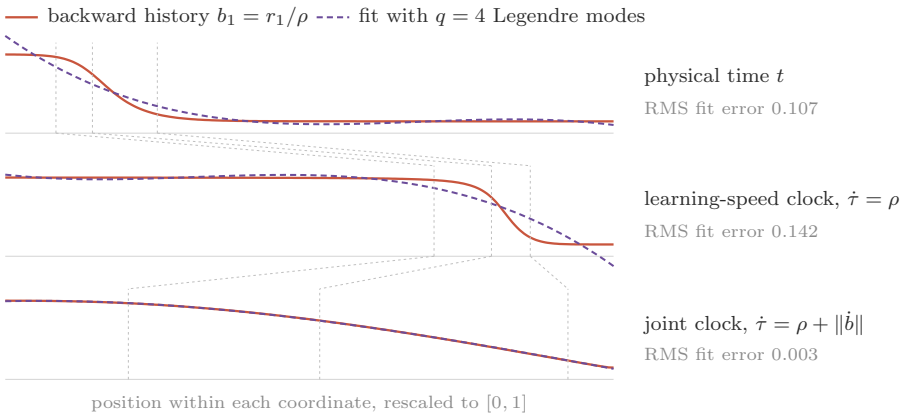


Figure 4: Illustrative four-mode fits of a normalized backward history in physical time and two alternative clocks. The displayed fit errors describe this example, not a general ordering of clock accuracy.

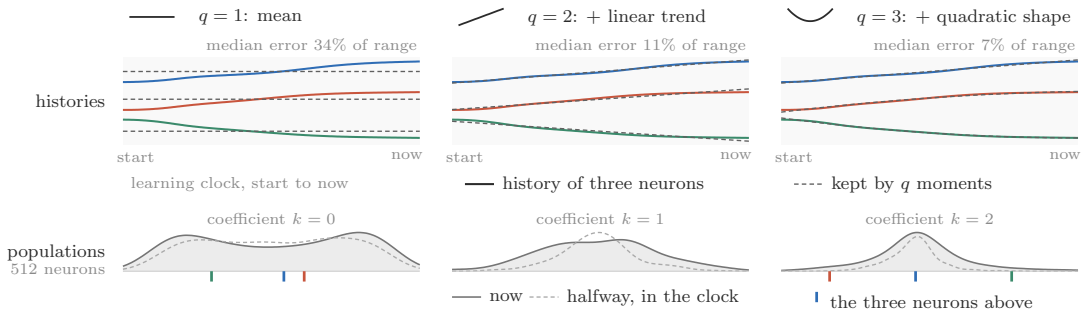


Figure 5: Three neuron histories reconstructed with one, two, and three Legendre coefficients, with coefficient distributions below. Errors are normalized by each history's range. This older nonzero-readout example illustrates memory coordinates, not the canonical initialization theorem.

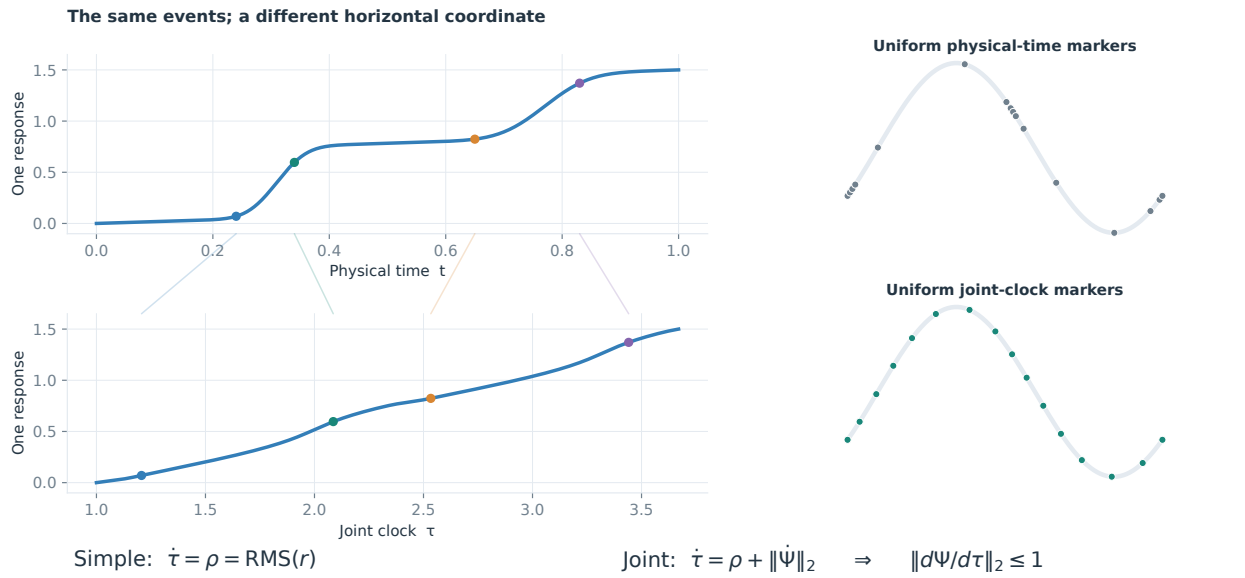


Figure 6: A constructed smooth response path in physical time and a clock that also tracks response motion. Matching colors identify the same events; this is a coordinate-change schematic, not a measured training trajectory or a constant-speed assertion.

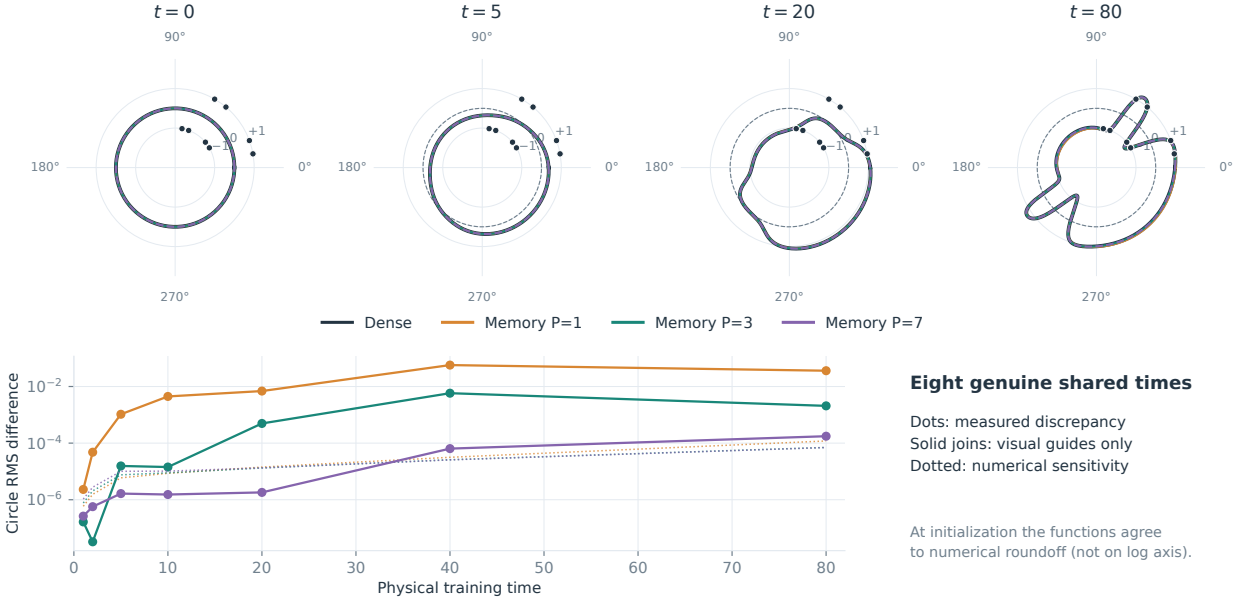


Figure 7: Earlier eight-checkpoint circle comparison: two tanh hidden layers, $n = 2048$, and $q = 1, 3, 7$. Snapshots share physical times; discrepancies use 2048 query angles. Dotted numerical sensitivities and connecting lines do not certify the unobserved times.

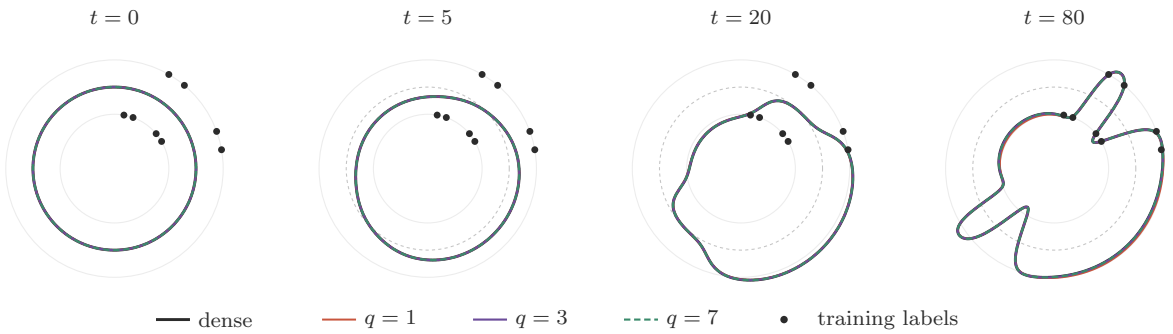


Figure 8: Paired-label circle predictions at common times $0, 5, 20, 80$: two tanh hidden layers, $n = 2048$, $q = 1, 3, 7$. Radius is $3+f(\theta)$ and black points are training labels. This is the later 39-checkpoint, two-tolerance Heun replay.

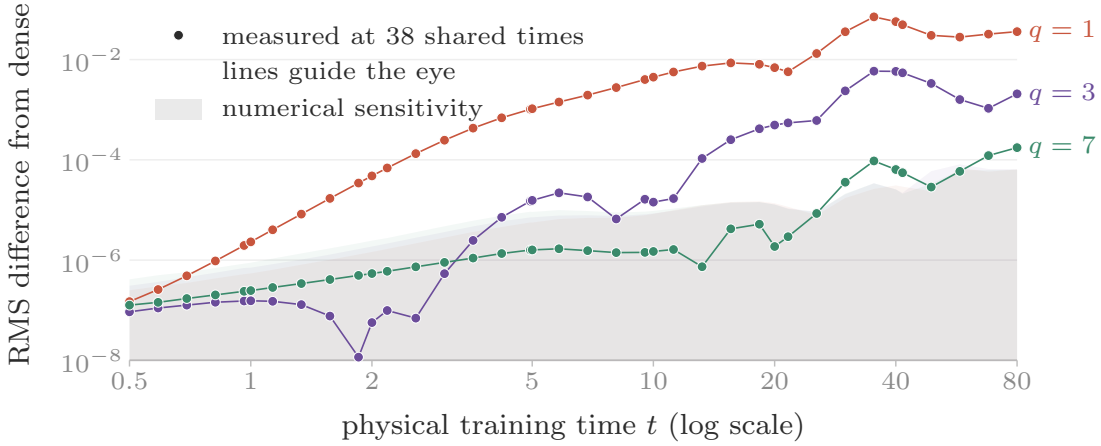


Figure 9: Companion dense-versus-memory RMS at the 38 positive common times of that replay. The coarse/fine sensitivity scale is an empirical numerical diagnostic, not a continuous-time bound.

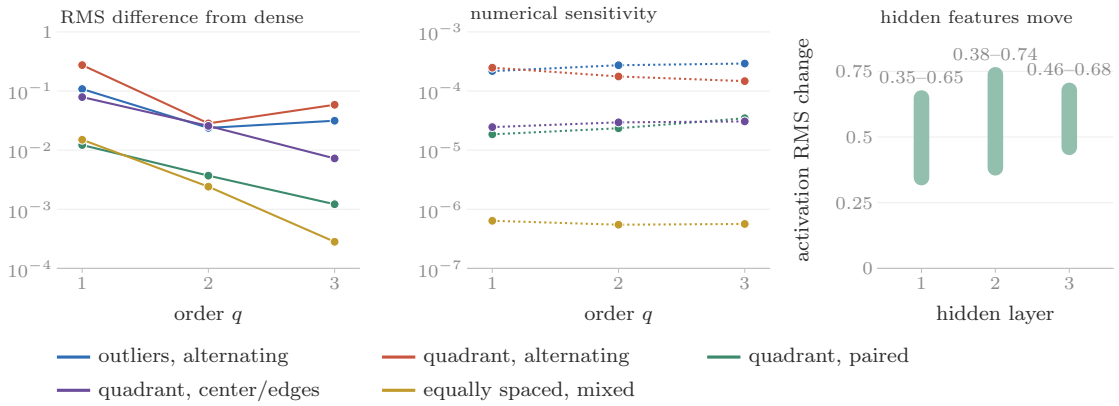


Figure 10: Deep-circle endpoint discrepancies, coarse/fine sensitivity, and hidden-feature motion for $q = 1, 2, 3$. These three-hidden-layer tanh runs have $n = 4096$ and individual training-MSE 10^{-3} endpoints; the sensitivity ranges are not confidence intervals.

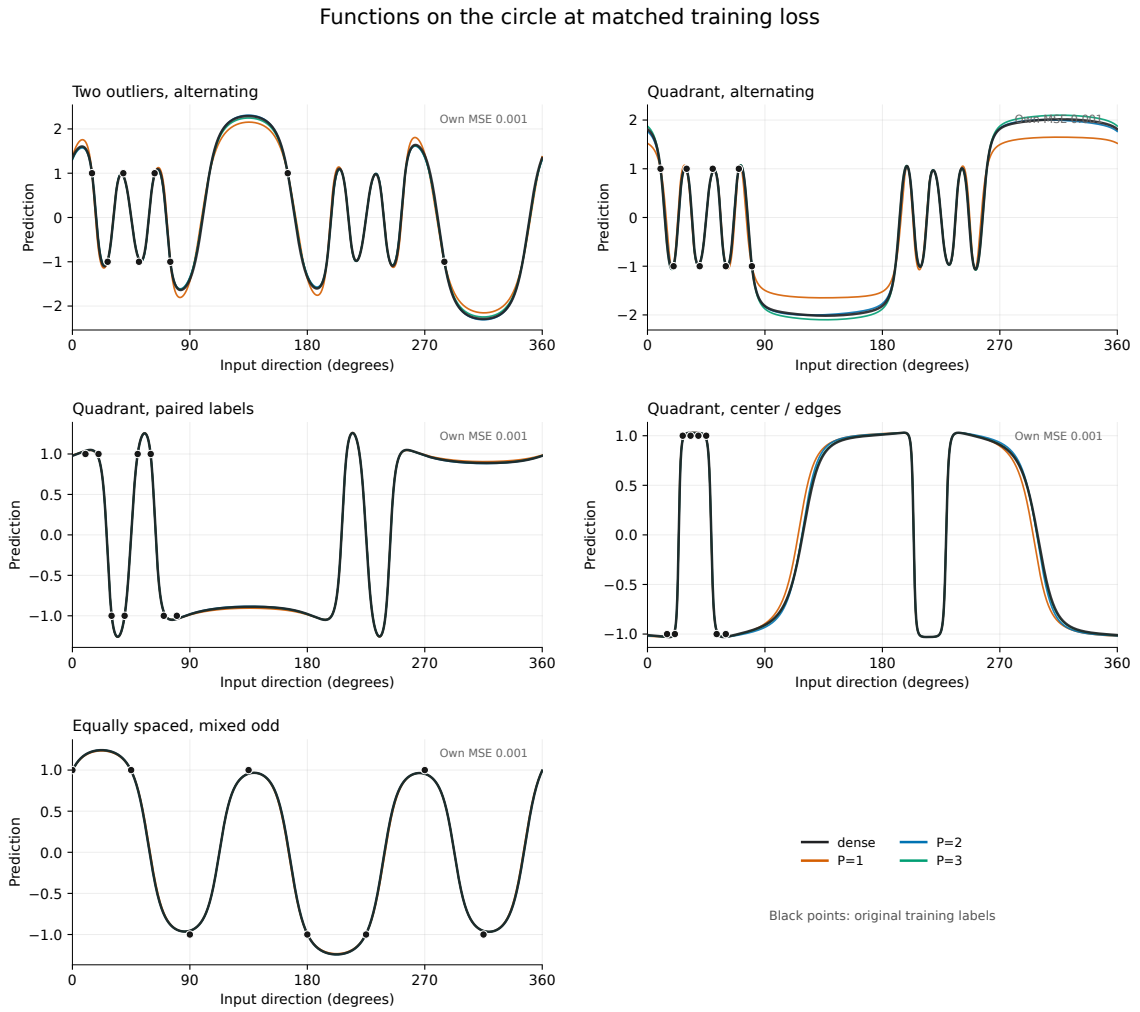


Figure 11: Five circle tasks with three tanh hidden layers, $n = 4096$, and $q = 1, 2, 3$. Each curve uses its own training-MSE 10^{-3} endpoint: the comparison is between fitted functions, not common physical times.

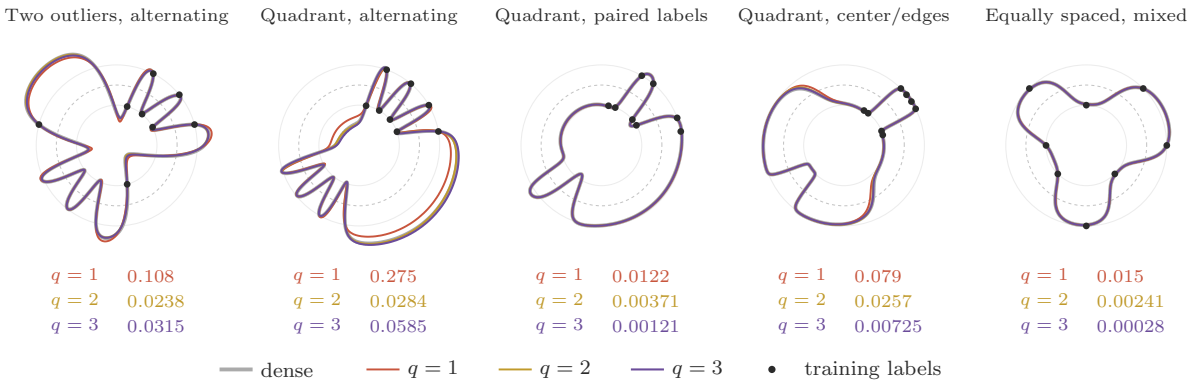


Figure 12: Compact radial view of the deep-circle family. Radius is $3 + f(\theta)$; the printed values are RMS differences from dense predictions on 8192 circle queries. This is an alternate layout, not an additional experiment.

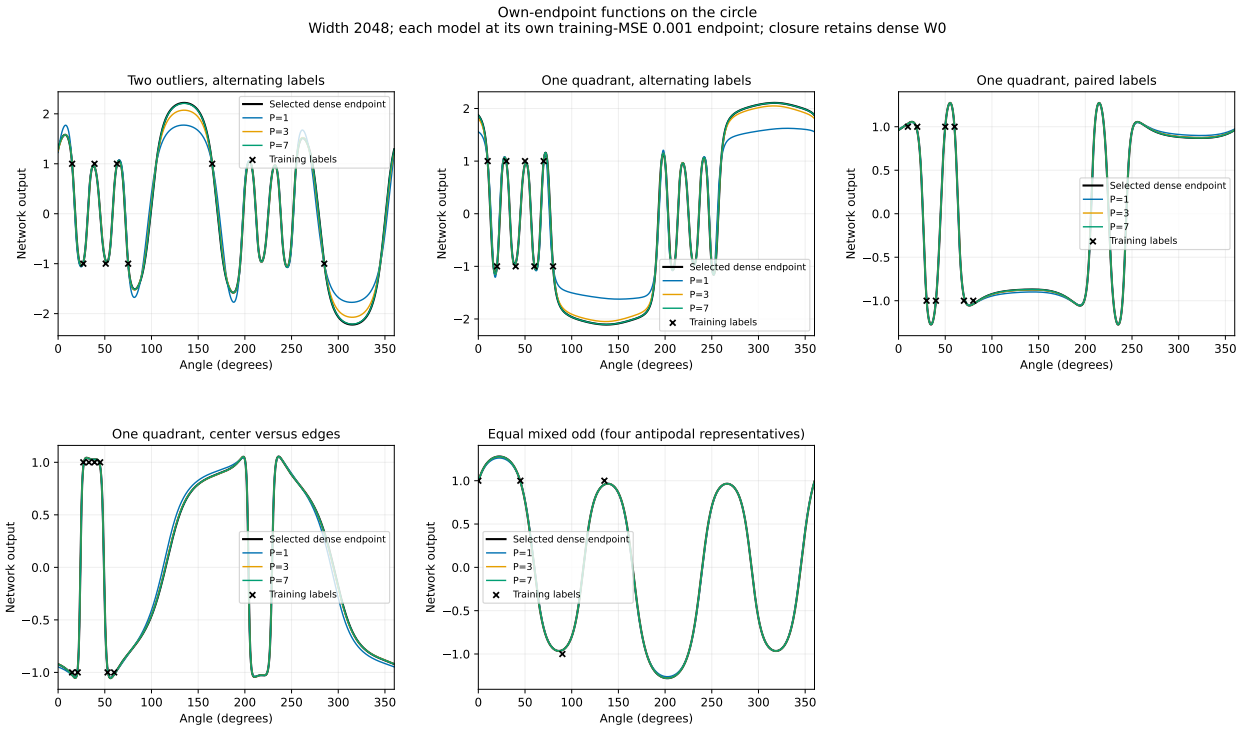


Figure 13: Earlier Cartesian circle comparison with two tanh hidden layers, $n = 2048$, and $q = 1, 3, 7$, at individual training-MSE 10^{-3} endpoints. This export uses archived dense references, unlike the selected references in the radial views.

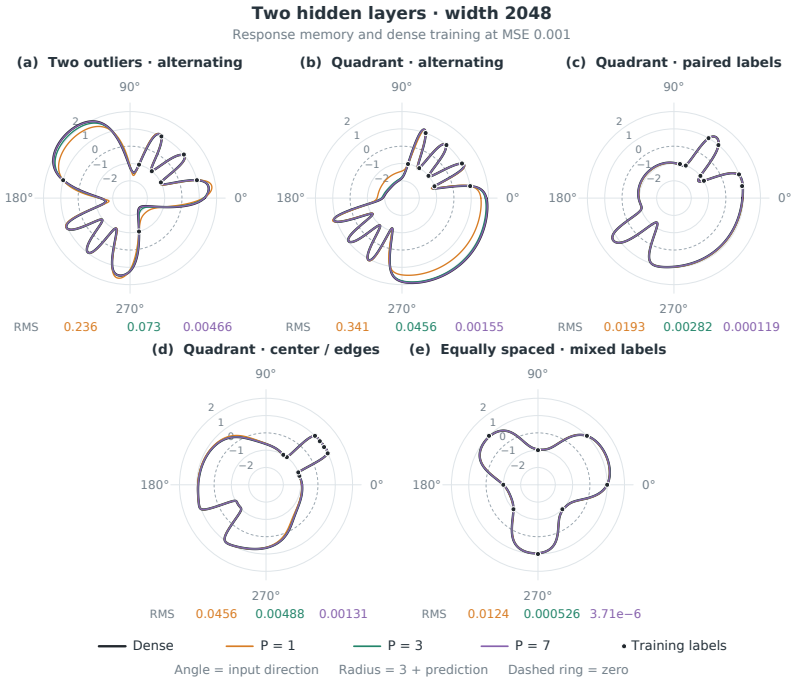


Figure 14: Shallow-circle radial comparison at the same loss threshold. Radius is $3 + f(\theta)$; RMS uses 8192 queries against four fresh dense references and one refined archived reference. Black points mark training labels.

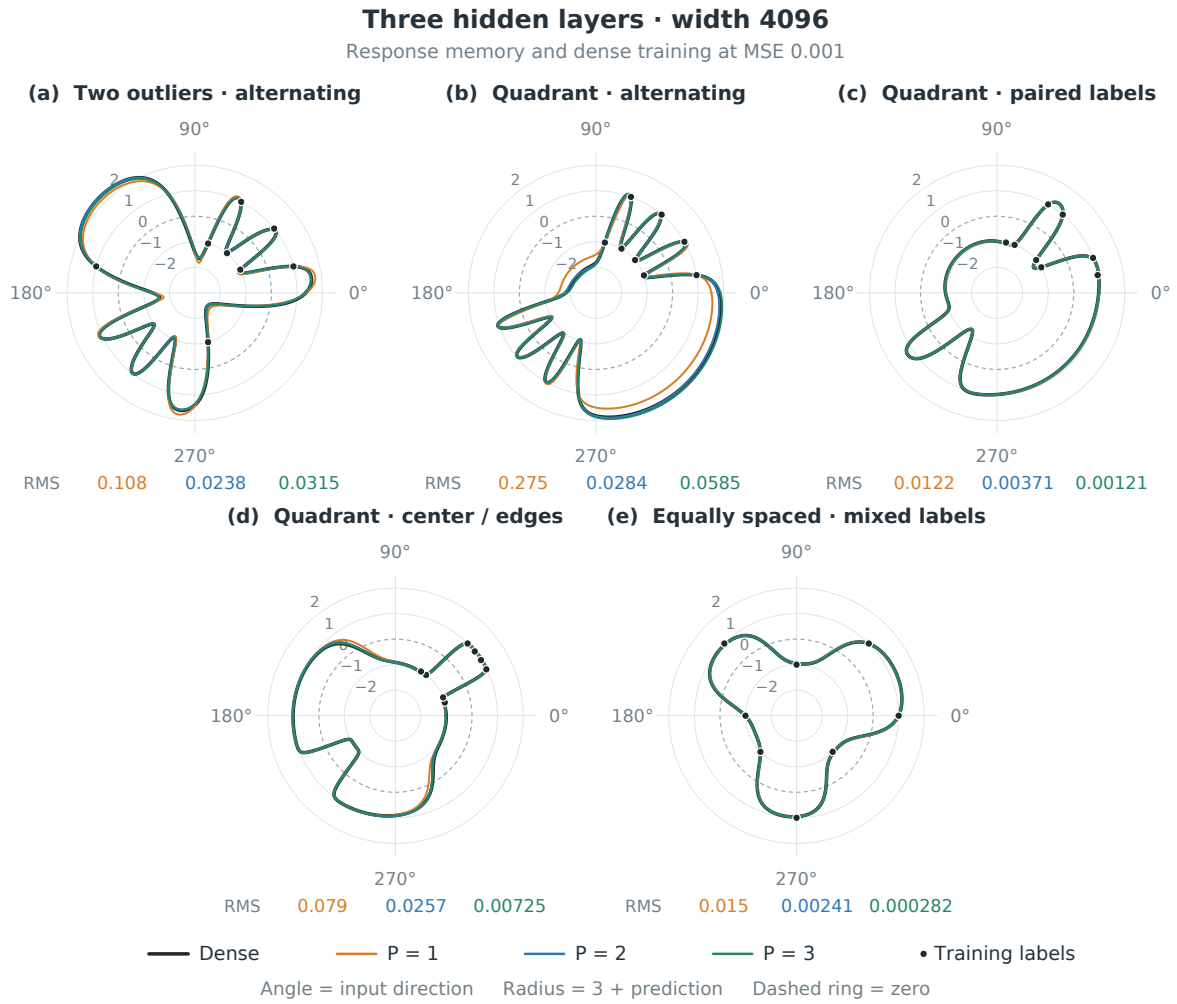


Figure 15: Larger radial panels for the deep-circle family: three tanh hidden layers, $n = 4096$, $q = 1, 2, 3$, and separately fitted training-MSE 10^{-3} endpoints. The printed discrepancies use 8192 circle queries; fixed dense mixers remain additional storage.

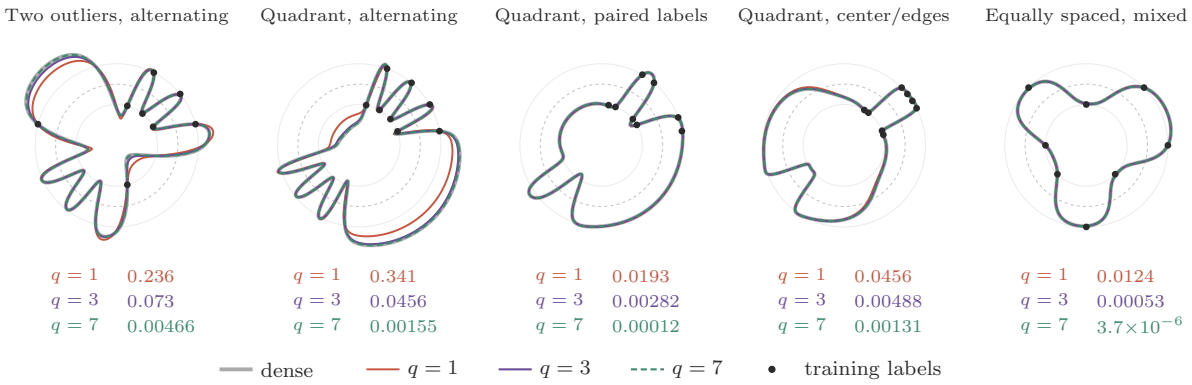


Figure 16: Compact radial layout of the shallow-circle family, using the same selected references as its larger radial view. The printed RMS values measure fitted-function disagreement, not label risk.

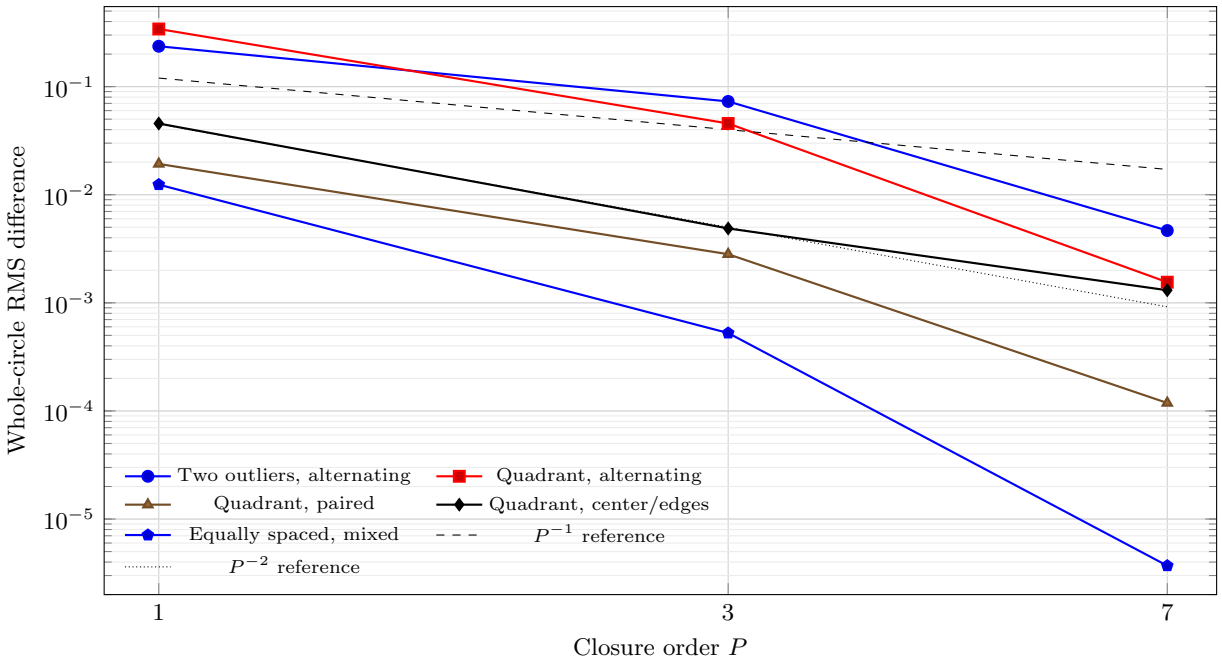


Figure 17: Observed circle discrepancy versus history order in five tasks. The inverse-order reference lines are visual comparisons, not fitted or certified convergence rates.

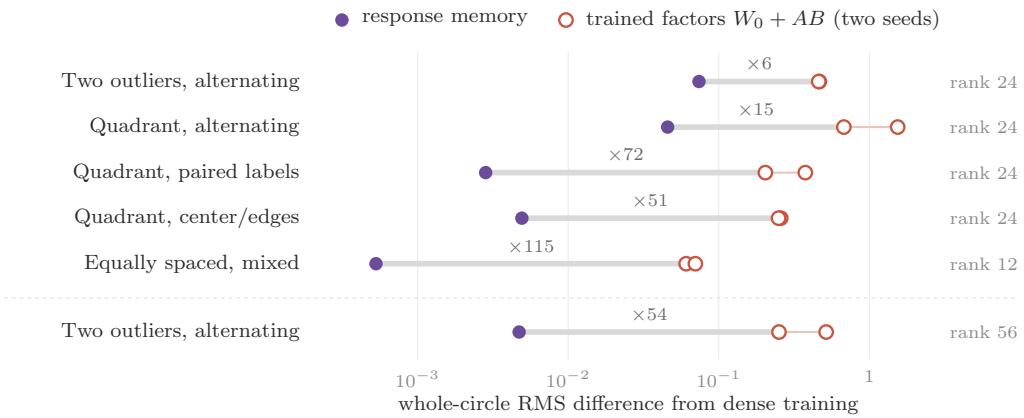


Figure 18: Selected circle endpoints compare response memory with two directly trained factor seeds at the same correction-rank bound. Equal correction rank does not imply equal total storage or a comparison with every low-rank optimizer.

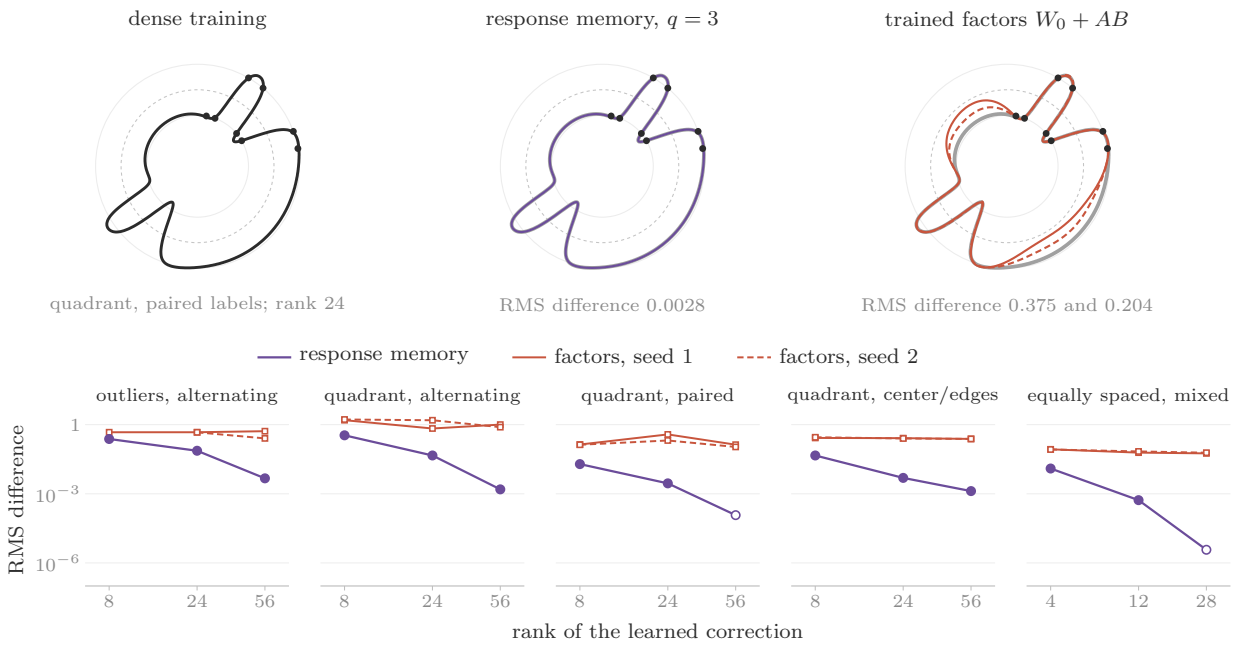


Figure 19: Paired-label fitted functions and five-task rank sweeps for two tanh hidden layers of width 2048. RMS uses 8192 queries at individual training-MSE 10^{-3} endpoints; fixed mixers are additional, and omitted or precision-limited runs are not exact successes.

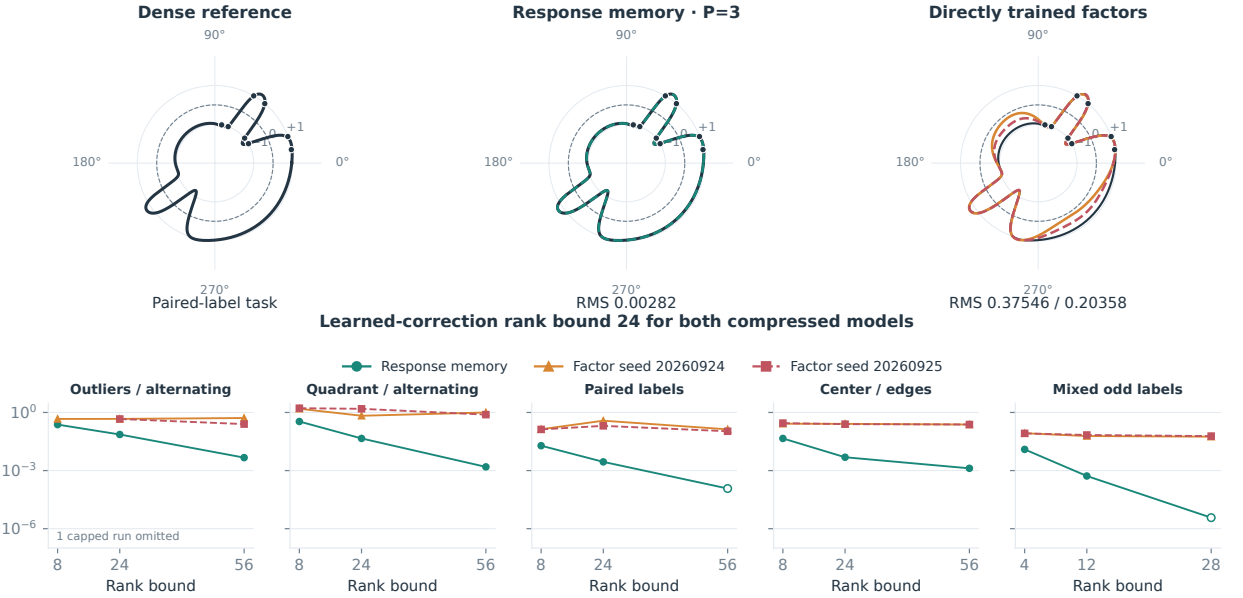


Figure 20: Alternate presentation of the factor comparison, including the rank-24 paired-label example and five-task sweep. Despite the filename, these are separately fitted endpoints, not common-time dynamics. Hollow points have limited numerical precision; one capped run is omitted.

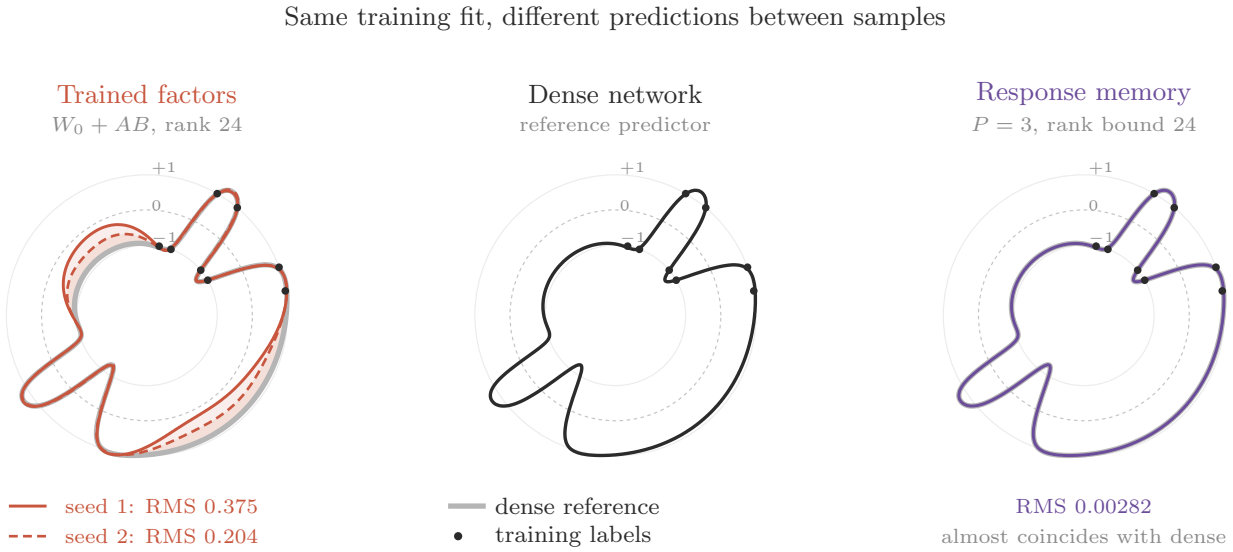


Figure 21: Enlarged paired-label comparison: dense training, order-3 memory, and two factor seeds with correction-rank bound 24. Radius is $3 + f(\theta)$; RMS uses 8192 angles at individual training-MSE 10^{-3} endpoints.

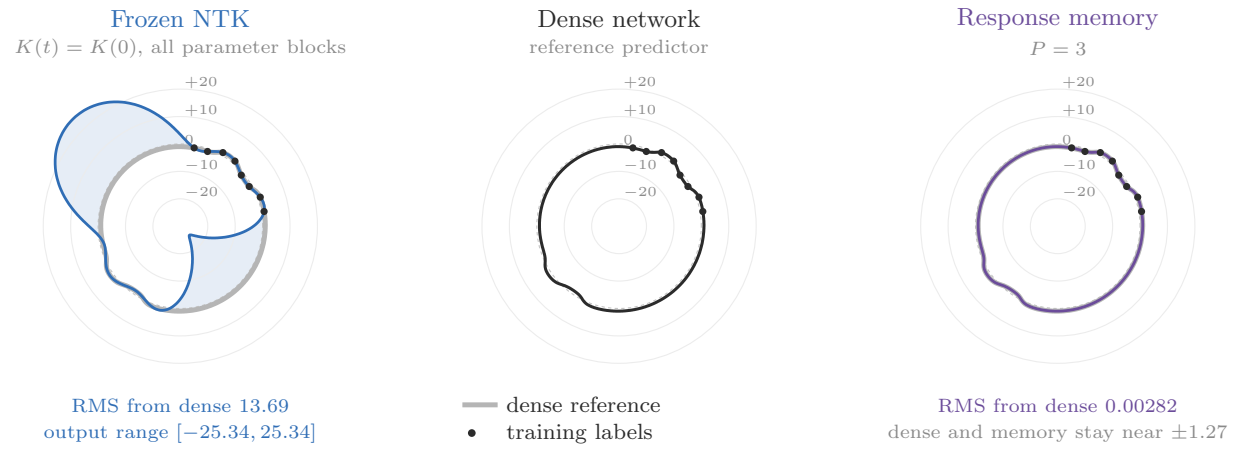


Figure 22: Dense training, its full frozen initial empirical NTK, and order-3 memory from the same width-2048 initialization. Each reaches training MSE 10^{-3} at its own time. Radius is $30 + f(\theta)$ here, preserving the large signed NTK excursions.

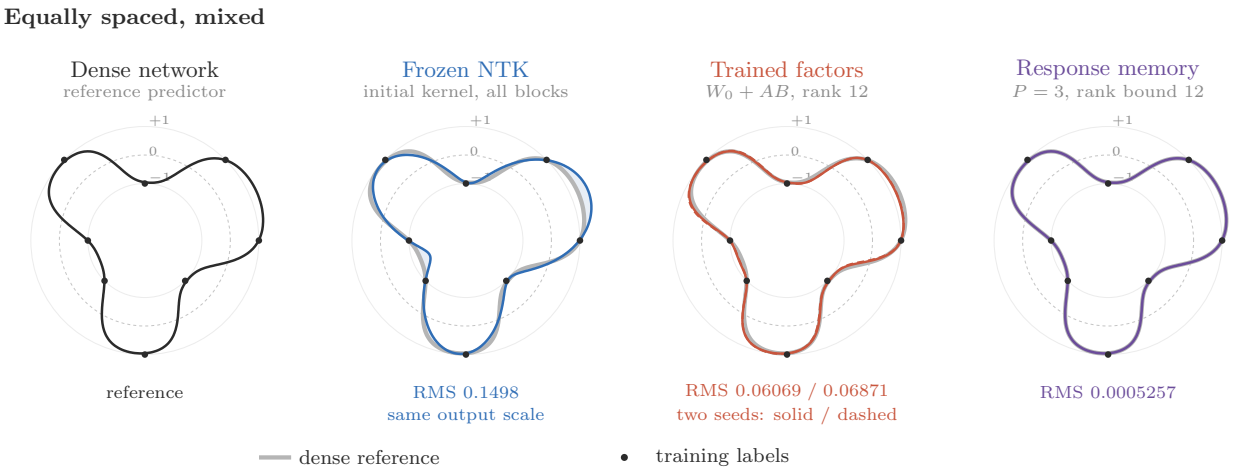


Figure 23: Equally spaced mixed-label control: dense training, frozen initial NTK, two factor seeds, and order-3 memory. RMS uses 8192 queries at individual training-MSE 10^{-3} endpoints. The correction-rank bound is 12 after the antipodal quotient.

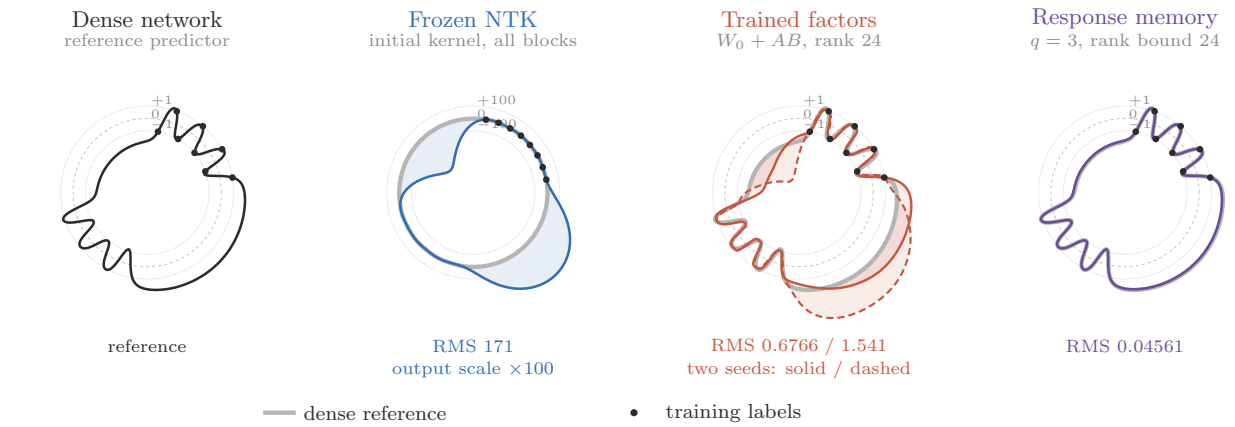


Figure 24: Quadrant alternating labels, with the same width-2048 two-hidden-layer tanh control protocol. The factor correction-rank bound is 24; the NTK panel explicitly uses a 100-times-wider output scale. RMS is disagreement with dense, not ground-truth error.

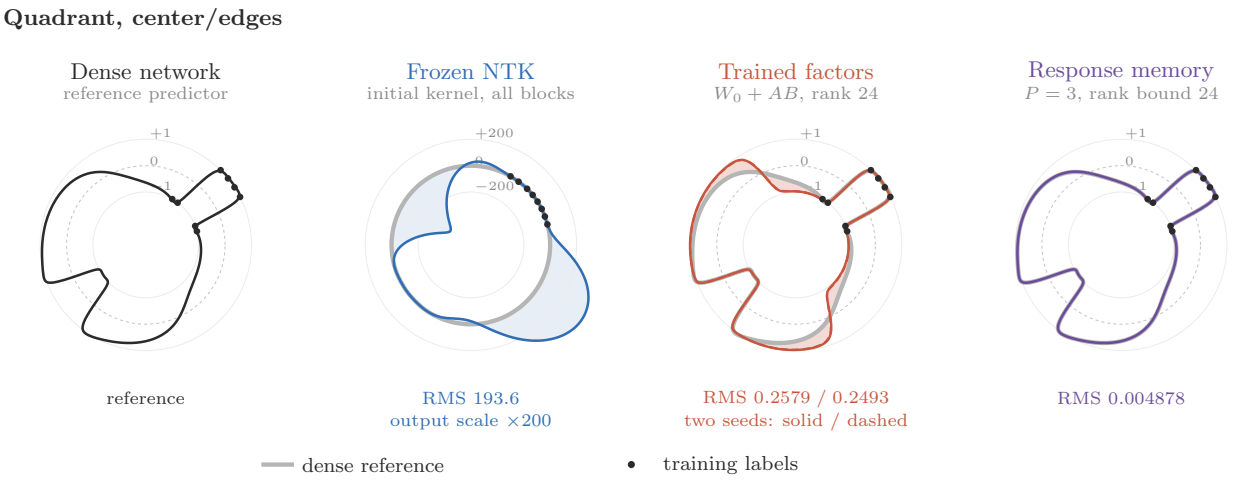


Figure 25: Quadrant center-versus-edges labels under the same endpoint protocol. The NTK panel alone uses a 200-times-wider output scale, with signed ticks in original units. All reported RMS discrepancies use 8192 circle queries.

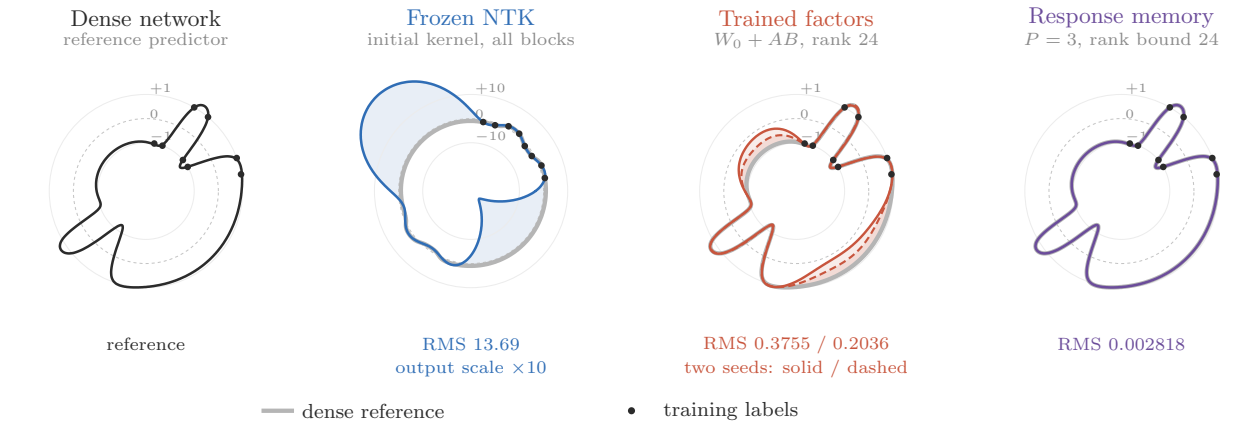


Figure 26: Quadrant paired labels: the same dense, frozen-NTK, rank-24 factor, and order-3 memory comparison. Each model fits separately to training MSE 10^{-3}; the NTK output scale is explicitly widened tenfold.

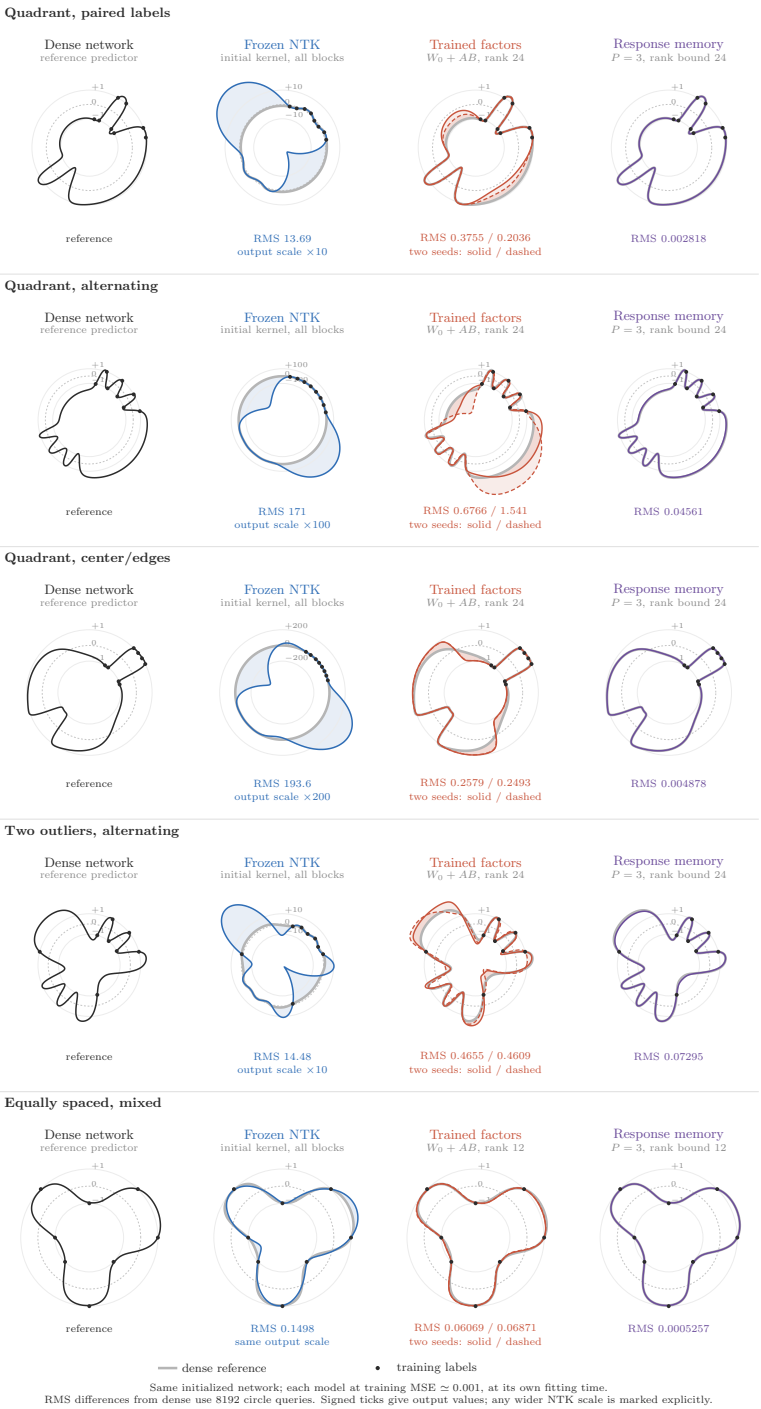


Figure 27: Overview of all five preceding control tasks. This combines the same observations, with marked NTK scale changes and correction-rank bounds 24 or 12; it is not another replicate.

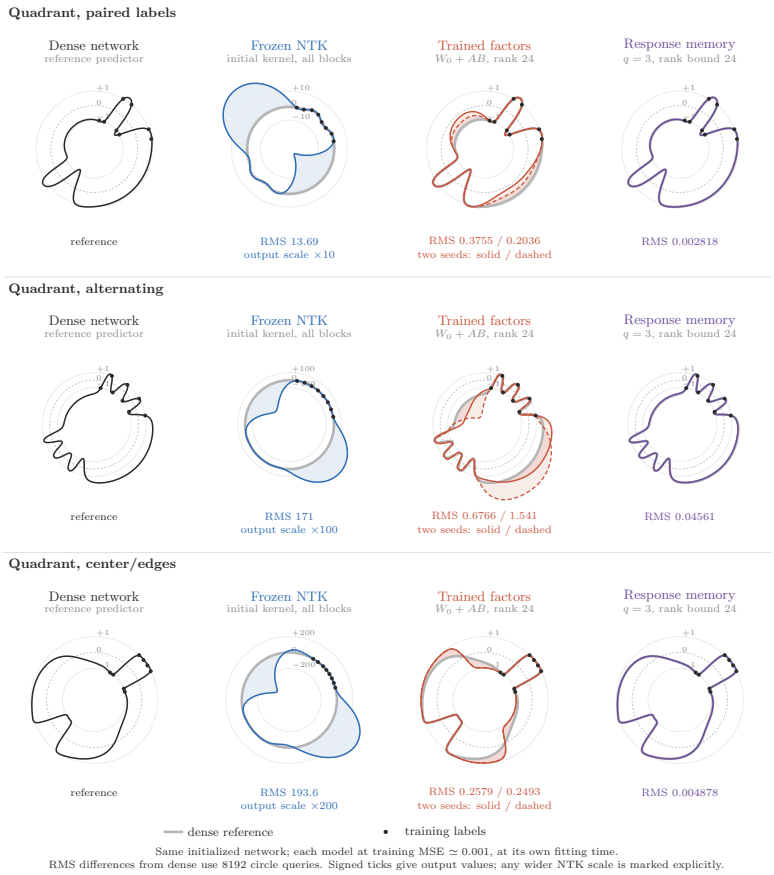


Figure 28: Page-friendly assembly of the three quadrant control tasks. The protocols, separate fitted endpoints, and marked NTK output-scale changes are unchanged.

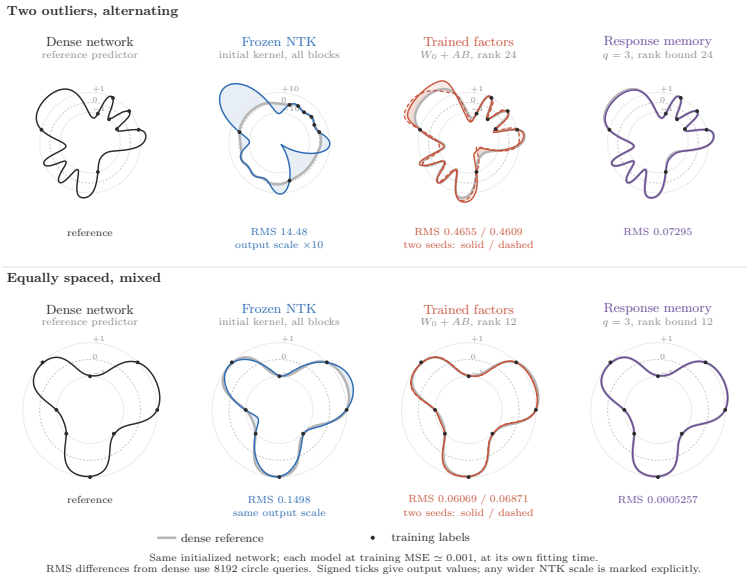


Figure 29: Companion assembly of the outlier and equally spaced mixed-label tasks. These are the remaining two rows of the full control gallery, not additional experiments.

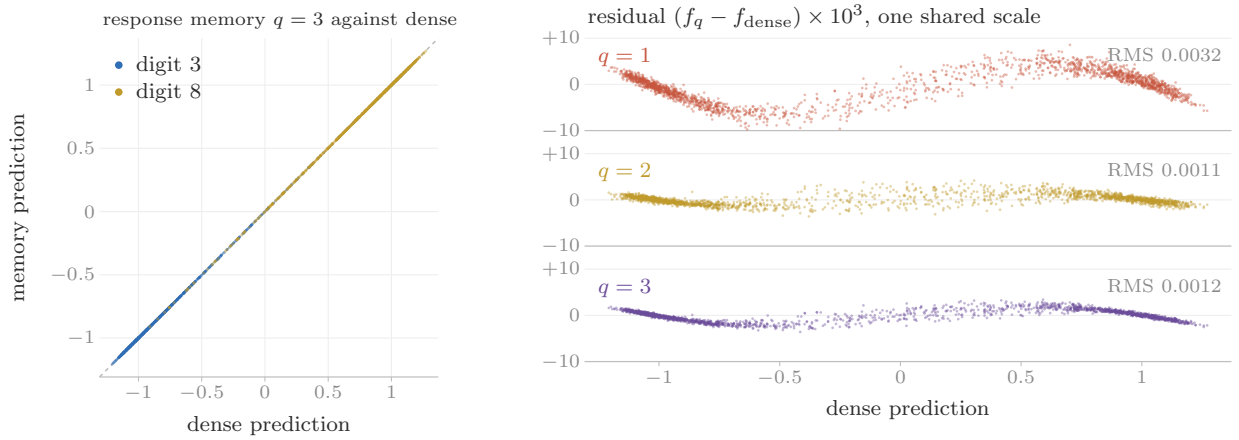


Figure 30: Earlier MNIST 3-versus-8 comparison: 100 training and 1984 held-out images, two tanh hidden layers, $n = 4096$. The order-3 agreement plot and order-1–3 differences use individual training-MSE 10^{-3} endpoints. RMS is disagreement with dense predictions.

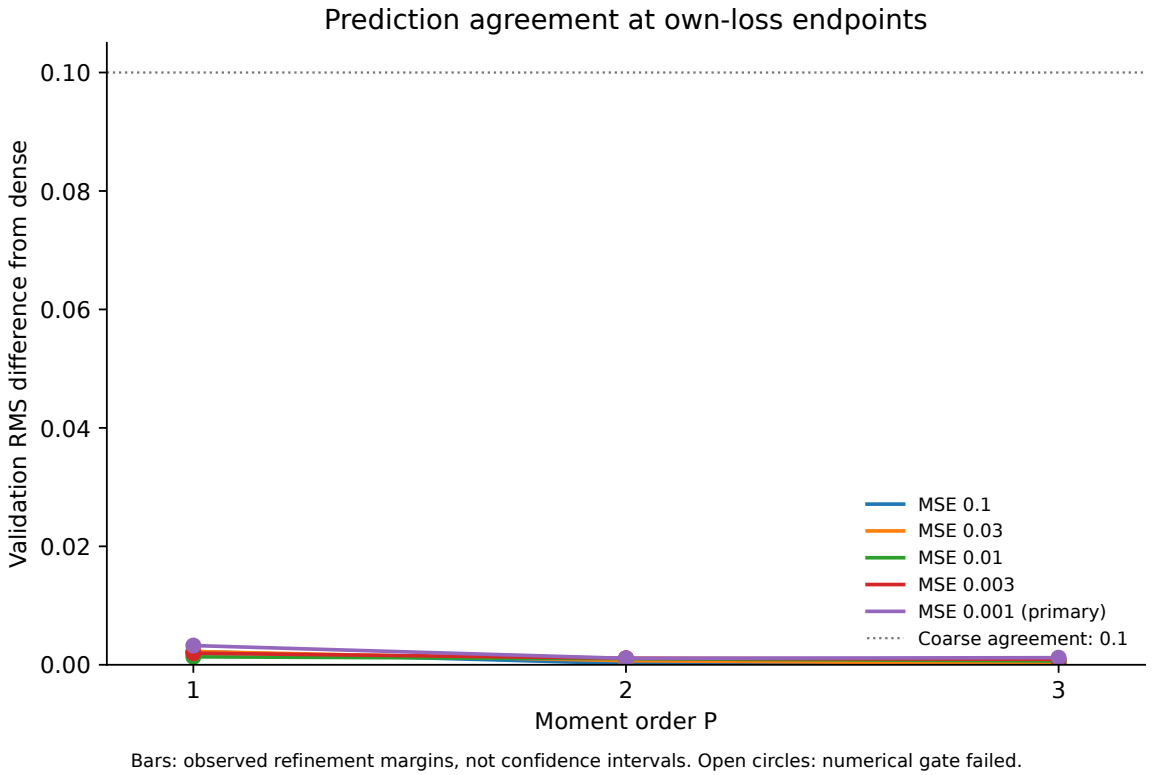


Figure 31: MNIST validation discrepancy across history orders and the displayed training-loss endpoints. Bars show observed refinement margins, not confidence intervals; open markers identify failures of the numerical gate.

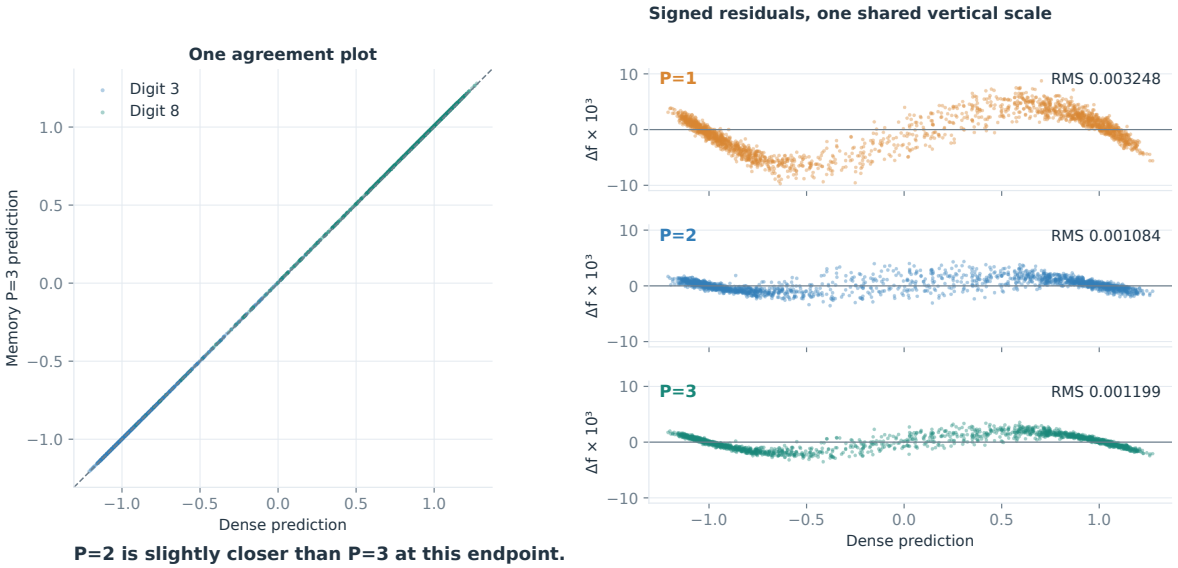


Figure 32: Alternate MNIST endpoint presentation. Signed memory-minus-dense differences share one vertical scale; order 2 is slightly closer than order 3 in this run. These are function-agreement measurements, not classification accuracies.

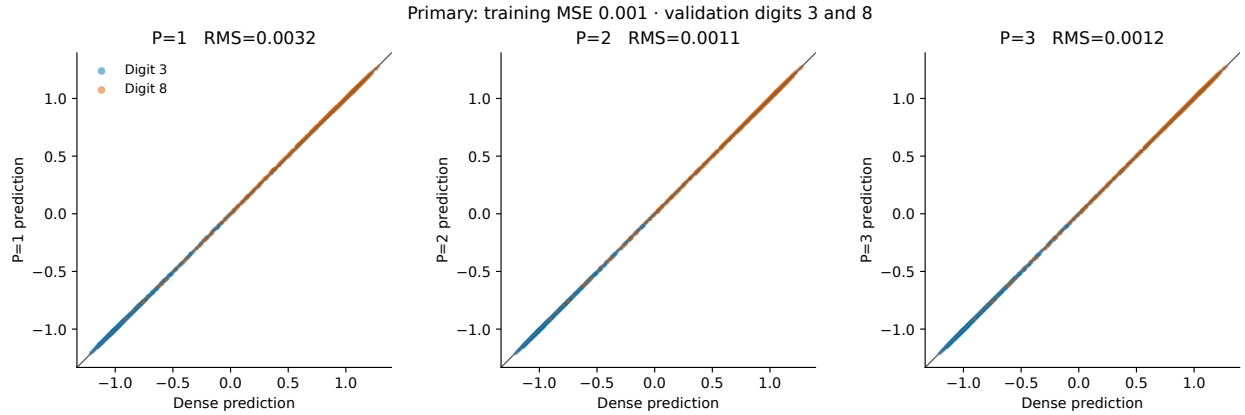


Figure 33: Held-out MNIST predictions for $q = 1, 2, 3$ against dense predictions at the primary training-MSE 10^{-3} endpoints. This scatter layout uses the same endpoint data, not another replicate.

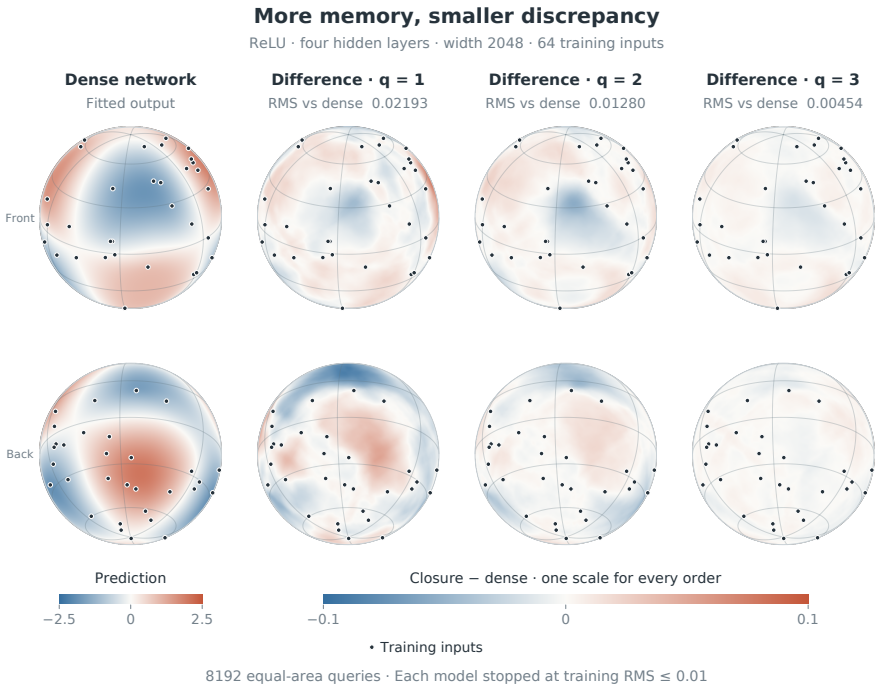


Figure 34: Sphere illustration with four ReLU hidden layers, $n = 2048$, $m = 64$, and 8192 equal-area queries. Both hemispheres and a shared error scale are shown. Each model stops at training RMS 0.01; the single-seed Euler run uses step 1/128. ReLU is outside the analytic activation theorem.

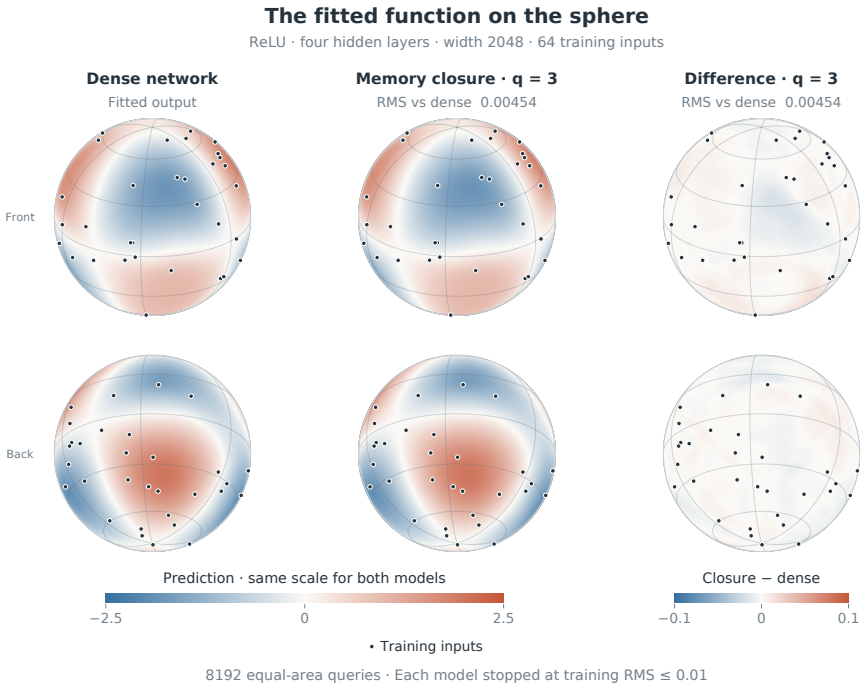


Figure 35: Dense output, order-3 memory output, and their signed difference for the same sphere run. Black points are training locations. This fitted-endpoint illustration has no continuous-flow refinement certificate.

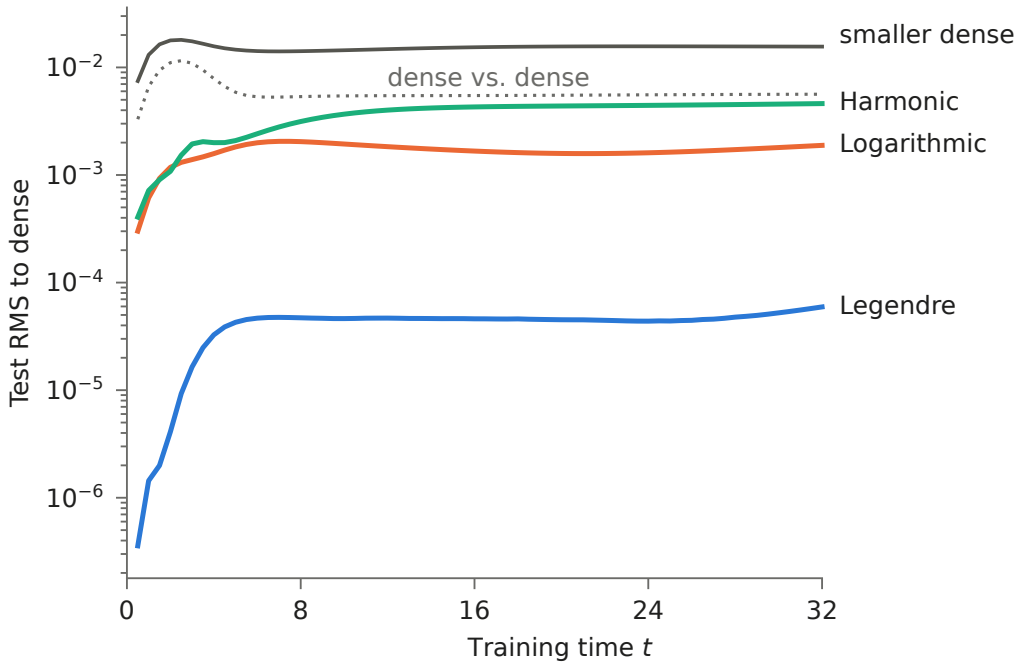


Figure 36: Saved sphere-3 query RMS from a width-4096 dense reference over recorded times, with eight training and 30 test inputs. The export's "Logarithmic" family is called Taylor by its adjacent plotting script, not the seeded decoder below. Dense-control curves aggregate three small-network seeds and two dense pairs by geometric means.

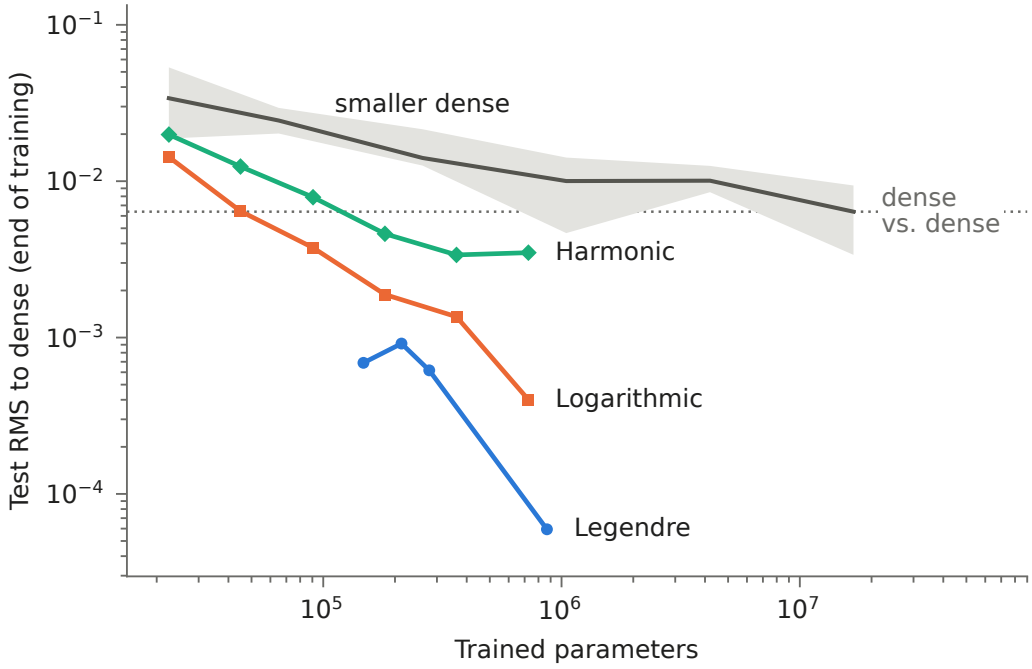


Figure 37: Sphere-3 fitted-endpoint RMS versus moving parameter count. Smaller-dense summaries are medians and observed ranges over three seeds. "Logarithmic" retains the same historical Taylor-family label; fixed storage and offline setup are additional.

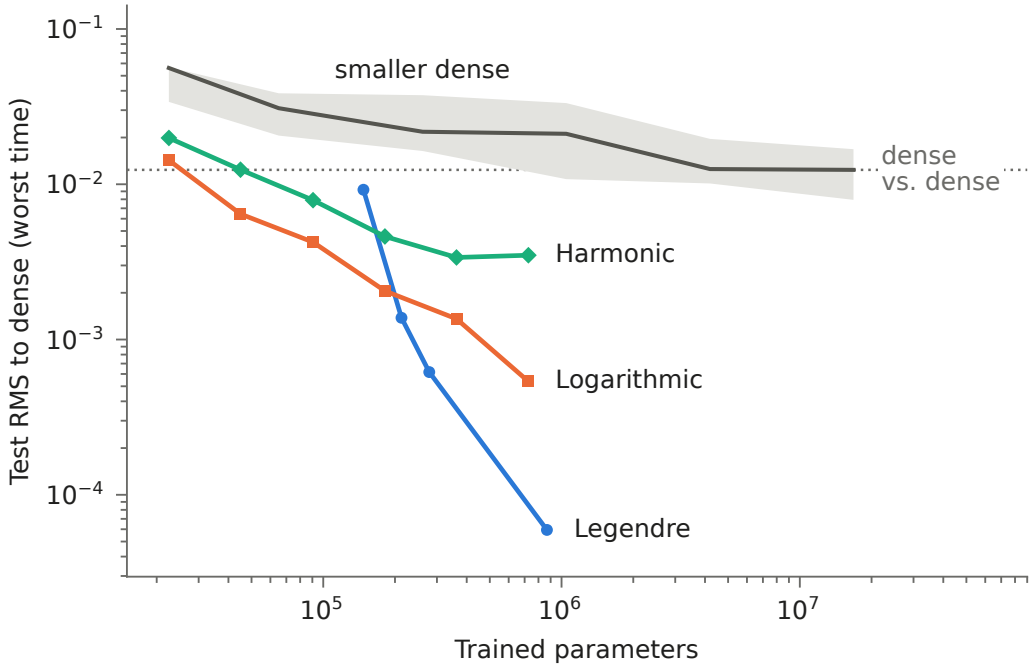


Figure 38: Companion sphere-3 tradeoff for maximum RMS over recorded times, rather than endpoint RMS. The shaded small-network range is not a confidence interval; the horizontal axis still excludes fixed storage and source-construction costs.

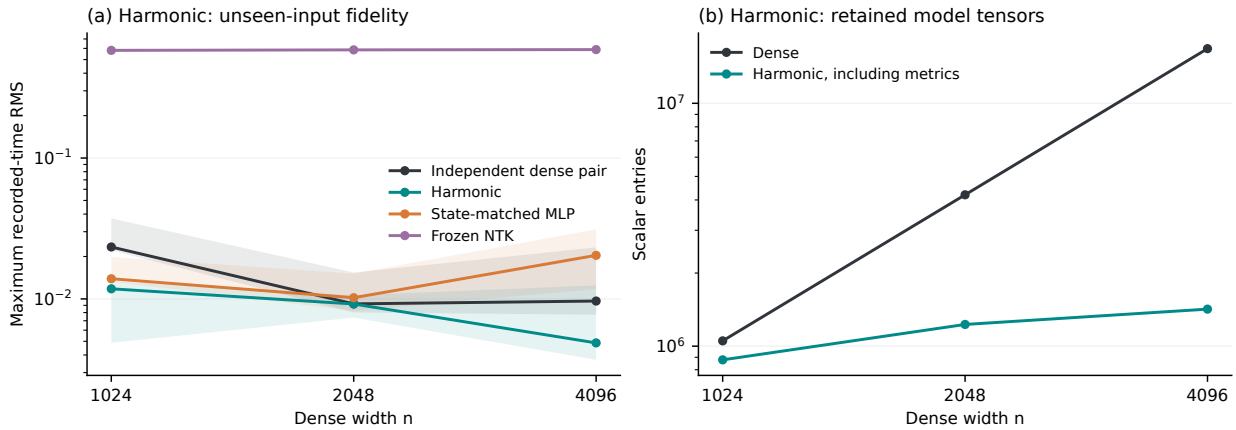


Figure 39: Earlier Harmonic circle confirmation at widths 1024, 2048, and 4096, with three held-out seeds per width. Curves show maximum-recorded-time query RMS and retained tensor scalars; shading is the observed seed range. Source setup and query workspace are additional.

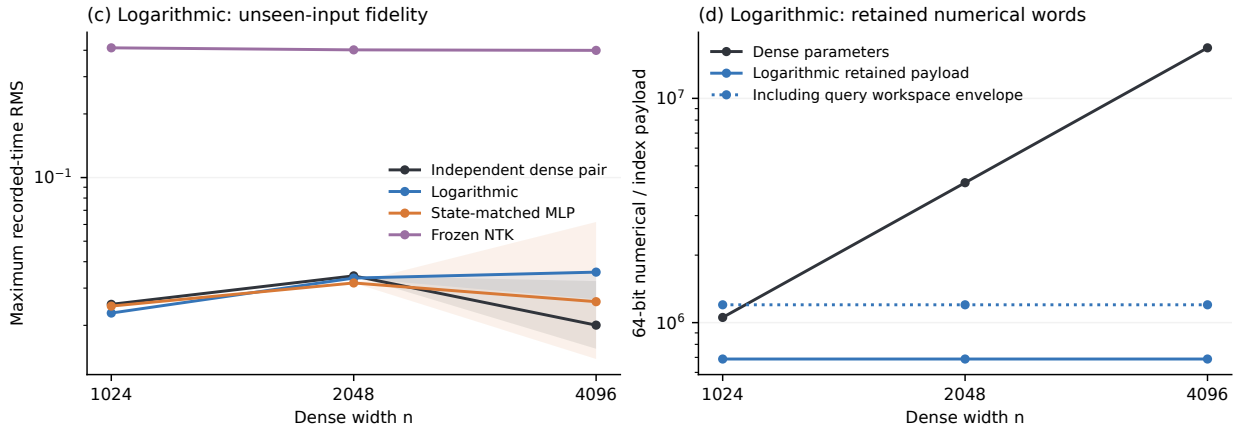


Figure 40: Earlier dimension-five *seeded decoder*, labelled "Logarithmic" in this artwork: finite-program query fidelity and persistent numerical payload across three widths. This is not Taylor. The dotted storage curve adds query workspace; flat payload in these tests is not an asymptotic scaling estimate.

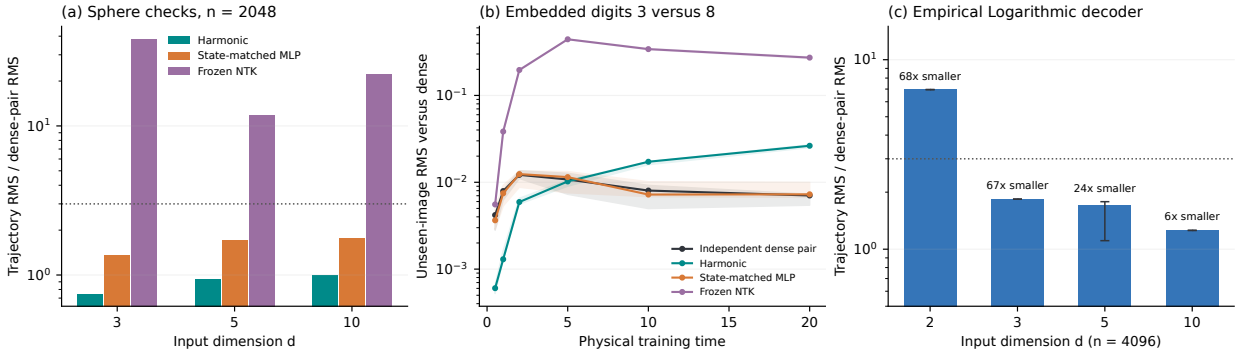


Figure 41: Further Harmonic dimension and PCA-based digits checks, followed by seeded-decoder dimension checks. The decoder's circle failure remains visible. Seed counts vary; ranges and the empirical factor-three threshold are not confidence statements. The digits task differs from the MNIST100 figures above.

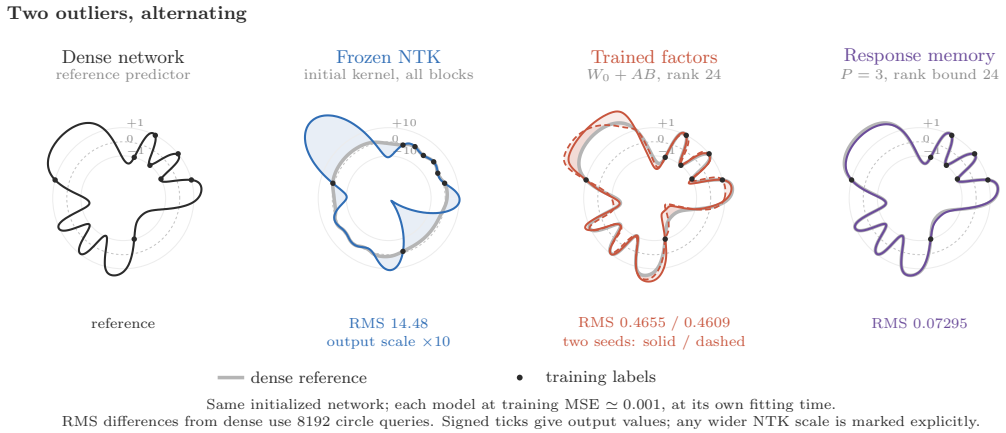


Figure 42: Remaining circle control: alternating labels with two outliers, under the earlier width-2048 tanh endpoint protocol. The NTK output scale is widened tenfold; RMS measures fitted-function disagreement on 8192 queries.

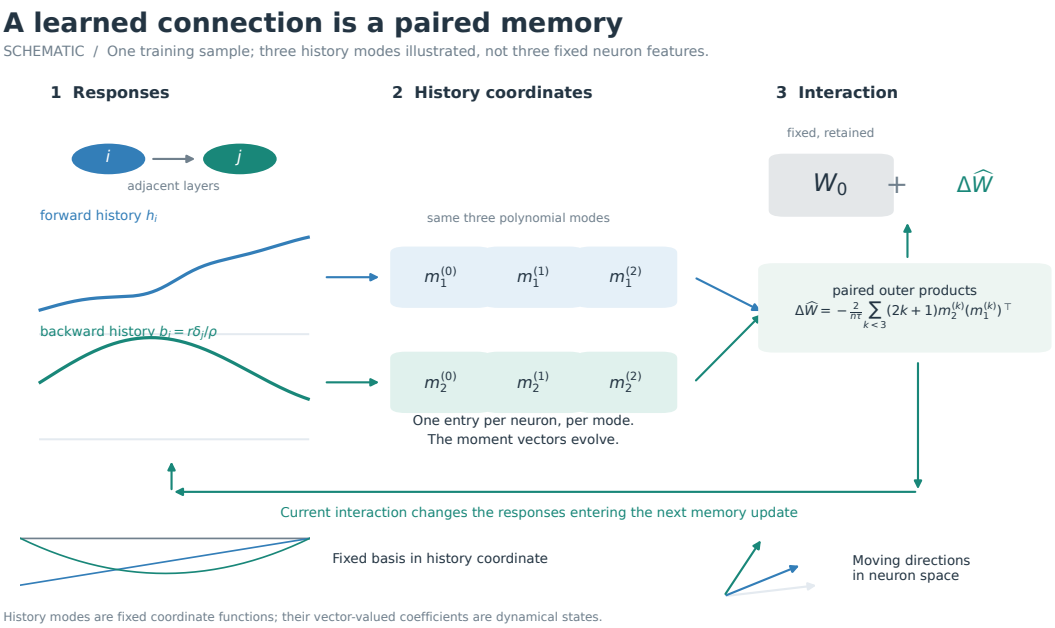


Figure 43: Original illustrated gallery, page 1: the three-mode memory mechanism. The next six plates preserve the remaining pages, including their explanatory headings; they reuse earlier illustrations rather than add experiments.

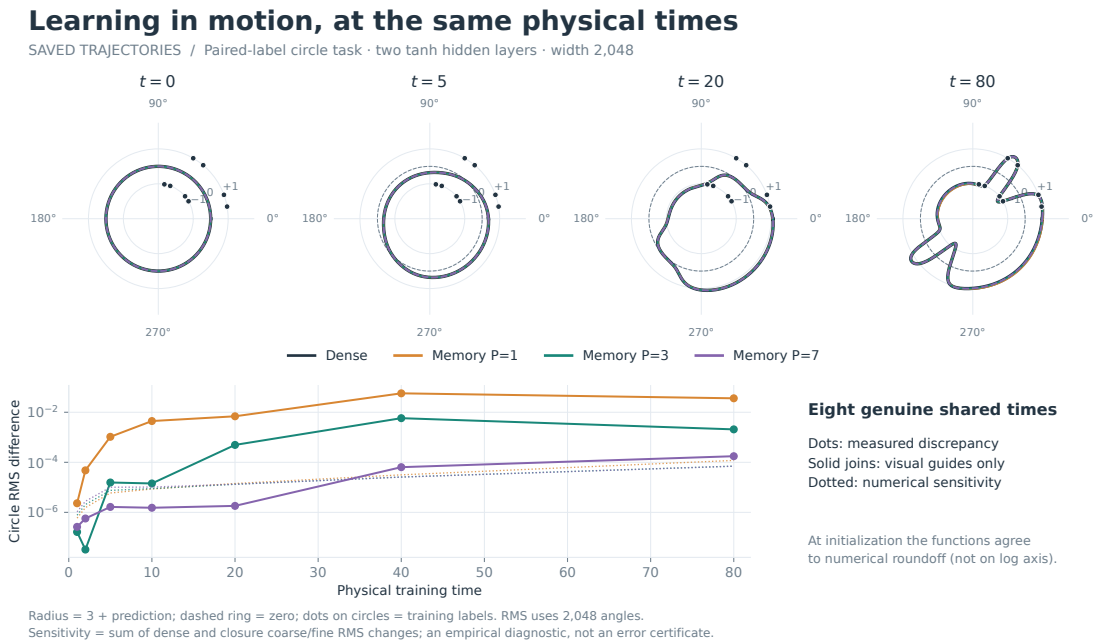


Figure 44: Gallery page 2: the earlier eight-checkpoint common-time circle comparison, not the later 39-checkpoint replay. Its numerical sensitivities are diagnostics between saved runs.

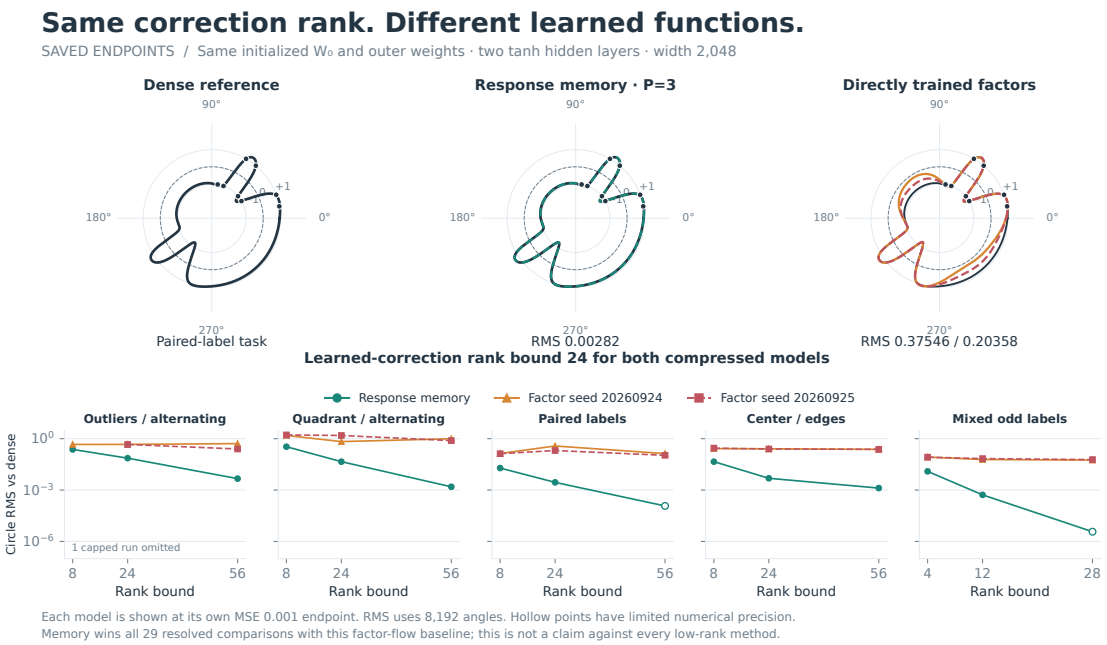


Figure 45: Gallery page 3: same-correction-rank factor controls at separately fitted endpoints. Rank equality is not an equal-total-storage claim.

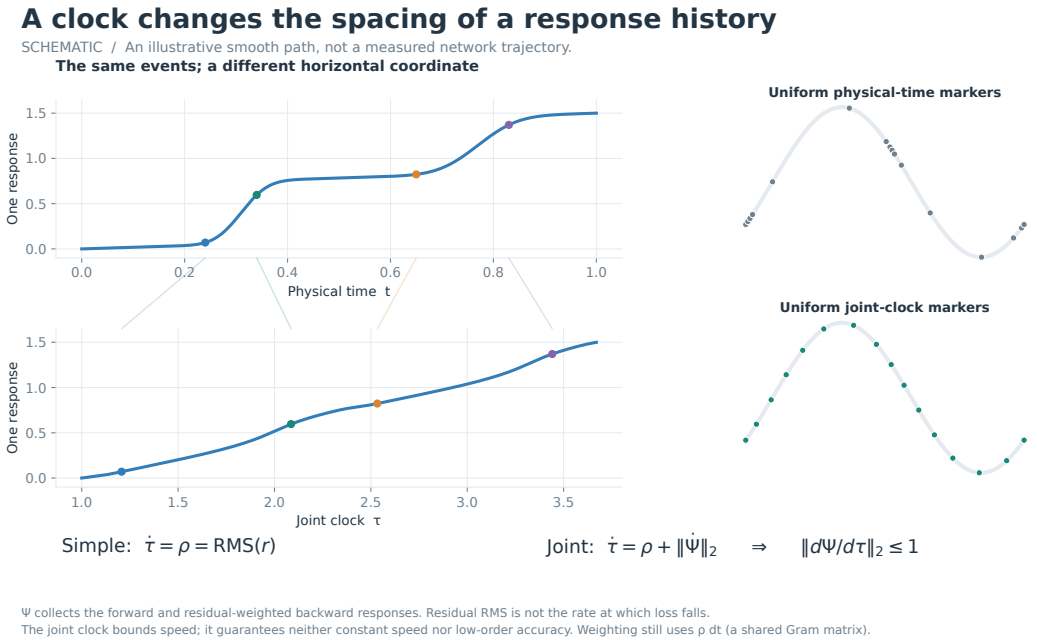


Figure 46: Gallery page 4: the constructed smooth-path clock illustration. The change of coordinate preserves events, not necessarily their speed.

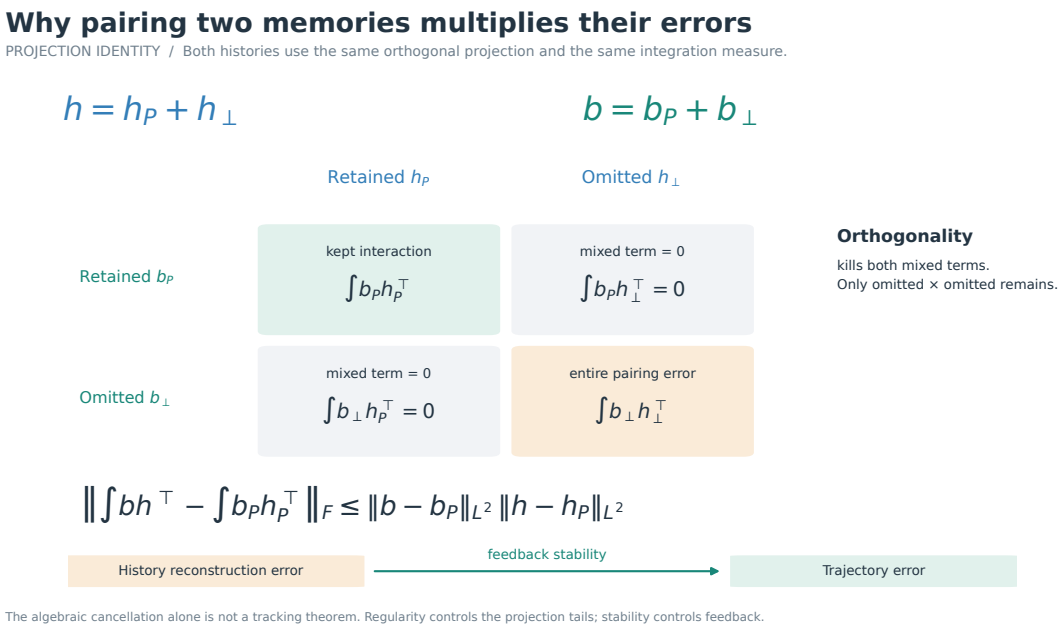


Figure 47: Gallery page 5: orthogonal pairing of two histories leaves a product of omitted tails. Feedback stability is a separate requirement for controlling the evolving network.

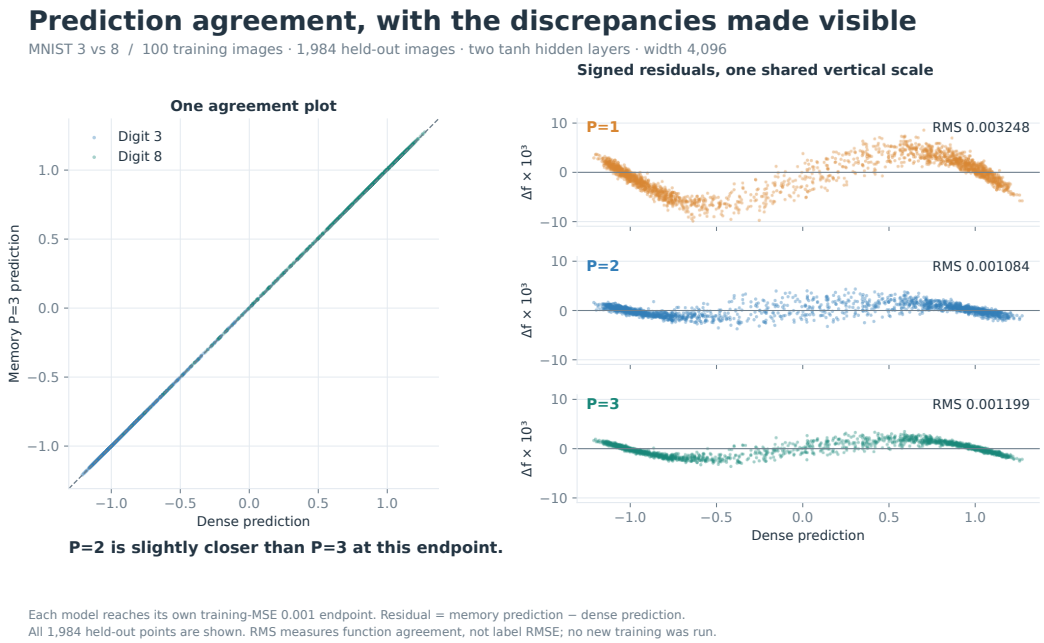


Figure 48: Gallery page 6: the earlier MNIST100 fitted-prediction comparison and signed differences. RMS measures disagreement with dense predictions, not error against labels.

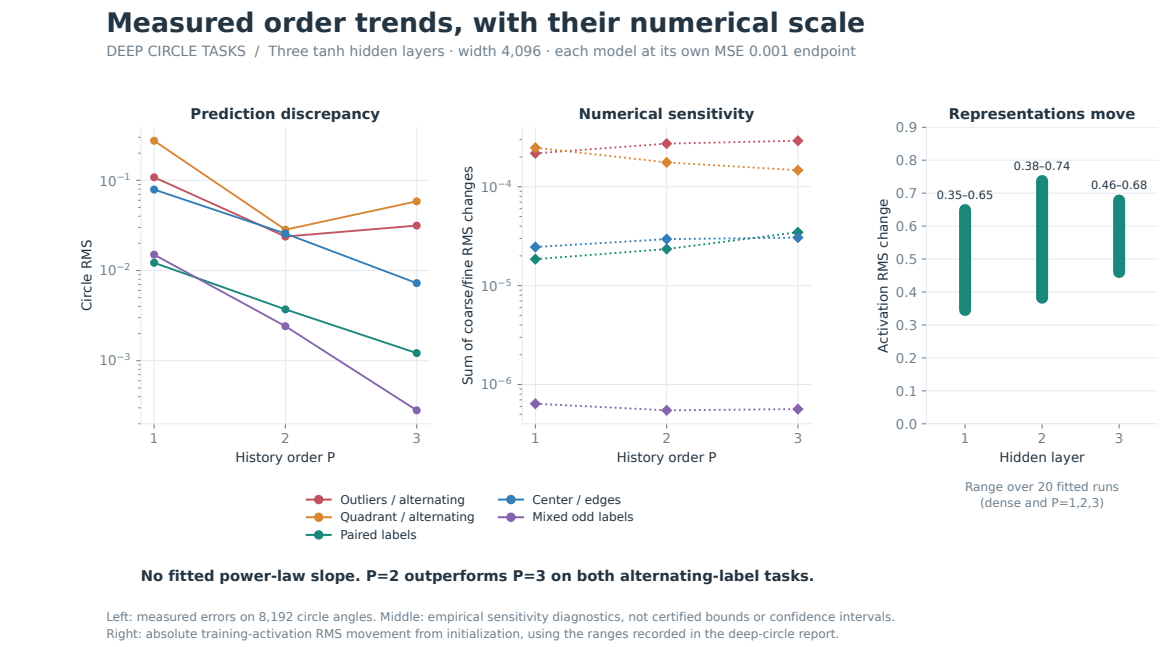


Figure 49: Gallery page 7: deep-circle order sensitivity and hidden-feature motion. Refinement scales are empirical diagnostics, and the observed order trend need not be monotone.

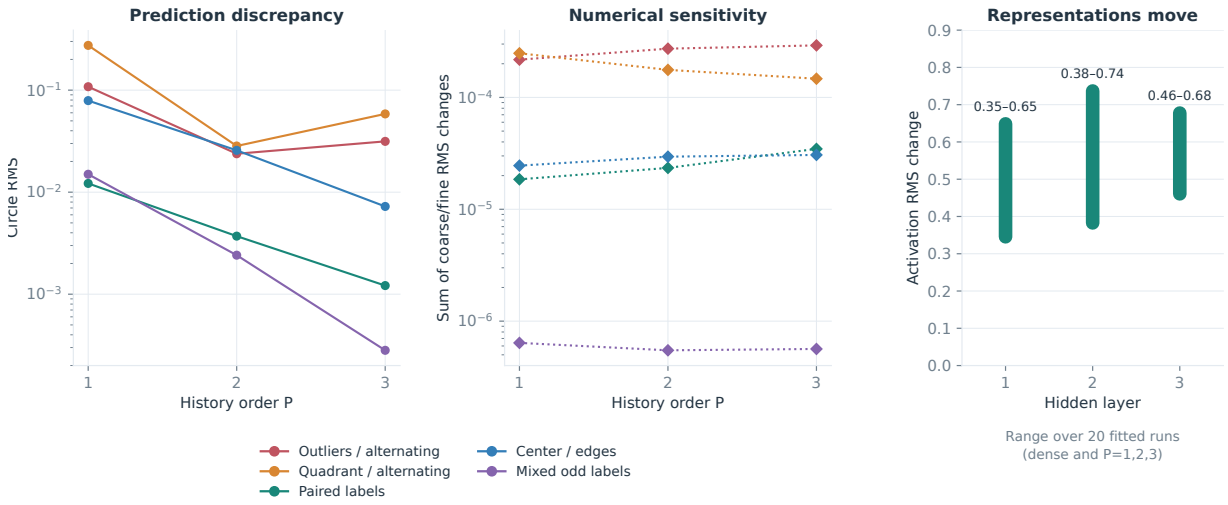


Figure 50: Standalone view of the deep-circle order, numerical-sensitivity, and feature-motion comparison. Each model uses its own training-MSE 10^{-3} endpoint; order 2 is closer than order 3 on the two alternating-label tasks, so the observed order trend is not uniformly monotone.